Class 9

Mathematics

(Solutions)

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1. Real Numbers

Solutions

LEVEL – I

1. Correct option: (B)

Step 1: Write the prime factorization of both numbers.

$$a = (2^2 \times 3^3 \times 5^4)$$

$$b = (2^3 \times 3^2 \times 5)$$

Step 2: Identify the common prime factors and their minimum powers in both numbers.

Common prime factors: 2, 3, and 5

Minimum power of 2 in a and b = $2^2 = 4$

Minimum power of 3 in a and b = $3^2 = 9$

Minimum power of 5 in a and b = $5 = 5$

Step 3: Multiply the common prime factors with their minimum powers to find the HCF.

$$HCF(a, b) = (2^2 \times 3^2 \times 5)$$

$$HCF(a, b) = 4 \times 9 \times 5 = 180$$

So, the correct answer is (b) 180.

2. Correct option: (C)

Step 1: Write the prime factorization of given numbers.

$$a = (2^3 \times 3^2 \times 5)$$

$$b = (2^2 \times 3^3 \times 5^2)$$

$$c = (2^4 \times 3 \times 5^3 \times 7)$$

Step 2: Identify the common prime factors and their minimum powers in both numbers.

Common prime factors: 2, 3, and 5

Minimum power of 2 in a, b and c = $2^2 = 4$

Minimum power of 3 in a, b and c = $3 = 3$

Minimum power of 5 in a, b and c = $5 = 5$

Step 3: Multiply the common prime factors with their minimum powers to find the HCF.

$$HCF(a, b, c) = 4 \times 3 \times 5 = 60$$

So, the correct answer is (c) 60.

3. Correct option: (D)

To find the LCM (Least Common Multiple) of two numbers, we need to find the highest power of each prime factor that appears in either number and multiply them together.

Let's find the LCM of the given numbers: $(2^3 \times 3 \times 5)$, and $(2^4 \times 5 \times 7)$,

Now, we'll find the highest power of each prime factor that appears in either number:

Prime factor 2: The highest power is 2^4 (from the second number).

Prime factor 3: The highest power is 3 (from the first number).

Prime factor 5: The highest power is 5 (common to both numbers).

Prime factor 7: The highest power is 7 (from the second number).

Now, we multiply these highest powers together to get the LCM:

LCM = $2^4 \times 3 \times 5 \times 7$

LCM = $16 \times 3 \times 5 \times 7$

LCM = 1680

So, the LCM of $(2^3 \times 3 \times 5)$ and $(2^4 \times 5 \times 7)$, is 1680.

4. Correct option: (C)

Given information:

HCF (Highest Common Factor) = 27

LCM (Least Common Multiple) = 162

One of the numbers = 54

We want to find the other number, denoted as "x."

Step 1: Use the relationship between HCF and LCM

HCF × LCM = Product of the two numbers

Step 2: Substitute the given values into the formula

Let's use the information provided:

$27 \times 162 = 54 \times x$

Step 3: Solve for "x"

Now, let's solve for "x":

$x = \frac{(27 \times 162)}{54}$

$x = 9$

Therefore, the other number is 9.

5. Correct option: (D)

To find the simplest form of the fraction $\frac{1095}{1168}$, we need to find the greatest common divisor (GCD) of the numerator and denominator and then divide both the numerator and denominator by the GCD.

Step 1: Find the GCD of 1095 and 1168.

Using a common method to find the GCD, we have:

$$1095 = 3 \times 5 \times 73$$
$$1168 = 2 \times 2 \times 2 \times 2 \times 73$$

The common factor between them is only 73.

Step 2: Divide both the numerator and denominator by the GCD (73).

$$1095 \div 73 = 15$$
$$1168 \div 73 = 16$$

Therefore, the simplest form of the fraction $\frac{1095}{1168}$ is $\frac{15}{16}$.

6. Correct option: (D)

When a number (denoted by "x") is divided by 143, it leaves 31 as the remainder. This can be expressed as: $x = 143 * q + 31$

where "q" is the quotient obtained when dividing x by 143.

Let's divide both sides of the equation $x = 143 * q + 31$ by 13:

$$\frac{x}{13} = \frac{143 * q + 31}{13}$$

The left side represents the division of x by 13, and the right side can be rewritten as:

$$\frac{x}{13} = 11q + 2 + \frac{5}{13}$$

So, the remainder obtained when dividing "x" by 13 is the remaining fraction $\frac{5}{13}$.

The remainder when the number is divided by 13 is 5.

7. Correct option: (D)

An irrational number is a number that cannot be expressed as a fraction of two integers and has non-repeating, non-terminating decimal expansions.

Let's analyze the given options:

a) $\frac{22}{7}$ - This is a rational number since it can be expressed as a fraction.

b) 3.1416 - This is a rational number because it is a finite decimal.

c) $3.\overline{1416}$ - This represents a repeating decimal, but it is still rational because it can be expressed as a fraction.

d) 3.141141114... - This is an irrational number. It represents an infinite non-repeating decimal, as there is no pattern that repeats. It cannot be expressed as a fraction.

So, the irrational number among the given options is:

d) 3.141141114....

8. Correct option: (C)

A rational number can be expressed as a terminating decimal if and only if its denominator is a power of 10. In other words, the denominator should have only 2's and 5's in its prime factorization.

Let's analyze the given options:

a) $\frac{124}{165}$ - The denominator (165) is not a power of 10, so this is not a terminating decimal.

b) $\frac{131}{30}$ - The denominator (30) is not a power of 10, so this is not a terminating decimal.

c) $\frac{2027}{625}$ - The denominator (625) is a power of 10 ($625 = 5^4$), so this is a terminating decimal.

d) $\frac{1625}{462}$ - The denominator (462) is not a power of 10, so this is not a terminating decimal.

So, the rational number that is expressible as a terminating decimal among the given options is: c) $\frac{2027}{625}$

9. Correct option: (B)

To determine if the decimal expansion of the rational number $\frac{37}{2^2 \times 5}$ will terminate, we need to analyze its denominator.

The denominator is $2^2 \times 5$, which simplifies to 4×5, equaling 20. Now, we observe the highest power of 2 and 5 in the denominator.

Highest power of 2 in the denominator: $2^2 = 4$

Highest power of 5 in the denominator: $5 = 5$

Since both the highest powers of 2 and 5 are finite (4 and 5, respectively), the decimal expansion will terminate after a number of decimal places equal to the highest power among them, which is 2 (from 2^2).

Therefore, the decimal expansion of $\frac{37}{2^2 \times 5}$ will terminate after two decimal places, and the value will be 1.85.

10. Correct option: (C)

The number $\frac{1}{\sqrt{2}}$ is:

c) An irrational number

The square root of 2 is an irrational number, and when we take the reciprocal of an irrational number $\frac{1}{\sqrt{2}}$ the result is also an irrational number. Rational numbers are those that can be expressed as a fraction of two integers, but $1/\sqrt{2}$ cannot be expressed in such a form. Therefore, it is an irrational number.

11. Correct option: (D)

Let's analyze the given expression:

$(-1)^n = -1$ (if n is odd)

$(-1)^n = 1$ (if n is even)

This pattern continues up to $(-1)^{2009}$, which is -1 (odd power), and finally, $(-1)^{2010}$, which is 1 (even power).

Now, let's group the terms in pairs:

$$(-1) + 1 + (-1) + 1 + \ldots + (-1) + 1$$

Each pair consists of one odd power (-1) and one even power (1).

The sum of each pair is 0.

Since there are 5 pairs in total (from 2001 to 2010), the overall sum of the expression is:

$$0 + 0 + 0 + 0 + 0 = 0$$

Therefore, the correct answer is: d) 0

12. Correct option: (B)

$$(\frac{-2}{3})^3 = \frac{-2^3}{3^3} = -\frac{8}{27}$$
$$(\frac{-4}{5})^2 = \frac{(-4)^2}{5^2} = \frac{16}{25}$$

Now, find the product of these two results:

$$\left(-\frac{8}{27}\right) * \left(\frac{16}{25}\right) = \frac{-8 * 16}{27 * 25} = \frac{-128}{675}$$

Therefore, the correct answer is: b) $\frac{-128}{675}$

13. Correct option: (C)

Let's simplify the expression $\left[(3^2 - 2^2) \div (\frac{1}{5})^2\right]$:

$$\left[(3^2 - 2^2) \div (\frac{1}{5})^2\right] = [(9 - 4) \div (\frac{1}{25})] = [5 \div (1/25)].$$

Now, to divide by a fraction, we multiply by its reciprocal:

$$[5 \div (\frac{1}{25})] = 5 * (\frac{25}{1}) = 5 * 25 = 125.$$

Therefore, the simplified value of the expression is:

$$\left[(3^2 - 2^2) \div (\frac{1}{5})^2\right] = 125.$$

14. Correct option: (B)

The exponential form of 3125 is: b) 5^5

3125 is equal to 5 raised to the power of 5 (5^5), which is 5 × 5 × 5 × 5 × 5 = 3125.

15. Correct option: (C)

Let's solve the given equation for the value of n: $5^{n-2} \times 3^{2n-3} = 135$

First, we can factor 135 as 3^3 × 5:

$$5^{n-2} \times 3^{2n-3} = 3^3 \times 5^1$$

Now, we can equate the powers of 3 and 5 on both sides:

For 3:

$$2n - 3 = 3$$
$$n = \frac{3 + 3}{2} = 3$$

For 5:

$$n - 2 = 1$$
$$n = 3$$

So, the correct value of n = 3

16. Correct option: (D)

To simplify the expression, let's factor out the common term of 5^{n+1} from the numerator and denominator:

$$\frac{10 \times 5^{n+1} + 25 \times 5^n}{3 \times 5^{n+2} + 10 \times 5^{n+1}} =$$

Now, cancel out the common term 5^{n+1}:

$$= \left[\frac{5^{n+1} \times \left(10 + \frac{25}{5}\right)}{5^{n+1} \times (3 \times 5 + 10)}\right] => \frac{15}{25} = \frac{3}{5}$$

So correct answer is option (d)

17. Correct option: (B)

To find the value of $\log_{10} 8$ we can use the property of logarithms that states:

$$\log_b(a^c) = c \times \log_b a$$

Since 8 can be written as 2^3, we can rewrite $\log_{10} 8$ as:

$$\log_{10} 8 = \log_{10} 2^3 = 3 \times \log_{10} 2$$

Given that $\log_{10} 2$ is approximately 0.3010 (as given in the question):

$$\log_{10} 8 = 3 * 0.3010$$
$$\log_{10} 8 = 0.9030$$

So, $\log_{10} 8$ is approximately 0.9030.

18. Correct option: (C)

To find the value of $\log_{30} 32$ in terms of a and b, we can use the properties of logarithms.
We know that $\log_{30} 3 = a$ and $\log_{30} 5 = b$.
Step 1: Express 32 as a power of 2:

$$\log_{30} 32 = \log_{30} 2^5$$
$$\log_{30} 32 = 5 \log_{30} 2$$

Step 2: Now, we need to express $\log_{30} 2$ in terms of a and b.

$$\log_{30} 2 = \log_{30} 30 - \log_{30} 15 = \log_{30} 30 - \log_{30} 3 - \log_{30} 5$$
$$\log_{30} 2 = 1 - a - b$$

Step 3: Finally, substitute $\log_{30} 2$ into the expression:

$$\log_{30} 32 = 5 \times \log_{30} 2 = 5(1 - a - b)$$

So, the correct option is: c) 5(1-a-b)

19. Correct option: (C)

To solve for x in the given equation

$$\log_{\sqrt{2}}[\log_2\{\log_4(x-15)\}] = 0$$
$$\log_2\{\log_4(x-15)\} = \sqrt{2}^0 = 1$$
$$\log_2\{\log_4(x-15)\} = 1$$

Rewrite the equation in exponential form:

$$\log_4(x-15) = 2^1$$

Solving further

$$(x-15) = 4^2$$
$$(x-15) = 16 => x = 31$$

Therefore, the value of x is 31.

20. Correct option: (C)

To solve for $\frac{1}{a} - \frac{1}{b}$
Given: $(3.7)^a = (0.37)^b = 1000$
Step 1: Since$(3.7)^a = 1000$, we can express a in terms of $\log_{3.7} 1000$:

$$a = \log_{3.7} 1000$$

Step 2: Now, since $(0.37)^b = 1000$, we can express b in terms of $\log_{0.37} 1000$:

$$b = \log_{0.37} 1000$$

Step 3: Calculate the values of a and b using a calculator:

$$a = \log_{3.7} 1000$$
$$b = \log_{0.37} 1000$$

Step 4: Now, find $\frac{1}{a}$ and $\frac{1}{b}$:

$$\frac{1}{a} = \frac{1}{\log_{3.7} 1000} = \log_{1000} 3.7$$
$$\frac{1}{b} = \frac{1}{\log_{0.37} 1000} = \log_{1000} 0.37$$

Finally, calculate $\frac{1}{a} - \frac{1}{b}$:

$$\frac{1}{a} - \frac{1}{b} = \log_{1000} 3.7 - \log_{1000} 0.37 = \log_{1000}\left(\frac{3.7}{0.37}\right) = \log_{1000} 10 = \frac{1}{3}\log_{10} 10 = \frac{1}{3}$$

Therefore, the value of $\frac{1}{a} - \frac{1}{b}$ is $1/3$.

LEVEL – II

1. Correct option: (B)

Step 1: Find the first multiple of 7 less than or equal to 150. We can do this by dividing 150 by 7 and taking the floor value:

$$150 \div 7 = 21.428\ldots$$

The largest multiple of 7 that is less than or equal to 150 is $21 * 7 = 147$.

Step 2: Now, we count the number of multiples of 7 from 7 to 147 (inclusive):

$$7, 14, 21, \ldots, 147$$

Step 3: To find the count

$$147 \div 7 = 21$$

So, there are 21 natural numbers up to 150 that are divisible by 7.

2. Correct option: (A)

Number of numbers between 11 and 111 that are multiples of both 2 and 5:
LCM(2, 5) = 2 * 5 = 10
Count = (110 ÷ 10) - (10 ÷ 10) = 11 - 1 = 10
Answer: 10

3. Correct option: (B)

Step 1: Find the unit digits of the first 50 odd natural numbers:

$$1, 3, 5, 7, 9, 11, 13, 15, \ldots, 95, 97, 99$$

The unit digit repeats in a cycle of 5: $1, 3, 5, 7, 9$.

Step 2: We need to find the unit digit of the product of these odd numbers:

Product = $1 * 3 * 5 * 7 * 9 * 11 * 13 * \ldots * 95 * 97 * 99$

Step 3: Since we are only interested in the unit digit of the product, we can focus on the unit digits of these numbers:
Unit digits = $1, 3, 5, 7, 9, 1, 3, \ldots, 5, 7, 9$
Step 4: Notice that we have 10 multiples of 5 (5, 15, 25, ..., 95) in the product, and each multiple of 5 contributes a unit digit of 5.
Step 5: All the other unit digits from 1 to 9 will multiply together, but their product will not change the unit digit as we have 5 in this.
Step 6: So, the unit digit in the product will be determined by the product of the 10 fives (5 * 5 * 5 * 5 * 5 * 5 * 5 * 5 * 5 * 5) which unit place is also 5:
Unit digit = 5
So , the unit digit in the product of the first 50 odd natural numbers is 5.

4. Correct option: (D)

Step 1: Let the three numbers be x, 2x, and 3x (since they are in the ratio 1:2:3).
Step 2: The LCM of three numbers is given as 12.
Step 3: Since the LCM of the numbers is 12, we can write the following equation:

$$LCM(x, 2x, 3x) = 12$$

Step 4: Let's find the LCM of these number

$$LCM(x, 2x, 3x) = 2 \times 3 \times x$$

Step 5: Now, equate the LCM

$$LCM(x, 2x, 3x) = 2 \times 3 \times x = 12$$

$$x = \frac{12}{6} = 2$$

Step 6: Since the value of the x is 2, we can conclude that the numbers are 2, 4, and 6
Therefore, the correct option is:
d) 2, 4, 6

5. Correct option: (B)

Solution: Step 1: Use the identity:

Product of two numbers = LCM×HCF

Let the two numbers be a and b, and assume a>b

So: $ab = 105 \times 1 = 105$

$a - b = 8$

Step 2: Use substitution

Let: $a = b+8$

Substitute into the product equation:

$(b + 8).b = 105 \Rightarrow b^2 + 8b = 105 \Rightarrow b^2 + 8b - 105 = 0$

Step 3: Solve the quadratic

$b^2 + 8b - 105 = 0$

Factor the quadratic:

$(b+15)(b-7)=0 \Rightarrow b=-15$ or $b=7$

We discard $b=-15$ $b = -15 b=-15$ (since number must be positive).

So: $b = 7$

$a = b + 8 = 15$

Step 4: Find the sum

$a+b=15+7=22$

6. Correct option: (C)

Step 1: Let's denote:

Let: $a = x^{\frac{1}{3}}$, $b = y^{\frac{1}{3}}$, $c = z^{\frac{1}{3}}$

Then: $a + b + c = 0$

Now cube both sides to help relate x,y,z and a,b,c.

Step 2: Use identity:

$(a + b + c)^3 = a^3 + b^3 + c^3 + 3(a + b)(b + c)(c + a)$

But since:

$a + b + c = 0 \Rightarrow (a + b + c)^3 = 0 \Rightarrow a^3 + b^3 + c^3 = 3abc$

So,

$x + y + z = a^3 + b^3 + c^3 = 3abc = 3(x^{1/3})(y^{1/3})(z^{1/3})$

$$x + y + z = 3(xyz)^{1/3}$$

Step 3: Plug into the expression

$$\log\left(\frac{x+y+z}{3}\right) = \log\left(\frac{3(xyz^{1/3})}{3}\right) = \log(xyz)^{1/3} = \frac{1}{3}\log(xyz)$$
$$= \frac{1}{3}(\log x + \log y + \log z)$$

7. Correct option: (B)

Let's use logarithm properties and algebraic manipulation to simplify the given equation:

Given equation: $\log_e(x-1) + \log_e(x) + \log_e(x+1) = 0$

Step 1: Use the logarithm property $\log_a x + \log_a y = \log_a xy$

$$\log_e((x-1)x(x+1))$$

Step 2: The logarithm equation can be rewritten as an exponential equation:

$$e^0 = x(x-1)(x+1)$$

Step 3: Since $e^0 = 1$, we have:

$$1 = x(x-1)(x+1)$$

Step 4: Expand the equation:

$$1 = x(x^2 - 1)$$
$$1 = (x^3 - x)$$
$$0 = x^3 - x - 1$$

Therefore, it match with option (b)

8. Correct options: (A, B, C)

To find which rational numbers lie between $-\frac{1}{2}$ and $\frac{1}{2}$, we need to compare the given options with these bounds.

Given range: $-\frac{1}{2} < x < \frac{1}{2}$

Let's check the options one by one:

a) $\frac{3}{10}$: $\frac{3}{10}$ is greater than $-\frac{1}{2}$ and less than $\frac{1}{2}$. So, it lies between $-\frac{1}{2}$ and $\frac{1}{2}$.

b) 0: 0 is greater than $-\frac{1}{2}$ and less than $\frac{1}{2}$. So, it lies between $-\frac{1}{2}$ and $\frac{1}{2}$.

c) $\frac{21}{100}$: $\frac{21}{100}$ is greater than $-\frac{1}{2}$ and less than $\frac{1}{2}$. So, it lies between $-\frac{1}{2}$ and $\frac{1}{2}$.

d) $\frac{2}{3}$: $\frac{2}{3}$ is greater than $\frac{1}{2}$, which is outside the given range. So, it does not lie between $-\frac{1}{2}$ and $\frac{1}{2}$.

Therefore, the rational numbers that lie between $-\frac{1}{2}$ and $\frac{1}{2}$ are:

$$a)\frac{3}{10}$$
$$b)\,0$$
$$c)\frac{21}{100}$$

9. Correct options: (A, B)

To determine which of the given fractions represent terminating decimals, we need to look at their denominators. A fraction represents a terminating decimal if and only if its denominator contains only the prime factors 2 or 5 or combination of both .

Let's analyze each option:

a) $\frac{5}{32}$: The denominator, 32, can be written as 2^5. It contains only the prime factors 2 , so 5/32 represents a terminating decimal.

b) $\frac{27}{80}$: The denominator, 80, can be written as $2^4 \times 5$. It contains both the prime factors 2 and 5, so $\frac{27}{80}$ represents a terminating decimal.

c) 11/24: The denominator, 24, can be written as $2^3 \times 3$. It contains the prime factor 3, so $\frac{11}{24}$ does not represent a terminating decimal.
d) 5/12: The denominator, 12, can be written as $2^2 \times 3$. It contains the prime factor 3, so $\frac{5}{12}$ does not represent a terminating decimal.

The correct option is a), b).

10. Correct options: (A, D)

Let's convert each repeating decimal to a fraction:

$$0.\overline{2} = 0.222\ldots$$
$$0.\overline{3} = 0.333\ldots$$
$$0.\overline{4} = 0.444\ldots$$
$$0.\overline{9} = 0.999\ldots$$
$$0.\overline{39} = 0.393939\ldots$$

Step 1: Write each repeating decimal as a fraction:

$$0.\overline{2} = \frac{2}{9}$$
$$0.\overline{3} = \frac{3}{9}$$
$$0.\overline{4} = \frac{4}{9}$$
$$0.\overline{9} = \frac{9}{9}$$
$$0.\overline{39} = \frac{39}{99}$$

Step 2: Now, add all the fractions together:

$$0.\overline{2} + 0.\overline{3} + 0.\overline{4} + 0.\overline{9} + 0.\overline{39} = \frac{2}{9} + \frac{1}{3} + \frac{4}{9} + 1 + \frac{39}{99}$$
$$\frac{22}{99} + \frac{33}{99} + \frac{44}{99} + \frac{99}{99} + \frac{39}{99} = 2 + \frac{39}{99} = \frac{237}{99}$$

Therefore, the value of $0.\overline{2} + 0.\overline{3} + 0.\overline{4} + 0.\overline{9} + 0.\overline{39}$ is: $\frac{237}{99}$

So $2\frac{13}{33}$ will also correct option as $2\frac{13}{33} = \frac{237}{99}$

11. Correct options: (A, B, C, D)

Let's simplify the equation $\log_a x = b \Rightarrow a^b = x$

a) If a and b are two irrational numbers, then x can be rational.
This statement is generally not correct. If both a and b are irrational numbers, it is unlikely that x will be rational. The logarithm of a rational number to an irrational base typically results in an irrational number. But in special case it can be rational
Example:
Let a = $\sqrt{10}$ (an irrational number) and b = $\log_{10} 4$(an irrational number).

$$x = \sqrt{10}^{\log_{10} 4}$$

Using log properties

$$q^{\log_q p} = p$$
$$x = 2$$

b) If a is rational and b is irrational numbers, then x can be rational.
This statement is not correct. When the base (a) is rational and the logarithm (b) is irrational, x will generally be irrational. But in special case it can be
Use a=10 in above equation
c) If a is irrational and b is rational numbers, then x can be rational.

This statement is correct. If the base (a) is irrational and the logarithm (b) is rational, x can be rational in some cases.
Example:
Let $a = \sqrt{2}$ (an irrational number) and $b = 2$ (a rational number).
Now, let's find x using the equation log_√2(x) = 2:

$$x = (\sqrt{2})^2 = 2$$

In this example, x is a rational number.
d) If both a and b are rational numbers, then x can be rational.
This statement is also correct. If both the base (a) and the logarithm (b) are rational numbers, then x can be rational.
Example:
Let a = 3 (a rational number) and b = 2 (a rational number).

$$x = 3^2 = 9$$

In this example, x is a rational number.

12. Correct option: (C, D)

Step 1: Simplify the inner expression

$$\left(\frac{-1}{5}\right)^{-2} = \left(\frac{5}{-1}\right)^2 = (-5)^2 = 25$$

Step 2: Now raise it to the power 2 again:

$$25^2 = 625$$

Step 3: Multiply with outer exponent (which is 1):

$$[25^2]^1 = 625$$

$$6^4 = 625$$

13. Correct option: (B)

Given the passage information:

$$(\frac{a}{b})^{-n} = (\frac{b}{a})^n \; or \; a^{-n} = \frac{1}{a^n}$$

Now, let's simplify the expression:

$$\frac{8^{-2} \times 5^3}{2^{-4}}$$

Step 1: Using the property $a^{-n} = \frac{1}{a^n}$:

$$8^{-2} = \frac{1}{8^2} = \frac{1}{64}$$

$$2^{-4} = \frac{1}{2^4} = \frac{1}{16}$$

Step 2: Simplify the expression further:

$$\frac{\left(\frac{1}{64} \times 5^3\right)}{\left(\frac{1}{16}\right)}$$

$$\frac{1}{64} \times 125 \times 16 = \frac{125}{4}$$

Therefore, the simplified expression is $\frac{125}{4}$.

14. Correct option: (C)

Given expression: $5^m \div 5^{-3} = 5^5$

Using the property, we can rewrite 5^{-3} as $\frac{1}{5^3}$

$$5^m \div \frac{1}{5^3} = 5^5$$

$$5^m \times 5^3 = 5^5$$

Using the property $a^m \times a^n = a^{m+n}$:

$$5^{m+3} = 5^5$$

Since the bases are the same (both are 5), the exponents must be equal:

$$m + 3 = 5$$

$$m = 2$$

Therefore, the value of m is 2.

15. Correct Matching is: a -Q, b – R, c – S, d - P

a) If $2^{a+1} \times 5^a = 200$, then a = ?

Step 1: Rewrite 200 as a product of its prime factors: $200 = 2^3 \times 5^2$.

Step 2: Now, equate the exponents of 2 and 5 on both sides of the equation:

$$a + 1 = 3 \quad a = 3 - 1$$

$$a = 2$$

So, a = 2.

a)--------→(Q)

b) If $\frac{2^{x-1} \times 4^{2x+1}}{8^{x-1}} = 64$, then x = ?

Step 1: Rewrite 64 as a power of 2: $64 = 2^6$.

Step 2: Simplify the equation using the properties of exponents and Combine the exponents:

$$2^{(x-1+4x+2-3(x-1))} = 2^6$$

Step 3: Since the bases (2 in this case) are the same, equate the exponents:

$$2^{(x-1+4x+2-3x+3)} = 2^6$$

$$2x + 4 = 6$$

$$x = 1$$

b)-------→(R)

R) If $\frac{2^1+2^0+2^{-1}}{2^{-2}+2^{-3}+2^{-4}}$ = ?

Step 1: Simplify both the numerator and denominator:

$$\frac{2^{-1}(2^2 + 2^1 + 1)}{2^{-4}(2^2 + 2^1 + 1)} = \frac{2^{-1}}{2^{-4}} = 2^3 = 8$$

So, the value of the expression is 8.

c)------→(S)

d) If $5^{-5} \times 5^{2n+1} = 5^{13} \div 5^5$, then n = ?

Step 1: Apply the rule of exponents and Simplify $5^{-5+2n+1} = \frac{5^{13}}{5^5} = 5^8$

Step 3: Since the bases are the same, equate the exponents:

$$2n - 4 = 8$$

$$2n = 8 + 4 = 12 => n = 6$$

So, n = 6.

d)------→(P)

16. Correct Answer is: 2

Given equation: $\log_2 \log_3(8 + \log_3 \log_2(x^2 + 4)) = 1$

Property used $\log_a b = c => b = a^c$

Eliminate the outermost logarithm

$$\log_3(8 + \log_3 \log_2(x^2 + 4)) = 2$$

Eliminate the next logarithm

$$8 + \log_3 \log_2(x^2 + 4) = 9$$
$$\log_3 \log_2(x^2 + 4) = 9 - 8 = 1$$
$$\log_2(x^2 + 4) = 3$$
$$(x^2 + 4) = 2^3 = 8$$
$$x^2 + 4 = 8 => x^2 = 4$$
$$x = \pm 2$$

So the positive value of $x = 2$

17. Correct Answer is: 2

Given equation: $2^{2x-1} = \frac{1}{8^{x-3}}$

$$2^{2x-1} = 2^{-3(x-3)}$$
$$2^{2x-1} = 2^{-3x+9}$$

Base value is same so exponent will be same

$$2x - 1 = -3x + 9$$
$$2x + 3x = 9 + 1$$
$$5x = 10$$
$$x = \frac{10}{5}$$
$$x = 2$$

So, the value of x is 2.

18. Correct Answer is:1

Given equation: $\frac{p}{q} = (\frac{4}{3})^2 \div (\frac{3}{4})^{-2}$

$$(\frac{4}{3})^2 = \frac{16}{9}$$
$$(\frac{3}{4})^{-2} = (\frac{4}{3})^2 = \frac{16}{9}$$
$$\frac{p}{q} = \left(\frac{16}{9}\right) \div \left(\frac{16}{9}\right) = 1$$

So, the value of $(\frac{p}{q})^{-10}$ is 1.

19. Correct Answer is: 81

Given expression: $\frac{3^{n+5} - 6.3^{n+2}}{3^{n-1} \times 7}$

$$\frac{3^{n+2}(3^3 - 6)}{3^{n-1} \times 7} = \frac{3^{n+2}(27 - 6)}{3^{n-1} \times 7} = \frac{3^{n+2}(21)}{3^{n-1} \times 7}$$
$$= \frac{3^{n+2} \times 3}{3^{n-1}} = \frac{3^{n-1}(3^3 \times 3)}{3^{n-1}} = 81$$

20. Correct Answer is: 3

Step 1-: $\frac{1}{9}$ can be written as 3^{-2}

Now, the equation becomes:

$$3^{x-5} = 3^{-2}$$

Since the bases are the same, the exponents must be equal:

$$x - 5 = -2$$
$$x = 3$$

STUDENT PRACTICE QUESTIONS

1. Correct option: (C)

To find the LCM of two numbers when their product and HCF are given, we can use the formula:

LCM = (Product of the two numbers) / HCF

Given information: Product of the two numbers = 1600
HCF = 5
Using the formula, we can find the LCM:

$$LCM = \frac{1600}{5}$$
$$LCM = 320$$

Therefore, the LCM of the two numbers is 320.

2. Correct option: (D)

Euclid's division lemma states that given any positive integers a and b (where b > 0), we can find unique integers q and r such that $a = bq + r$, where r is greater than or equal to 0 and less than b.
In other words, the remainder (r) obtained when dividing a by b is always positive and less than the divisor (b).
So r must satisfy the option d) inequality $0 < r < b$

3. Correct option: (B)

The number 3.24636363... is a rational number.
A rational number is a number that can be expressed as a fraction of two integers. The number 3.24636363... is a repeating decimal because the digits "63" repeat infinitely.
Let's represent the repeating decimal as follows: $x = 3.24636363\ldots$
Now, to convert the repeating decimal into a fraction, we can subtract it from a shifted version of itself: $100x = 324.636363\ldots$
Now, we subtract the two equations to eliminate the repeating part:

$$100x - x = 324.636363\ldots - 3.24636363\ldots$$
$$99x = 321$$

Divide both sides by 99 to find the value of x:

$$x = \frac{321}{99}$$

Since 321 and 99 are both integers, and 99 is not equal to 0, the number 3.24636363... is indeed a rational number.
Therefore, the correct answer is: b) A rational number.

4. Correct option: (B)

The number 1.732 is:
b) A rational number
The number 1.732 is a finite decimal and non-repeating.

5. Correct option: (B)

Let's separate -1 from the expression $(-2)^5 \times 3^2$:

$$(-2)^5 \times 3^2$$
$$= (-1) \times 2^5 \times 3^2$$

Now, calculate the expression:

$$-1 \times 32 \times 9 = -32 \times 9 = -288$$

Therefore, the correct answer is: $b) -288$

6. Correct option: (D)

Let's simplify the expression:

$$\frac{(3^5 \times 10^5 \times 25)}{5^7 \times 6^5}$$

First, let's do prime factor of the numbers:

$$10^5 = (2 \times 5)^5 = 2^5 \times 5^5$$
$$25 = 5 \times 5 = 5^2$$
$$6^5 = (2 \times 3)^5 = 2^5 \times 3^5$$

Now, substitute the values:

$$\frac{(3^5 \times 2^5 \times 3^5 \times 5^2)}{5^7 \times 3^5 \times 2^5}$$

Now, let's simplify the expression:

$$\frac{(3^5 \times 2^5 \times 5^5 \times 5^2)}{5^7 \times 3^5 \times 2^5}$$
$$= \frac{(3^5 \times 2^5 \times 5^7)}{5^7 \times 3^5 \times 2^5}$$

Therefore, the value of the expression is 1.

7. Correct option: (B)

To convert a logarithmic equation into its exponential form, follow this general formula:

$$\log_a b = c => a^c = b$$

Using the given logarithmic equation $\log_{\sqrt{8}} b = 3\frac{1}{3}$

Given: $\log_{\sqrt{8}} b = 3\frac{1}{3}$

The formula becomes: $(\sqrt{8})\text{^}(\frac{10}{3}) = b$

$$\sqrt{8} = 8^{\frac{1}{2}}$$
$$(8^{(\frac{1}{2})})^{\frac{10}{3}} = 8^{(\frac{1}{2} \times \frac{10}{3})} = 8^{(\frac{5}{3})}$$
$$8 = 2^3 => 8^{(\frac{5}{3})} = 2^5$$

So value of b = 32 if $\log_{\sqrt{8}} b = 3\frac{1}{3}$

8. Correct option: (A)

$$\frac{1}{\log_x xy} + \frac{1}{\log_y xy}$$

Using the property of logarithms, $\log_b mn = \log_b m + \log_b n$, we can split the terms:

$$\frac{1}{\log_x xy} = \log_{xy} x$$
$$\frac{1}{\log_y xy} = \log_{xy} y$$

Now, the expression becomes:

$$\log_{xy} x + \log_{xy} y$$
$$\log_{xy} x + \log_{xy} y = \log_{xy} xy$$
$$\log_{xy} xy = 1$$

So, the simplified expression is: $\frac{1}{\log_x xy} + \frac{1}{\log_y xy} = 1$

9. Correct option: (C)

Given: $\log_4(\log_3 x) = \frac{1}{2}$

Step 1: Rewrite the equation in exponential form:

$$\log_4(\log_3 x) = \frac{1}{2} => 4^{\frac{1}{2}} = \log_3 x$$

Step 2: Simplify the left-hand side: $4^{\frac{1}{2}} = 2$

Step 3: Now, the equation becomes: $2 = \log_3 x$

Step 4: Rewrite the equation in exponential form: $3^2 = x$

Step 5: Calculate the value of x: $x = 3^2 = 9$

10. Correct option: (D)

To find the value of x in the given equation

$$\frac{1}{\log_x 10} = \frac{1}{\log_a 10^{-2}}$$

Using laws of logarithms

$$\log_{10} x = \frac{1}{-2\log_a 10} = \frac{\log_{10} a}{-2}$$
$$-2\log_{10} x = \log_a 10 = \log_{10} x^{-2}$$
$$\log_{10} a = \log_{10} x^{-2} => a = x^{-2}$$
$$x = a^{-1/2}$$

11. Correct option: (A)

Step 1: Remove logs using one-to-one property

Since logs are equal, their arguments must be equal:

$$x + y = \frac{3x-3y}{2}$$

Step 2: Eliminate the fraction

Multiply both sides by 2:

$$2(x+y)=3x-3y \Rightarrow 2x+2y=3x-3y$$

Bring all terms to one side:

$$2x+2y-3x+3y=0 \Rightarrow -x+5y=0 \Rightarrow x=5y$$

Step 3: Use this in logx−logy

We know:

$$\log x - \log y = \log\left(\frac{x}{y}\right) \Rightarrow \log\left(\frac{5y}{y}\right) = \log 5$$

12. Correct option: (A)

Step 1: Simplify each term

$(-1)^3 = -1$
$(a^2)^3 = a^{2x3} = a^6$
$(5a)^2 = 5^2.a^2 = 25a^2$

Step 2: Multiply all parts

$(-1) \times a^6 \times 25a^2 = -25.a^{6+2} = -25a^8$

13. Correct option: (A)

Given equation: $x^{\log_x (x+3)^2} = 16$

Using the identity, we can rewrite the equation as:

$$(x+3)^2 = 16$$

Taking the square root of both sides:

$$x + 3 = 4 \; or \; x + 3 = -4$$

Now, we have two cases to consider:

$$x = 4 - 3 \; or \; x = -4 - 3$$
$$x = 1 \; or \; x = -7$$
$$a^{\log_a b} = b, b > a > 0, a \neq 1$$

So according given rule $x \neq 1, x > 0$ but $x = 1 \; or \; x = -7$ founded by above equation will contradict the rule .

Therefore, there are no valid solutions to the equation $x^{\log_x (x+3)^2} = 16$

So, the correct answer is option a) 0.

14. Correct option: (C)

15. Correct option: (B)

We have the equation:

$2^{\log_6(-4x)} = \log_7 2401$

Step 1: Convert 2401 to exponential form as 7^4

$2^{\log_6(-4x)} = \log_7 7^4 = 4$

Step 2: Convert 4 to exponential form as 2^2

Now, the equation becomes:

$2^{\log_6(-4x)} = 2^2$

Step 3: Solve for x:

$\log_6(-4x) = 2$

$6^2 = -4x => x = -9$

So, the equation has one solution: x = -9.

The correct answer is (b) 1.

16. Correct Matching is: a – p, r, s b - q, t c – q, s d – p, t

a) $\log_{0.5} 0.0625$

$0.5^4 = 0.0625$

So, $\log_{0.5} 0.0625 = 4$

a)------→(P, R, S)

b) $\log_2 8$

$2^3 = 8$

So, $\log_2 8 = 3$

b)------→(Q, T)

R) $\log_4 3 \log_5 4 \log_3 5$

To evaluate these expressions, use the properties of logarithms and try to express the given numbers in terms of the different bases:

$\log_3 5 = 1/\log_5 3$

$\log_5 4 \log_3 5 = \log_5 4 / \log_5 3 = \log_3 4$

So we can write $\log_5 4 \log_3 5$ as $\log_3 4$

Now equation => $\log_4 3 \log_3 4$

$$\log_4 3 \log_3 4 = \frac{\log_4 3}{\log_4 3} = 1$$

Therefore, the value is 1:

c)------→(Q, S)

S) $\log_3 \log_2 512$

To evaluate this expression

$\log_2 512$ can be simplified as follows:

So $\log_2 512 = 9$.

Now, evaluate $\log_3 9$

$\log_3 \log_2 512 = 2$

d)------→(P,T)

17. Correct Answer is: 2

Given equations:

1. $3^{x-1} = 27$

2. $4^{y+2} = 256$

Solve Equation 1 for x:

$$3^{x-1} = 27 = 3^3$$

$$x - 1 = 3 => x = 4$$

Solve Equation 2 for y:

$$4^{y+2} = 256 = 4^4$$

$$y + 2 = 4 => y = 2$$

Calculate $\frac{x}{y}$:

$$\frac{x}{y} = \frac{4}{2} = 2$$

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2. Quadratic Equations and Complex Numbers

Solutions

LEVEL – I

1. Correct option: (C)

Given equation

$$6x^2 - 5x + 1 = 0$$

Using quadratic formula for roots

$$x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4 \times 6 \times 1}}{2 \times 6} = \frac{5 \pm \sqrt{1}}{12} = \frac{6}{12}, \frac{4}{12}$$

Therefore, roots of the given quadratic equation are

$$x = \frac{1}{2}, \frac{1}{3}$$

Then the correct option is C

2. Correct option: (B)

Given equation

$$x^2 + 6x + 36 = 0$$

Using the formula of discriminant

$$D = (6^2) - 4 \times 1 \times 36 = -3 \times 36 = -108$$

Then the correct option is B)

3. Correct option: (B)

Let the first number is x then the second consecutive number will be $x + 1$

According the question we can write

$$x^2 + (x + 1)^2 = 613$$

Therefore our equation will be

$$x^2 + x^2 + 1 + 2x = 613 => 2x^2 + 2x - 612 = 0$$
$$=> x^2 + x - 306 = 0$$

Using quadratic formula for x

$$x = \frac{-(1) \pm \sqrt{(1)^2 - 4 \times 1 \times (-306)}}{2} = \frac{-1 \pm \sqrt{1225}}{2} = \frac{-1 \pm 35}{2} = 17, -18$$

We are given that the x is positive integer so only

Therefore roots of the given quadratic equation are

$$x = 17, -18$$

Then the correct option is B)

4. Correct option: (C)

Our equation

$$12x^2 + mx + 5 = 0$$

Using quadratic formula for x

$$x = \frac{-(m) \pm \sqrt{(m)^2 - 4 \times 12 \times (5)}}{24} = \frac{-m \pm \sqrt{m^2 - 240}}{24}$$

So roots are

$$\frac{-m + \sqrt{m^2 - 240}}{24}, \frac{-m - \sqrt{m^2 - 240}}{24}$$

According given information

$$\frac{\frac{-m + \sqrt{m^2 - 240}}{24}}{\frac{-m - \sqrt{m^2 - 240}}{24}} = \frac{3}{2}$$

$$\frac{-m+\sqrt{m^2-240}}{-m-\sqrt{m^2-240}}=\frac{3}{2}$$
$$=> -2m+2\sqrt{m^2-240}=-3m-3\sqrt{m^2-240}$$
$$=> 5\sqrt{m^2-240}=-m$$

Squaring both side

$$25(m^2-240)=m^2 => 24m^2=6000$$

Solving for m

$$m=\pm\sqrt{250}=\pm5\sqrt{10}$$

Then the correct option is C)

5. Correct option: (D)

Given equation

$$x^2+px+4=0$$

We are also given that both of the roots are equal, so value of discriminant is zero.
Therefore

$$D=p^2-4\times4=p^2-16=0 => p=\pm4$$

Then the correct option is D)

6. Correct option: (B)

Given equation

$$(3m+1)x^2+2(m+1)x+m=0$$

We are also given that both of the roots are equal, so value of discriminant is zero.
Therefore

$$D=\big(2(m+1)\big)^2-4\times(3m+1)\times m=0 => 4m^2+4+8m-12m^2-4m=0$$
$$-8m^2+4m+4=0$$

Solving for m

$$m=\frac{-4\pm\sqrt{16-4\times-8\times4}}{-16}=\frac{-4\pm12}{-16}=1,-\frac{1}{2}$$

Then the correct option is B

7. Correct option: (D)

Given equation

$$(q-r)x^2+(r-p)x+(p-q)=0$$

Using quadratic formula for roots

$$x=\frac{-(r-p)\pm\sqrt{(r-p)^2-4\times(q-r)\times(p-q)}}{2\times(q-r)}=\frac{p-r\pm\sqrt{r^2+p^2-2pr-4pq+4pr+4q^2-4qr}}{2\times(q-r)}$$
$$=\frac{p-r\pm\sqrt{r^2+p^2+2pr-4pq+4q^2-4qr}}{2\times(q-r)}$$
$$=\frac{p-r\pm\sqrt{(p+r-2q)^2}}{2\times(q-r)}=\frac{p-r\pm(p+r-2q)}{2\times(q-r)}$$
$$x=\frac{2p-2q}{2(q-r)},\frac{2r-2q}{2(q-r)}, => x=\frac{2p-2q}{2(q-r)},-1$$

Therefore roots of the given quadratic equation are

$$x=\frac{2p-2q}{2(q-r)},-1$$

Then the correct option is D

8. Correct option: (A)

Given equation

$$x^2+2x+3=0$$

Given roots of the equation are α,β

We have to find the value of $\alpha^2 + \beta^2$

$$=> \alpha^2 + \beta^2 = \alpha^2 + \beta^2 + 2\alpha\beta - 2\alpha\beta = (\alpha+\beta)^2 - 2\alpha\beta$$

So we can say that

$$\alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta = (sum\ of\ roots)^2 - 2(product\ of\ roots)$$

As we know that for a quadratic equation $ax^2 + bx + c = 0$

$$sum\ of\ roots = -\frac{b}{a}, product\ of\ roots = \frac{c}{a}$$

Therefore we can write

$$\alpha + \beta = -\frac{2}{1}, \alpha\beta = \frac{3}{1}$$

So the value of $\alpha^2 + \beta^2$

$$= (\alpha+\beta)^2 - 2\alpha\beta = (-2)^2 - 2 \times 3 = 4 - 6 = -2$$

Then the correct option is A

9. Correct option: (C)

Given equation

$$ax^2 + bx + c = 0$$

Given roots of the equation are α, β

We have to find the value of

$$(a\alpha + b)^{-2} + (a\beta + b)^{-2}$$

$$=> \frac{1}{(a\alpha+b)^2} + \frac{1}{(a\beta+b)^2} = \frac{(a\beta+b)^2 + (a\alpha+b)^2}{(a\alpha+b)^2(a\beta+b)^2}$$

$$= \frac{a^2(\alpha^2+\beta^2) + 2ab(\alpha+\beta) + 2b^2}{(a^2\alpha\beta + ab(\alpha+\beta) + b^2)^2} = \frac{a^2((\alpha+\beta)^2 - 2\alpha\beta) + 2ab(\alpha+\beta) + 2b^2}{(a^2\alpha\beta + ab(\alpha+\beta) + b^2)^2}$$

So as we know that

$$\alpha + \beta = sum\ of\ roots, \alpha\beta = (product\ of\ roots)$$

As for a quadratic equation $ax^2 + bx + c = 0$

$$sum\ of\ roots = \alpha + \beta = -\frac{b}{a}, product\ of\ roots = \alpha\beta = \frac{c}{a}$$

Therefore we can write

$$\frac{a^2((\alpha+\beta)^2 - 2\alpha\beta) + 2ab(\alpha+\beta) + 2b^2}{(a^2\alpha\beta + ab(\alpha+\beta) + b^2)^2} = \frac{a^2\left(\left(-\frac{b}{a}\right)^2 - \frac{2c}{a}\right) + 2ab\left(-\frac{b}{a}\right) + 2b^2}{\left(\frac{a^2c}{a} + ab\left(-\frac{b}{a}\right) + b^2\right)^2}$$

$$= \frac{a^2\left(\frac{b^2}{a^2} - \frac{2c}{a}\right) - 2b^2 + 2b^2}{(ac - b^2 + b^2)^2}$$

$$= \frac{b^2 - 2ac}{(ac)^2}$$

So the value of $(a\alpha + b)^{-2} + (a\beta + b)^{-2}$

$$= \frac{b^2 - 2ac}{(ac)^2}$$

Then the correct option is C

10. Correct option: (B)

Given equation

$$8x^2 - 3x + 27 = 0$$

Given roots of the equation are α, β

We have to find the value of

$$\left(\frac{\alpha^2}{\beta}\right)^{\frac{1}{3}} + \left(\frac{\beta^2}{\alpha}\right)^{\frac{1}{3}}$$

$$=> \frac{\alpha+\beta}{(\alpha\beta)^{\frac{1}{3}}}$$

So as we know that

$$\alpha+\beta = sum\ of\ roots, \alpha\beta = (product\ of\ roots)$$

As for a quadratic equation $8x^2-3x+27=0$

$$sum\ of\ roots = \alpha+\beta = \frac{3}{8}, product\ of\ roots = \alpha\beta = \frac{27}{8}$$

Therefore we can write

$$\frac{\alpha+\beta}{(\alpha\beta)^{\frac{1}{3}}} = \frac{\frac{3}{8}}{\left(\frac{27}{8}\right)^{\frac{1}{3}}} = \frac{3}{8}\times\frac{2}{3} = \frac{1}{4}$$

$$= 0$$

So the value of

$$\left(\frac{\alpha^2}{\beta}\right)^{\frac{1}{3}} + \left(\frac{\beta^2}{\alpha}\right)^{\frac{1}{3}} = \frac{1}{4}$$

Then the correct option is B

11. Correct option: (A)

Given equation

$$ax^2+bx+c=0$$

Given roots of the equation are α, β

We have to find the value of

$$\alpha^5\beta^8 + \alpha^8\beta^5$$

$$=> \alpha^5\beta^5(\alpha^3+\beta^3) = (\alpha\beta)^5(\alpha+\beta)((\alpha+\beta)^2-3\alpha\beta)$$

So as we know that

$$\alpha+\beta = sum\ of\ roots, \alpha\beta = (product\ of\ roots)$$

As for a quadratic equation $ax^2+bx+c=0$

$$sum\ of\ roots = \alpha+\beta = -\frac{b}{a}, product\ of\ roots = \alpha\beta = \frac{c}{a}$$

Therefore we can write

$$(\alpha\beta)^5(\alpha+\beta)((\alpha+\beta)^2-3\alpha\beta) = \left(\frac{c}{a}\right)^5\left(-\frac{b}{a}\right)\left(\left(-\frac{b}{a}\right)^2-\frac{3c}{a}\right)$$

$$= \frac{c^5(3abc-b^3)}{a^8}$$

So the value of

$$\alpha^5\beta^8 + \alpha^8\beta^5 = \frac{c^5(3abc-b^3)}{a^8}$$

Then the correct option is A

12. Correct option: (C)

Given equation

$$x^2+kx+2=0$$

Given roots of the equation are α, β

We have to find the value of k for given condition $\alpha-\beta=1$

So as we know that

$$\alpha+\beta = sum\ of\ roots, \alpha\beta = (product\ of\ roots)$$

As for a quadratic equation $x^2+kx+2=0$

$$sum\ of\ roots = \alpha+\beta = -k, product\ of\ roots = \alpha\beta = 2$$

Therefore we can write

$$(\alpha-\beta)^2 = 1 = (\alpha+\beta)^2 - 4\alpha\beta = k^2-8$$

$$k^2 = 9 => k = \pm 3$$

Then the correct option is C

13. Correct option: (C)

Given equation $x^2 - 3kx + 2.e^{2\log k} - 1 = 0$

Given product of roots of the equation are 7

We have to find the value of k for given condition

As for a quadratic equation $x^2 - 3kx + 2.e^{2\log k} - 1 = 0$

$$product\ of\ roots = 2.e^{2\log k} - 1 = 7$$

$$2.e^{2\log k} = 8$$

Using log property we can write $2k^2 = 8 => k^2 = 4 => k = \pm 2$

We are given that $k > 0 => k = 2$

Therefore we can write

$$sum\ of\ the\ roots = 3k = 6$$

Then the correct option is C

14. Correct option: (C)

Given roots of the equation are $2\sqrt{3} - 5, -2\sqrt{3} - 5$

We have to find the equation for the given roots

So we can say that $sum\ of\ roots = -10, product\ of\ roots = 13$

As we know that quadratic equation for the given sum of roots and product of roots

$$x^2 - (sum\ of\ roots)x + product\ of\ roots = 0$$

Therefore we can write $x^2 - (-10)x + 13 = 0 => x^2 + 10x + 13 = 0$

Then the correct option is C

15. Correct option: (B)

Given expression

$$2 - \frac{1}{2 - \frac{1}{2 - \cdots}}$$

Let

$$2 - \frac{1}{2 - \frac{1}{2 - \cdots}} = x$$

Then we can write

$$2 - \frac{1}{2 - \frac{1}{2 - \cdots}} = 2 - \frac{1}{x} = x$$

So our expression become

$$2 - \frac{1}{x} = x => x^2 - 2x + 1 = 0 => (x-1)^2 = 0 => x = 1$$

Therefore value of our expression will be 1.

Then the correct option is B)

16. Correct option: (C)

Given equation $x^2 - 2(5 + 2k)x + 3(7 + 10k) = 0$

Given that roots are equal

We have to find the value of k for given condition

From above condition we can write

$$D = 0 = \left(-2(5 + 2k)\right)^2 - 12(7 + 10k) = 0$$

Solving for k

$$100 + 16k^2 + 80k - 84 - 120k = 0 => 16k^2 - 40k + 16 = 0$$

$$4k^2 - 10k + 4 = 0$$

Using quadratic formula for k

$$k = \frac{10 \pm \sqrt{100 - 64}}{8} = \frac{10 \pm 6}{8} = 2, \frac{1}{2}$$

Then the correct option is C)

17. Correct option: (C)

Given equation

$$x^{\frac{2}{3}} + x^{\frac{1}{3}} - 2 = 0$$

We have to find the solution set for the above equation

Let $x^{\frac{1}{3}} = y$ then our equation in terms of $y \Rightarrow y^2 + y - 2 = 0$

Using quadratic formula for y

$$y = \frac{-1 \pm \sqrt{1+8}}{2} = \frac{-1 \pm 3}{2} = 1, -2$$

Therefore $x \Rightarrow x^{\frac{1}{3}} = y \Rightarrow x = y^3 \Rightarrow x = 1, -8$

Then the correct option is C)

18. Correct option: (A)

Given equation

$$2^x + 27(2^{-x}) = 12$$

We have to find the solution set for the above equation

Let $2^x = y$ then our equation in terms of $y \Rightarrow y + \frac{27}{y} - 12 = 0$

$\Rightarrow y^2 - 12y + 27 = 0$

Using quadratic formula for y

$$y = \frac{12 \pm \sqrt{144 - 108}}{2} = \frac{12 \pm 6}{2} = 9, 3$$

Therefore $2^x = y \Rightarrow x \Rightarrow x = \log_2 y \Rightarrow x = \log_2 9, \log_2 3$

Then the correct option is A)

19. Correct option: (D)

Let's first number as x then other number will be $x + 1$

Using above condition we can write

$$\frac{1}{x} + \frac{1}{x+1} = \frac{23}{132}$$

Therefore

$$(x + 1 + x)132 = 23(x + 1)x \Rightarrow 264x + 132 = 23x^2 + 23x$$
$$\Rightarrow 0 = 23x^2 - 241x - 132$$

Using quadratic formula for x

$$x = \frac{241 \pm \sqrt{70225}}{46} = \frac{241 \pm 265}{46} = 11, -\frac{12}{23}$$

Then the numbers are $11, 12$

Then the correct option is D)

20. Correct option: (C)

We have to find the equation with given condition as

Let roots are $\alpha + \beta$

Given that $\alpha + \beta = 12, \alpha^3 + \beta^3 = 468$

We can write

$$\alpha^3 + \beta^3 = (\alpha + \beta)((\alpha + \beta)^2 - 3\alpha\beta) = 12(144 - 3\alpha\beta)$$

$$144 - 3\alpha\beta = \frac{468}{12} = 39 \Rightarrow \alpha\beta = \frac{144 - 39}{3} = 35$$

As we know that quadratic equation for the given sum of roots and product of roots

$$x^2 - (sum\ of\ roots)x + product\ of\ roots = 0$$

Therefore we can write

$$x^2 - 12x + 35 = 0$$

Then the correct option is C)

LEVEL-II

1. Correct option: (B)

From the given equation only option B) is quadratic equation which has a variable with power 2 and no other variable with power greater than 2 or less than 0.

Then the correct option is B)

2. Correct option: (A)

Given equation

$$3a^2x^2 + 8abx + 4b^2 = 0$$

Using quadratic formula for roots

$$x = \frac{-(8ab) \pm \sqrt{(8ab)^2 - 4 \times 3a^2 \times 4b^2}}{2 \times 3a^2} = \frac{-8ab \pm 4ab}{2 \times 3a^2} = -\frac{2b}{a}, \frac{2b}{3a}$$

Then the correct option is A)

3. Correct option: (D)

Given equation

$$25x^2 + 30x + 7 = 0$$

Using quadratic formula for roots

$$x = \frac{-(30) \pm \sqrt{(30)^2 - 4 \times 25 \times 7}}{2 \times 25} = \frac{-30 \pm \sqrt{200}}{50} = \frac{-3 + \sqrt{2}}{5}, \frac{-3 - \sqrt{2}}{5}$$

Then the correct option is D)

4. Correct option: (A)

Given equation

$$2^{2x+3} - 57 = 65(2^x - 1)$$

$$2^{2x} \times 8 - 57 = 65 \times 2^x - 65 \Rightarrow 8 \times 2^{2x} - 57 + 65 - 65 \times 2^x = 0$$

$$\Rightarrow 8 \times 2^{2x} + 8 - 65 \times 2^x = 0$$

Let's take $2^x = p$

$$\Rightarrow 8p^2 - 65p + 8 = 0$$

Using quadratic formula for roots

$$p = \frac{-(-65) \pm \sqrt{(65)^2 - 4 \times 8 \times 8}}{2 \times 8} = \frac{65 \pm 63}{16} = 8, \frac{1}{8}$$

Therefore

$$2^x = p \Rightarrow 2^x = 8 \Rightarrow x = 3 \; and \; 2^x = \frac{1}{8} \Rightarrow x = -3$$

Then the correct option is A)

5. Correct option: (B)

Given that $\alpha + \beta = 2, \alpha\beta = 3$

We have to find the value of $\alpha^3 + \beta^3$

So we can write

$$\alpha^3 + \beta^3 = (\alpha + \beta)((\alpha + \beta)^2 - 3\alpha\beta)$$

Putting the given value to find the unknown value

$$\Rightarrow \alpha^3 + \beta^3 = (2)(2^2 - 9) = -10$$

Then the correct option is B)

6. Correct option: (B)

Given equation $12x^2 + mx + 5 = 0$

We are also given that the roots are in ration of 5:4 and we have to find the value of m for given condition.

Using quadratic formula for roots

$$x = \frac{-(m) \pm \sqrt{(m)^2 - 240}}{24}$$

Then we can write

$$\frac{\frac{-(m)+\sqrt{(m)^2-240}}{24}}{\frac{-(m)-\sqrt{(m)^2-240}}{24}}=\frac{5}{4}$$

$$=>\frac{-(m)+\sqrt{(m)^2-240}}{-(m)-\sqrt{(m)^2-240}}=\frac{5}{4}$$

Cross multiplication

$$-4m+4\sqrt{(m)^2-240}=-5m-5\sqrt{(m)^2-240}$$

$$m=-9\sqrt{(m)^2-240}$$

Squaring both side

$$m^2=81(m^2-240)=>m^2=81\times\frac{240}{80}=243$$

Therefore $m=\pm\sqrt{243}=\pm 9\sqrt{3}$

Then the correct option is B)

7. Correct option: (C)

Given equation $x^2-10x+3(7+10k)=0$

Let's one root of the equation is p then other root will be 2p as given in question.

Then we can write

$$p+2p=sum\ of\ roots=10=>p=\frac{10}{3}$$

Then one root will be $\frac{10}{3}$ and other one is $\frac{20}{3}$

Therefore product of the root

$$\frac{200}{9}=3(7+10k)=>\frac{11}{270}=k$$

Then the correct option is C)

8. Correct option: (D)

We are given perimeter and area of a rectangle as

$$P=28, A=40$$

Let's the length of the rectangle l

Then the breadth of the rectangle $\frac{P}{2}-l=>14-l$

So we can write

$$area=l(14-l)=40=>l^2-14l+40=0$$

Solving for l

$$l=\frac{(14)\pm\sqrt{196-160}}{2}=10,4$$

Therefore breadth for the given length

$$14-l=>b=4,10$$

Then the correct option is D)

9. Correct option: (D)

Let's the altitude of the triangle is L then its base will be L+4

According question we can write

$$\frac{1}{2}L(L+4)=48=>L^2+4L-96=0$$

Solving for L

$$L=\frac{(-4)\pm\left(\sqrt{16+386}\right)}{2}=-2\pm 10=-12,8$$

Only valid value for L is 8 so base will be $8+4=12$

Then the correct option is D)

10. Correct options: (A, C, D)

Given equation

$$x^2 + x + 2 = 0$$

Given roots of the equation α, β

From the equation we can write

$$\alpha + \beta = -1, \alpha\beta = 2$$

Let's take the option

A) $\alpha^3 + \beta^3 = 5$

$$\alpha^3 + \beta^3 = (\alpha + \beta)((\alpha + \beta)^2 - 3\alpha\beta)$$
$$=> (-1)(1 - 3 \times 2) = (-1)(-5) = 5$$

Option A) is correct.

B) $\alpha^3\beta^3 + \alpha^2\beta^3 + \alpha^3\beta^2 = 3$

$$\alpha^3\beta^3 + \alpha^2\beta^3 + \alpha^3\beta^2 = (\alpha^2\beta^2)(\alpha\beta + \alpha + \beta)$$
$$=> (2^2)(2 + (-1)) = 4$$

Option B) is incorrect.

C) $\alpha - \beta = \sqrt{-7}$

$$(\alpha - \beta)^2 => (\alpha + \beta)^2 - 4\alpha\beta = 1 - 4(2) = -7$$

From the given one we can prove $(\alpha - \beta)^2 = -7$

Option C) is correct.

D) $\alpha^2 + \beta^2 = -3$

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta => \alpha^2 + \beta^2 = -3$$

Option D) is correct.

Then the correct option is A, C, D)

11. Correct option: (A, B, C, D)

Given equation

$$6\left(x^2 + \frac{1}{x^2}\right) - 25\left(x - \frac{1}{x}\right) + 12 = 0$$

Let's the

$$x - \frac{1}{x} = p => \left(x - \frac{1}{x}\right)^2 = p^2 => x^2 + \frac{1}{x^2} = p^2 + 2$$

Therefore our equation in form of p

$$6(p^2 + 2) - 25p + 12 = 0 => 6p^2 - 25p + 24 = 0$$

Using quadratic formula

$$p = \frac{25 \pm \sqrt{625 - 576}}{12} = \frac{25 \pm 7}{12} = \frac{8}{3}, \frac{3}{2}$$

So we can write

$$x - \frac{1}{x} = \frac{8}{3} => 3x^2 - 3 = 8x => 3x^2 - 8x - 3 = 0$$

Using quadratic formula

$$x = \frac{8 \pm \sqrt{64 + 36}}{6} = 3, -\frac{1}{3}$$

For $p = \frac{3}{2} =>$

$$x - \frac{1}{x} = \frac{3}{2} => 2x^2 - 2 = 3x => 2x^2 - 3x - 2 = 0$$

Using quadratic formula

$$x = \frac{3 \pm \sqrt{9 + 16}}{4} = 2, -\frac{1}{2}$$

Then the correct option is A, B, C ,D)

12. Correct options: (A)

Given equation

$$\frac{1}{k}(5x^2 - 4x + 38) + (-4x^2 - 2x - 1) = 0$$
$$=> (5 - 4k)x^2 + (-4 - 2k)x + (38 - k) = 0$$

Given that product of the roots is -5

We can write using the equation as

$$product\ of\ root = -5 = \frac{38 - k}{5 - 4k}$$

Solving for k

$$-25 + 20k = 38 - k => 21k = 63 => k = 3$$

Then the correct option is A)

COMPREHENSION TYPE

Given equation

$$ax^2 + bx + c = 0$$

Given roots are α, β

From the equation we can write

$$\alpha + \beta = -\frac{b}{a}, \alpha\beta = \frac{c}{a}$$

13. Correct option: (C)

We have to find the value of

$$\frac{\alpha^3 + \beta^3}{\alpha^{-3} + \beta^{-3}}$$

So we can write

$$\frac{\alpha^3 + \beta^3}{\alpha^{-3} + \beta^{-3}} = \frac{\alpha^3 + \beta^3}{\frac{\alpha^3 + \beta^3}{\alpha^3\beta^3}} = \alpha^3\beta^3 = (\alpha\beta)^3 = \left(\frac{c}{a}\right)^3$$

Then the correct option is C)

14. Correct option: (A)

We have to find the value of

$$\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha}$$

So we can write

$$\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha} = \frac{\alpha^3 + \beta^3}{\alpha\beta} = \frac{(\alpha + \beta)((\alpha + \beta)^2 - 3\alpha\beta)}{\alpha\beta} = \frac{\left(-\frac{b}{a}\right)\left(\left(-\frac{b}{a}\right)^2 - \frac{3c}{a}\right)}{\frac{c}{a}} = \frac{(-b)(b^2 - 3ac)}{ca^2}$$
$$= \frac{3abc - b^3}{a^2c}$$

Then the correct option is A)

15. Correct option: (B)

We have to find the value of

$$\frac{1}{\alpha^2} + \frac{1}{\beta^2}$$

So we can write

$$\frac{1}{\alpha^2}+\frac{1}{\beta^2}=\frac{\alpha^2+\beta^2}{(\alpha\beta)^2}=\frac{((\alpha+\beta)^2-2\alpha\beta)}{(\alpha\beta)^2}=\frac{\left(\left(-\frac{b}{a}\right)^2-\frac{2c}{a}\right)}{\left(\frac{c}{a}\right)^2}=\frac{b^2-2ac}{c^2}$$

Then the correct option is B)

Given equation

$$ax^2+bx+c=0$$

Given roots are α,β and also given that

$$\alpha+\beta=-\frac{b}{a},\alpha\beta=\frac{c}{a}$$

$$\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta,(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta$$

MATRIX MATCHING TYPE

16. Correct Matching is: a-QT,b-R,c-P,d-

a) $12+\sqrt{12+\sqrt{12+\sqrt{12+\cdots}}}$

Let's the $12+\sqrt{12+\sqrt{12+\sqrt{12+\cdots}}}=x$ then we can write

$$x=12+\sqrt{x}\Rightarrow x-12=\sqrt{x}$$

$$\Rightarrow x^2+144-24x=x\Rightarrow x^2-25x+144=0$$

Using quadratic equation

$$x=\frac{25\pm\sqrt{625-4\times144}}{2}=\frac{25\pm7}{2}=16,9$$

Then the matching option is (Q), (T)

b) $\sqrt{3-\sqrt{3-\sqrt{3-\cdots}}}$

Let's the $\sqrt{3-\sqrt{3-\sqrt{3-\cdots}}}=x$ then we can write

$$x=\sqrt{3-x}$$

$$\Rightarrow x^2=3-x\Rightarrow x^2+x-3=0$$

Using quadratic equation

$$x=\frac{-1\pm\sqrt{1-(-12)}}{2}=\frac{-1\pm\sqrt{13}}{2}$$

Then the matching option is (R)

c) The value of

$$1+\frac{1}{1+\frac{1}{1+\frac{1}{1+\cdots}}}$$

Let's

$$1+\frac{1}{1+\frac{1}{1+\frac{1}{1+\cdots}}}=x$$

Then we can write

$$x=1+\frac{1}{x}\Rightarrow x^2-x-1=0$$

Using quadratic equation

$$x=\frac{1\pm\sqrt{1-(-4)}}{2}=\frac{-1\pm\sqrt{5}}{2}$$

Then the matching option is (P)

d) $\sqrt{6+\sqrt{6+\sqrt{6+\cdots}}}$

Let's the $\sqrt{6+\sqrt{6+\sqrt{6+\cdots}}} = x$ then we can write

$$x = \sqrt{6+x}$$
$$=> x^2 = 6 + x => x^2 - x - 6 = 0$$

Using quadratic equation

$$x = \frac{1 \pm \sqrt{1-(-24)}}{2} = \frac{-1 \pm 5}{2} = -3, 2$$

No matching option.

17. Correct Answer is: (1)

Given equation

$$(a^2-1)x^2 + (a-1)x + (a^2 - 4a + 3) = 0$$

Also given that above equation is an identity in x means given equation is valid for all value of x

So we can conclude that

$$a^2 - 1 = 0 \; and \; a - 1 = 0 \; and \; a^2 - 4a + 3$$
$$=> a = -1, 1 \; and \; a = 1 \; and \; a = 3, 1$$

So we can write that only solution for the above statement is

$$a = 1$$

Then the correct answer for $a = 1$.

18. Correct Answer is: (9)

Given equation

$$x^2 + x - 20 = 0$$

We are given that the roots are α, β

We have to find the value of $|\alpha^2 - \beta^2|$

Using the given equation we can write

$$\alpha + \beta = -1, \alpha\beta = -20$$

So we can also write

$$|\alpha^2 - \beta^2| = |(\alpha+\beta)(\alpha-\beta)| = \left|(\alpha+\beta)\sqrt{(\alpha+\beta)^2 - 4(\alpha\beta)}\right|$$
$$=> \left|(-1) \times \sqrt{1+80}\right| = 9$$

Then the value of the $|\alpha^2 - \beta^2|$ is 9.

19. Correct Answer is: (1)

Let's the first number is x then second number is $x + 1$

We can write

$$x^2 + (x+1)^2 = 613 => 2x^2 + 2x - 612 = 0$$
$$=> x^2 + x - 306$$

Using quadratic formula for x

$$x = \frac{-1 \pm \sqrt{1+1224}}{2} = 17, -18$$

For $x = 17 => second\ number = 18, x = -18 => seocond\ number\ is = -17$

Then absolute difference is 1.

20. Correct Answer is: (2)

Given equation

$$2^{2x^2-7x+5} = 1$$

We can write

$$2^{2x^2-7x+5} = 2^0$$

Equating the power for same base

$$2x^2 - 7x + 5 = 0$$

Using quadratic formula for solution

$$x = \frac{7 \pm \sqrt{49 - 40}}{4} = \frac{7 \pm 3}{4} = \frac{10}{4}, 1$$

So number of real roots for the given equation is 2.

STUDENT PRACTICE QUESTIONS

1. Correct option: (C)

Given equation $\sqrt{3}x^2 + 10x - 8\sqrt{3} = 0$

Using quadratic formula for roots

$$x = \frac{-(10) \pm \sqrt{(10)^2 - 4 \times \sqrt{3} \times (-8\sqrt{3})}}{2 \times \sqrt{3}} = \frac{-10 \pm \sqrt{196}}{2\sqrt{3}} = \frac{-10 \pm 14}{2\sqrt{3}} = \frac{4}{2\sqrt{3}}, -\frac{24}{2\sqrt{3}}$$

Therefore roots of the given quadratic equation are

$$x = \frac{2}{\sqrt{3}}, -4\sqrt{3}$$

Then the correct option is C)

2. Correct option: (B)

Given equation $x^2 - 5x + k = 0$

Given that 2 is one of the root of the equation so we can write

$$2^2 - 5 \times 2 + k = 0 \Rightarrow k = 6$$

Now our equation become

$$x^2 - 5x + 6 = 0$$

Therefore

$$x = \frac{-(-5) \pm \sqrt{25 - 24}}{2} = \frac{5 \pm 1}{2} = 3, 2$$

So other root will be 3

Then the correct option is B)

3. Correct option: (A)

Given equation

$$\frac{3}{x+2} - \frac{2}{x-1} = 5$$

Taking LCM and cross multiplication

$$3(x-1) - 2(x+2) = 5(x-1)(x+2)$$
$$\Rightarrow 3x - 3 - 2x - 4 = 5x^2 - 5x + 10x - 10$$
$$\Rightarrow 5x^2 + 4x - 3 = 0$$

Using quadratic formula for roots

$$x = \frac{-(4) \pm \sqrt{(4)^2 - 4 \times (5) \times (-3)}}{2 \times (5)} = \frac{-4 \pm \sqrt{76}}{10}$$
$$= \frac{-4 \pm 2\sqrt{19}}{10} = \frac{-2 \pm \sqrt{19}}{5}$$

Therefore roots of the given quadratic equation are

$$x = \frac{-2 \pm \sqrt{19}}{5}$$

Then the correct option is A

4. Correct option: (A)

Given roots of the equation are 3,5

We have to find the equation for the given roots

So we can say that $sum\ of\ roots = 8, product\ of\ roots = 15$

As we know that quadratic equation for the given sum of roots and product of roots

$$x^2 - (sum\ of\ roots)x + product\ of\ roots = 0$$

Therefore we can write

$$x^2 - 8x + 15 = 0$$

Then the correct option is A

5. Correct option: (D)

Given equation $ax^2 + bx + c = 0$

Given roots of the equation are α, β

We have to find the value of

$$(a\alpha + b)^{-3} + (a\beta + b)^{-3}$$

$$=> \frac{1}{(a\alpha + b)^3} + \frac{1}{(a\beta + b)^3} = \frac{(a\beta + b)^3 + (a\alpha + b)^3}{(a\alpha + b)^3(a\beta + b)^3}$$

$$\frac{(a(\alpha + \beta) + 2b)\left(a^2((\alpha + \beta)^2 - 2\alpha\beta) + 2ab(\alpha + \beta) + 2b^2 - (a^2\alpha\beta + ab(\alpha + \beta) + b^2)\right)}{(a^2\alpha\beta + ab(\alpha + \beta) + b^2)^3}$$

So as we know that

$$\alpha + \beta = sum\ of\ roots, \alpha\beta = (product\ of\ roots)$$

As for a quadratic equation $ax^2 + bx + c = 0$

$$sum\ of\ roots = \alpha + \beta = -\frac{b}{a}, product\ of\ roots = \alpha\beta = \frac{c}{a}$$

Therefore we can write

$$\frac{(a(\alpha + \beta) + 2b)\left(a^2((\alpha + \beta)^2 - 2\alpha\beta) + 2ab(\alpha + \beta) + 2b^2 - (a^2\alpha\beta + ab(\alpha + \beta) + b^2)\right)}{(a^2\alpha\beta + ab(\alpha + \beta) + b^2)^3}$$

$$= \frac{\left(a\left(-\frac{b}{a}\right) + 2b\right)\left(a^2\left(\left(-\frac{b}{a}\right)^2 - \frac{2c}{a}\right) + 2ab\left(-\frac{b}{a}\right) + 2b^2 - \left(\frac{a^2c}{a} + ab\left(-\frac{b}{a}\right) + b^2\right)\right)}{\left(\frac{a^2c}{a} + ab\left(-\frac{b}{a}\right) + b^2\right)^3}$$

$$= \frac{(b)(b^2 - 2ac - ac)}{(ac)^3}$$

$$= \frac{b^3 - 3abc}{(ac)^3}$$

So the value of $(a\alpha + b)^{-3} + (a\beta + b)^{-3}$

$$= \frac{b^3 - 3abc}{(ac)^3}$$

Then the correct option is D

6. Correct option: (A)

Given equation $ax^2 + bx + c = 0$

Given roots of the equation are α, β

We have to find the value of

$$\left(\frac{\alpha}{a\beta + b}\right)^3 - \left(\frac{\beta}{a\alpha + b}\right)^3$$

$$=> \left(\frac{\alpha}{a\beta + b} - \frac{\beta}{a\alpha + b}\right)\left(\left(\frac{\alpha}{a\beta + b}\right)^2 + \left(\frac{\beta}{a\alpha + b}\right)^2 + \left(\frac{\alpha}{a\beta + b}\right)\left(\frac{\beta}{a\alpha + b}\right)\right)$$

$$= \left((\alpha - \beta)(a(\alpha + \beta) + b)\right)\left(\left(\frac{\alpha}{a\beta + b}\right)^2 + \left(\frac{\beta}{a\alpha + b}\right)^2 + \left(\frac{\alpha}{a\beta + b}\right)\left(\frac{\beta}{a\alpha + b}\right)\right)$$

Let

$$\left(\frac{\alpha}{a\beta+b}\right)^2+\left(\frac{\beta}{a\alpha+b}\right)^2+\left(\frac{\alpha}{a\beta+b}\right)\left(\frac{\beta}{a\alpha+b}\right)=\gamma$$

Then

$$=\big((\alpha-\beta)(a(\alpha+\beta)+b)\big)\gamma$$

So as we know that

$$\alpha+\beta=sum\ of\ roots, \alpha\beta=(product\ of\ roots)$$

As for a quadratic equation $ax^2+bx+c=0$

$$sum\ of\ roots=\alpha+\beta=-\frac{b}{a}, product\ of\ roots=\alpha\beta=\frac{c}{a}$$

Therefore we can write

$$\big((\alpha-\beta)(a(\alpha+\beta)+b)\big)\gamma=\left((\alpha-\beta)\left(a\left(-\frac{b}{a}\right)+b\right)\right)\gamma=\big((\alpha-\beta)(0)\big)\gamma$$

$$=0$$

So the value of

$$\left(\frac{\alpha}{a\beta+b}\right)^3-\left(\frac{\beta}{a\alpha+b}\right)^3=0$$

Then the correct option is A

7. Correct option: (C)

Given equation $9x^2+kx+1=0$

Given that roots are equal

We have to find the value of k for given condition

From above condition we can write $D=0=k^2-4\times9\times1 => k^2=36 => k=\pm6$

Then the correct option is C)

8. Correct option: (*)

Given equation $3^{2x}-3x^{x+1}+1=0$

We have to find the solution set for the above equation

Let's use the given option

A) $3^0-0+1=2\neq0$

B) $3^{14}-3\times7^8+1=4782969-17294403+1=-12511433\neq0$

C) $3^2-3\times7^2+1=9-147+1=-137\neq0$

D) $3^6-3\times7^4+1=729-7203+1=-6473\neq0$

None of the given option is correct.

9. Correct option: (B)

Given equation $7^{1+x}+7^{1-x}=50$

We have to find the solution set for the above equation

Let $7^x=y$ then our equation in terms of $y => 7y+\frac{7}{y}-50=0$

$=> 7y^2-50y+7=0$

Using quadratic formula for y

$$y=\frac{50\pm\sqrt{2500-196}}{14}=\frac{50\pm48}{14}=7,\frac{1}{7}$$

Therefore $7^x=y => x=1,-1$

Then the correct option is B)

10. Correct options: (A, B)

Given equation $ax^2+bx+c=0$

Given roots are α,β

We have to find the value of

$$\sqrt{\left(-\frac{b}{a}\right)^2-\frac{4c}{a}}$$

From the equation we can write

$$\alpha+\beta=-\frac{b}{a},\alpha\beta=\frac{c}{a}$$

Putting the value

$$\sqrt{\left(-\frac{b}{a}\right)^2-\frac{4c}{a}}=\sqrt{(\alpha+\beta)^2-4\alpha\beta}=\sqrt{(\alpha-\beta)^2}=|\alpha-\beta|=|\beta-\alpha|$$

Then the correct option is A, B)

11. Correct options: (C, D)

Given equation $x^2+x+1=0$

Given roots are α,β

We can write using the equation as $\alpha+\beta=-1,\alpha\beta=1$

For the given

$$\frac{\alpha}{\beta}+\frac{\beta}{\alpha}$$

We can write

$$\frac{\alpha}{\beta}+\frac{\beta}{\alpha}=\frac{(\alpha+\beta)^2-2\alpha\beta}{\alpha\beta}=\frac{(-1)^2-2}{1}=-1$$

So we can write

$$\frac{\alpha}{\beta}+\frac{\beta}{\alpha}=-1$$

For the given

$$\frac{1}{\alpha^2}+\frac{1}{\beta^2}$$

We can write

$$\frac{1}{\alpha^2}+\frac{1}{\beta^2}=\frac{(\alpha+\beta)^2-2\alpha\beta}{\alpha^2\beta^2}=\frac{(-1)^2-2}{1}=-1$$

So we can write

$$\frac{1}{\alpha^2}+\frac{1}{\beta^2}=-1$$

Then the correct option is C, D)

12. Correct option: (B)

Given that α,β are roots of the equation $x^2+7x+12=0$

We have to find the equation whose roots are $(\alpha+\beta)^2,(\alpha-\beta)^2$

Form the given equation we can write

$$\alpha+\beta=-7,\alpha\beta=12$$

Then

$$(\alpha+\beta)^2=49,(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta=49-48=1$$

Now we can write for the required equation

$$sum\ of\ roots=50,product\ of\ roots=49$$

So the required equation is

$$x^2-(sum\ of\ roots)x+(product\ of\ roots)=0 \Rightarrow x^2-50x+49=0$$

Then the correct option is B)

13. Correct option: (A)

Given that $3p^2=5p+2,3q^2=5q+2$

So we can write a common equation for these as

$$3x^2-5x-2=0$$

Where p,q are roots of the above equation now we have to find the value of the equation whose roots are $3p-2q,3q-2p$

So we can write using above equation that

$$p+q=\frac{5}{3}.pq=-\frac{2}{3}$$

For the required equation whose roots are $3p-2q,3q-2p$ we can write

$$sum\ of\ the\ roots = (3p - 2q) + (3q - 2p) = p + q$$

$$product\ of\ the\ roots = (3p - 2q) + (3q - 2p) = -6((p+q)^2 - 2pq) + 13pq = -\frac{50}{3} - \frac{50}{3} = -\frac{100}{3}$$

Therefore the required equation

$$x^2 - \left(\frac{5}{3}\right) + \left(-\frac{100}{3}\right) = 0 \Rightarrow 3x^2 - 5x - 100 = 0$$

Then the correct option is A)

14. Correct option: (B)

Given that α, β are roots of the equation $ax^2 + bx + c = 0$

We have to find the equation whose roots are $\alpha + \beta, \alpha\beta$

Form the given equation we can write $\alpha + \beta = -\frac{b}{a}, \alpha\beta = \frac{c}{a}$

Now we can write for the required equation

$$sum\ of\ roots = (\alpha + \beta) + \alpha\beta = \frac{c - b}{a},$$

$$product\ of\ roots = (\alpha + \beta)\alpha\beta = -\frac{bc}{a^2}$$

So the required equation is

$$x^2 - (sum\ of\ roots)x + (product\ of\ roots) = 0 \Rightarrow x^2 - \frac{c-b}{a}x - \frac{bc}{a^2} = 0$$

$$\Rightarrow a^2x^2 + a(b - c)x - bc = 0$$

Then the correct option is B)

15. Correct Matching is: a-P,b-S,c_QR,d-P

a) Roots of $3x^2 + 17 = 0$

To find the type of roots we have to find the value of discriminant of the equation as

$$D = b^2 - 4ac = 0 - 4 \times 3 \times 17 < 0$$

So we can conclude that the roots are Imaginary.

Then the matching option is P)

b) Roots of $25x^2 - 30x + 9 = 0$

To find the type of roots we have to find the value of discriminant of the equation as

$$D = b^2 - 4ac = 900 - 4 \times 25 \times 9 = 0$$

So we can conclude that the roots are Real and Equal.

Then the matching option is S)

c) Roots of $x^2 = 16x$

To find the type of roots we have to find the value of discriminant of the equation as

$$D = b^2 - 4ac = 16^2 > 0$$

So we can conclude that the roots are Real, distinct, Rational.

Then the matching option is Q), R)

d) Roots of $x^2 + 5x + 12 = 0$

To find the type of roots we have to find the value of discriminant of the equation as

$$D = b^2 - 4ac = 25 - 48 < 0$$

So we can conclude that the roots are Imaginary.

Then the matching option is P)

16. Correct Answer is: (1)

Given that sum of two numbers is 27 and product of these numbers are 182.

Let's the first number is x then second number is $27 - x$

We can write

$$x(27 - x) = 182 \Rightarrow x^2 - 27x + 182 = 0$$

Using quadratic formula for x

$$x = \frac{27 \pm \sqrt{1}}{2} = 14, 13$$

For $x = 14 => second\ number = 13, x = 13 => seocond\ number\ is = 14$
Then absolute difference is 1.

17. Correct Answer is: (3)

Given equation

$$(5x^2 - 4x + 38) + k(-4x^2 - 2x - 1) = 0$$
$$=> (5 - 4k)x^2 + (-4 - 2k)x + (38 - k) = 0$$

Given that product of the roots is -5
We can write using the equation as

$$product\ of\ root = -5 = \frac{38 - k}{5 - 4k}$$

Solving for k

$$-25 + 20k = 38 - k => 21k = 63 => k = 3$$

So value of k is 3.

COMPLEX NUMBERS
LEVEL- I

1. Correct option: (B)

We are given that i^{243}
Using the complex number $i^2 = -1, i^3 = -i$
So we can write

$$i^{243} = (i^3)^{81} = (-i)^{81} = -i \times (i^2)^{40} = -i$$

Then the correct option is B)

2. Correct option: (B)

We are given that

$$i^{57} + \frac{1}{i^{25}}$$

Using the complex number $i^2 = -1, i^3 = -i$
So we can write

$$i^{57} + \frac{1}{i^{25}} = (i^{56})i + \frac{i^{25}}{i^{50}} = i + i(i^{24}) = i + i = 2i$$

Then the correct option is B)

3. Correct option: (B)

We are given that $(3 + 4i)(2 + 3i)$
We have to find the conjugate complex of the given complex number

$$(3 + 4i)(2 + 3i) = 6 + 8i + 9i - 12 = 17i - 6$$

We know that conjugate complex of a number is

$$C(a + ib) = a - ib$$

Therefore the conjugate complex of the given number is $-17i - 6$
Then the correct option is B)

4. Correct option: (A)

Multiplicative inverse of the

$$M\left(\frac{50}{(3 - 4i)}\right) = x + iy$$

We have to find the value of x for above condition

$$M\left(\frac{50}{(3 - 4i)}\right) = M\left(\frac{50}{25}(3 + 4i)\right) = M(6 + 8i)$$

We know that multiplicative inverse of a complex number is

$$M(a+ib)=\frac{a-ib}{a^2+b^2}=\frac{C(a+ib)}{a^2+b^2}$$

Therefore we can write

$$M\left(\frac{50}{3-4i}\right)=\frac{6-8i}{100}=\frac{3-4i}{50}=x+iy$$

So value of x is $\frac{3}{50}$

Then the correct option is A)

5. Correct option: (A)

We have to find the value of

$$\frac{i^{4n+1}-i^{4n-1}}{2}$$

So we can write

$$i^{4n+1}-i^{4n-1}=i(i^{4n})-\frac{i^{4n}}{i}=>i-\frac{1}{i}=i+i=2i$$

Therefore

$$\frac{i^{4n+1}+i^{4n-1}}{2}=\frac{2i}{2}=i$$

Then the correct option is A)

6. Correct option: (C)

We are given that

$$\frac{(3-4i)}{2i}$$

We have to find the conjugate complex of the given complex number

$$\frac{(3-4i)}{2i}=\left(\frac{3-4i}{-2}\right)i=\frac{3i-4i^2}{-2}=\frac{-4-3i}{2}$$

We know that conjugate complex of a number is

$$C(a+ib)=a-ib$$

Therefore the conjugate complex of the given number is $\frac{-4+3i}{2}$

Then the correct option is C)

7. Correct option: (A)

We are given that

$$\frac{7-24i}{3+4i}$$

We have to find the modulus of the given complex number

$$\frac{7-24i}{3+4i}=\frac{(7-24i)(3-4i)}{(3+4i)(3-4i)}=\frac{(7-24i)(3-4i)}{9+16}=\frac{21-72i-28i-96}{25}$$

$$\frac{-75-100i}{25}=-3-4i$$

Therefore

$$\left|\frac{7-24i}{3+4i}\right|=|-3-4i|=\sqrt{9+16}=5$$

Then the correct option is A)

8. Correct option: (B)

We are given that

$$\frac{1}{1-\cos\theta+i\sin\theta}$$

We have to find the modulus of the given complex number

$$\frac{1}{1-\cos\theta+i\sin\theta}=\frac{1-\cos\theta-i\sin\theta}{(1-\cos\theta+i\sin\theta)(1-\cos\theta-i\sin\theta)}$$

$$=\frac{1-\cos\theta-i\sin\theta}{(1-\cos\theta)^2+\sin^2\theta}=\frac{1-\cos\theta-i\sin\theta}{1+\cos^2\theta+\sin^2\theta-2\cos\theta}=\frac{1-\cos\theta-i\sin\theta}{2-2\cos\theta}=\frac{1-\cos\theta-i\sin\theta}{2(1-\cos\theta)}$$

$$=\frac{1}{2}-\frac{i\sin\theta}{2(1-\cos\theta)}$$

Therefore

$$\left|\frac{1}{2}-\frac{i\sin\theta}{2(1-\cos\theta)}\right|=\sqrt{\left(\frac{1}{2}\right)^2+\left(\frac{\sin\theta}{2(1-\cos\theta)}\right)^2}=\frac{1}{2}\sqrt{1+\frac{\sin^2\theta}{\left((1-\cos\theta)\right)^2}}=\frac{1}{2}$$

$$=\frac{1}{2}\sqrt{\frac{1+\cos^2\theta-2\cos\theta+\sin^2\theta}{(1-\cos\theta)^2}}=\frac{1}{2}\sqrt{\frac{2-2\cos\theta}{(1-\cos\theta)^2}}=\frac{1}{2}\sqrt{\frac{2(1-\cos\theta)}{(1-\cos\theta)^2}}=\frac{1}{2}\sqrt{\frac{2}{1-\cos\theta}}$$

$$=\frac{1}{2}\sqrt{\frac{2}{2\sin^2\frac{\theta}{2}}}=\frac{\cos ec\theta}{2}$$

Then the correct option is B)

9. Correct option: (B)

We are given that $(-11-60i)$

We have to find the square root of the given complex number

So let's take the square root of complex number as

$$x+iy=\sqrt{-11-60i}$$

Squaring both of the side we get

$$(x+iy)^2=(-11-60i)$$

$$=>x^2-y^2+2xyi=-11-60i$$

Comparing complex numbers

$$x^2-y^2=-11, 2xy=-60$$

Solving both of the equation give us the values of the x, y which can be used to find the square root.

We can easily get the square roots of the given complex number using the direct method as

$$\sqrt{z}=\sqrt{r}\frac{(z+r)}{|z+r|}, r=|z|$$

Putting the values as

$$z=-11-60i, r=\sqrt{121+3600}=\sqrt{3721}=61$$

So

$$\sqrt{z}=\pm\sqrt{61}\frac{(50-60i)}{|50-60i|}=\pm\sqrt{61}\frac{(50-60i)}{\sqrt{6100}}=\pm\frac{(50-60i)}{10}=\pm(5-6i)$$

Then the correct option is B)

10. Correct option: (B)

We have to find the value of given expression in simple form $\sqrt{i}+\sqrt{-i}$

We can write

$$\sqrt{i}=\pm\frac{1(i+1)}{\sqrt{2}}, \sqrt{-i}=\pm\frac{1(-i+1)}{\sqrt{2}}$$

So

$$=\pm\frac{(i+1)}{\sqrt{2}}\pm\frac{(-i+1)}{\sqrt{2}}$$

Then the correct option is B)

11. Correct option: (C)

Given $x=1+2i$

We have to find the value of $x^3 + 7x^2 - x + 16$

We can write

$$(1 + 2i)^3 = x^3 => 1 + 8i^3 + 6i + 12i^2 = 1 - 12 - 8i + 6i = -11 - 2i$$

$$x^2 = (1 + 2i)^2 = 1 + 4i^2 + 4i = -3 + 4i$$

So our equation after putting the value of x

$$-11 - 2i - 21 + 28 - 1 - 2i + 16 = 0$$

$$=> (-17 + 24i)$$

Then the correct option is C)

12. Correct option: (D)

Given $3 + i$ is one of the roots of the given equation

$$x^2 + ax + b = 0$$

We have to find the value of a, b

We also know that for a quadratic equation if one root is a complex number then other root is also complex number.

So other root is $3 - i$

Using the roots of the equation we can write

$$sum\ of\ the\ roots = -a = (3 + i) + (3 - i) = 6 => a = -6$$

$$product\ of\ the\ roots = (3 + i)(3 - i) = 9 - i^2 = 10 = b$$

So value of $a = -6\,, b = 10$

Then the correct option is D)

13. Correct option: (A)

Given that

$$(1 + i)z = (1 - i)\bar{z}$$

We have to find the value of z

So we can write

$$z = a + ib$$

So our equation become

$$(1 + i)(a + ib) = (1 - i)(a - ib)$$

$$=> (a + ia + ib + i^2 b = a - ia - ib + i^2 b$$

$$=> a + b = -a - b => a = -b$$

So we can write

$$z = a - ia = a(1 - i), a \in R$$

Then the correct option is A)

14. Correct option: (C)

Given that $1, \omega, \omega^2$ is cube root of unity

We have to find the value of

$$(1 - \omega + \omega^2)(1 - \omega^2 + \omega^4)(1 - \omega^4 + \omega^8)(1 - \omega^8 + \omega^{16})$$

We can simplify the given equation using $\omega^3 = 1$ as

$$(1 - \omega + \omega^2)(1 - \omega^2 + \omega^4)(1 - \omega^4 + \omega^8)(1 - \omega^8 + \omega^{16})$$

$$=> (1 - \omega + \omega^2)(1 - \omega^2 + \omega)(1 - \omega + \omega^2)(1 - \omega^2 + \omega)$$

Using sum of roots we can write

$$=> (-2\omega)^2(-2\omega^2)^2 = 16\omega^6 = 16$$

Then the correct option is C)

15. Correct option: (B)

Given that $1, \omega, \omega^2$ is cube root of unity

We have to find the value of

$$(x + y + z)(x + y\omega + z\omega^2)(x + y\omega^2 + z\omega)$$

We can simplify the given equation using $\omega^3 = 1, 1 + \omega + \omega^2 = 0$ as

$$(x+y+z)(x+y\omega+z\omega^2)(x+y\omega^2+z\omega)$$
$$=(x+y+z)(x+xy\omega+xz\omega^2+xy\omega^2+y\omega y\omega^2+z\omega^2 y\omega^2+xz\omega+y\omega z\omega+z\omega^2 z\omega)$$
$$=>(x+y+z)(x^2+y^2+z^2-xy-zy-zx)$$
$$=>(x^3+y^3+z^3-3xyz)$$

Therefore

$$(x+y+z)(x+y\omega+z\omega^2)(x+y\omega^2+z\omega)=(x^3+y^3+z^3-3xyz)$$

Then the correct option is B)

16. Correct option: (B)

Given that $1,\omega,\omega^2$ is cube root of unity

We have to find the value of

$$\frac{a+b\omega+c\omega^2}{a\omega+b\omega^2+c}$$

We can simplify the given equation using $\omega^3=1, 1+\omega+\omega^2=0$ as

$$\frac{a+b\omega+c\omega^2}{a\omega+b\omega^2+c}=\frac{1}{\omega}\times\frac{a\omega+b\omega^2+c}{a\omega+b\omega^2+c}=\frac{\omega^2}{\omega^3}=\omega$$

Therefore

$$\frac{a+b\omega+c\omega^2}{a\omega+b\omega^2+c}=\omega$$

Then the correct option is B)

17. Correct option: (C)

We have to find the value of

$$\left(\frac{-1+i\sqrt{3}}{2}\right)^{17}+\left(\frac{-1-i\sqrt{3}}{2}\right)^{17}$$

We can write

$$\frac{-1+i\sqrt{3}}{2}=\omega,\frac{-1-i\sqrt{3}}{2}=\omega^2,\omega^3=1,1+\omega+\omega^2=0$$

Therefore we have to find the value of $\omega^{17}+\omega^{34}=\omega^2+\omega$

$$\omega^2+\omega=-1$$
$$\left(\frac{-1+i\sqrt{3}}{2}\right)^{17}+\left(\frac{-1-i\sqrt{3}}{2}\right)^{17}=-1$$

Then the correct option is C)

18. Correct option: (C)

Given that

$$x=a+b, y=a\omega+b\omega^2, z=a\omega^2+b\omega$$

We have to find the value of xyz

Putting the values

$$xyz=(a+b)(a\omega+b\omega^2)(a\omega^2+b\omega)=(a+b)(a+ab\omega+ab\omega^2+b)$$
$$=>(a+b)(a^2+b^2-ab)=(a^3+b^3)$$

Then the correct option is C)

19. Correct option: (B)

We are given that

$$\left(\frac{3+i\sqrt{3}}{2}\right)^{50}=3^{25}(x+iy)$$

We have to find the value of x and y for above equation

We can write

$$\left(\frac{3+i\sqrt{3}}{2}\right)^{50}=\left(\sqrt{3}\right)^{50}(x+iy)=>\left(\frac{3+i\sqrt{3}}{2\sqrt{3}}\right)^{50}=x+iy$$

$$=> \left(\frac{\sqrt{3}+i}{2}\right)^{50} = x+iy = \left(\frac{(-i^2)\sqrt{3}+i}{2}\right)^{50} = \left((-i)\frac{i\sqrt{3}-1}{2}\right)^{50}$$

$$=> (-i)^{50}(\omega)^{50} = -\omega^2 = -\left(\frac{-i\sqrt{3}-1}{2}\right) = \frac{i\sqrt{3}+1}{2}$$

Comparing the complex number

$$x+iy = \frac{i\sqrt{3}+1}{2} => x = \frac{1}{2}, y = \frac{\sqrt{3}}{2}$$

Then the correct option is B)

20. Correct option: (C)

We are given that

$$(x+iy) = \left(1-i\sqrt{3}\right)^{100}$$

We have to find the value of x and y for above equation

We can write

$$(x+iy) = \left(1-i\sqrt{3}\right)^{100} = \frac{2^{100}\left(1-i\sqrt{3}\right)^{100}}{2^{100}}$$

$$=> 2^{100}\frac{\left(1-i\sqrt{3}\right)^{100}}{2^{100}} = 2^{100}\left(\frac{1-i\sqrt{3}}{2}\right)^{100}$$

$$=> 2^{100}(-\omega)^{100} = 2^{100}\omega = 2^{99}\left(-1+i\sqrt{3}\right) = -2^{99}+2^{99}\sqrt{3}i$$

Comparing the complex number

$$x+iy == -2^{99}+2^{99}\sqrt{3}i => x = -2^{99}, y = 2^{99}\sqrt{3}$$

Then the correct option is C)

LEVEL - II

1. Correct option: (A)

We are given that

$$(1+i)^6+(1-i)^6$$

We can write

$$((1+i)^2)^3+((1-i)^2)^3$$

$$=> (2i)^3+(-2i)^3 = 0$$

Then the correct option is A)

2. Correct option: (B)

We are given that

$$\frac{i^{592}+i^{590}+i^{588}+i^{586}+i^{584}}{i^{582}+i^{580}+i^{576}+i^{574}+i^{578}}$$

We can write

$$\frac{i^{10}(i^{582}+i^{580}+i^{576}+i^{574}+i^{578})}{i^{582}+i^{580}+i^{576}+i^{574}+i^{578}} = i^{10} = (i^2)^5 = -1$$

Then the correct option is B)

3. Correct option: (D)

We have to find the value of the given

$$i^2+i^4+i^6+\cdots+i^{2(2n+1)}$$

Putting the values

$$-1+1+(-1)+\cdots+(-1) = -1$$

Then the correct option is D)

4. Correct option: (A)

We have to find the conjugate complex of the given complex number

$$=> -\frac{5i}{7+i}$$

$$=> -\frac{5i}{7+i} = -\frac{(5i)(7-i)}{7^2-i^2} = \frac{i^2-7i}{10} = \frac{-1-7i}{10}$$

So we have to find the conjugate complex of the modified complex number

$$\bar{z} = \frac{-1+7i}{10}$$

Then the correct option is A)

5. Correct option: (A)

We have to find the conjugate complex of the given complex number

$$=> \frac{2-i}{(1-2i)^2}$$

$$=> \frac{2-i}{(1-2i)^2} = \frac{2-i}{1+4i^2-4i} = \frac{2-i}{-3-4i} = \frac{(2-i)(-3+4i)}{9-16i^2} = \frac{-2+11i}{25}$$

So we have to find the conjugate complex of the modified complex number

$$\bar{z} = \frac{-2-11i}{25}$$

Then the correct option is A)

6. Correct option: (D)

We have to find the value of x for which given complex number will be conjugate each other.

$$\sin x + i\cos 2x\,, \cos x - i\sin 2x$$

We can write that

$$\sin x = \cos x \;\; and \;\; \cos 2x = \sin 2x$$

$$\sin x = \cos x => x = \frac{\pi}{4}, \frac{5\pi}{4} \ldots$$

$$\cos 2x = \sin 2x => x = \frac{\pi}{8}, \frac{5\pi}{8} \ldots$$

So the value for x

$$x = \frac{\pi}{4}, \frac{5\pi}{4} \ldots \cap x = \frac{\pi}{8}, \frac{5\pi}{8} \ldots = (x \in \emptyset)$$

So no value of x satisfies the above equation.

Then the correct option is D)

7. Correct option: (C)

We have to find the value of $x^2 + y$ for which given complex number will be conjugate each other.

$$3 + ix^2y, x^2 + y + 4i$$

We can write that

$$3 = x^2 + y$$

Then the correct option is C)

8. Correct option: (A)

We have to find the value of

$$\frac{1}{2+\omega} - \frac{1}{1+\omega} + \frac{1}{1+2\omega}$$

Where ω is cube root of unity.

So we can write

$$\frac{1}{1+1+\omega} - \frac{1}{1+\omega} + \frac{1}{1+2\omega} = \frac{1}{1-\omega^2} - \frac{1}{1+\omega} + \frac{1}{1+2\omega} = \frac{\omega}{1-\omega^2} + \frac{1}{1+2\omega}$$

$$\frac{2(1+\omega+\omega^2)}{(1-\omega^2)(1+2\omega)} = 0$$

Then the correct option is A)

9. Correct option: (A)

We have to find the value of

$$(1-2\omega+\omega^2)^6$$

So we can write

$$(1+\omega^2-2\omega)=(-\omega-2\omega)=-3\omega$$

Therefore

$$(-3\omega)^6=3^6\omega^6=3^6=729$$

Then the correct option is A)

10. Correct option: (A)

We have to find the value of x for which given complex number will be conjugate each other.

$$\frac{2-i}{(1-2i)^2},-\frac{x+11i}{25}$$

We can write that

$$\frac{2-i}{(1-2i)^2}=\frac{2-i}{1+4i^2-4i}=\frac{2-i}{-3-4i}=-\left(\frac{2-i}{3+4i}\right)\frac{3-4i}{3-4i}=-\left(\frac{6-3i-8i+4i^2}{25}\right)=-\frac{2-11i}{25}$$

Therefore

$$-\frac{2-11i}{25}=-\frac{x+11i}{25}=>x=2$$

Then the correct option is A)

11. Correct option: (C, D)

We have to find the conjugate complex of

$$\frac{7+i}{2-i}$$

So we can write

$$\frac{7+i}{2-i}=\frac{7+i}{2-i}\times\frac{2+i}{2+i}=\frac{14+2i+7i+i^2}{4+1}=\frac{13+9i}{5}$$

Therefore

$$\bar{z}=\frac{13-9i}{5}$$

Then the correct option is C, D)

12. Correct option: (A, B, C, D)

We have to find the modulus of complex number which is equal to modulus of $(-5,12)$

Modulus of the $(-5,12)=13$

Checking the options

A) $5i-12$

$$|5i-12|=\sqrt{25+144}=13$$

Option A) is correct.

B) $-5i-12$

$$|-5i-12|=\sqrt{25+144}=13$$

Option B) is correct.

C) $5+12i$

$$|5+12i|=\sqrt{25+144}=13$$

Option C) is correct.

D) $5-12i$

$$|5-12i|=\sqrt{25+144}=13$$

Option D) is correct.

Then the correct option is A, B, C, D)

13. Correct option: (*)

We have to find the value of

$$(1+\omega-\omega^2)^5+(1-\omega+\omega^2)^5$$

We can write

$$(1+\omega-\omega^2)=(-\omega^2-\omega^2)=-2\omega^2$$
$$(1-\omega+\omega^2)=(-\omega-\omega)=-2\omega$$

So we can write

$$(-2\omega^2)^5+(-2\omega)^5=-2^5(\omega^{10}+\omega^5)=-2^5(\omega+\omega^2)=-2^5(-1)=32$$

14. Correct option: (C)

We have to find the value of $(x+iy)(x-iy)$

Where $|x|+|y|\neq 0$

Therefore we can write

$$(x+iy)(x-iy)=x^2-i^2y^2=x^2+y^2$$

Then the correct option is C)

15. Correct option: (C)

We are given that $(a-ib)^2=x+iy$

We have to find the value of $x-y$

So we can write $a^2-b^2=x, y=-2ab$

Therefore

$$x-y=a^2-b^2-2ab=\left(a+b+\sqrt{2}b\right)(a+b-\sqrt{2}b)$$

Then the correct option is C)

16. Correct option: (A)

We have to find the value of additive inverse of

$$\frac{1+2i}{1+i}=\frac{1+2i}{1+i}\times\frac{1-i}{1-i}=\frac{1+2i-i-2i^2}{2}=\frac{3+i}{2}$$

So additive inverse for

$$\frac{1+2i}{1+i}$$

Will be equal to

$$\frac{-3-i}{2}$$

Then the correct option is A)

17. Correct Matching is: a-R, b-Q, c-S, d-P

a)

We have to find the value of

$$\frac{1}{i-1}+\frac{1}{i+1}$$

So we can write

$$\frac{(i+1)+(i-1)}{i^2-1}=\frac{2i}{-2}=-i$$

a) --→(R)

We have to find the value of

$$1+i^2+\cdots.+i^{2n}$$

So we can write

$$1+(-1)+(1)+(-1)+\cdots(1)=1$$

b) --→(Q)

We have to find the value of

$$\frac{i^3+i^6+i^9}{i^2+i^4-i^6}$$

So we can write

$$\frac{i^3+i^6+i^9}{i^2+i^4-i^6}=\frac{-i+i^2+i}{-1+1-i^2}=\frac{-1}{-i^2}=-1$$

c) --→(S)

We have to find the value of

$$i+i^2+i^3+i^4$$

So we can write

$$i+(-1)+(-i)+1=0$$

d) --→(P)

18. Correct Answer is: (0)

We have to find the value of $1+i^2+i^4+i^6$

So we can write as

$$1+(-1)+1+(-1)=0$$

Therefore the value of above expression after putting the real values is 0.

19. Correct Answer is: (5)

We have to find the value of $x^2-4x+18$

If we are given that $x=2-3i$

So we can write as

$$x^2-4x+18=(2-3i)^2-4(2-3i)+18=4+9i^2-12i-8+12i+18$$
$$=5$$

Therefore the value of the given equation is equal to 5.

20. Correct Answer is: (1)

We have to find the value of

$$(1+\omega)(1+\omega^2)(1+\omega^4)(1+\omega^8)\ldots(2n)factors$$

So we can write as

$$(1+\omega)(1+\omega^2)(1+\omega^4)(1+\omega^8)\ldots(1+\omega^{2n})=(-\omega^2)(-\omega)(-\omega^2)\ldots.2n\ factor$$

Therefore we can write

$$=(\omega)^{2n+n}=\omega^{3n}=1$$

So value of above expression is 1.

STUDENT PRACTICE QUESTIONS

1. Correct option: (C)

We have to find the value of

$$(1+i)^4+(1-i)^4$$

So we can write

$$((1+i)^2)^2+((1-i)^2)^2=(1+i^2+2i)^2+(1+i^2-2i)^2$$
$$=>4i^2+4i^2=-8$$

Then the correct option is C)

2. Correct option: (A)

We are given that

$$z=5+3\sqrt{2}i$$

We have to find the value of $z\bar{z}$

Then we can write $\bar{z}=5-3\sqrt{2}i$

Therefore the value of

$$z\bar{z}=\left(5+3\sqrt{2}i\right)\left(5-3\sqrt{2}i\right)=\left(25-\left(3\sqrt{2}i\right)^2\right)=25+18=43$$

Then the correct option is A)

3. Correct option: (A)

We are given that

$$\left|\frac{\left(3-i\sqrt{2}\right)^2}{1+2i}\right|$$

We have to find the modulus of the given complex number

$$\left|\frac{\left(3-i\sqrt{2}\right)^2}{1+2i}\right| = \left|\frac{\left(9-2-6\sqrt{2}i\right)(1-2i)}{1+4}\right|$$

$$\left|\frac{\left(7-6\sqrt{2}i\right)(1-2i)}{5}\right| = \left|\frac{7-6\sqrt{2}i-14i-12\sqrt{2}}{5}\right| = \left|\frac{7-12\sqrt{2}-(6\sqrt{2}+14)i}{5}\right|$$

Therefore

$$\left|\frac{\left(3-i\sqrt{2}\right)^2}{1+2i}\right| = \left|\frac{7-12\sqrt{2}-(6\sqrt{2}+14)i}{5}\right|$$

$$= \sqrt{\frac{1}{25}\left(49+288-168\sqrt{2}+72+196+168\sqrt{2}\right)}$$

$$= \sqrt{\frac{1}{25}(121+484)} = \sqrt{\frac{605}{25}} = \sqrt{\frac{121}{5}} = \frac{11}{\sqrt{5}}$$

Then the correct option is A)

4. Correct option: (A)

We are given that

$$\frac{1}{(1-i)^2} - \frac{1}{(1+i)^2}$$

We have to find the modulus of the given complex number

$$\frac{1}{(1-i)^2} - \frac{1}{(1+i)^2} = \frac{(1+i)^2-(1-i)^2}{(1-i)^2(1+i)^2}$$

$$\frac{2i-(-2i)}{(2i)(-2i)} = \frac{4i}{4} = i$$

Therefore

$$\left|\frac{1}{(1-i)^2} - \frac{1}{(1+i)^2}\right| = |i|$$

$$= 1$$

Then the correct option is A)

5. Correct option: (C)

We have to find the value of given expression in simple form

$$\sqrt{x^2+2ix-1}$$

We can write

$$x^2+2ix-1 = (x^2+2ix+i^2) = (x+i)^2$$

So $$\sqrt{x^2+2ix-1} = \sqrt{(x+i)^2} = x+i$$

Then the correct option is C)

6. Correct option: (A)

We have to find the value of following complex number

$$\frac{1}{2i}\left[\sqrt{-3+4i} - \sqrt{-3-4i}\,\right]$$

We can easily get the square roots of the given complex number using the direct method as

$$\sqrt{z} = \sqrt{r}\frac{(z+r)}{|z+r|}, r = |z|$$

Putting the values for $-3+4i$

$$z = -3 + 4i, r = \sqrt{25} = 5$$

So

$$\sqrt{z} = \sqrt{5}\frac{(2+4i)}{|2+4i|} = \frac{(2+4i)}{2} = (1+2i)$$

Same as

$$\sqrt{z} = \sqrt{(-3-4i)} = 1 - 2i$$

So we can write

$$\frac{1}{2i}\left[\sqrt{-3+4i} - \sqrt{-3-4i}\,\right] = \frac{1}{2i}[1+2i-(1-2i)] = \frac{1}{2i}[4i] = 2$$

Then the correct option is A)

7. Correct option: (C)

Given that ω is cube root of unity

We have to find the value of $(1+\omega-\omega^2)^5 + (1-\omega+\omega^2)^5$

For equation

$$x^3 = 1 \Rightarrow x^3 - 1 = 0$$

We can write sum of the roots = 0

So $1 + \omega + \omega^2 = 0 \Rightarrow 1 + \omega = -\omega^2, 1 + \omega^2 = -\omega$

Putting the values

$$1 + \omega - \omega^2 = -2\omega^2, 1 - \omega + \omega^2 = -2\omega$$

Substituting the values

$$(1+\omega-\omega^2)^5 + (1-\omega+\omega^2)^5 = (-2\omega^2)^5 + (-2\omega)^5 = -32(\omega^{10} + \omega^5) = -32(\omega^9\omega + \omega^3\omega^2)$$
$$= -32(\omega + \omega^2) = -32(-1) = 32$$

Then the correct option is C)

8. Correct option: (C)

Given that $1, \omega, \omega^2$ is cube root of unity

We have to find the value of $(1+\omega)(1+\omega^2)(1+\omega^4)\ldots(1+\omega^{2n})$

We can simplify the given equation using $\omega^3 = 1, 1 + \omega + \omega^2 = 0$ as

$$(1+\omega)(1+\omega^2)(1+\omega^4)\ldots(1+\omega^{2n})$$
$$\Rightarrow (1+\omega)(1+\omega^2)(1+\omega)(1+\omega^2)\ldots(1+\omega)(1+\omega^2)$$
$$\Rightarrow (1+\omega)^n(1+\omega^2)^n = (-\omega^2)^n(-\omega)^n = (-\omega^3)^{2n} = 1$$

Therefore

$$(1+\omega)(1+\omega^2)(1+\omega^4)\ldots(1+\omega^{2n}) = 1$$

Then the correct option is C)

9. Correct option: (C)

Given that $1, \omega, \omega^2$ is cube root of unity

We have to find the value of $(2-\omega)(2-\omega^2)(2-\omega^{10})(2-\omega^{11})$

We can simplify the given equation using $\omega^3 = 1, 1 + \omega + \omega^2 = 0$ as

$$(2-\omega)(2-\omega^2)(2-\omega)(2-\omega^2) = (2-\omega)^2(2-\omega^2)^2$$
$$\Rightarrow (2-\omega)^2(2-\omega^2)^2 = (4+\omega^2-4\omega)(4+\omega^4-4\omega^2)$$
$$\Rightarrow (4-1-\omega-4\omega)(4-1-\omega^2-4\omega^2) = (3-5\omega)(3-5\omega^2)$$
$$\Rightarrow 9 - 15\omega - 15\omega^2 + 25 = 9 - 15(\omega+\omega^2) + 25 = 34 + 15 = 49$$

Therefore

$$(2-\omega)(2-\omega^2)(2-\omega^{10})(2-\omega^{11}) = 49$$

Then the correct option is C)

10. Correct option: (B)

We have to find the value of $\left(\frac{-1+i\sqrt{3}}{2}\right)^{-8} + \left(\frac{-1-i\sqrt{3}}{2}\right)^{-8}$

We can write $\frac{-1+i\sqrt{3}}{2} = \omega, \frac{-1-i\sqrt{3}}{2} = \omega^2, \omega^3 = 1, 1 + \omega + \omega^2 = 0$

Therefore we have to find the value of $\omega^{-8} + \omega^{-16}$

$$= \frac{\omega^9}{\omega^8} + \frac{\omega^{18}}{\omega^{16}} = \omega + \omega^2$$

$$\omega^2 + \omega = -1$$

$$\left(\frac{-1 + i\sqrt{3}}{2}\right)^{-8} + \left(\frac{-1 - i\sqrt{3}}{2}\right)^{-8} = -1$$

Then the correct option is B)

11. Correct option: (A)

We have to find the square root of $4ab - 2i(a^2 - b^2)$

So we can write

$$(x + iy)^2 = 4ab - 2i(a^2 - b^2)$$

Comparing the real and imaginary terms

$$x^2 - y^2 = 4ab$$

$$xy = a^2 - b^2$$

Comparing the magnitude

$$x^2 + y^2 = 2a^2 + 2b^2$$

Solving both of the equations

$$x^2 = a^2 + b^2 + 2ab, y^2 = a^2 + b^2 - 2ab$$

So only valid answer

$$\pm((a + b) - i(a - b))$$

Then the correct option is A)

12. Correct option: (A, C)

We have to find the value of x, y for equation

$$\left(\frac{1}{3}x - \frac{1}{4}y\right) + \frac{3}{4}yi = -3 + 5i$$

So we can write

$$\frac{3}{4}y = 5 \Rightarrow y = \frac{20}{3}$$

Therefore

$$\frac{x}{3} - \frac{y}{4} = -3 \Rightarrow x = -4$$

Then the correct option is A, C)

13. Correct option: (A)

We are given that

$$\left(\frac{1 + i}{1 - i}\right)^n = 1$$

We have to find the minimum positive integer value of n

So we can write

$$\left(\frac{1 + i}{1 - i}\right) = \frac{(1 + i)}{1 - i} \times \frac{1 + i}{1 + i} = \frac{(1 + i)^2}{2} = i$$

Therefore we can write

$$i^n = 1 \Rightarrow n = 4$$

Then the correct option is A)

14. Correct option: (C)

We are given that $z = \frac{\sqrt{3}+i}{2}$

We can write $(z^{101} + i^{103})^{105}$

We can use the cube root of unity as

$$z = \frac{\sqrt{3}(-i^2) + i}{2} = -\frac{i(-1 + \sqrt{3}i)}{2} = -i\omega$$

So we can write $(z^{101} + i^{103})^{105} = ((-i\omega)^{101} + i^{103})^{105} = (-i^{101}\omega^{101} + i^{103})^{105}$

$$=> (-i\omega^2 - i)^{105} = (-i)^{105}\omega^{105} = -i$$

So we can write $(z^{101} + i^{103})^{105} = -i = (z)^3 = (-i\omega)^3 = -i$

Then the correct option is C)

15. Correct option: (A)

We have to find the value of $\omega^5 + \omega^4 + 1$

So we can write as$\omega^3 = 1$ $\omega^2 + \omega + 1 = 0$

Then the correct option is A)

16. Correct Matching is: a-S, b-P, c-T, d-Q

a)

The multiplicative inverse of $(3 + 2i)^2$

Let 's find $(3 + 2i)^2 = 9 + 4i^2 + 12i = 5 + 12i$

So multiplicative inverse will be $\frac{5-12i}{169}$

a) ---→(S)

We have to find the value of $(3 + 2i)^3$

So we can write

$$(3 + 2i)^3 = 3^3 + (2i)^3 + 18i(3 + 2i) = 27 - 8i + 54i - 36 = -9 + 16i$$

b) ---→(P)

We have to find the value of

We are given that $\left(\frac{1+i}{1-i}\right)^n = 1$

We have to find the minimum positive integer value of n

So we can write $\left(\frac{1+i}{1-i}\right) = \frac{(1+i)}{1-i} \times \frac{1+i}{1+i} = \frac{(1+i)^2}{2} = i$

Therefore we can write $i^n = 1 => n = 4$

c) ---→(T)

We have to find the value of $\sqrt{i} + \sqrt{-i}$

So we can write

$$\sqrt{i} = \pm\frac{1}{\sqrt{2}}(1 + i), \sqrt{-i} = \pm\frac{1}{\sqrt{2}}(1 - i)$$

$$\sqrt{i} + \sqrt{-i} = \pm\frac{1}{\sqrt{2}}(1 + i) + \pm\frac{1}{\sqrt{2}}(1 - i) = \pm\sqrt{2}$$

d) ---→(Q)

17. Correct Answer is: (5)

We have to find the value of $mod(3 - 4i)$

So we can write as $|3 - 4i| = \sqrt{3^2 + 4^2} = 5$

Therefore the modulus of the given complex number is equal to 5.

18. Correct Answer is: (-1)

We have to find the value of

$$\left(\left(\frac{(-1 + i\sqrt{3})}{2}\right)^n + \left(\frac{(-1 - i\sqrt{3})}{2}\right)^n\right)$$

So we can write as

$$\omega = \frac{(-1 + i\sqrt{3})}{2}, \omega^2 = \frac{(-1 - i\sqrt{3})}{2}$$

Therefore we can write

$$\left(\frac{(-1 + i\sqrt{3})}{2}\right)^n + \left(\frac{(-1 - i\sqrt{3})}{2}\right)^n = \omega^n + \omega^{2n}$$

For any integer value n the equation is valid

$$1 + \omega^n + \omega^{2n} = 0$$

So we can write $\omega^n + \omega^{2n} = -1$

3. Co-ordinate Geometry

Solutions

Distance and Section Formulae

LEVEL – I

1. Correct option: (D)

Given points $(4, 2), (3, 4)$

The distance between two points can be calculated using the distance formula:

$$Distance = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Putting the values in formula

$$= \sqrt{(3-4)^2 + (4-2)^2} = \sqrt{5}$$

Then the correct option is D)

2. Correct option: (A)

Given points $A = (0, 0), B = (36, 15)$

The distance between two points can be calculated using the distance formula:

$$Distance = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Putting the values in formula

$$= \sqrt{(36)^2 + (15)^2} = \sqrt{1296 + 225} = 39$$

Then the correct option is A)

3. Correct option: (D)

The distance between two points can be calculated using the distance formula:

$$Distance = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Let's calculate the distances between these points:

1. Distance between (2, -2) and (-1, 2):

$$Distance = \sqrt{(-1-2)^2 + (2-(-2))^2}$$
$$Distance = \sqrt{9 + 16}$$
$$Distance = \sqrt{25}$$
$$Distance = 5$$

2. Distance between (2, -2) and (3, 5):

$$Distance = \sqrt{(3-2)^2 + (5-(-2))^2}$$
$$Distance = \sqrt{1 + 49}$$
$$Distance = \sqrt{50}$$

3. Distance between (-1, 2) and (3, 5):

$$Distance = \sqrt{(3-(-1))^2 + (5-2)^2}$$
$$Distance = \sqrt{16+9}$$
$$Distance = \sqrt{25}$$
$$Distance = 5$$

The distances between the points are $5, \sqrt{50} and\ 5$.

Based on these distances, we can conclude that the triangle formed by these points is an isosceles triangle, since two sides have equal lengths.

An isosceles right triangle is characterized by having two sides of equal length, while the sum of the squares of these two sides is equivalent to the square of the remaining side.

Here $5^2 + 5^2 = \sqrt{50}^2$

So the correct option is D)

4. Correct option: (C)

We have one point given as (3, 2) and the ordinate (y-coordinate) of the second point is given as -1. We also know that the distance between these two points is 5.

Given one point as (3, 2) and the y-coordinate of the second point as -1, let's plug in the values and solve for the possible x-coordinates:

Let the other point P = (x, -1)

We have: $Distance = \sqrt{(3-x)^2 + (2+1)^2}$

Simplifying further: $25 = (3-x)^2 + (3)^2 => 9 + (3-x)^2$

$$9 + 9 + x^2 - 6x = 25 => x^2 - 6x = 7$$

$$x^2 - 6x - 7 = 0$$

Solving this using quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-6 \pm \sqrt{6^2 - 4(-7)}}{2} = \frac{-6 \pm 8}{2} = 1, -7$$

Then the correct option is C)

5. Correct option: (D)

Given two opposite vertices A (1, -2) and B (4, 1) of a square, we need to calculate the length of the side of the square.

1. Calculate the distance between the two given points (diagonal length):

$$Distance\ AB = \sqrt{(4-1)^2 + \left(1-(-2)\right)^2}$$

$$Distance\ AB = \sqrt{3^2 + 3^2}$$

$$Distance\ AB = \sqrt{18}$$

$$Distance\ AB = 3\sqrt{2}$$

2. Now, in a square, the diagonal divides the square into two congruent right-angled triangles. The diagonal of the square are hypotenuses of these triangles.

We can write: $Diagonal^2 = Side^2 + Side^2$

$$\left(3\sqrt{2}\right)^2 = 2 \times Side^2$$

$$18 = 2 \times Side^2$$

$$Side^2 = \frac{18}{2}$$

$$Side^2 = 9$$

$$Side = 3$$

So, the correct answer is: D) 3

6. Correct option: (C)

Given points A (5, 3) and B (11, -5), and we are given that angle APB is 90°, meaning triangle APB is a right-angled triangle. We need to find the value of λ for the point P (12, λ).

In a right-angled triangle, the sum of the squares of the lengths of the two shorter sides (legs) is equal to the square of the length of the longest side (hypotenuse).

Let's use this property to solve for λ:

Length of side AB = $\sqrt{(11-5)^2 + (-5-3)^2}$

$$= \sqrt{36 + 64}$$

$$= \sqrt{100}$$

$$= 10$$

Length of side AP = $\sqrt{(12-5)^2 + (\lambda - 3)^2}$

$$= \sqrt{49 + (\lambda - 3)^2}$$

Length of side BP = $\sqrt{(12-11)^2 + \left(\lambda - (-5)\right)^2}$

$$= \sqrt{1 + (\lambda + 5)^2}$$

According to the property: $AP^2 + BP^2 = AB^2$

$$\left(\sqrt{49 + (\lambda - 3)^2}\right)^2 + \left(\sqrt{1 + (\lambda + 5)^2}\right)^2 = 10^2$$

$$49 + (\lambda - 3)^2 + 1 + (\lambda + 5)^2 = 100$$

$$(\lambda - 3)^2 + (\lambda + 5)^2 = 50$$

$$\lambda^2 - 6\lambda + 9 + \lambda^2 + 10\lambda + 25 = 50$$

$$2\lambda^2 + 4\lambda - 16 = 0$$

$$\lambda^2 + 2\lambda - 8 = 0$$

Factor the quadratic equation: $(\lambda + 4)(\lambda - 2) = 0$

This gives us two possible values for λ: $\lambda = -4 \, or \, \lambda = 2$

So, the correct answer is: C) 2 or -4

7. Correct option: (D)

Given points: (2,5), (0,3), (2,1), (4,3)

1. Sides

Calculate the distances between consecutive points:

- Distance between (2, 5) and (0, 3): $\sqrt{(0-2)^2 + (3-5)^2} = \sqrt{8}$
- Distance between (0, 3) and (2, 1): $\sqrt{(2-0)^2 + (1-3)^2} = \sqrt{8}$
- Distance between (2, 1) and (4, 3): $\sqrt{(4-2)^2 + (3-1)^2} = \sqrt{8}$
- Distance between (4, 3) and (2, 5): $\sqrt{(2-4)^2 + (5-3)^2} = \sqrt{8}$

All sides are equal in length (√8), which satisfies the criterion for a square, rhombus.

2. Angles

- Calculate the slopes of consecutive sides:
  - Slope of AB: $\frac{3-5}{0-2} = 1$
  - Slope of BC: $\frac{1-3}{2-0} = -1$
  - Slope of CD: $\frac{3-1}{4-2} = 1$
  - Slope of DA: $\frac{5-3}{4-2} = -1$

Since the product of the slopes of AB and BC is -1, lines AB and BC are perpendicular. Similarly, the product of the slopes of CD and DA is -1, indicating that lines CD and DA are perpendicular.

This confirms that all angles are right angles (90°), which satisfies the criterion for a square.

Based on both the equal sides and the right angles, we can conclude that the points (2, 5), (0, 3), (2, 1), and (4, 3) form a square.

So, the correct answer is indeed: D) Square

8. Correct option: (C)

Given that distance of (4,0) from (a, b) is the double the distance of point (a, b) from the x-axis.

Let's calculate the distances using the formula and then equate them as given. The distance of the point (a, b) from the x-axis can be directly determined, as the distance of a point from the x-axis is equal to its y-coordinate.

Now equation become

$$2b = \sqrt{\left(b - (4)\right)^2 + (b - 0)^2}$$

Solving further $4b^2 = (a-4)^2 + b^2$

$$3b^2 = (a-4)^2$$

So the correct option is C)

9. Correct option: (B)

To find the coordinates of point R, we can use the external division formula:

For a point P(x_1, y_1) dividing the line segment joining Q$(x_2, y_2)$and R(x_3, y_3)externally in the ratio $m:n$, the coordinates of point P can be calculated as:

$$x_1 = \frac{mx_3 - nx_2}{m - n}$$
$$y_2 = \frac{my_3 - ny_2}{m - n}$$

Let point R (x, y)

Given points:

$$Q(-2, -3), R(x, y)$$

Ratio: 1:2 (P divides QR externally)

Using the formula:

$$-5 = \frac{x - 2 \times (-2)}{-1} = -(x + 4) => x = 1$$
$$4 = \frac{y - 2 \times (-3)}{-1} = -(y + 6) => y = -10$$

So, the coordinates of point R are $(1, -10)$.

Then the correct option is B)

10. Correct option: (C)

Given points are

$$A = (x_1, y_1), B = (x_2, y_2), C = (x_1 + t(x_2 - x_1), y_1 + t(y_2 - y_1))$$

We have to find the ration in which point C divides the line AB

Distance from A to C

$$AC = \sqrt{((x_1 + t(x_2 - x_1) - x_1)^2 + (y_1 + t(y_2 - y_1) - y_1)^2}$$
$$= t\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

1. Distance from C to B

$$CB = \sqrt{((x_1 + t(x_2 - x_1) - x_2)^2 + (y_1 + t(y_2 - y_1) - y_2)^2}$$
$$= \sqrt{\left((1 - t)x_1 + (1 - t)x_2\right)^2 + \left((1 - t)y_1 + (1 - t)y_2\right)^2}$$
$$= (1 - t)\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

So the ration in which the point C divide the AB

$$k = \frac{AC}{BC} = \frac{t\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}}{(1 - t)\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}} = \frac{t}{1 - t}$$

Then the correct option is C)

11. Correct option: (C)

For a point P(x_1, y_1) dividing the line segment joining Q$(x_2, y_2)$and R(x_3, y_3) internally in the ratio $m:n$, the coordinates of point P can be calculated as:

$$x_1 = \frac{mx_3 + nx_2}{m + n}$$
$$y_2 = \frac{my_3 + ny_2}{m + n}$$

Value of y co-ordinate equal to zero on the x-axis

Given points

$$P = (-3,5), Q = (5,1), R = (3,2)$$

Putting the values in formulas for division

$$-3 = \frac{3m + 5n}{m + n} => \frac{3 \times \frac{m}{n} + 5}{\frac{m}{n} + 1}$$

$$-3\left(\frac{m}{n}+1\right)=3\times\frac{m}{n}+5 \Rightarrow -6\frac{m}{n}=8$$
$$\frac{m}{n}=-\frac{4}{3}$$

We can say that point P divide the QR in ration of $\frac{4}{3}$ externally

Then the correct option is C)

12. Correct option: (B)

For a point P(x_1, y_1) dividing the line segment joining Q$(x_2, y_2)$and R(x_3, y_3) internally in the ratio $m:n$, the coordinates of point P can be calculated as:

$$x_1 = \frac{mx_3 + nx_2}{m+n}$$
$$y_2 = \frac{my_3 + ny_2}{m+n}$$

Given points

$$P=(x,0), Q=(2,-3), R=(5,7)$$

Putting the values in formulas for division

$$0 = \frac{m\times 7 + n\times(-3)}{m+n} \Rightarrow \frac{7\times\frac{m}{n}-3}{\frac{m}{n}+1}$$
$$0=7\times\frac{m}{n}-3 \Rightarrow \frac{m}{n}=\frac{3}{7}$$

We can say that point x axis divide the QR in ration of $\frac{3}{7}$ internally.

Then the correct option is B)

13. Correct option: (B)

For a point P(x_1, y_1) dividing the line segment joining Q$(x_2, y_2)$and R(x_3, y_3) internally in the ratio $m:n$, the coordinates of point P can be calculated as:

$$x_1 = \frac{mx_3 + nx_2}{m+n}$$
$$y_2 = \frac{my_3 + ny_2}{m+n}$$

Let the point P (x, y)

Given points

$$P=(x,y), Q=(5,7), R=(-1,3)$$

Given ration is $\frac{2}{-3}=\frac{m}{n} \Rightarrow m=2, n=-3$

Putting the values in formulas for co-ordinate of P

$$x = \frac{2*(-1)+(-3)*5}{2-3} = -(-17) = 17$$
$$y = \frac{2*3+(-3)*7}{2-3} = -(-15) = 15$$

So the co-ordinates of the point which divide the given line segment in ratio of $\frac{2}{-3}$ are (17,15).

Then the correct option is B)

14. Correct option: (A)

Let's take the point A, B, C which divides the line PQ into four equal parts.

So B will be the midpoint of PQ , A will be the midpoint of PB and C will be the midpoint of BQ.

Given points $P=(-2,2), Q=(2,8)$

Point B

$$x = \frac{-2+2}{2} = 0$$

$$y = \frac{2+8}{2} = 5$$

So $B = (0,5)$

Point A

$$x = \frac{-2+0}{2} = -1$$

$$y = \frac{2+5}{2} = \frac{7}{2}$$

So $A = \left(-1, \frac{7}{2}\right)$

Point C

$$x = \frac{0+2}{2} = 1$$

$$y = \frac{5+8}{2} = \frac{13}{2}$$

So $C = \left(1, \frac{13}{2}\right)$

Therefore the points which divide the line segment into four equal parts are A, B, C

Then the correct option is A).

15. Correct option: (C)

Let the point P (x, y)

Given points $(-4,3), (5,7)$

Given that the point P cut line segment in $\frac{2}{3}$ and $\frac{1}{3}$ so point P cut line segment in $2:1$. Then Coordinate of the point P

$$x = \frac{5 \times 2 + (-4 \times 1)}{3} => 2$$

$$y = \frac{7 \times 2 + 3 \times 1}{3} => \frac{17}{3}$$

Then the correct option is C).

16. Correct option: (D)

For a point P(x_1, y_1) dividing the line segment joining Q$(x_2, y_2)$and R(x_3, y_3), into two equal half then coordinates of point P can be calculated as:

$$x_1 = \frac{x_3 + x_2}{2}$$

$$y_1 = \frac{y_3 + y_2}{2}$$

Let the point P (x, y)

Given points

$$P = \left(\frac{3}{2}, \frac{5}{2}\right), Q = (x, y+1), R = (x+1, y+2)$$

Given that point P is midpoint of the line segment QR.

Midpoint formula

$$\frac{3}{2} = \frac{x + x + 1}{2} => x = 1$$

$$\frac{5}{2} = \frac{y + 1 + y + 2}{2} => y = 1$$

We can now identify the points for which we want to find the midpoint, as they are given in the form of coordinates (x and y).

Now the points are $A = (x-1, y+1) = (0,2), B = (x+1, y-1) = (2,0)$

So midpoints of the AB line segment $x = \frac{0+2}{2} = 1, y = \frac{2+0}{2} = 1$

Then the correct option is D).

17. Correct option: (D)

For a point P(x_1, y_1) dividing the line segment joining Q$(x_2, y_2)$and R(x_3, y_3) internally in the ratio $m: n$, the coordinates of point P can be calculated as:

$$x_1 = \frac{mx_3 + nx_2}{m + n}$$
$$y_2 = \frac{my_3 + ny_2}{m + n}$$

Value of y co-ordinate equal to zero on the x-axis

Given points $P = (2,3), Q = (4,8), R = (-2,-7)$

Putting the values in formulas for division

$$2 = \frac{m \times (-2) + n \times (4)}{m + n} => \frac{-2 \times \frac{m}{n} + 4}{\frac{m}{n} + 1}$$
$$2\left(\frac{m}{n} + 1\right) = -2 \times \frac{m}{n} + 4 => 4\frac{m}{n} = 2$$
$$\frac{m}{n} = \frac{1}{2}$$

We can say that point P divide the QR in ration of $\frac{1}{2}$ internally

Then the correct option is D)

18. Correct option: (C)

Let the point P (x, y)

Given points $P = (x, y), A = (1,-2), B = (-5,6)$

For internal division

As we have to find the co-ordinate of two points which divide the QR into 3 equal sections, then the ration for the first point will be $\frac{1}{2}$ and for the second point the ratio will be $\frac{2}{1}$.

For first point P

Given ration is $\frac{1}{2} = \frac{m}{n} => m = 1, n = 2$

Putting the values in formulas for co-ordinate of P

$$x = \frac{1 \times (-5) + 2 \times 1}{1 + 2} = -\frac{3}{3} = -1$$
$$y = \frac{1 \times 6 + 2 \times (-2)}{1 + 2} = \frac{2}{3}$$

For second point Q

Given ration is $\frac{2}{1} = \frac{m}{n} => m = 2, n = 1$

Putting the values in formulas for co-ordinate of S

$$x = \frac{2 \times (-5) + 1 \times 1}{1 + 2} = \frac{-9}{3} = -3$$
$$y = \frac{2 \times 6 + 1 \times (-2)}{1 + 2} = \frac{10}{3}$$

So the co-ordinates of the points which divide the given line segment trisection is $P = \left(-1, \frac{2}{3}\right)$ and $S = (-3, \frac{10}{3})$

Then the value of PQ $\sqrt{\left(-3 - (-1)\right)^2 + \left(\frac{10}{3} - \frac{2}{3}\right)^2} = \sqrt{4 + \frac{64}{9}} = \sqrt{\frac{36+64}{9}} = \frac{10}{3}$

Then the correct option is C)

19. Correct option: (A)

For a point P(x_1, y_1) dividing the line segment joining Q$(x_2, y_2)$and R(x_3, y_3), into two equal half then coordinates of point P (mid point) can be calculated as:

$$x_1 = \frac{x_3 + x_2}{2}$$

$$y_1 = \frac{y_3 + y_2}{2}$$

We have given end points of a circle diameter as

$$A = (3, -4), B = (7,2)$$

Then, the midpoint of this diameter will be the center of the circle, as we know that the center cuts the diameter in half.

Putting the values

$$x = \frac{7+3}{2} = \frac{10}{2} = 5$$

$$y = \frac{2+(-4)}{2} = -\frac{2}{2} = -1$$

Coordinate of the center will be

$$P = (5, -1)$$

Then the correct option is A)

20. Correct option: (C)

First solve for the ratio:

$$2(m+n) = -2m + 4n$$

$$4m = 2n => \frac{m}{n} = \frac{1}{2}$$

The harmonic conjugate will divide the line segment in the ratio -1:2 (opposite sign because it's the conjugate).

Use the weighted average formula to find the harmonic conjugate:

$$Harmonic\ Conjugate = \left[\frac{(-2)\times(-1)+(4\times 2)}{1}, \frac{-7\times(-1)+(8)\times(2)}{1}\right]$$

Simplify: $Harmonic\ Conjugate = (10, 23)$

Then the correct option is C)

LEVEL – II

1. Correct option: (B)

Given point $P = (2,3)$

We need to calculate the distance of point P from the x-axis.

The distance of a point from the x-axis is equal to its y-coordinate:

$$Distance = P_y$$

Similarly, the distance of a point from the y-axis is equal to its x-coordinate:

$$Distance = P_x$$

Then the distance of point P will be 3.

Then the correct option is B)

2. Correct option: (B)

Given points $A = (0,6), B = (0, -2)$

Distance between two points

$$= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Putting the values in formula

$$= \sqrt{(0-0)^2 + (-2-6)^2} = \sqrt{64} = 8$$

Then the correct option is B)

3. Correct option: (C)

We know that the coordinate of origin will be$(0, 0)$

Then given points

$$P = (-6,8), O = (0,0)$$

Distance between two points

$$= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Putting the values in formula

$$= \sqrt{(0 - (-6))^2 + (0 - 8)^2} = \sqrt{36 + 64} = 10$$

Then the correct option is C)

4. Correct option: (B)

Given points

$$A = (0, 5), B = (-5, 0)$$

Distance between two points

$$= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

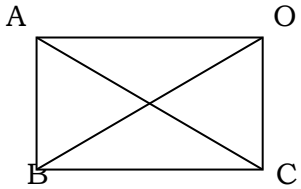
Putting the values in formula

$$= \sqrt{((-5) - 0)^2 + (0 - 5)^2} = \sqrt{50} = \sqrt{2 \times 25} = 5\sqrt{2}$$

Then the correct option is B)

5. Correct option: (A)

Given a rectangle $AOBC$ as



Given points $A = (0, 3), O = (0, 0), B = (5, 0)$

We know that the lengths of the diagonals are the same for a rectangle.

$$AC = OB$$

Distance between two points

$$= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

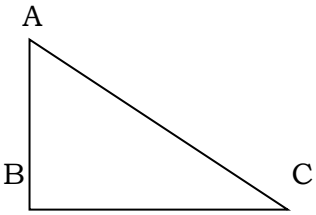
Putting the values in formula

$$OB = \sqrt{(5 - 0)^2 + (0 - 0)^2} = \sqrt{20} = 5$$

Then the correct option is A)

6. Correct option: (B)

Given a triangle as



Given points

$$A = (0, 4), B = (0, 0), C = (3, 0)$$

The perimeter of a triangle will be the sum of all its side lengths.

$$P = AB + BC + AC$$

Putting the values

$$P = \sqrt{(0 - 0)^2 + (4 - 0)^2} + \sqrt{(3 - 0)^2 + (0 - 0)^2} + \sqrt{(3 - 0)^2 + (4 - 0)^2}$$

$$P = \sqrt{16} + \sqrt{9} + \sqrt{25} = 4 + 3 + 5 = 12$$

Then the correct option is B)

7. Correct option: (C)

If point A (0, 2) is equidistant from points B (3, p) and C (p, 5), then the value of p is:

To be equidistant from two points, the distance from A to B should be equal to the distance from A to C.

Given points:

$$A(0,2), B(3,p), C(p,5)$$

Distance from A to B:

$$\sqrt{(3-0)^2+(p-2)^2} = \sqrt{9+(p-2)^2}$$

Distance from A to C:

$$\sqrt{(p-0)^2+(5-2)^2} = \sqrt{p^2+9}$$

Setting up the equation for equidistance:

$$\sqrt{9+(p-2)^2} = \sqrt{p^2+9}$$
$$9+(p-2)^2 = p^2+9$$
$$(p-2)^2 = p^2$$
$$p^2-4p+4 = p^2$$
$$-4p+4 = 0$$
$$4p = 4$$
$$p = 1$$

So, the value of p is 1.

The correct option is C) 1.

8. Correct option: (*)

Given points

$$A = (-4,0), B = (4,0), C = (3,0)$$

A degenerate triangle is a geometric concept referring to a triangle that has collapsed into a lower-dimensional shape. In a degenerate triangle, the sides might overlap or be of zero length, causing the triangle to become a line segment or even a single point. In essence, a degenerate triangle doesn't have the typical appearance of a triangle with distinct sides and angles.

In a non-degenerate triangle, the sum of the lengths of any two sides is always greater than the length of the third side (Triangle Inequality Theorem). In a degenerate triangle, this inequality is not satisfied.

In our case, the points (-4, 0), (4, 0), and (3, 0) are collinear and form a degenerate triangle because all three points lie on the x-axis, and therefore the triangle have zero area.

9. Correct option: (D)

To find the point that divides the line segment joining (7, -6) and (3, 4) in the ratio 1:2 internally, we can use the section formula. The section formula states that if a point divides the line segment joining two points (x_1, y_1) and (x_2, y_2) in the ratio of m:n internally, then the coordinates of the point can be calculated as:

$$x = \frac{mx_2 + nx_1}{m+n}$$
$$y = \frac{my_2 + ny_2}{m+n}$$

In this case, m represents the ratio 1 and n represents the ratio 2.

Given points:

$$A(7,-6)$$
$$B(3,4)$$

Let's calculate the coordinates of the point that divides the segment internally in the ratio 1:2.

$$x = \frac{3+14}{3} = \frac{17}{3}$$
$$y = \frac{4-12}{3} = -\frac{8}{3}$$

Now, let's determine in which quadrant this point lies:

- The x-coordinate is positive, so the point is not in the II or III quadrant.

- The y-coordinate is negative, so the point is in the IV quadrant.
So, the correct answer is D. IV quadrant.

10. Correct option: (B)

Let's use the slope method to find the fourth vertex D of the parallelogram ABCD.
Given vertices:

$$A(-2,3)$$
$$B(6,7)$$
$$C(8,3)$$

Let the fourth vertex D = (x, y)
Using midpoint property of diagonals of the parallelogram

$$M = \left[\frac{-2+8}{2}, \frac{3+3}{2}\right] = \left[\frac{6+x}{2}, \frac{7+y}{2}\right]$$
$$6 = 6x \Rightarrow x = 0, 7 + y = 6 \Rightarrow y = -1$$

Therefore the coordinates of the point D is

$$D = (0,-1)$$

Then the correct option is B)

11. Correct option: (B, D)

Given points

$$A = (4,2), B = (8,4), P = (2,1)$$

According the given option we have to find the values of AB, BC, AP, BP

$$AB = \sqrt{(4-2)^2 + (8-4)^2} = \sqrt{4+16} = \sqrt{20}$$
$$BP = \sqrt{(1-4)^2 + (2-8)^2} = \sqrt{9+36} = \sqrt{45}$$
$$AP = \sqrt{(1-2)^2 + (2-4)^2} = \sqrt{5}$$

Let's check the given options

A) Given $AP = \frac{1}{3}AB$

Putting the values

$$AP = \sqrt{5} \Rightarrow \frac{1}{3}AB = \frac{1}{3}\sqrt{20} \Rightarrow AP \neq \frac{1}{3}AB$$

Option A) is wrong

B) Given $BP = 3\sqrt{5}$

Putting the values

$$BP = \sqrt{45} \Rightarrow BP = \sqrt{9 \times 5} \Rightarrow BP = 3\sqrt{5}$$

Option B) is right

C) Given $PB = \frac{1}{3}AB$

Putting the values

$$BP = \sqrt{45} \Rightarrow \frac{1}{3}AB = \frac{1}{3}\sqrt{20} \Rightarrow PB \neq \frac{1}{3}AB$$

Option C) is wrong

D) Given $AP = \frac{1}{2}AB$

Putting the values

$$AP = \sqrt{5} \Rightarrow \frac{1}{2}AB = \frac{1}{2}\sqrt{4 \times 5} = \sqrt{5} \Rightarrow AP = \frac{1}{2}AB$$

Option D) is right

12. Correct option: (A, C)

To find the coordinates (x, y) where the perpendicular bisector of the line segment AB cuts the y-axis, follow these steps:
Given points:$A(1,5), B(4,6)$
1. Midpoint of AB:

$$Midpoint = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right) = \left(\frac{1 + 4}{2}, \frac{5 + 6}{2}\right) = (2.5, 5.5)$$

2. Slope of AB:

$$Slope\ of\ AB = \frac{y_2 - y_1}{x_2 - x_1} = \frac{6 - 5}{4 - 1} = \frac{1}{3}$$

3. Negative reciprocal of the slope of AB (slope of perpendicular bisector):
Slope of perpendicular bisector

$$= -\frac{1}{\frac{1}{3}} = -3$$

4. Equation of perpendicular bisector using the midpoint:
Using the point-slope form: $y - y_1 = m(x - x_1)$

$$y - 5.5 = -3(x - 2.5)$$
$$y - 5.5 = -3x + 7.5$$
$$y = -3x + 13$$

5. Finding the point of intersection with the y-axis:
When x = 0 (since the point is on the y-axis): $y = -3(0) + 13$

$$y = 13$$

So, the point of intersection with the y-axis is (x, y) = (0, 13).
Then the correct option is A, C)

13. Correct option: (A, D)

Given one end of the line segment is (2, -3) and the abscissa (x-coordinate) of the other end is 10. Let's denote the coordinates of the other end as (10, y).
The length of the line segment can be calculated using the distance formula:

$$Length = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Given points: $A(2, -3)\ B(10, y)$
Length = 10 (given)
Using the distance formula:

$$10 = \sqrt{(10 - 2)^2 + \left(y - (-3)\right)^2}$$
$$10 = \sqrt{64 + (y + 3)^2}$$
$$100 = 64 + (y + 3)^2$$
$$(y + 3)^2 = 100 - 64$$
$$(y + 3)^2 = 36$$

Taking the square root of both sides: $y + 3 = \pm 6$
Solving for y: $y = -3 \pm 6$
This gives two possible values for y: -9 and 3.
Then the correct option is A, D)

14. Correct option: (B)

The value of ordinate at x-axis will be zero. So let point P= $(x, 0)$
Now given points $P = (x, 0), A = (2,3), B = (-1,2)$
Putting the values in formulas for division

$$0 = \frac{m \times 2 + n \times (3)}{m + n} => \frac{2 \times \frac{m}{n} + 3}{\frac{m}{n} + 1}$$
$$0 = 2 \times \frac{m}{n} + 3 => \frac{m}{n} = \frac{-3}{2}$$

We can say that x axis divide the AB in ration of $\frac{3}{2}$ externally.
Then the correct option is B)

15. Correct option: (C)

The value of abscissa at y-axis will be zero. So let point P= $(0, y)$

Now given points $P = (0, y), A = (3,2), B = (-6,5)$

Putting the values in formulas for division

$$0 = \frac{m \times (-6) + n \times 3}{m + n} => \frac{-6 \times \frac{m}{n} + 3}{\frac{m}{n} + 1}$$

$$0 = -6 \times \frac{m}{n} + 3 => \frac{m}{n} = \frac{1}{2}$$

We can say that y-axis divide the AB in ration of $\frac{1}{2}$ internally.

Then the correct option C)

16. Correct option: (D)

Let the point P (x, y)

Given points $P = (x, y), A = (1,2), R = (3,5)$

Given ration is $\frac{1}{2} = \frac{m}{n} => m = 1, n = 2$

Putting the values in formulas for co-ordinate of P

$$x = \frac{3 \times 1 + 1 \times 2}{1 + 2} = \frac{5}{3}$$

$$y = \frac{5 \times 1 + 2 \times 2}{1 + 2} = \frac{9}{3} = 3$$

So the co-ordinates of the point which divide the given line segment in ratio of $\frac{1}{2}$ are $\left(\frac{5}{3}, 3\right)$.

Then the correct option is D)

17. Correct Matching is: a-T, b-P, c-S, d-Q

a) --→ (T)

Given points are

$$A = (-2,5), B = (2,5)$$

Applying the midpoint formula

$$x = \frac{2+(-2)}{2} = 0$$

$$y = \frac{5 + 5}{2} = 5$$

So the midpoint of the points A and B is (0,5)

b) --→ (P)

Let the midpoint of the line PQ is R

So according the question

$$R = (2, -5), Q = (4,0)$$

As we are aware that point P lies on the Y-axis, its abscissa will be zero.

Let point P be represented as P = (0, y).

Putting the values in the midpoint formula

$$-5 = \frac{0+y}{2} => y = -10$$

Then the point P will be $(0, -10)$.

c) --→ (S)

We are given a point P on the X-axis, denoted as P(x, 0), and the distance from this point P to another point (7, -4) is $2\sqrt{5}$ units. To find the value of x, we can apply the distance formula.

Putting the given values into the distance formula will allow us to determine the exact value of x.

$$2\sqrt{5} = \sqrt{(x-7)^2 + (0-(-4))^2} = \sqrt{(x-7)^2 + 16}$$

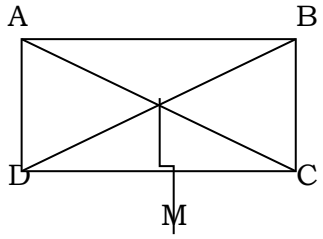
$$20 = (x-7)^2 + 16$$

$$4 = (x-7)^2 => x - 7 = \pm 2 => x = 9{,}5$$

So from the given option (5, 0) is matching.

d) --→ (Q)

Given that the points form a rectangle, we can leverage the concept of the midpoint of diagonals in a rectangle to readily determine the coordinates of the unknown vertex.
Let our rectangle be as follows:



The coordinates of point M can be found from points A and C. Then, by using point M, we can determine the coordinates of point D using point C.
Given points $A = (2, -2), B = (7,3), C = (11, -1)$
Then

$$M = \left(\frac{11+2}{2}, \frac{-1-2}{2}\right)$$

Let the unknown point D = (x, y)
Then

$$M = \left(\frac{7+x}{2}, \frac{3+y}{2}\right)$$

Equating the coordinates values

$$\frac{13}{2} = \frac{7}{2} + \frac{x}{2} => x = 6$$

$$-\frac{3}{2} = \frac{3}{2} + \frac{y}{2} => y = -6$$

18. Correct Answer is: 2

Using the midpoint formula:

$$(a, b) = \left(\frac{10 + k}{2}, \frac{-6 + 4}{2}\right)$$

$$(a, b) = \left(\frac{10 + k}{2}, -1\right)$$

Given that $a - 2b = 18$:
We find that b = -1, then

$$a - 2(-1) = 18$$

$$a + 2 = 18$$

$$a = 16$$

So, the coordinates of the midpoint (a, b) are (16, -1).
Solving for k:

$$16 = \frac{10 + k}{2}$$

$$32 = 10 + k$$

$$k = 32 - 10$$

$$k = 22$$

Then $\frac{k}{11}$

$$\frac{k}{11} = \frac{22}{11} = 2$$

19. Correct Answer is: 3

To find the least value of a, we can use the information provided.

The center-radius form of the equation of a circle is:

$$(x - h)^2 + (y - k)^2 = r^2$$

Where $(h, k)$the center of the circle and r is its radius

Given:

Center = $(2a, a - 7)$

Point on circle = $(11, -9)$

Diameter = $10\sqrt{2} => r = 5\sqrt{2}$

Then

$$(11 - 2a)^2 + (-9 - (a - 7))^2 = (5\sqrt{2})^2$$

$$(11 - 2a)^2 + (-9 - a + 7)^2 = 50$$

$$(11 - 2a)^2 + (-a - 2)^2 = 50$$

$$121 - 44a + 4a^2 + a^2 + 4a + 4 = 50$$

$$5a^2 - 40a + 75 = 0$$

$$a^2 - 8a + 15 = 0$$

$$(a - 5)(a - 3) = 0$$

The least value of a for the given conditions is 3.

20. Correct Answer is: 0

Given that the line segment AB, joining points A(3, 2) and B(5, 1), is divided by point P in the ratio 1:2, and it lies on the line 3x - 18y + k = 0, we can find the value of k.

Let's first find the coordinates of point P using the section formula.

Given: $A(3, 2)$

$B(5, 1)$

$Ratio\ m{:}n = 1{:}2$

Calculating the coordinates of point P:

$$x = \frac{5 + 6}{3} = \frac{11}{3}$$

$$y = \frac{1 + 4}{3} = \frac{5}{3}$$

So, the coordinates of point P are $P\left(\frac{11}{3}, \frac{5}{3}\right)$.

Since point P lies on the line $3x - 18y + k = 0$, we can substitute its coordinates into the equation:

$$11 - 30 + k = 0$$

$$k = 19$$

Therefore, $k - 19 = 0$.

STUDENT PRACTICE QUESTIONS

1. Correct option: (B)

Given points $A = (1, 4), B = (5, 1)$

The distance between two points can be calculated using the distance formula:

$$Distance = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Putting the values in formula $= \sqrt{(5 - 1)^2 + (1 - 4)^2} = \sqrt{16 + 9} = \sqrt{25} = 5$

Then the correct option is B)

2. Correct option: (*)

To determine the type of triangle formed by the given points $(\sqrt{3}-1,\sqrt{3}+1),(\sqrt{3},1),(\sqrt{3}+1,\sqrt{3}-1)$ We can calculate the distances between these points and see if they satisfy the conditions for any of the given types of triangles.

The distance between two points can be calculated using the distance formula:

$$Distance = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

Let's calculate the distances between these points:

1. Distance between $(\sqrt{3}-1,\sqrt{3}+1),(\sqrt{3},1)$:

$$Distance = \sqrt{(\sqrt{3}-(\sqrt{3}-1))^2+(1-(\sqrt{3}+1))^2}$$

$$Distance = \sqrt{1+3}$$

$$Distance = \sqrt{4}$$

$$Distance = 2$$

2. Distance between $(\sqrt{3},1),(\sqrt{3}+1,\sqrt{3}-1)$:

$$Distance = \sqrt{(\sqrt{3}+1-\sqrt{3})^2+(\sqrt{3}-1-(1))^2}$$

$$Distance = \sqrt{1+3+4-4\sqrt{3}}$$

$$Distance = \sqrt{8-4\sqrt{3}}$$

3. Distance between $(\sqrt{3}-1,\sqrt{3}+1)$, $(\sqrt{3}+1,\sqrt{3}-1)$:

$$Distance = \sqrt{(\sqrt{3}+1-(\sqrt{3}-1))^2+(\sqrt{3}-1-(\sqrt{3}+1))^2}$$

$$Distance = \sqrt{4+4}$$

$$Distance = \sqrt{8}$$

The distances between the points are $2, \sqrt{8-4\sqrt{3}}$ *and* $\sqrt{8}$.

None of the option is correct.

3. Correct option: (A)

To find the centroid of a triangle with vertices (x_1, y_1), (x_2, y_2), and (x_3, y_3), we can use the following formula: $Centroid = ((x_1+x_2+x_3)/3, (y_1+y_2+y_3)/3)$

In our case, the vertices of the triangle are (9, 3), (7, -1), and (-1, 4). Plug these coordinates into the formula:

$$Centroid = \left(\frac{9+7-1}{3}, \frac{3-1+4}{3}\right)$$

$$Centroid = \left(\frac{15}{3}, \frac{6}{3}\right)$$

$$Centroid = (5,2)$$

Then the correct option is A)

4. Correct option: (B)

Given points: $(7,1),(4,4),(-2,-2),(1,-5)$

1. Sides

Calculate the distances between consecutive points:

- Distance between (7, 1) and (4, 4): $\sqrt{(4-7)^2+(4-1)^2} = \sqrt{18}$
- Distance between (4, 4) and (-2,-2): $\sqrt{(-2-4)^2+(-2-4)^2} = \sqrt{72}$
- Distance between (-2,-2) and (1,-5): $\sqrt{(1-(-2))^2+(-5-(-2))^2} = \sqrt{18}$

- Distance between (1,-5) and (7, 1): $\sqrt{(7-1)^2 + \left(1-(-5)\right)^2} = \sqrt{72}$

Two sides are equal to $= \sqrt{18}$ and two other sides are $= \sqrt{72}$

Then it may be a rectangle or a Parallelogram.

2. Angles

- Calculate the slopes of consecutive sides:

Slope of AB: $\frac{4-1}{4-7} = -1$

Slope of BC: $\frac{-2-4}{-2-4} = 1$

Slope of CD: $\frac{-5-(-2)}{1-(-2)} = -1$

Slope of DA: $\frac{1-(-5)}{7-1} = 1$

Since the product of the slopes of AB and BC is -1, lines AB and BC are perpendicular. Similarly, the product of the slopes of CD and DA is -1, indicating that lines CD and DA are perpendicular.

This confirms that all angles are right angles (90°), which satisfies the criterion for a Rectangle.

Based on both the equal sides and the right angles, we can conclude that the points $(7,1), (4,4), (-2,-2), (1,-5)$ form a Rectangle.

So, the correct answer is indeed: B) Rectangle

5. Correct option: (C)

Given points: $(7,8), (1,6), (-1,0), (5,2)$

1. Sides

Calculate the distances between consecutive points:

Distance between (7, 8) and (1, 6):

$$\sqrt{(1-7)^2 + (6-8)^2} = \sqrt{36+4} = \sqrt{40}$$

Distance between (1, 6) and (-1, 0):

$$\sqrt{(-1-1)^2 + (0-6)^2} = \sqrt{4+36} = \sqrt{40}$$

Distance between (-1, 0) and (5, 2):

$$\sqrt{\left(5-(-1)\right)^2 + (2-0)^2} = \sqrt{36+4} = \sqrt{40}$$

Distance between (5, 2) and (7, 8):

$$\sqrt{(7-5)^2 + (8-2)^2} = \sqrt{36+4} = \sqrt{40}$$

All sides have equal lengths ($\sqrt{40}$), which satisfies the criterion for a square or rhombus.

2. Angles

- Calculate the slopes of consecutive sides:

Slope of (7, 8) and (1, 6):

$$\frac{6-8}{1-7} = \frac{-2}{-6} = \frac{1}{3}$$

Slope of (1, 6) and (-1, 0):

$$\frac{0-6}{-1-1} = \frac{-6}{-2} = 3$$

Slope of (-1, 0) and (5, 2):

$$\frac{2-0}{5-(-1)} = \frac{2}{6} = \frac{1}{3}$$

Slope of (5, 2) and (7, 8):

$$\frac{8-2}{7-5} = \frac{6}{2} = 3$$

Since the product of the slopes of AB and BC is 1, lines AB and BC are not perpendicular. Similarly, the product of the slopes of CD and DA is 1.
Let's find the slopes of diagonal $AC = using\ (7,8), (-1,0)\ and\ BD = using\ (1,6), (5,2)$
Slope of (7, 8) and (-1, 0):

$$\frac{0\ -\ 8}{-1\ -\ 7} = \frac{-8}{-8} = 1$$

Slope of (1, 6) and (5, 2):

$$\frac{2\ -\ 6}{5\ -\ 1} = -\frac{4}{4} = -1$$

As product of slopes of the diagonals $= -1$, we can say that diagonals are perpendicular to each other which satisfy the property of the Rhombus.
So, the correct answer is indeed: C) Rhombus

6. Correct option: (D)

For a point P(x_1, y_1) dividing the line segment joining Q$(x_2, y_2)$and R(x_3, y_3) internally in the ratio $m:n$, the coordinates of point P can be calculated as:

$$x_1 = \frac{mx_3 + nx_2}{m + n}$$
$$y_2 = \frac{my_3 + ny_2}{m + n}$$

Value of x co-ordinate equal to zero on y axis
Given points $P = (0, y), Q = (3,5), R = (-4,7)$
Putting the values in formulas for division

$$0 = \frac{m \times (-4) + n \times 3}{m + n} => \frac{-4 \times \frac{m}{n} + 3}{\frac{m}{n} + 1}$$
$$0 = -4 \times \frac{m}{n} + 3 => \frac{m}{n} = \frac{3}{4}$$

We can say that point x axis divide the QR in ration of $\frac{3}{4}$ internally.
Then the correct option is D)

7. Correct option: (C)

For a point P(x_1, y_1) dividing the line segment joining Q$(x_2, y_2)$and R(x_3, y_3) internally in the ratio $m:n$, the coordinates of point P can be calculated as:

$$x_1 = \frac{mx_3 + nx_2}{m + n}$$
$$y_2 = \frac{my_3 + ny_2}{m + n}$$

Let the point P (x, y)
Given points $P = (x, y), Q = (1,7), R = (6, -3)$
For internal division
Given ration is $\frac{2}{3} = \frac{m}{n} => m = 2, n = 3$
Putting the values in formulas for co-ordinate of P

$$x = \frac{2 \times 6 + 3 \times 1}{2 + 3} = 3$$
$$y = \frac{2 \times -3 + 3 \times 7}{2 + 3} = 3$$

So the co-ordinates of the point which divide the given line segment in ratio of $2/3$ internally are $(3, 3)$.
For external division
Given ration is $-\frac{2}{3} = \frac{m}{n} => m = 2, n = -3$
Putting the values in formulas for co-ordinate of P

$$x = \frac{2 \times 6 - 3 \times 1}{2 - 3} = -9$$

$$y = \frac{2 \times (-3) - 3 \times 7}{2 - 3} = 27$$

So the co-ordinates of the point which divide the given line segment in ratio of $2/3$ externally are $(-9, 27)$.

Then the correct option is C)

8. Correct option: (B)

For a point P(x_1, y_1) dividing the line segment joining Q$(x_2, y_2)$and R(x_3, y_3) internally in the ratio $m: n$, the coordinates of point P can be calculated as:

$$x_1 = \frac{mx_3 + nx_2}{m + n}$$

$$y_2 = \frac{my_3 + ny_2}{m + n}$$

Let the point P (x, y)

Given points

$$P = (x, y), Q = (-5,2), R = (3,6)$$

For internal division

As we have to find the co-ordinate of two points which divide the QR into 3 equal sections, then the ration for the first point will be $\frac{1}{2}$ and for the second point the ratio will be $\frac{2}{1}$.

For first point

Given ration is $\frac{1}{2} = \frac{m}{n} => m = 1, n = 2$

Putting the values in formulas for co-ordinate of P

$$x = \frac{1 \times 3 + 2 \times (-5)}{1 + 2} = -\frac{7}{3}$$

$$y = \frac{1 \times 6 + 2 \times 2}{1 + 2} = \frac{10}{3}$$

For second point

Given ration is $\frac{2}{1} = \frac{m}{n} => m = 2, n = 1$

Putting the values in formulas for co-ordinate of S

$$x = \frac{2 \times 3 + 1 \times (-5)}{1 + 2} = \frac{1}{3}$$

$$y = \frac{2 \times 6 + 1 \times 2}{1 + 2} = \frac{14}{3}$$

So the co-ordinates of the points which divide the given line segment trisection is $P = \left(-\frac{7}{3}, \frac{10}{3}\right)$ and $S = \left(\frac{1}{3}, \frac{14}{3}\right)$

Then the correct option is B).

9. Correct option: (C)

For a point P(x_1, y_1) dividing the line segment joining Q$(x_2, y_2)$and R(x_3, y_3) internally in the ratio $m: n$, the coordinates of point P can be calculated as:

$$x_1 = \frac{mx_3 + nx_2}{m + n}$$

$$y_2 = \frac{my_3 + ny_2}{m + n}$$

Value of y co-ordinate equal to zero on the x-axis

Given points $P = (-3,5), Q = (5,1), R = (3,2)$

Putting the values in formulas for division

$$-3 = \frac{m \times (3) + n \times (5)}{m + n} => \frac{3 \times \frac{m}{n} + 5}{\frac{m}{n} + 1}$$

$$-3\left(\frac{m}{n} + 1\right) = 3 \times \frac{m}{n} + 5 => -6\frac{m}{n} = 8$$

$$\frac{m}{n} = -\frac{8}{6} = -\frac{4}{3}$$

We can say that point P divide the QR in ration of $\frac{4}{3}$ externally.

10. Correct option: (C)

To find the harmonic conjugate of (4, -2) with respect to the points (2, -4) and (7, 1), first, determine the ratio in which (4, -2) divides the line segment connecting (2, -4) and (7, 1).
Solve for the ratio:

$$4(m + n) = 7m + 2n$$

$$2n = 3m => \frac{m}{n} = \frac{2}{3}$$

The harmonic conjugate will divide the line segment in the ratio -2:3 (opposite sign because it's the conjugate).
Use the weighted average formula to find the harmonic conjugate:

$$Harmonic\ Conjugate = \left[\frac{7 \times (-2) + (2 \times 3)}{-2 + 3}, \frac{1 \times (-2) + (3) \times (-4)}{-2 + 3}\right]$$

Simplify: $Harmonic\ Conjugate = (-8, -14)$

Then the correct option is C)

11. Correct option: (B, D)

We have given points $P = \left(\frac{a}{3}, 4\right), Q = (-6,5), R = (-2,3)$

Then, the midpoint of these points can be calculated as
Putting the values

$$\frac{a}{3} = \frac{-2 + (-6)}{2} = \frac{-8}{2} => a = -12$$

So the correct options are B), D).

12. Correct option: (A, C)

Given points: $A(4, p) B(1, 0)$
The distance between A and B is given as 5:

$$5 = \sqrt{(1 - 4)^2 + (0 - p)^2}$$

$$25 = (3)^2 + p^2$$

$$25 = 9 + p^2$$

$$p^2 = 16$$

Taking the square root of both sides: $p = \pm 4$

So, the possible values of p are 4 and -4.
Then the correct option A, C)

13. Correct option: (A)

Given points are $A = (0,1), B = (0, -10), C = (0, \frac{1}{3})$

To accurately determine the shape formed by the points, we need to calculate the distances between them.
First of all, by observation, we can see that all three vertices lie on the x-axis. Therefore, we can easily conclude that these points are vertices of a straight line.
So the correct option will be straight line.
Then the correct option is A)

14. Correct option: (B)

Given points are $A = (a+b, a), B = (b, a+b)$

We have to find the value of AB^2

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} => AB^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

Putting the values $AB^2 = (b - (a+b))^2 + (a + b - (a))^2 => a^2 + b^2$

Then the correct option is B)

15. Correct Matching is: a-Q, b-T, c-RS, d-P

a) --→ (Q)

Given points are $(6,2), (0p), (4 - 7)$

Applying the centroid formula $-\frac{5}{3} = \frac{2+p+(-7)}{3} => -5 = p - 5 => p = 0$

b) --→ (T)

Distance between given point is 9

Then we can write $9^2 = (-3 - a)^2 + (-14 + 5)^2 => 9^2 = (a+3)^2 + 9^2$

$$(a+3)^2 = 0 => a + 3 = 0 => a = -3$$

c) --→ (R, S)

Distance between given point is 10

Then we can write $10^2 = (2 - p)^2 + (-3 - 5)^2 => 10^2 = (p-2)^2 + 8^2$

$$(p-2)^2 = 6^2 => p - 2 = \pm 6 => p = 8, -4$$

d) --→ (P)

Given that A, B, C are collinear

So line of equation AB is also satisfied by point C.

Find the equation of line using A and B

$$y = 2x$$

Then for C $x = 2 \times \frac{1}{2} = x = 1$

16. Correct Answer is: 4

Given points are $(-1, k-1), (6, -3), (-3,6)$

Applying the centroid formula for y coordinate

$$2 = \frac{k - 1 + (-3) + 6}{3} => 6 = k + 2 => k = 4$$

Then the value of k = 4.

17. Correct Answer is: 2

Given points $A(2,9), B(a,5), C(7,5)$

The sides of the triangle are AB, BC, and AC. Since B is the right angle, then the hypotenuse is AC: $AC^2 = AB^2 + BC^2$

Using the distance formula to calculate the lengths of the sides:

$$(7 - 2)^2 + (5 - 9)^2 = (a - 2)^2 + (5 - 9)^2 + (7 - a)^2 + (5 - 5)^2$$

$$25 = (a - 2)^2 + (7 - a)^2 => a^2 + 4 - 4a + 49 + a^2 - 14a = 25$$

$$2a^2 - 18a + 28 = 0 => a^2 - 9a + 14 = 0$$

$$a^2 - 9a + 14 = 0 => a = 2 \text{ or } 7$$

If we consider the value a = 7, then the coordinates of point B and point C would coincide, which would result in a degenerate triangle where all three points are collinear or coincide. In a degenerate triangle, the sides have zero length, and it's not a meaningful geometric configuration.

In the case of a = 2, the points (2, 9), (2, 5), and (7, 5) form a right-angled triangle with the right angle at vertex B. This satisfies the criteria of a non-degenerate right-angled triangle, and the value a = 2 is indeed the correct and only considerable value.

18. Correct Answer is: 2

Given that points A, B, and C are collinear, we have the following coordinates:

$$Point\ A: A = (k + 1, 2k)$$
$$Point\ B: B = (3k, 2k + 3)$$
$$Point\ C: C = (5k - 1, 5k)$$

We are given that the points are collinear, which means that the slope of AB is equal to the slope of BC.

$$slope\ of\ AB = \frac{2k + 3 - 2k}{3k - k + 1} = \frac{3}{2k - 1}$$
$$slope\ of\ BC = \frac{5k - 2k - 3}{5k - 1 - 3k} = \frac{3k - 3}{2k - 1}$$

Equating the slopes $\frac{3}{2k-1} = \frac{3k-3}{2k-1} => k = 2$

So, the integral value of k that results in the given points being collinear is 2.

19. Correct Answer is: 2

The points (-2, 0), (-2, -1), and (-2, 2) are 2 units away from the Y-axis towards the west direction.

To prove that the points (-2, 0), (-2, -1), and (-2, 2) are 2 units away from the Y-axis towards the west direction, we need to calculate the distance of these points from the Y-axis and demonstrate that the distance is indeed 2 units.

The distance between a point (x, y) and the Y-axis can be calculated using the formula $|x|$. The absolute value of x gives the distance of the point from the Y-axis.

Given points:

$$Point\ A: (-2, 0)$$
$$Point\ B: (-2, -1)$$
$$Point\ C: (-2, 2)$$

1. Distance of point A from Y-axis:

$$Distance_A = |x_A| = |-2| = 2\ units$$

2. Distance of point B from Y-axis:

$$Distance_B = |x_B| = |-2| = 2\ units$$

3. Distance of point C from Y-axis:

$$Distance_C = |x_C| = |-2| = 2\ units$$

All three points have a distance of 2 units from the Y-axis, and they lie on the left side of the Y-axis (west direction), as their x-coordinates are negative.

Therefore, it is proven that the points (-2, 0), (-2, -1), and (-2, 2) are 2 units away from the Y-axis towards the west direction.

20. Correct Answer is: 7

The points (-5, 7), (0, 7), (5, 7), and (7, 7) are 7 units away from the X-axis.

The distance of a point (x, y) from the X-axis is given by $|y|$.

Given points:

$$Point\ A: (-5, 7)$$
$$Point\ B: (0, 7)$$
$$Point\ C: (5, 7)$$
$$Point\ D: (7, 7)$$

1. $Distance\ of\ point\ A\ from\ X - axis: |y_A| = |7| = 7\ units$
2. $Distance\ of\ point\ B\ from\ X - axis: |y_B| = |7| = 7\ units$
3. $Distance\ of\ point\ C\ from\ X - axis: |y_C| = |7| = 7\ units$
4. $Distance\ of\ point\ D\ from\ X - axis: |y_D| = |7| = 7\ units$

All four points have a distance of 7 units from the X-axis, which is their y-coordinate.

Thus, it is confirmed that the points (-5, 7), (0, 7), (5, 7), and (7, 7) are 7 units away from the X-axis.

4. Trigonometry and Properties of Triangles

Solutions

Trigonometric Ratios and Identities

LEVEL-I

1. Correct option: (A)

As we know, $\pi\, rad\ is\ equal\ to\ 180°$,

Hence,

$1\, rad = \frac{180}{\pi} degree$

$\Rightarrow -\frac{13\pi}{6} rad = -\frac{13\pi}{6} \times \frac{180}{\pi} = -13 \times 30° = -390°.$

2. Correct option: (A)

θ lies in the first quadrant

$5tan\theta = 4$

$\Rightarrow tan\theta = \frac{5}{4}.$

$\Rightarrow \frac{5sin\theta - 3cos\theta}{sin\theta + 2cos\theta} = \frac{cos\theta(5tan\theta - 3)}{cos\theta(tan\theta + 2)} = \frac{4-3}{\frac{4}{5}+2} = \frac{1}{\frac{14}{5}} = \frac{5}{14}$

Hence, the answer is $\frac{5}{14}$.

3. Correct option: (A)

Using $sin^2\theta + cos^2\theta = 1$,

$\left(-\frac{7}{25}\right)^2 + cos^2\,\theta = 1$

$cos^2\,\theta = 1 - \frac{49}{625}$

$cos^2\,\theta = \frac{625}{625} - \frac{49}{625} = \frac{576}{625}$

$cos\,\theta = \pm\frac{24}{25}.$

As θ is in third quadrant, so cosθ will be $-\frac{24}{25}$.

Cos θ is negative in third quadrant,

Now,

$tan\theta = \frac{sin\theta}{cos\theta}$

$tan\theta = \frac{-\frac{7}{25}}{-\frac{24}{25}} = \frac{7}{24}.$

$cot\theta = \frac{1}{tan\,\theta}$

$cot\theta = \frac{24}{7}.$

Now,

$$\frac{7\,cot\theta - 24tan\theta}{7cot\theta + 24tan\theta} = \frac{7 \times \frac{24}{7} - 24 \times \frac{7}{24}}{7 \times \frac{24}{7} + 24 \times \frac{7}{24}} = \frac{24-7}{7+24} = \frac{17}{31}$$

Therefore $\frac{7\,cot\theta - 24tan\theta}{7cot\theta + 24tan\theta} = \frac{17}{31}$.

4. Correct option: (C)

$$\frac{cot\theta + cosec\theta - 1}{cot\theta - cosec\theta + 1} = \frac{cot\theta + cosec\theta - (cosec^2\theta - cot^2\theta)}{cot\theta - cosec\theta + 1}$$

$$= \frac{cot\theta + cosec\theta - (cot\theta - cosec\theta)(cot\theta + cosec\theta)}{cot\theta - cosec\theta + 1}$$

$$= \frac{(cot\theta + cosec\theta)(1 - cosec\theta + cot\theta)}{cot\theta - cosec\theta + 1}$$
$$= cot\theta + cosec\theta$$
$$= \frac{cos\theta}{sin\theta} + \frac{1}{sin\theta}$$
$$= \frac{1 + cos\theta}{sin\theta}$$

Therefore, $\frac{cot\theta + cosec\theta - 1}{cot\theta - cosec\theta + 1} = \frac{1 + cos\theta}{sin\theta}$

5. Correct option: (A)

$x = acos^3\,\theta, \quad y = asin^3\,\theta$

$\frac{x}{a} = cos^3\,\theta, \quad \frac{y}{a} = sin^3\,\theta$

$\left(\frac{x}{a}\right)^{\frac{2}{3}} + \left(\frac{y}{a}\right)^{\frac{2}{3}} = (cos^3\,\theta + sin^3\,\theta)^{\frac{2}{3}}$

$\left(\frac{x}{a}\right)^{\frac{2}{3}} + \left(\frac{y}{a}\right)^{\frac{2}{3}} = cos^2\,\theta + sin^2\,\theta$

$\left(\frac{x}{a}\right)^{\frac{2}{3}} + \left(\frac{y}{a}\right)^{\frac{2}{3}} = 1.$

$x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}.$

Therefore, $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}.$

6. Correct option: (C)

$cos(n+1)\alpha cos(n-1)\alpha + sin(n+1)\alpha sin(n-1)\alpha$

$Assuming$

$(n+1)\alpha = A, (n-1)\alpha = B$

$cos(n+1)\alpha cos(n-1)\alpha + sin(n+1)\alpha sin(n-1)\alpha = cosAcosB + sinAsinB$

$= cos(A - B) = cos\big((n+1)\alpha - (n-1)\alpha\big)$

$= cos2\alpha$

Therefore, $cos(n+1)\alpha cos(n-1)\alpha + sin(n+1)\alpha sin(n-1)\alpha = cos2\alpha.$

7. Correct option: (A)

$Sin\,\theta + Sin\,(120° + \theta) - Sin\,(120° - \theta)$

$= Sin\,\theta + [\,Sin\,120°\,Cos\,\theta + Cos120°\,Sin\,\theta\,] - [\,Sin\,120°\,Cos\,\theta - Cos\,120°\,Sin\,\theta\,]$

$= Sin\,\theta + Sin\,120°\,Cos\,\theta + Cos120°\,Sin\,\theta - Sin\,120°\,Cos\,\theta + Cos\,120°\,Sin\,\theta$

$= Sin\,\theta + 2\,Cos\,120°\,Sin\,\theta$

$= Sin\,\theta + 2\left(-\frac{1}{2}\right)Sin\,\theta$

$= Sin\,\theta - Sin\,\theta$

$= 0$

8. Correct option: (A)

$cosA = \frac{3}{5}$

$sin\,B = \frac{5}{13}$

$cos^2A + sin^2\,A = 1$

$sin^2\,A = 1 - \frac{9}{5}$

$sin^2\,A = \frac{16}{25}$

$sinA = +\frac{4}{5} or\ -\frac{4}{5}$

$But\ SinA\ = +\frac{4}{5}\ (Hence, it\ is\ in\ Quadrant\ I)$

$cos^2\ B\ = 1\ -\ sin^2\ B$

$cos^2\ B\ = 1\ -\ 25/169$

$cos^2\ B\ = \frac{144}{169}$

$cosB\ = +\frac{12}{13}\ or\ -\frac{12}{13}$

$But\ Cos\ B\ = +\frac{12}{13}\ (Hence, it\ is\ in\ quadrant\ I)$

$$\frac{tanA + tanB}{1 - tanA \times tanB}$$

$$= \frac{\left(\frac{SinA}{CosA}\right) + \left(\frac{SinB}{CosB}\right)}{1 - \left(\frac{Sin\ A}{Cos\ A}\right)\left(\frac{SinB}{CosB}\right)}$$

$$= \frac{\frac{\left(\frac{4}{5}\right)}{\left(\frac{3}{5}\right)} + \frac{\left(\frac{5}{13}\right)}{\left(\frac{12}{13}\right)}}{1 - \frac{\left(\frac{4}{5}\right)}{\left(\frac{3}{5}\right)} \times \frac{\left(\frac{5}{13}\right)}{\left(\frac{12}{13}\right)}}$$

$$= \frac{\left(\frac{4}{3}\right) + \left(\frac{5}{12}\right)}{1 - \left(\frac{20}{36}\right)}$$

$$= \frac{\left(\frac{21}{12}\right)}{\left(\frac{4}{9}\right)}$$

$$= \frac{189}{48}$$

$$\frac{tanA + tanB}{1 - tanA \times tanB} = \frac{63}{16}$$

Therefore, $\frac{tanA+tanB}{1-tanA\times tanB} = \frac{63}{16}$.

9. Correct option: (B)

$tanA + tanB = p\ and$

$cotA + cotB = q$

$$\Rightarrow \frac{1}{tanA} + \frac{1}{tanB} = q$$

$$\Rightarrow \frac{tanA + tanB}{tanA \times tanB} = q$$

$$\Rightarrow \frac{p}{tanA \times tanB} = q$$

$$\Rightarrow tanA \times tanB = \frac{p}{q}$$

$$cot(A + B) = \frac{1}{tan(A + B)}$$

$$= \frac{1}{\frac{1 - tanA \times tanB}{tanA + tanB}}$$

$$= \frac{1 - tanA \times tanB}{tanA + tanB}$$

$= \frac{1-\frac{p}{q}}{p}$

$cotA + cotB = \frac{q-p}{pq}$

Therefore, $cotA + cotB = \frac{q-p}{pq}$.

10. Correct option: (A)

$sin^2 75° - sin^2 15°$

$= sin^2(90° - 15°) - sin^2 15°$

$= cos^2 15° - sin^2 15°$

$= cos(2 \times 15°)$

$= cos30°$

$= \frac{\sqrt{3}}{2}$

11. Correct option: (A)

As we know that

$cos(A+B)cos(A-B) = cos^2 A - sin^2 B$

$cos^2\left(\frac{\pi}{6}+\theta\right) - sin^2\left(\frac{\pi}{6}-\theta\right) = cos\left(\frac{\pi}{6}+\theta+\frac{\pi}{6}-\theta\right)cos\left(\frac{\pi}{6}+\theta-\frac{\pi}{6}+\theta\right)$

$= cos\frac{\pi}{3}cos2\theta$

$= \frac{1}{2}cos2\theta$

12. Correct option: (C)

We know that,

$sin^2 A - sin^2 B = sin(A+B) \cdot sin(A-B)$

$\frac{sin\theta}{sin\left(\frac{\pi}{8}+\frac{\theta}{2}+\frac{\pi}{8}-\frac{\theta}{2}\right) \times sin\left(\frac{\pi}{8}+\frac{\theta}{2}-\frac{\pi}{8}+\frac{\theta}{2}\right)} = \frac{sin\theta}{sin\left(\frac{2\pi}{8}\right) \times sin\left(\frac{2\theta}{2}\right)}$

$= \frac{sin\theta}{sin\left(\frac{\pi}{4}\right) \times sin(\theta)} = \frac{1}{sin\left(\frac{\pi}{4}\right)}$

$= \frac{1}{\left(\frac{1}{\sqrt{2}}\right)}$ *Since* $sin\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$

$\frac{sin\theta}{sin\left(\frac{\pi}{8}+\frac{\theta}{2}+\frac{\pi}{8}-\frac{\theta}{2}\right) \times sin\left(\frac{\pi}{8}+\frac{\theta}{2}-\frac{\pi}{8}+\frac{\theta}{2}\right)} = \sqrt{2}$

13. Correct option: (C)

$sin40° - sin80° + sin160° = 2sin\left(\frac{40°-80°}{2}\right)cos\left(\frac{40°+80°}{2}\right) + sin(180°-20°)$

$= -2sin20°cos60° + sin20°$

$= -sin20° + sin20° = 0$

$sin40° - sin80° + sin160° = 0$

14. Correct option: (D)

$cos40° + coss80° + cos160° + cos240°$

$= 2cos\left(\frac{40°+80°}{2}\right)cos\left(\frac{40°-80°}{2}\right) + cos160° - cos(180°+60°)$

$[\because cosA + cosB = 2cos\left(\frac{A+B}{2}\right)cos\left(\frac{A-B}{2}\right)]$

$= 2cos60°\, cos(-20°) + cos160° - \frac{1}{2}$

$= 2 \times 12cos20° + cos160° - \frac{1}{2}$

$= -cos(180 - 20)° + cos160° - \frac{1}{2}$

$cos40° + coss80° + cos160° + cos240° = -\frac{1}{2}$.

15. Correct option: (D)

$$tan\left(\frac{\pi}{4} + \theta\right) \times tan\left(\frac{\pi}{4} - \theta\right) = \frac{tan\frac{\pi}{4} + tan\theta}{1 - tan\frac{\pi}{4} \times tan\theta} \times \frac{tan\frac{\pi}{4} + tan\theta}{1 + tan\frac{\pi}{4} \times tan\theta}$$

$$= \frac{1 + tan\theta}{1 - tan\theta} \times \frac{1 + tan\theta}{1 + tan\theta} = 1$$

$tan\left(\frac{\pi}{4} + \theta\right) \times tan\left(\frac{\pi}{4} - \theta\right) = 1.$

16. Correct option: (D).

$Given \Rightarrow cotAcotB = 2$

$So,\ cosAcosB = 2sinAsinB \quad \ldots(1)$

$Given\ cos(A + B) = \frac{3}{5}.$

$So,\ cosAcosB - sinAsinB = \frac{3}{5} \quad \ldots.(2)$

$Fromeq.(1) and eq.(2)$

$sinAsinB = \frac{3}{5}.$

17. Correct option: (A).

$Given,\ tan8A - tan5A - tan3A = Ktan8Atan5Atan3$

$$\boldsymbol{tan8A = tan(5A + 3A) = \frac{tan5A + tan3A}{1 - tan5A \times tan3A}}$$

$$\boldsymbol{tan8A(1 - tan5A \times tan3A) = tan5A + tan3A}$$

$$\boldsymbol{tan5Atan3Atan8A = tan8A - tan5A - tan3A}$$

$$\boldsymbol{tan8A - tan5A - tan3A = tan8Atan5Atan3}$$

on comparing, we get $K = 1$ hence, (A) is the correct option.

18. Correct option: (D)

To simplify the expression, let's use the half-angle identities for sine and cosine:

$$sin\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1 - \cos\theta}{2}}$$

$$cos\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1 + \cos\theta}{2}}$$

Let's substitute θ = 45 degrees ($22\frac{1}{2}$ degrees * 2) into these identities:

$$sin\left(\frac{45}{2}\right) = \pm\sqrt{\frac{1 - \cos 45}{2}} = \pm\sqrt{\frac{1 - \frac{\sqrt{2}}{2}}{2}} = \pm\sqrt{\frac{2 - \sqrt{2}}{4}} = \pm\frac{\sqrt{2 - \sqrt{2}}}{2}$$

$$cos\left(\frac{45}{2}\right) = \pm\sqrt{\frac{1 + \cos 45}{2}} = \pm\sqrt{\frac{1 + \frac{\sqrt{2}}{2}}{2}} = \pm\sqrt{\frac{2 + \sqrt{2}}{4}} = \pm\frac{\sqrt{2 + \sqrt{2}}}{2}$$

Now we can substitute these values into the original expression:

$$\frac{cos\ 22\ \frac{1}{2} - sin22\ \frac{1}{2}}{cos\ 22\ \frac{1}{2} + sin22\ \frac{1}{2}} = \frac{\frac{\sqrt{2+\sqrt{2}}}{2} - \frac{\sqrt{2-\sqrt{2}}}{2}}{\frac{\sqrt{2+\sqrt{2}}}{2} + \frac{\sqrt{2-\sqrt{2}}}{2}}$$

To simplify further, let's multiply the numerator and denominator by the conjugate of the denominator:

$$\frac{cos\ 22\ \frac{1}{2} - sin22\ \frac{1}{2}}{cos\ 22\ \frac{1}{2} + sin22\ \frac{1}{2}} = \frac{\left(\frac{\sqrt{2+\sqrt{2}}}{2} - \frac{\sqrt{2-\sqrt{2}}}{2}\right) \times \left(\frac{\sqrt{2-\sqrt{2}}}{2} - \frac{\sqrt{2+\sqrt{2}}}{2}\right)}{\left(\frac{\sqrt{2+\sqrt{2}}}{2} + \frac{\sqrt{2-\sqrt{2}}}{2}\right) \times \left(\frac{\sqrt{2-\sqrt{2}}}{2} + \frac{\sqrt{2+\sqrt{2}}}{2}\right)}$$

Expanding and simplifying the numerator and denominator:

$$= \frac{\left(\left(\sqrt{2+\sqrt{2}}\right)^2 - \left(\sqrt{2-\sqrt{2}}\right)^2\right)}{\left(\left(\left(\sqrt{2+\sqrt{2}}\right)^2 - \left(\sqrt{2-\sqrt{2}}\right)^2\right)\right)}$$

$$= \frac{2+\sqrt{2}-(2-\sqrt{2})}{2+\sqrt{2}-(2-\sqrt{2})}$$

$$= \frac{2+\sqrt{2}-2+\sqrt{2}}{2+\sqrt{2}-2+\sqrt{2}}$$

$$= \frac{2\sqrt{2}}{2\sqrt{2}}$$

$$= 1$$

Therefore, the value of the expression is 1.

19. Correct option: (C)

$sin(120 - \alpha) = sin(120 - \beta), 0 < \alpha\ and\ \beta < \pi$

$\Rightarrow either\ 120 - \alpha = 120 - \beta \Rightarrow \alpha = \beta] or\ 120 - \alpha = 180 - (120 - \beta)$

$\therefore\ 120 - \alpha = 60 + \beta$

$\therefore\ \alpha + \beta = 60° = \left(\frac{\pi}{3}\right)$

$\therefore\ \alpha = \beta\ or\ \alpha + \beta = \left(\frac{\pi}{3}\right).$

20. Correct option: (C).

$A + B + C = 180$

$A + B = 180 - C$

$tan(A + B) = -tanC$

$$\frac{tanA + tanB}{1 - tanAtanB} = -tanC$$

$tanA + tanB + tanC = tanA\ tanB\ tanC = 6$

$Now,\ tanC = 3.\ \ So, the\ angle\ C\ is\ acute.$

$tanA + tanB = 3$

$tanA.tanB = 2$

$\therefore tanA = 2, tanB = 1$

$A\ and\ B\ are\ also\ acute.$

Therefore, it is acute angled triangle.

LEVEL -II

1. Correct option: (C)

$$\frac{tan225° - cot81°.cot69°}{cot261° + tan21°} = \frac{tan(270 - 45) - cot(90 - 9).cot(90 - 21)}{cot(270 - 9) + tan21°}$$

$$= \frac{cot45 - tan9.tan21}{cot(270 - 9) + tan21°} = \frac{1 - tan9.tan21}{tan9 + tan21}$$

$= \frac{1}{\frac{1 - tan9.tan21}{tan9 + tan21}}$

$= \frac{1}{tan30}$

$\frac{tan225° - cot81°.cot69°}{cot261° + tan21°} = \sqrt{3}$

Therefore, $\frac{tan225° - cot81°.cot69°}{cot261° + tan21°} = \sqrt{3}$

2. Correct option: (C)

Given $A + B = 135°$

$Tan(A + B) = Tan\,135°$

$But\ Tan(A + B) = \frac{TanA + TanB}{1 - TanA\,TanB}$;

$Tan\,135° = Tan(90° + 45°) = -Cot45° = -1$

$\frac{TanA + TanB}{1 - TanA\,TanB} = -1$

$TanA + TanB = -1 + TanATanB$

$TanATanB - TanA - TanB = 1$

$Add\ one\ on\ both\ sides.$

$TanATanB - TanA - TanB + 1 = 1 + 1$

$(1 - TanB) - TanA(1 - TanB) = 2$

$(1 - TanB)\,(1 - TanA) = 2.$

3. Correct option: (B)

$tan2\alpha = tan(\alpha + \beta + \alpha - \beta) = \frac{tan(\alpha + \beta) + tan(\alpha - \beta)}{1 - tan(\alpha + \beta)tan(\alpha - \beta)}$

$= \frac{\frac{3}{4} + \frac{5}{12}}{1 - \frac{3}{4} \times \frac{5}{12}} = \frac{\frac{9 + 5}{12}}{\frac{48 - 15}{48}}$

$tan2\alpha = \frac{56}{33}.$

4. Correct option: (A)

$Given\ sinxcosy = \frac{1}{4}$ 1

$3tanx = 4tany$

$3sinxcosy = 4sinycosx$

$from\ 1$

$3 \times \frac{1}{4} = 4sinycosx$

$sinycosx = \frac{3}{16}$ ……2

$\therefore sin(x - y) = sinxcosy - sinycosx$

$= \frac{1}{4} - \frac{3}{16} = \frac{1}{16}.$

Therefore, $sin(x - y) = \frac{1}{16}$.

5. Correct option: (B).

Given that,

$SinA = \frac{1}{\sqrt{10}}, \quad Sin\,B = \frac{1}{\sqrt{5}}$

Using $sin^2\theta + cos^2\theta = 1$,

$cos\,A = \sqrt{1 - sin^2A}$

$cos\ A = \sqrt{1 - \left(\frac{1}{\sqrt{10}}\right)^2} = \frac{1}{\sqrt{3}}$

Similarly,

$cos\ B = \frac{2}{\sqrt{5}}$.

$sin(A + B) = sin\ A\ cosB + cosA\ sinB$

$sin(A + B) = \frac{1}{\sqrt{10}} \cdot \frac{2}{\sqrt{5}} + \frac{3}{\sqrt{10}} \cdot \frac{1}{\sqrt{5}}$

$sin(A + B) = \frac{2 + 3}{\sqrt{50}} = \frac{1}{\sqrt{2}}$

$sin(A + B) = \frac{1}{\sqrt{2}}$

$(A + B) = \sin^{-1}\left(sin\frac{\pi}{4}\right)$

$A + B = \frac{\pi}{4}$.

6. Correct option: (A)

Given A + B + C = $\frac{\pi}{2}$;

$A + B = 90° - C;\ tan(A + B) = tan(90° - C)$

$\frac{tanA + tanB}{1 - tanA \times tanB} = c\ o\ t\ c$

$\Rightarrow tanA\ tanC + tanB\ tanC = 1 - tanA \cdot tanB$

$\Rightarrow tan\ A\ tanB + tanB\ tanC + tanC\ tanA = 1.$

7. Correct option: (A)

$cos^2 45^0 - sin^2 15^0$

We know that $cos^2 A - sin^2 B = cos(A + B)cos(A - B)$

$cos(45^0 + 15^0)cos(45^0 - 15^0) = cos60^0 cos30^0 = \frac{1}{2} \times \frac{\sqrt{3}}{2}$

$cos^2 45^0 - sin^2 15^0 = \frac{\sqrt{3}}{4}$.

8. Correct option: (B)

$tan4x + tan5x - tan9x = ktan4xtan5xtan9x$

$we\ can\ wirte\ tan9x = tan(4x + 5x) = \frac{tan(4x) + tan(5x)}{1 - tan4xtan5x}$

$tan4x + tan5x = tan9x(1 - tan4xtan5x)$

$tan4x + tan5x - tan9x = -tan4xtan5xtan9x$

$\Rightarrow k = -1$

$Hence,\ (B)\ is\ the\ correct\ of\ option.$

9. Correct option: (B)

$tan\alpha = \frac{sin\theta - cos\theta}{sin\theta + cos\theta}$

$sin\alpha = \frac{sin\theta - cos\theta}{\sqrt{sin^2\theta + cos^2\theta + sin^2\theta + cos^2\theta}}$

$sin\alpha = \pm\frac{sin\theta - cos\theta}{\sqrt{1 + 1}} = \pm\frac{sin\theta - cos\theta}{\sqrt{2}}$

$sin\alpha = \pm\frac{sin\theta - cos\theta}{\sqrt{2}}$.

10. Correct option: (D)

$sin20cos40 + cos20sin40$

$sinAcosB + cosAsinB = sin(A + B)$

$sin20cos40 + cos20sin40 = sin(40 + 20)$

$sin20cos40 + cos20sin40 = sin60 = \frac{\sqrt{3}}{2}.$

$sin20cos40 + cos20sin40 = \frac{\sqrt{3}}{2}.$

11. Correct option: (A, B)

$cosx + cosy + cosz = 0$

$and\ sinx + siny + sinz = 0$

$\Rightarrow cosx + cosy = -cosz$

$and\ sinx + siny = -sinz$

$\Rightarrow 2cos\left(\frac{x+y}{2}\right)cos\left(\frac{x-y}{2}\right) = -cosz \quad \cdots(1)$

$and\ 2sin\left(\frac{x+y}{2}\right)cos\left(\frac{x-y}{2}\right) = -sinz \quad \cdots(2)$

$(1)^2 + (2)^2\ gives,$

$4\ cos^2\left(\frac{x-y}{2}\right)\ cos^2\left(\frac{x+y}{2}\right) + 4\ cos^2\left(\frac{x-y}{2}\right)\ sin^2\left(\frac{x+y}{2}\right) = cos^2z + sin^2z$

$\Rightarrow 4\ cos^2\left(\frac{x-y}{2}\right)[\ cos^2\left(\frac{x+y}{2}\right) + sin^2\left(\frac{x+y}{2}\right)] = cos^2z + sin^2z$

$\Rightarrow 4\ cos^2\left(\frac{x-y}{2}\right) = 1$

$\therefore\ cos^2\left(\frac{x-y}{2}\right) = \left(\frac{1}{4}\right)$

$\Rightarrow cos\left(\frac{x-y}{2}\right) = \pm\left(\frac{1}{2}\right)$

12. Correct option: (B)

$tanA = \frac{1}{35}\ , \qquad tanB = \frac{18}{17}$

$tan(A-B) = \frac{tanA - tanB}{1 + tanA \times tanB}$

$tan(A-B) = \frac{\frac{1}{35} - \frac{18}{17}}{1 + \frac{1}{35} \times \frac{18}{17}} = \frac{17 - 18(35)}{35(17) + 18} = \frac{-613}{613}$

$tan(A-B) = -1$

13. Correct option: (A, D)

Expressing each tangent in terms of sine and cosine:

$$tanA = \frac{sinA}{cosA}$$

$$tanB = \frac{sinB}{cosB}$$

$$tanC = \frac{sinC}{cosC}$$

Substituting these expressions into the equation, we have:

$\frac{sin\ A}{cos\ A} + \frac{sin\ B}{cosB} + \frac{sin\ C}{cos\ C} - \frac{sin\ A}{cos\ A} \times \frac{sin\ B}{cos\ B} \times \frac{sin\ C}{cos\ C} = 1\ .$

To simplify the left-hand side, we can find a common denominator for all the terms, which is $cosA \times cosB \times cosC$:

$$\frac{[sinA \times cosB \times cosC + sinB \times cosA \times cosC + sinC \times cosA \times cosB - sinA \times sinB \times sinC]}{(cosA \times cosB \times cosC)}$$

$$= 1$$

Using the trigonometric identity $sin(A + B + C) = sinA \times cosB \times cosC + sinB \times cosA \times cosC + sinC \times cosA \times cosB - sinA \times sinB \times sinC$, we can rewrite the numerator:

$sin(A + B + C) - (sinA \times sinB \times sinC) = (cosA \times cosB \times cosC)$
Since A + B + C = 180 degrees in a triangle, $sin(A + B + C) = sin(180\ degrees) = 0$.
Therefore, the equation becomes:
$0 - (sinA \times sinB \times sinC) = (cosA \times cosB \times cosC)$
Simplifying further:
$-sinA \times sinB \times sinC = cosA \times cosB \times cosC$
Using the trigonometric identity sin(-x) = -sin(x), we can rewrite the equation as:
$sinA \times sinB \times sinC = -cosA \times cosB \times cosC$
Now, we know that in a triangle, the angles A, B, and C are related by the identity $cosA \times cosB \times cosC = sinA \times sinB \times sinC$. Therefore, we can substitute this identity:
$sinA \times sinB \times sinC = -sinA \times sinB \times sinC$
Both sides of the equation are equal, verifying the given statement.
Hence, we have proven that if A, B, C are angles of a triangle, and if none of them is equal to 90 degrees, then $tanA + tanB + tanC - tanA \times tanB \times tanC = sin(90°)\ or\ cos(0°)$.

14. Correct option: (D)

$A + B = \left(\frac{5\pi}{4}\right)$

$tan(A + B) = tan\left(\frac{5\pi}{4}\right)$

$\frac{tanA+tanB}{1-tanAtanB} = 1 \qquad Since, \left(\pi + \frac{\pi}{4}\right) = 1.$

$tanA + tanB = 1 + 1 - tanA\ tanB$

$tanA + tanB + tanA\ tanB = 2$

$(1 + tanA)(1 + tanB) = 2.$

15. Correct option: (B)

$A + C = B$

$tan(A + C) = tan\text{B}$

$\frac{tanA+tanC}{1-tanAtanC} = tan\text{B}$

$tanA + tanC = tan\text{B}(1 - tanA\ tanC)$

$tanA + tanC + tan\text{B} = -tanA\ tanCtan\text{B}$

$tanB - tanC - tan\text{A} = tanA\ tanCtan\text{B}$

$tanA\ tanCtan\text{B} = tanB - tanC - tan\text{A}.$

16. Correct Matching is: $a \rightarrow P$, $b \rightarrow$ Q, c$\rightarrow R$, d$\rightarrow P$

a) $cos20° + cos100° + cos140° = 2cos\left(\frac{20°+100°}{2}\right) cos\left(\frac{20°-100°}{2}\right) + cos140°$

$\left[\because cosA + cosB = 2cos\left(\frac{A+B}{2}\right) cos\left(\frac{A-B}{2}\right)\right]$

$= 2 \times cos60° \times cos(-40°) + cos140°$

$= 2 \times \frac{1}{2} \times cos25° + cos155°$

$= cos40° + cos140°$

$= 2cos\left(\frac{40° + 140°}{2}\right) cos\left(\frac{40° - 140°}{2}\right)$

$= 2 \times cos90° \times cos50° = 0.$

$cos20° + cos100° + cos140° = 0.$

b) $(1 + tanA)(1 + tanB) = 2 + tanA + tanB - \frac{tanA+tanB}{\tan(A+B)}.$

$(1 + tan20°)(1 + tan25°) = 2 + tan20° + tan25° - \frac{tan20° + tan25°}{\tan(20° + 25°)}.$

$(1 + tan20°)(1 + tan25°) = 2 + tan20° + tan25° - \frac{tan20° + tan25°}{\tan(45°)}.$

$(1 + tan20°)(1 + tan25°) = 2 + tan20° + tan25° - \frac{tan20° + tan25°}{\tan(45°)}.$

$(1+tan20°)(1+tan25°) = 2 + tan20° + tan25° - \frac{tan20° + tan25°}{1}$.

$(1+tan20°)(1+tan25°) = 2 + tan20° + tan25° - tan20° - tan25°$.

$(1+tan20°)(1+tan25°) = 2$.

c) From given, we have,

$20° + 32° + 38° = 90°$

$S = tanAtanB + tanBtanC + tanCtanA \ldots\ldots (1)$

$\Rightarrow S = tanAtanB + tanC(tanA + tanB)$

$where\ A + B + C = 90°, and\ hence,\ C = 90° - (A + B)$

$tanC = tan(90° - A - B) = cot(A + B) = \frac{1}{tan(A+B)}$

$tanC = \frac{1 - tanAtanB}{tanA + tanB}$

$tanC(tanA + tanB) = 1 - tanAtanB \ldots (2)$

$from\ (1)\ and\ (2)$

$S = tanAtanB + tanC(tanA + tanB)$

$= tanAtanB + (1 - tanAtanB) = 1$

d) $\sin^2 10^0 + \sin^2 110^0 + \sin^2 130^0 = \sin^2(10^0 + 110^0 + 130^0)$

$= \sin^2(250^0)$

$= \sin^2(90^0 + 160^0)$

$= \cos^2(160^0) = \cos^2(90^0 + 70^0)$

$= -\sin^2 70^0 = 0.$

17. Correct Answer is: 0

$\sum \frac{sin(A+B)sin(A-B)}{cos^2Acos^2B} = \sum \frac{(sinAcosB + cosAsinB)(sinAcosB - cosAsinB)}{cos^2Acos^2B}$

$= \sum \frac{(sin^2Acos^2B - cos^2Asin^2B)}{cos^2Acos^2B}$

$= \sum \frac{(sin^2Acos^2B)}{cos^2Acos^2B} - \sum \frac{(cos^2Asin^2B)}{cos^2Acos^2B}$

$= \sum \frac{(sin^2A)}{cos^2A} - \sum \frac{(sin^2B)}{cos^2B}$

$= \sum tan^2A - tan^2B$

$= tan^2A - tan^2B + tan^2B - tan^2C + tan^2C - tan^2A = 0$

18. Correct Answer is: 2

$\frac{tan80° - tan10°}{cos70°} = \frac{\frac{sin80°}{cos80°} - \frac{sin10°}{cos10°}}{\frac{sin70°}{cos70°}} = \frac{(sin80°cos10° - sin10°cos80°)}{\frac{cos80°cos10°}{\frac{sin70°}{cos70°}}}$

$= \frac{sin(80° - 10°).cos70°}{cos80°cos10°sin70°}$

$= \frac{sin70°cos70°}{cos80°cos10°sin70°}$

$= \frac{cos70°}{cos80°cos10°}$

$= \frac{cos(80° - 10°)}{cos80°cos10°}$

$= \frac{cos80°cos10° + sin80°sin10°}{cos80°cos10°}$

$= 1 + tan80°tan10°$

$= 1 + tan(90° - 10°)tan10°$

$= 1 + cot10°tan10° = 1 + 1 = 2.$

19. Correct Answer is: 1.

$cot\left(\frac{\pi}{4}-\theta\right).cot\left(\frac{\pi}{4}+\theta\right)$

$=\frac{1}{tan\left(\frac{\pi}{4}-\theta\right)}\times\frac{1}{tan\left(\frac{\pi}{4}+\theta\right)}$

$=\frac{1-tan\theta}{1+tan\theta}\times\frac{1+tan\theta}{1-tan\theta}$

$=1$

20. Correct Answer is: 0

$cosA+cos(240°+A)+cos(240°-A)$

$=cosA+cos240°cosA-sin240°sinA+cos240°cosA+sin240°sinA$

$=cosA+cos240°cosA$

$=cosA(1+2cos240°)$

$=cosA\big(1+2sin(180°+60°)\big)$

$=cosA\big(1+2(-sin60°)\big)$

$=cosA\left(1-2\left(\frac{1}{2}\right)\right)=0.$

STUDENT PRACTICE QUESTIONS

1. Correct option: (C)

$tan\theta=\frac{p}{q}$.

$$\frac{psin\theta-qcos\theta}{p\,sin\theta+qcos\theta}=\frac{p\frac{sin\theta}{cos\theta}-q\frac{cos\,\theta}{cos\theta}}{p\,\frac{sin\theta}{cos\theta}+q\frac{cos\,\theta}{cos\theta}}$$

$$=\frac{ptan\theta-q}{p\,tan\theta+q}=\frac{p\times\frac{p}{q}-q}{p\,\times\frac{p}{q}+q}$$

$$=\frac{p^2-q^2}{p^2+q^2}$$

Therefore, $\frac{psin\theta-qcos\theta}{p\,sin\theta+qcos\theta}=\frac{p^2-q^2}{p^2+q^2}$.

2. Correct option: (A)

$sin\theta+cos\theta=\sqrt{3}cos\theta$

$\Rightarrow sin\theta=\sqrt{3}cos\theta-cos\theta$

$\Rightarrow sin\theta=(\sqrt{3}-1)cos\theta$

$\Rightarrow\frac{sin\theta}{cos\theta}=(\sqrt{3}-1)$

$\Rightarrow\frac{sin\theta}{cos\theta}=\frac{(\sqrt{3}-1)(\sqrt{3}+1)}{\sqrt{3}+1}$

$\Rightarrow\frac{sin\theta}{cos\theta}=\frac{\left(\sqrt{3}\right)^2-(1)^2}{\sqrt{3}+1}$

$\Rightarrow\frac{sin\theta}{cos\theta}=\frac{2}{\sqrt{3}+1}$

$\Rightarrow 2cos\theta=\sqrt{3}sin\theta+sin\theta$

$\Rightarrow 2cos\theta-sin\theta=\sqrt{3}sin\theta$

Therefore, $2cos\theta-sin\theta=\sqrt{3}sin\theta$.

3. Correct option: (B)

Given, B and A+B is an acute angle then A is also acute angle.

$sinB = \frac{5}{13}, \quad cosB = \frac{12}{13}.$

$sin(A+B) = \frac{12}{13}, \quad cos(A+B) = \frac{5}{13}.$

Now,

$sin(A+B) = \frac{12}{13}$

$sinA\ cosB + cos\ A\ sinB = \frac{12}{13}$

$sinA\ \times\frac{12}{13} + cos\ A \times \frac{5}{13} = \frac{12}{13}$

$12sinA + 5cosA = 12. - - - - - - - - - - - - - - - (1)$

Now, $cos(A+B) = \frac{5}{13}$

$cosA\ cosB - sin\ A\ sinB = \frac{5}{13}$

$cosA\ \times\frac{12}{13} - sin\ A \times \frac{5}{13} = \frac{5}{13}$

$12cosA - 5sinA = 5.$

$cosA = \frac{5+5sinA}{12} - - - - - - - - - - - - - - - (2)$

Substitute $cosA = \frac{5+5sinA}{12}$ in (1).

$12sinA + \ 5 \times \frac{5+5sinA}{12} = 12$

$144sinA + 25sinA + 25 = 144.$

$169sinA = 119$

$sinA = \frac{119}{169}.$

Therefore, $sinA = \frac{119}{169}.$

4. Correct option: (D)

$tan135° = tan(90° + 45°) = \ -tan45° = -1$

$tan(69° + 66°) = \frac{tan69° + tan66°}{1 - tan69°tan66°}$

$\Rightarrow -1 = \frac{tan69° + tan66°}{1 - tan69°tan66°}$

$\Rightarrow tan69° + tan66° - tan69°tan66° = -1$

$Therefore,$

$2k = -1$

$\Rightarrow k = -\frac{1}{2}.$

5. Correct option: (B)

$secA + tanA = 3 \rightarrow (1)$

$secA - tanA = \frac{1}{3} \rightarrow (2)$

Add (1) and (2)

$2secA = \frac{10}{3}$

$secA = \frac{10}{6} = \frac{5}{3}$

Therefore, $secA = \frac{5}{3}.$

6. Correct option: (B)

$sinx\ cosy + cos\ x\ siny = sin(x + y)$

$Assuming \quad x = 45° - A, y = 45° - B$

$sin(45° - A)cos(45° - B) + cos(45° - A)sin(45° - B)$

$= sin(45° + 45° - A - B)$

$= sin\big(90° - (A + B)\big)$

$= \cos(A + B)$

Therefore, $sin(45° - A)cos(45° - B) + cos(45° - A)sin(45° - B) = cos(A + B)$.

7. Correct option: (B)

$sinx\ siny\ +\ cosx\ cosy\ =\ cos(x - y)$

$Substituting \quad x = (n + 1)A$

$y = (n + 2)A$

$we\ get \quad sin(n + 1)A\ sin(n + 2)A\ +\ cos(n + 1)A\ cos(n + 2)A = cos[(n + 1)A - (n + 2)A]$

$= cos[-A]$

$= cosA.$

Therefore, $sin(n + 1)Asin(n + 2)A + cos(n + 1)Acos(n + 2)A = cosA.$

8. Correct option: (B)

$Using\ the\ formulae -$

$CosA.CosB - SinASinB = Cos(A + B)$

$Here\ A = 28°, B = 32°$

$then, Cos(A + B) = Cos(28° + 32°)$

$Cos60° = \frac{1}{2}.$

Therefore, $Cos60° = \frac{1}{2}.$

9. Correct option: (A)

$tan75° = tan(45° + 30°)$

$tan(45° + 30°) = \frac{tan45° + tan30°}{1 - tan45°tan30°}$

Since, $tan45° = 1\ and\ tan30° = \frac{1}{\sqrt{3}}.$

Now,

$$tan(45° + 30°) = \frac{1 + \frac{1}{\sqrt{3}}}{1 - 1 \times \frac{1}{\sqrt{3}}} = \frac{1 + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}} = \frac{\frac{\sqrt{3} + 1}{\sqrt{3}}}{\frac{\sqrt{3} - 1}{\sqrt{3}}} = \frac{\sqrt{3} + 1}{\sqrt{3} - 1}$$

$$tan(45° + 30°) = tan75° = \frac{\sqrt{3} + 1}{\sqrt{3} - 1}$$

$cot75° = \frac{\sqrt{3}-1}{\sqrt{3}+1}$ (Cotangent is reciprocal of tangent.)

Now,

$$tan75° - cot75° = \frac{\sqrt{3} + 1}{\sqrt{3} - 1} - \frac{\sqrt{3} - 1}{\sqrt{3} + 1}$$

$$tan75° - cot75° = \frac{\left(\sqrt{3} + 1\right)^2 - \left(\sqrt{3} - 1\right)^2}{\left(\sqrt{3}\right)^2 - (1)^2}$$

$$tan75° - cot75° = \frac{\left[\left(\sqrt{3}\right)^2 + 2\sqrt{3}(1) + (1)^2\right] - \left[\left(\sqrt{3}\right)^2 - 2\sqrt{3}(1) + (1)^2\right]}{3 - 1}$$

$$tan75° - cot75° = \frac{\left[3 + 2\sqrt{3} + 1\right] - \left[3 - 2\sqrt{3} + 1\right]}{2} = \frac{4 + 2\sqrt{3} - 4 + 2\sqrt{3}}{2} = \frac{4\sqrt{3}}{2} = 2\sqrt{3}.$$

Therefore, $tan75° - cot75° = 2\sqrt{3}.$

10. Correct option: (A)

$We\ know\ that,$

$sin^2A - sin^2B = sin(A+B) \cdot sin(A-B)$

$sin^2 52\frac{1^\circ}{2} - sin^2 22\frac{1^\circ}{2} = sin75^\circ \times sin30^\circ = sin30^\circ \times sin(45^\circ + 30^\circ)$

$= sin30^\circ(sin45^\circ cos30^\circ + sin30^\circ cos45^\circ)$

$sin^2 52\frac{1^\circ}{2} - sin^2 22\frac{1^\circ}{2} = \frac{1}{2} \times \frac{\sqrt{3}+1}{2\sqrt{2}} = \frac{\sqrt{3}+1}{4\sqrt{2}}$

Therefore, $sin^2 52\frac{1^\circ}{2} - sin^2 22\frac{1^\circ}{2} = \frac{\sqrt{3}+1}{4\sqrt{2}}$.

11. Correct option: (B)

$tan(A+B) = \dfrac{tanA + tanB}{1 - tanA \times tanB}$

Here $A = 100^\circ$, $B = 125^\circ$.

$tan(100^\circ + 125^\circ) = \dfrac{tan100^\circ + tan125^\circ}{1 - tan100^\circ \times tan125^\circ}$

$tan(225^\circ) = \dfrac{tan100^\circ + tan125^\circ}{1 - tan100^\circ \times tan125^\circ}$

$1 = \dfrac{tan100^\circ + tan125^\circ}{1 - tan100^\circ \times tan125^\circ}$

Since $tan(225^\circ) = 1$.

$\dfrac{tan100^\circ + tan125^\circ}{1 - tan100^\circ \times tan125^\circ} = 1$

$tan100^\circ + tan125^\circ = 1 - tan100^\circ \times tan125^\circ$

$tan100^\circ + tan125^\circ + tan100^\circ \times tan125^\circ = 1$

Therefore, $tan100^\circ + tan125^\circ + tan100^\circ \times tan125^\circ = 1$.

12. Correct option: (A)

$cos35^\circ + cos85^\circ + cos155^\circ = 2cos\left(\dfrac{35^\circ + 85^\circ}{2}\right) cos\left(\dfrac{35^\circ - 85^\circ}{2}\right) + cos155^\circ$

$\left[\because cosA + cosB = 2cos\left(\dfrac{A+B}{2}\right) cos\left(\dfrac{A-B}{2}\right)\right]$

$= 2 \times cos60^\circ \times cos(-25^\circ) + cos155^\circ$

$= 2 \times \dfrac{1}{2} \times cos25^\circ + cos155^\circ$

$= cos25^\circ + cos155^\circ$

$= 2cos\left(\dfrac{25^\circ + 155^\circ}{2}\right) cos\left(\dfrac{25^\circ - 155^\circ}{2}\right)$

$= 2 \times cos90^\circ \times cos65^\circ = 0$

$cos35^\circ + cos85^\circ + cos155^\circ = 0.$

13. Correct option: (A)

$Cos\ 42^\circ + cos\ 78^\circ + cos\ 162^\circ = cos(60^\circ - 18^\circ) + cos(60^\circ + 18^\circ) + cos(180^\circ - 18^\circ)$

$= cos\ 60^\circ \times cos\ 18^\circ + sin\ 60^\circ \times sin\ 18^\circ + cos\ 60^\circ \times cos\ 18^\circ - sin60^\circ \times sin\ 18^\circ - cos\ 18^\circ$

$= 2cos\ 60^\circ \times cos\ 18^\circ - cos\ 18^\circ$

$= 2 \times \dfrac{1}{2} \times cos\ 18^\circ - cos\ 18^\circ$

$Cos\ 42^\circ + cos\ 78^\circ + cos\ 162^\circ = 0$

14. Correct option: (B)

$We\ have, cosA = \dfrac{5}{13}$

$Then,\ \ sinA = \frac{12}{13}$

$Since, tanB = -\dfrac{15}{8}$

$Then,\ \ cosB = \frac{8}{17}, sinB = \frac{15}{17}$

$Therefore,$

$sin(A+B) = sinA.cosB + sinB.cosA$

$sin(A+B) = \frac{12}{13}.\frac{8}{17} + \frac{15}{17}.\frac{5}{13}$

$sin(A+B) = \frac{96}{221} + \frac{75}{221}$

$sin(A+B) = \frac{171}{221}$

15. Correct option: (A)

$\boldsymbol{tan25°.tan31° + tan31°.tan34° + tan34°.tan25°}$

Here, $a = 25, b = 31, c = 34$

$a + b + c = 25 + 31 + 34 = 90$

$\boldsymbol{tan}(a+b+c) = \frac{tan\ a + tan\ b + tanc - (tan\ a.tan\ b.tan\ c)}{1-(tan\ a \times tan\ b + tan\ b \times tan\ c + tan\ c \times tan\ a)}$

$\boldsymbol{tan}(90) = \frac{tana + tanb + tanc - (tana.tanb.tanc)}{1-(tana \times tanb + tanb \times tanc + tanc \times tana)}$

$\frac{1}{0} = \frac{tana + tanb + tanc - (tana.tanb.tanc)}{1-(tana \times tanb + tanb \times tanc + tanc \times tana)}$

$1 - (tan\ a \times tan\ b + tan\ b \times tan\ c + tan\ c \times tan\ a) = 0$

$1 = (tana \times tanb + tanb \times tanc + tanc \times tana)$

$tan25°.tan31° + tan31°.tan34° + tan34°.tan25° = 1$

16. Correct option: (C)

$\frac{cos15° + sin15°}{cos15° - sin15°} = \frac{1 + \frac{sin15°}{cos15°}}{1 - \frac{sin15°}{cos15°}} = \frac{tan45° + tan15°}{1 - tan45°tan15°} = tan60°$

$\frac{cos15° + sin15°}{cos15° - sin15°} = \sqrt{3}.$

17. Correct option: (A)

$\Rightarrow cotB - cotA = y$

$\therefore \frac{1}{(tanB)} - \frac{1}{(tanA)} = y$

$\Rightarrow \frac{tanA - tanB}{tanAtanB} = y$

$\Rightarrow \frac{x}{y} = tanAtanB$

$Now,$

From the given option let take,

$cot(A-B) = \frac{1}{tan(A-B)} = \frac{1 + tanAtanB}{tanA - tanB} = \frac{1 + \frac{x}{y}}{x} = \frac{1}{x} + \frac{x}{y}.$

Therefore, $\frac{1}{x} + \frac{x}{y} = cot(A-B)$.

18. Correct option: (A, B)

$We\ have$

$sin(\alpha - \beta) = sin(\alpha - \theta + \theta - \beta)$

$= sin[(\theta - \beta) - (\theta - \alpha)]$

$= sin(\theta - \beta)cos(\theta - \alpha) - cos(\theta - \beta)sin(\theta - \alpha)$

$= ba - \sqrt{1-b^2}\sqrt{1-a^2}$

$and\ cos(\alpha - \beta) = cos[(\theta - \beta) - (\theta - \alpha)]$

$= cos(\theta - \beta)cos(\theta - \alpha) + sin(\theta - \beta)sin(\theta - \alpha)$

$= a\sqrt{1-b^2} + b\sqrt{1-a^2}$

Substituting these values in the given expression, we get

$cos^2(\alpha-\beta) + 2absin(\alpha-\beta)$

$= \left(a\sqrt{1-b^2} + b\sqrt{1-a^2}\right) + 2ab\left[ab - \sqrt{1-a^2}\sqrt{1-b^2}\right]$

$= a^2(1-b^2) + b^2(1-a^2) + 2ab\sqrt{1-a^2}\sqrt{1-b^2} + 2a^2b^2 - 2ab\sqrt{1-a^2}\sqrt{1-b^2}$

$= a^2 + b^2 = (a+b)^2 - 2ab.$

19. Correct option: (B)

tanA = $-\frac{15}{8}$

sinB = $-\frac{7}{25}$

Given,

$8tanA = -15 \Rightarrow tanA = -\frac{15}{8}$

Here, tanA is negative and A does not lie in 4th quadrant, so A lies in 2nd quadrant.

In a right angled triangle, $tanA = \frac{opposite\ side}{adjacent\ side}$

So, length of opposite side is 15 and adjacent is 8.

So, hypotenuse side=$\sqrt{15^2 + 8^2} = \sqrt{289} = 17$

So, sinA=$\frac{opposite\ side}{hypotenuse\ side}$=$\frac{15}{17}$

∴sinA=$\frac{15}{17}$ ∵sinA is positive in 2nd quadrant

and cosA=$\frac{adjacent\ side}{hypotenuse\ side}$=$\frac{8}{17}$

∴cosA=-$\frac{8}{17}$ ∵$cosA$ is negative in 2nd quadrant

Also, given that $25sinB = -7 \Rightarrow sinB = -\frac{7}{25}$

Here, sinB is negative and B does not lie in 4th, so B lies in 3rd quadrant.

We know that $cos\,B = \sqrt{1 - sin^2\,B}$

$$cos\,B = \sqrt{1 - \frac{49}{625}} = \sqrt{\frac{576}{625}} = \pm\frac{24}{25}$$

$\therefore cos\,B = -\frac{24}{25}$ ∵cos B is negative in 3rd quadrant

Now, $sinAcosB + CosAsinB = \frac{15}{17} \times \left(\frac{-24}{25}\right) + \left(\frac{-8}{17}\right) \times \left(\frac{-7}{25}\right) = \frac{-304}{425}$

$sin(A+B) = sinAcosB + CosAsinB = \frac{-304}{425}$.

20. Correct option: (A)

We have $acos\theta = c - bsin\theta$

Squaring both sides

$\Rightarrow a^2(1 - sin^2\theta) = c^2 - 2bcsin\theta + b^2sin^2\theta$

$\Rightarrow (a^2 + b^2)sin^2\theta - 2bcsin\theta + (c^2 - a^2) = 0$

Since α and β are the values of θ as given,

∴ roots of the above equation are $sin\alpha\ and\ sin\beta$.

$\therefore sin\alpha + sin\beta = Sum\ of\ roots = \frac{2bc}{a^2+b^2}$

and $sin\alpha \times sin\beta = Product\ of\ roots = \frac{c^2-a^2}{a^2+b^2}$.

21. Correct option: (A)

We have $acos\theta - c = bsin\theta$

Squaring both sides

$\Rightarrow a^2(\cos^2\theta) + c^2 - 2ac\,coss\theta = b^2sin^2\theta$

$\Rightarrow a^2cos^2\theta + c^2 - 2accos\theta = b^2(1 - cos^2\theta)$

$\Rightarrow cos^2\theta(a^2+b^2) - 2ac\,cos\theta + c^2 - b^2 = 0.$

Since α and β are the values of θ as given,

∴ roots of the above equation are $cos\alpha$ and $cos\beta$.

$\therefore \cos\alpha + \cos\beta = Sum\ of\ roots = \frac{2ac}{a^2+b^2}$

and $cos\alpha \times \cos\beta = Product\ of\ roots = \frac{c^2-b^2}{a^2+b^2}$.

22. Correct option: (C)

We have $acos\theta = c - bsin\theta$

Squaring both sides

$\Rightarrow a^2(1 - sin^2\theta) = c^2 - 2bcsin\theta + b^2sin^2\theta$

$\Rightarrow (a^2 + b^2)sin^2\theta - 2bcsin\theta + (c^2 - a^2) = 0$

Since α and β are the values of θ as given,

∴ roots of the above equation are $sin\alpha$ and $sin\beta$.

$\therefore sin\alpha + sin\beta = Sum\ of\ roots = \frac{2bc}{a^2+b^2}$

and $sin\alpha \times sin\beta = Product\ of\ roots = \frac{c^2-a^2}{a^2+b^2}$.----------(1)

We have $acos\theta - c = bsin\theta$

Squaring both sides

$\Rightarrow a^2(\cos^2\theta) + c^2 - 2ac\,coss\theta = b^2sin^2\theta$

$\Rightarrow a^2cos^2\theta + c^2 - 2accos\theta = b^2(1 - cos^2\theta)$

$\Rightarrow cos^2\theta(a^2+b^2) - 2ac\,cos\theta + c^2 - b^2 = 0.$

Since α and β are the values of θ as given,

∴ roots of the above equation are $cos\alpha$ and $cos\beta$.

$\therefore \cos\alpha + \cos\beta = Sum\ of\ roots = \frac{2ac}{a^2+b^2}$

and $cos\alpha \times \cos\beta = Product\ of\ roots = \frac{c^2-b^2}{a^2+b^2}$.----------(2)

Subtracting (2) from (1),

$cos\alpha \times \cos\beta - sin\alpha \times sin\beta = \frac{c^2-b^2}{a^2+b^2} - \frac{c^2-a^2}{a^2+b^2}$

$cos(\alpha+\beta) = \frac{a^2-b^2}{a^2+b^2}$.

23. Correct Matching is: $a \rightarrow Q$, $b \rightarrow$ R, c→ S, d→ R.

a) $Taking\ cosine\ on\ both\ sides, we\ get$

$cos(A+B) = cos\frac{\pi}{2}$

$cosAcosB - sinAsinB = 0$

$cosAcosB = sinAsinB$

$tanA = tanB$

$A = n\pi + B$

$If\ n = 0,$

$A = B - - - - - - - - -1$

$LHS = tanA$

$= tanB - - - - - - - 2\ \ (By\ 1)$

$RHS = tanB + 2tan(A-B)$

$= tanB\ + 2tan(B-B) - - - -By\ 1$

$= tanB - - - - - - - - 3$

$from\ 2\ and\ 3,$

$LHS = RHS$

Therefore, $tanA = tanB + 2tan(A-B)$.

b) $A+B=\left(\frac{\pi}{4}\right)$

$tan(A+B)=tan\left(\frac{\pi}{4}\right)$

$\frac{tanA+tanB}{1-tanAtanB}=1$ $Since, \tan\left(\frac{\pi}{4}\right)=1.$

$tanA+tanB=1+1-tanA\,tanB$

$tanA+tanB+tanA\,tanB=2$

$(1+tanA)(1+tanB)=2.$

c) $Given\ A+B=\frac{5\pi}{4}$

$Taking\ both\ side\ 'cot'\ and\ we\ get \quad cot(A+B)=cot\frac{5\pi}{4}$

$\frac{cotAcotB-1}{cotB+cotA}=cot(\pi+\frac{\pi}{4})$

$\therefore cot(A+B)=\frac{cotAcotB-1}{cotB+cotA}$

$\Rightarrow \frac{cotAcotB-1}{cotB+cotA}=cot\frac{\pi}{4}\ \therefore cot(\pi+\theta)=cot\theta$

$\Rightarrow \frac{cotAcotB-1}{cotB+cotA}=1\ \therefore cot\frac{\pi}{4}=1$

$\Rightarrow cotAcotB-1=cotB+cotA$

$on\ adding\ both\ side(cotAcotB+1)$

$\Rightarrow cotAcotB-1+cotAcotB+1=cotB+cotA+cotAcotB+1$

$\Rightarrow 2cotAcotB=cotB+cotAcotB+cotA+1$

$\Rightarrow 2cotAcotB=cotB(1+cotA)+1(1+cotA)$

$\Rightarrow 2cotAcotB=(1+cotA)(1+cotB)$

$\Rightarrow (1+cotA)(1+cotB)=2cotAcotB$

$\Rightarrow \frac{cotAcotB}{(1+cotA)(1+cotB)}=2$

$On\ reciprocal\ that$

$\Rightarrow \frac{(1+cotA)(1+cotB)}{cotAcotB}=\frac{1}{2}.$s

d) $A+B=\left(\frac{3\pi}{4}\right)$

$tan(A+B)=tan\left(\frac{3\pi}{4}\right)$

$\frac{tanA+tanB}{1-tanAtanB}=-1$ $Since, \tan\left(\frac{3\pi}{4}\right)=-1.$

$tanA+tanB=-1(1-tanA\,tanB)$

$tanA+tanB+tanA\,tanB=2$

$(1+tanA)(1+tanB)=2.$

24. Correct Answer is: 7

$tan\left(\frac{\pi}{4}+\theta\right)+\ tan\left(\frac{\pi}{4}-\theta\right)=3$

$Squaring\ both\ sides$

$=>\left(tan\left(\frac{\pi}{4}+\theta\right)+\ tan\left(\frac{\pi}{4}-\theta\right)\right)^2=3^2$

$Using\ identity\ (a+b)^2=a^2+b^2+2ab$

$=>\ tan^2\left(\frac{\pi}{4}+\theta\right)+\ tan^2\left(\frac{\pi}{4}-\theta\right)+2\ tan\left(\frac{\pi}{4}+\theta\right)tan\left(\frac{\pi}{4}-\theta\right)=9$

$tan(\theta)=cot\left(\frac{\pi}{2}-\theta\right)$

$=>\ tan\left(\frac{\pi}{4}+\theta\right)=cot\left(\frac{\pi}{2}-\left(\frac{\pi}{4}-\theta\right)\right)=cot\left(\frac{\pi}{4}-\theta\right)$

$tanx=1/cotx$

$=> tanx\,.cotx = 1$

$=> \tan\left(\frac{\pi}{4}+\theta\right)\tan\left(\frac{\pi}{4}-\theta\right) = \cot\left(\frac{\pi}{4}-\theta\right) \times \tan\left(\frac{\pi}{4}-\theta\right) = 1$

$\tan^2\left(\frac{\pi}{4}+\theta\right) + \tan^2\left(\frac{\pi}{4}-\theta\right) + 2\tan\left(\frac{\pi}{4}-\theta\right)\tan\left(\frac{\pi}{4}-\theta\right) = 9$

$=> \tan^2\left(\frac{\pi}{4}+\theta\right) + \tan^2\left(\frac{\pi}{4}-\theta\right) + 2 \times 1 = 9$

$=> \tan^2\left(\frac{\pi}{4}+\theta\right) + \tan^2\left(\frac{\pi}{4}-\theta\right) = 7.$

25. Correct Answer is: 2

$A+B+C=\frac{\pi}{2}$

$\sum\frac{cos(B+C)}{cosbcosC}$

$=\sum\frac{cosBcosC-sinBsinC}{cosBcosC}$

$=\sum 1-tanBtanC$

$=3-[tanAtanB+tanBtanC+tanCtanA]$

$=3-1=2$

26. Correct Answer is: 3

$cot\alpha, cot\beta, cot\gamma \text{ are in } A.P.$

$2cot\beta = cot\alpha + cot\gamma = \frac{tan\alpha + tan\gamma}{tan\alpha tan\gamma}$

$2cot\beta = \frac{tan(\alpha+\gamma)(1-tan\alpha tan\gamma)}{tan\alpha tan\gamma}$

$2cot\beta = cot\beta\frac{(1-tan\alpha tan\gamma)}{tan\alpha tan\gamma}$

$Let\ tan\alpha tan\gamma = k$

$2=\frac{1-k}{k}$

$3k=1$

$tan\alpha tan\gamma = k = \frac{1}{3}$

$cot\alpha cot\gamma = 3.$

27. Correct Answer is: $\frac{\pi}{4}$

$sinA=\frac{1}{\sqrt{10}} \Rightarrow A=\sin^{-1}\left(\frac{1}{\sqrt{10}}\right)$

$sinB=\frac{1}{\sqrt{5}} \Rightarrow B=\sin^{-1}\left(\frac{1}{\sqrt{5}}\right)$

$Now,\quad A+B=\sin^{-1}\left(\frac{1}{\sqrt{10}}\right)+\sin^{-1}\left(\frac{1}{\sqrt{5}}\right)$

$As\ we\ know\ that,\qquad \sin^{-1}A+\sin^{-1}B=\sin^{-1}\left(A\sqrt{1-B^2}+B\sqrt{1-A^2}\right)$

$\therefore A+B=\sin^{-1}\frac{1}{\sqrt{10}}\left(1-\frac{1}{\sqrt{5}}\right)+\frac{1}{\sqrt{5}}\left(1-\frac{1}{\sqrt{10}}\right)$

$A+B=\sin^{-1}\left(\frac{2}{\sqrt{10}}+\frac{3}{\sqrt{50}}\right)$

$\Rightarrow A+B=\sin^{-1}\left(\frac{5}{\sqrt{50}}\right)$

$\Rightarrow A+B=\sin^{-1}\left(\frac{1}{\sqrt{2}}\right).$

$\Rightarrow A+B=\frac{\pi}{4}(\because A \text{ and } B \text{ are positive acute angles})$

$Hence\ the\ correct\ answer\ is\frac{\pi}{4}.$

MULTIPLE AND SUBMULTIPLE ANGLES - TRANSFORMATIONS

LEVEL -I

1. Correct option: (C)

$\sin 2\theta = 2\sin\theta\cos\theta$

$1 + \cos 2\theta = 2\cos^2\theta$

$\frac{\sin 2\theta}{1+\cos 2\theta} = \frac{2\sin\theta\cos\theta}{2\cos^2\theta} = \frac{\sin\theta}{\cos\theta} = \tan\theta.$

2. Correct option: (D)

$\sqrt{2+\sqrt{2+2\cos 2\theta}} = \sqrt{2+\sqrt{2(1+\cos 2\theta)}} = \sqrt{2+\sqrt{2(2\cos^2\theta)}}$

$= \sqrt{2+2\cos^2\theta} = 2\cos\frac{\theta}{2}.$ [Since $1 + \cos 2\theta = 2\cos^2\theta$]

3. Correct option: (C)

$\cos\alpha = \frac{15}{17}$

We know, $\cos 2\theta = 2\cos^2\theta - 1$

$\cos 2\alpha = 2\cos^2\alpha - 1 = 2\left(\frac{15}{17}\right)^2 - 1 = 2\left(\frac{225}{289}\right) - 1 = \frac{450-289}{289} = \frac{161}{289}.$

4. Correct option: (B)

$\frac{sinA}{1+cosA} + \frac{sinA}{1-cosA}$

$= sinA\left(\frac{1-cosA+1+cosA}{(1+cosA)(1-cosA)}\right)$

$= sinA\left(\frac{2}{sin^2A}\right) = \frac{2}{sinA} = 2cosecA$

$\frac{sinA}{1+cosA} + \frac{sinA}{1-cosA} = 2cosecA.$

5. Correct option: (A)

Given, $\sin\frac{A}{2} = \frac{3}{5}$

Now, $sin3A = 3sinA - 4sin^3A,$ $(Since\ sin3\theta = 3sin\theta - 4sin^3\theta)$

$sin3A = 3\left(\frac{3}{5}\right) - 4\left(\frac{3}{5}\right)^3 = \frac{9}{5} - 4\left(\frac{27}{125}\right) = \frac{9}{5} - \frac{108}{125} = \frac{117}{125}$

$sin3A = \frac{117}{125}.$

6. Correct option: (B)

$\left(\frac{\sin A}{1-\cos A}\right) - \left(\frac{1-\cos A}{\sin A}\right) = \frac{\sin^2 A - (1+\cos^2 A - 2\cos A)}{\sin A(1-\cos A)}$

$= \frac{\sin^2 A - \cos^2 A - (\sin^2 A - \cos^2 A) - 2\cos A}{\sin A(1-\cos A)} = \frac{2\cos^2 A - 2\cos A}{\sin A(1-\cos A)}$

$= \frac{2\cos A\,(1-\cos A)}{\sin A(1-\cos A)} = \frac{2\cos A}{\sin A} = 2\cot A.$

7. Correct option: (B)

$\frac{\sin\alpha}{a} = \frac{\cos\alpha}{b}$

$sin\alpha = \frac{acos\alpha}{b}$

$Since, = asin2\alpha + bcos2\alpha$

$= 2asin\alpha cos\alpha + b(1 - 2sin2\alpha)$

$= \frac{2a^2cos^2\alpha}{b} + b\left(1 - \frac{2a^2cos^2\alpha}{b^2}\right)$

$= \frac{2a^2cos^2\alpha}{b} + b - \frac{2a^2cos^2\alpha}{b} = b.$

8. Correct option: (A)

$3sin6° - 4cos^3 84°$

$= 3sin6° - 4sin^3 6°$

$= sin3(6°)$

$= sin18°$

$3sin6° - 4cos^3 84° = \frac{\sqrt{5}-1}{4}$.

9. Correct option: (B)

$\sin 22\frac{1^0}{2} = \sin\left(\frac{45}{2}\right)^0$.

We know that

$2\sin^2\theta = 1 + \cos\theta$

$2\sin^2\left(\frac{45}{2}\right)^0 = \frac{1+\cos 45°}{2}$

$2\sin^2\left(\frac{45}{2}\right)^0 = \frac{1-\frac{1}{\sqrt{2}}}{2}$

$\sin^2\left(\frac{45}{2}\right)^0 = \frac{1-\frac{\sqrt{2}}{2}}{2} = \frac{1}{2} - \frac{\sqrt{2}}{4}$

$\sin\left(\frac{45}{2}\right)^0 = \sqrt{\frac{1}{2} - \frac{\sqrt{2}}{4}} = \sqrt{\frac{\sqrt{2}-1}{2\sqrt{2}}}$

$\sin 22\frac{1^0}{2} = \sqrt{\frac{\sqrt{2}-1}{2\sqrt{2}}}$.

10. Correct option: (C)

$Given\ cos\theta = -\frac{3}{5}$

$\frac{1-\tan^2\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}} = -\frac{3}{5}$

$5 - 5\tan^2\frac{\theta}{2} = -3 - 3\tan^2\frac{\theta}{2}$

$8 = 2\tan^2\frac{\theta}{2}$

$4 = \tan^2\frac{\theta}{2}$

$tan\frac{\theta}{2} = -2$

11. Correct option: (A)

$(cos\alpha + cos\beta)^2 + (sin\alpha - sin\beta)^2$

$= (cos^2\alpha + cos^2\beta + 2cos\alpha cos\beta) + (sin^2\alpha + sin^2\beta + 2sin\alpha sin\beta)$

$= (cos^2\alpha + sin^2\alpha) + (cos^2\beta + sin^2\beta) + 2(cos\alpha cos\beta + sin\alpha sin\beta)$

$= 1 + 1 + 2[cos(\alpha - \beta)]$ $[\ Since,\ cos(A-B) = cosAcosB + sinAsinB\ ]$

$= 2 + 2[cos(\alpha - \beta\ ]$ $= 2[1 + cos(\alpha - \beta)]$

$= 2 \times 2cos^2\frac{(\alpha-\beta)}{2}$

$(cos\alpha + cos\beta)^2 + (sin\alpha - sin\beta)^2 = 4cos^2\frac{(\alpha-\beta)}{2}$

12. Correct option: (B)

$$\frac{1}{\sin 10^0} - \frac{\sqrt{3}}{\cos 10^0} = 2\left[\frac{1}{\sin 10^0} - \frac{\sqrt{3}}{\cos 10^0}\right] = 2\left[\frac{\sin 30^0}{\sin 10^0} - \frac{\cos 10^0}{\cos 10^0}\right]$$

$$= 2\left[\frac{\sin 30^0 \cos 10^0}{\sin 10^0} - \frac{\cos 10^0 \sin 10^0}{\cos 10^0}\right]$$

$$= 2\left[\frac{\sin 30^0 \cos 10^0 - \cos 10^0 \sin 10^0}{\sin 10^0 \cos 10^0}\right]$$

$$= 2\left[\frac{\sin 20^0}{\sin 10^0 \cos 10^0}\right]$$

$$= 2 \times 2\left[\frac{\sin 10^0 \cos 10^0}{\sin 10^0 \cos 10^0}\right] = 4.$$

13. Correct option: (C)

$$\Rightarrow \frac{1}{2}[\cos(40^0 - 20^0) - \cos(40^0 + 20^0)] \cdot \sin 100^0$$

$$= \frac{1}{2}[\cos(20^0) - \cos(60^0)] \cdot \sin 100^0$$

$$= \frac{1}{2}\left[\cos(20^0) - \frac{1}{2}\right] \cdot \sin 100^0$$

$$= \frac{1}{4}[2\cos(20^0) - 1] \cdot \sin 100^0$$

$$= \frac{1}{4}[2\cos(20^0) \cdot \sin 100^0 - \sin 100^0]$$

$$= \frac{1}{4}[\sin(20^0 + 100^0) - \sin(20^0 - 100^0) - \sin 100^0]$$

$$= \frac{1}{4}[\sin(120^0) + \sin(80^0) - \sin 100^0]$$

$$= \frac{1}{4}\left[\frac{\sqrt{3}}{2} + \sin(80^0) - \sin 100^0\right]$$

$$= \left[\frac{\sqrt{3}}{8} + 2\cos\left(\frac{100^0 + 80^0}{2}\right)\sin\left(\frac{100^0 - 80^0}{2}\right)\right]$$

$$= \left[\frac{\sqrt{3}}{8} + 2\cos(90^0)\sin(10^0)\right]$$

$$= \left[\frac{\sqrt{3}}{8} + 2 \times\ 0 \times \sin(10^0)\right] = \frac{\sqrt{3}}{8}$$

14. Correct option: (D)

$$\cos 20 \times \cos 40 \times \cos 60 \times \cos 80 = \cos 60[\cos 20 \times \cos 40 \times \cos 80]$$

$$= \frac{1}{2 \times 2}[2\cos 20 \times \cos 40 \times \cos 80]$$

$$= \frac{1}{4}[(\cos(40 + 20) + \cos(40 - 20)) \times \cos 80]$$

$$= \frac{1}{4}[(\cos(60) + \cos(20)) \times \cos 80]$$

$$= \frac{1}{4} \times \frac{1}{2} \times \cos 80 + \frac{1}{4}[\cos(20) \times \cos 80]$$

$$= \frac{1}{8} \times \cos 80 + \frac{1}{4}[\cos(20) \times \cos 80]$$

$$= \frac{1}{8} \times \cos 80 + \frac{1}{4 \times 2}[2\cos 20 \times \cos 80]$$

$$= \frac{1}{8} \times \cos 80 + \frac{1}{8}[\cos(80 + 20) + \cos(80 - 20)]$$

$= \frac{1}{8} \times \cos 80 + \frac{1}{8}\cos(100) + \frac{1}{8}\cos(60)$

$= \frac{1}{8} \times \cos 80 + \frac{1}{8}\cos(180 - 80) + \frac{1}{8} \times \frac{1}{2}$

$= \frac{1}{8} \times \cos 80 - \frac{1}{8}\cos(80) + \frac{1}{16}$

$\cos 20 \times \cos 40 \times \cos 60 \times \cos 80 = \frac{1}{16}.$

15. Correct option: (A)

$\frac{\sin x}{\sin \frac{x}{8}} = \frac{\sin \frac{x}{2}}{\sin \frac{x}{8}} \times \cos \frac{x}{2} = \frac{4 \sin \frac{x}{4}}{\sin \frac{x}{8}} \times \cos \frac{x}{4} \times \cos \frac{x}{2} = \frac{8 \sin \frac{x}{8}}{\sin \frac{x}{8}} \times \cos \frac{x}{8} \times \cos \frac{x}{4} \times \cos \frac{x}{2}$

$\frac{\sin x}{\sin \frac{x}{8}} = 8 \cos \frac{x}{8} \times \cos \frac{x}{4} \times \cos \frac{x}{2}$

16. Correct option: (D)

$\sin 2x = \frac{1}{2}$

$sin\, 2x = sin \frac{\pi}{6}$

$2\, x = \frac{\pi}{6}.$

$\theta = n\pi + (-1)^n \frac{\pi}{6}, n \in z$

$2\, x = \frac{n\pi}{2} + (-1)^n \frac{\pi}{12}, n \in z$

Let $n = 1, x = 75^0 (45^0, 90^0)$

$sin3x = 3sinx - 4sin^3 x, \qquad (Since\ sin3\theta = 3sin\theta - 4sin^3\theta)$

$= \frac{4 \sin^3 x - 3 \sin 3x}{4}$

$\frac{sin\, 3x}{4} = \frac{sin\, 3(75^0)}{4} = \frac{sin(225^0)}{4} = \frac{sin(180^0 + 45^0)}{4} = \frac{-\, sin\, 45^0}{4} = -\frac{1}{4\sqrt{2}}.$

17. Correct option: (C)

$(2cos\theta - 1)(2cos2\theta - 1)(2cos2^2\theta - 1) \cdots (2cos2^{n-1}\theta - 1)$

multiplying and dividing by $2cos\theta + 1$

$= \frac{2cos\theta + 1}{2cos\theta + 1}(2cos\theta - 1)(2cos2\theta - 1)(2cos2^2\theta - 1) \cdots (2cos2^{n-1}\theta - 1)$

Now,

$(2cos\theta + 1)(2cos\theta - 1) = 4cos^2\theta - 1 = 2(1 + cos2\theta) - 1 = 2cos2\theta + 1 - - - - - - - - - - (i)$

similarly, $(2cos2\theta + 1)(2cos2\theta - 1) = 4cos^2 2\theta - 1 = 2(1 + cos2^2\theta) - 1 = 2cos2^2\theta + 1$

……

……

$(2cos2^{n-1}\theta + 1)(2cos2^{n-1}\theta - 1) = 4cos^2 2^{n-1}\theta - 1 = 2(1 + cos2^n\theta) - 1 = 2cos2^n\theta + 1\ - -(ii)$

Now,

From (i) and (ii)

$(2cos\theta - 1)(2cos2\theta - 1)(2cos2^2\theta - 1) \cdots (2cos2^{n-1}\theta - 1) = \frac{2cos2^n\theta + 1}{2cos\theta + 1}.$

18. Correct option: (B)

$\frac{sin(x + y)}{sin(x - y)} = \frac{a + b}{a - b}$

Using componedo and dividendo rule

$\frac{sin(x + y) + sin(x - y)}{sin(x - y) - sin(x - y)} = \frac{a + b + a - b}{a - b - a + b}$

$\frac{2\sin x\cos y}{2\cos x\sin y}=\frac{2a}{2b}$

$\frac{\sin x\cos y}{\cos x\sin y}=\frac{a}{b}$

$\tan x\cot y=\frac{a}{b}$

$\frac{\tan x}{\tan y}=\frac{a}{b}$.

19. Correct option: (A)

$$\frac{\cos 2B-\cos 2A}{\sin 2A+\sin 2B}=\frac{-2\sin\left(\frac{2A+2B}{2}\right)\cdot\sin\left(\frac{2B-2A}{2}\right)}{2\sin\left(\frac{2A+2\beta}{2}\right)\cdot\cos\left(\frac{2B-2A}{2}\right)}=\frac{\sin(A-B)}{\cos(A-B)}=\tan(A-B).$$

20. Correct option: (C)

$$\frac{\sin A+\sin 3A+\sin 5A+\sin 7A}{\cos A+\cos 3A+\cos 5A+\cos 7A}$$

$$=\frac{2\sin\left(\frac{A+3A}{2}\right)\cdot\cos\left(\frac{A-3A}{2}\right)+2\sin\left(\frac{5A+7A}{2}\right)\cdot\cos\left(\frac{5A-7A}{2}\right)}{2\cos\left(\frac{A+3A}{2}\right)\cdot\cos\left(\frac{A-3A}{2}\right)+2\cos\left(\frac{5A+7A}{2}\right)\cdot\cos\left(\frac{5A-7A}{2}\right)}$$

$$=\frac{2\sin(2A)\cdot\cos(-A)+2\sin(6A)\cdot\cos(-A)}{2\cos(2A)\cdot\cos(-A)+2\cos(6A)\cdot\cos(-A)}=\frac{2\sin(2A)+\sin(6A)}{2\cos(2A)+\cos(6A)}$$

$$=\frac{2\sin\left(\frac{2A+6A}{2}\right)\cdot\cos\left(\frac{2A-6A}{2}\right)}{2\cos\left(\frac{2A+6A}{2}\right)\cdot\cos\left(\frac{2A-6A}{2}\right)}=\frac{\sin(4A)}{\cos(4A)}=\tan(4A).$$

LEVEL-II

1. Correct option: (A)

Consider

$cos^4\theta - sin^4\theta = (cos^2\theta)^2 - (sin^2\theta)^2$

$\because a^2-b^2=(a-b)(a+b)$

$\therefore cos^4\theta - sin^4\theta = (cos^2\theta)^2 - (sin^2\theta)^2$

$= (cos^2\theta - sin^2\theta)(cos^2\theta + sin^2\theta)$

$= (cos^2\theta - sin^2\theta)\ [\because cos^2\theta + sin^2\theta = 1]$

$= cos^2\theta - (1 - cos^2\theta)\ [\because sin^2\theta = 1 - cos^2\theta]$

$= cos^2\theta - 1 + cos^2\theta$

$= 2\,cos^2\theta - 1$

$= cos\ 2\theta.$

2. Correct option: (D)

$$a\cos 2\theta + b\sin 2\theta = a\left(\frac{1-\tan^2\theta}{1+\tan^2\theta}\right)+b\left(\frac{2\tan\theta}{1+\tan^2\theta}\right)=a\left(\frac{1-\left(\frac{b}{a}\right)^2}{1+\left(\frac{b}{a}\right)^2}\right)+b\left(\frac{2\left(\frac{b}{a}\right)}{1+\left(\frac{b}{a}\right)^2}\right)$$

$= a\left(\frac{a^2-b^2}{a^2+b^2}\right)+b\left(\frac{2ab}{a^2+b^2}\right)=\frac{a^3-ab^2+2ab^2}{a^2+b^2}=\frac{a^3+ab^2}{a^2+b^2}=a.$

3. Correct option: (D)

$8\,cos^3\ 20° - 6\,cos\ 20°$

$Taking\ 2\ common, we\ get = 2(4\,cos^3 20° - 3\,cos\ 20°)\ \ldots(i)$

$We\ know\ that, \cos 3x = 4cos^3x - 3\,cosx$

$Here, x = 20°$

$So, eq.(i)\ becomes$

$2[cos\ 3(20°)] = 2[cos\ 60°] = 2\times\frac{1}{2}=1.$

4. Correct option: (B)

$-8\ sin^3\ 15° + \ 6\ sin\ 15°$

$Taking\ 2\ common, we\ get\ = \ 2(-4sin^3\ 15° + \ 3\ sin\ 15°)\ \ldots(i)$

$We\ know\ that, \sin 3x = \ -4sin^3 x \ - \ 3\ sinx$

$Here, x \ = \ 15°$

$So, eq.(i)\ becomes$

$2[sin\ 3(15°)] = \ 2[sin\ 45°] = 2 \times \frac{1}{\sqrt{2}} = \sqrt{2}.$

5. Correct option: (B)

$$\frac{(1 - cos2A)}{(1 + cos2A)} = \frac{2\sin^2 A}{2\cos^2 A} = \frac{\sin^2 A}{\cos^2 A} = \tan^2 A$$

6. Correct option: (A)

$$\frac{1 + cos\theta - sin\theta}{-1 + cos\theta + sin\theta}$$

$Dividing\ Nr\ and\ Dr\ by\ sin\theta\ we\ get,$

$$\frac{\frac{1}{\sin\theta} + \frac{\cos\theta}{\sin\theta} - \frac{\sin\theta}{\sin\theta}}{-\frac{1}{\sin\theta} + \frac{\cos\theta}{\sin\theta} + \frac{\sin\theta}{\sin\theta}} = \frac{\operatorname{cosec}\theta + \cot\theta - 1}{-\operatorname{cosec}\theta + \cot\theta + 1} = \frac{\operatorname{cosec}\theta + \cot\theta - (\operatorname{cosec}^2\theta - \cot^2\theta)}{-\operatorname{cosec}\theta + \cot\theta + 1}$$

$$= \frac{(\operatorname{cosec}\theta + \cot\theta)(1 - \operatorname{cosec}\theta + \cot\theta)}{-\operatorname{cosec}\theta + \cot\theta + 1}$$

$$= \operatorname{cosec}\theta + \cot\theta = \cot\frac{\theta}{2}.$$

7. Correct option: (C)

$$\frac{\sin 2\theta}{1 + \cos 2\theta} = \frac{2\sin\theta\cos\theta}{2\cos^2\theta} = \frac{\sin\theta}{\cos\theta} = \tan\theta$$

8. Correct option: (A)

$$\frac{\sin 2\theta}{1 - \cos 2\theta} = \frac{2\sin\theta\cos\theta}{2\sin^2\theta} = \frac{\cos\theta}{\sin\theta} = \cot\theta$$

9. Correct option: (A)

$$\cot A + \tan A = \frac{\cos A}{\sin A} + \frac{\sin A}{\cos A} = \frac{2\cos^2 A + \sin^2 A}{2\sin A\ cosA} = \frac{2(1)}{sin2\ A} = 2cosec\ 2A$$

10. Correct option: (B)

$$\sqrt{2 - 2\cos\theta} = \sqrt{2(1 - \cos\theta)} = \sqrt{2 \times 2\left(\frac{1 - \cos\theta}{2}\right)} = \sqrt{4\left(\frac{1 - \cos\theta}{2}\right)} = 2\sin\frac{\theta}{2}.$$

11. Correct option: (A)

$$\frac{1 + cos\theta - sin\theta}{-1 + cos\theta + sin\theta}$$

$Dividing\ Nr\ and\ Dr\ by\ sin\theta\ we\ get,$

$$\frac{\frac{1}{\sin\theta} + \frac{\cos\theta}{\sin\theta} - \frac{\sin\theta}{\sin\theta}}{-\frac{1}{\sin\theta} + \frac{\cos\theta}{\sin\theta} + \frac{\sin\theta}{\sin\theta}} = \frac{\operatorname{cosec}\theta + \cot\theta - 1}{-\operatorname{cosec}\theta + \cot\theta + 1} = \frac{\operatorname{cosec}\theta + \cot\theta - (\operatorname{cosec}^2\theta - \cot^2\theta)}{-\operatorname{cosec}\theta + \cot\theta + 1}$$

$$= \frac{(\operatorname{cosec}\theta + \cot\theta)(1 - \operatorname{cosec}\theta + \cot\theta)}{-\operatorname{cosec}\theta + \cot\theta + 1}$$

$$= \operatorname{cosec}\theta + \cot\theta = \cot\frac{\theta}{2}.$$

12. Correct option: (C)

$$\frac{\sin 2\theta}{1 + \cos 2\theta} = \frac{2\sin\theta\cos\theta}{2\cos^2\theta} = \frac{\sin\theta}{\cos\theta} = \tan\theta$$

13. Correct option: (A)

$$\cot A + \tan A = \frac{\cos A}{\sin A} + \frac{\sin A}{\cos A} = \frac{2\cos^2 A + \sin^2 A}{2\sin A\, cosA} = \frac{2(1)}{sin2\,A} = 2cosec\ 2A$$

14. Correct option: (B)

$We\ have\ \sin^3 10^0 + \sin^3 50^0 - \sin^3 70^0$

$= \frac{1}{4}[(3sin10° - sin30°) + (3sin50° - sin150°) - (3sin70° - sin210°)]$

$= \frac{1}{4}\left(3(sin10° + sin50° - sin70°) - \frac{3}{2}\right)$

$= \frac{1}{4}\left(3(sin10° - 2cos60°.sin10°) - \frac{3}{2}\right) = -\frac{3}{8}.$

15. Correct option: (C)

$\cos^3 20^0 + \cos^3 140^0 + \cos^3 260^0 = \cos^3 20^0 + \cos^3(120^0 + 20^0) + \cos^3(240^0 + 20^0)$

$= -\frac{3}{4}\cos 3(20^0) = -\frac{3}{4}\left(\frac{1}{2}\right) = -\frac{3}{8}.$

16. Correct option: (C)

$\cos^3 10^0 + \cos^3 130^0 + \cos^3 250^0 = \cos^3 10^0 + \cos^3(130^0 + 10^0) + \cos^3(250^0 + 10^0)$

$= -\frac{3}{4}\cos 3(10^0)$

$= -\frac{3}{4}\left(\frac{\sqrt{3}}{2}\right) = -\frac{3\sqrt{3}}{8}.$

17. Correct option: (a-Q, b-PT, c-S, d-R)

18. Correct Answer: 0

19. Correct Answer: 4

20. Correct Answer: 3

STUDENT PRACTICE QUESTIONS

1. Correct option: (D)

$\sin 2\theta = 2\sin\theta\cos\theta$

$1 + \cos 2\theta = 2\sin^2\theta$

$\frac{\sin 2\theta}{1+\cos 2\theta} = \frac{2\sin\theta\cos\theta}{2\sin^2\theta} = \frac{\cos\theta}{\sin\theta} = \cot\theta.$

2. Correct option: (A)

We have, $\therefore\ \frac{cos\ 3\theta}{2\,cos\ 2\,\theta - 1} = \frac{4\,cos^3\theta - 3\,cos\,\theta}{2(2\cos^2\theta - 1) - 1}[\because cos3\theta = 4cos^3\theta - 3cos\theta]$

$= \frac{4\ \cos^2\theta - 3\,cos\,\theta}{4\cos^2\theta - 2 - 1} = \frac{4\cos^2\theta\ - 3\,cos\,\theta}{4\cos^2\theta - 3}$

$= cos\,\theta\left(\frac{4\cos^2\theta - 3}{4\cos^2\theta - 3}\right) = cos\,\theta.$

3. Correct option: (D)

we know that $cos3\theta = 4cos^3\theta - 3cos\theta \Longrightarrow cos3\theta + 3cos\theta = 4cos^3\theta\ - - - - - (1)$

$and\ sin3\theta = 3sin\theta - 4sin^3\theta \Longrightarrow 3sin\theta - sin3\theta = 4sin^3\theta\ - - - - - - (2)$

$Therefore,\ \frac{3cos\theta + cos3\theta}{3sin\theta - sin3\theta} = \frac{4cos^3\theta}{4sin^3\theta} = cot^3\theta.$

4. Correct option: (A)

$$cosec\ 2A + cot2A = \frac{1}{\sin 2\,A} + \frac{\cos 2A}{\sin 2A} = \frac{1 + 2\cos^2 A - 1}{\sin 2A}$$

$$= \frac{2\cos^2 A}{2\sin A\cos A} = \frac{\cos A}{\sin A} = \cot A.$$

5. Correct option: (D)

$$\frac{sin3A}{1+cos2A}=\frac{3sinA-4sin3A}{1+2(1-2sin^2A)}=\frac{sinA(3-4sin^2A)}{3-4sin^2A}=sinA.$$

6. Correct option: (A)

$(\sin 3A+\sin A)\times\sin A+(\cos 3A-\cos A)\times cos\,A$

$=\sin 3A\times\sin A+\cos 3A\times cos\,A+\sin^2 A-\cos^2 A$

$=\sin 3A\times\sin A+\cos 3A\times cos\,A=\ cos\,(3A-A)-cos\,2A=cos\,2A-cos\,2A=0.$

7. Correct option: (B)

$Given,\ x+\frac{1}{x}=2cos\theta$

$\Rightarrow\left(x+\frac{1}{x}\right)^3=(2\cos\theta)^3$

$\Rightarrow x^3+\frac{1}{x^3}+3x\times\frac{1}{x}\left(x+\frac{1}{x}\right)=8\cos^3\theta$

$\Rightarrow x^3+\frac{1}{x^3}+3\times 2cos\theta=8\cos^3\theta$

$\Rightarrow x^3+\frac{1}{x^3}=2(4\cos^3\theta-3cos\theta)$

$\Rightarrow x^3+\frac{1}{x^3}=2cos3\theta.$

8. Correct option: (C)

$$\sin\theta=\frac{2\tan\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}.$$

$$cos\boldsymbol{\theta}=\frac{1-\tan^2\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}$$

Substituting this in the above equation we get,

$$\frac{1+\tan^2\frac{\theta}{2}+2\tan\frac{\theta}{2}-1+\tan^2\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}+2\tan\frac{\theta}{2}+1-\tan^2\frac{\theta}{2}}=\frac{2\tan\frac{\theta}{2}+2\tan^2\frac{\theta}{2}}{2+2\tan\frac{\theta}{2}}=\tan\frac{\theta}{2}.$$

9. Correct option:

$sin\alpha\,.\,sin(\mathbf{60}-\alpha).sin(\mathbf{60}+\alpha)=sin\alpha(sin^2\mathbf{60}-sin^2\alpha)$

$=sin\alpha\left(\frac{\mathbf{3}}{\mathbf{4}}-sin^2\alpha\right)=\left(\frac{\mathbf{3}sin\alpha-\mathbf{4}sin^3\alpha}{\mathbf{4}}\right)=\frac{\mathbf{1}}{\mathbf{4}}sin\mathbf{3}\alpha.$

10. Correct option: (B)

$\sin 20\times\sin 40\times\sin 60\times\sin 80=\sin 60[\sin 20\times\sin 60\times\sin 80]$

$=\sin 60[\sin 20\times\sin(60-20)\times\sin(60+20)]$

$=\frac{\sqrt{3}}{2}\left[\frac{\sin 3(20)}{4}\right]=\frac{\sqrt{3}}{2}\left[\frac{\sin 60}{4}\right]$

$=\frac{\sqrt{3}}{2}\left[\frac{\sqrt{3}}{2}\times 4\right]$

$=\frac{\sqrt{3}}{2}\times\frac{\sqrt{3}}{8}=\frac{3}{16}$

11. Correct option: (D)

$\cos\alpha\,.\,\cos(\mathbf{60}-\alpha).\cos(\mathbf{60}+\alpha)=\cos\alpha(cos^2\alpha-sin^260)$

$=\cos\alpha\left(cos^2\alpha-\frac{\mathbf{3}}{\mathbf{4}}\right)=\left(\frac{\mathbf{4}cos^3\alpha-\mathbf{3}\mathbf{cos}\alpha}{\mathbf{4}}\right)=\frac{\mathbf{1}}{\mathbf{4}}\cos\mathbf{3}\alpha.$

12. Correct option: (B)

$\cos^2\alpha + \cos^2(\alpha + 120^0) + \cos^2(\alpha - 120^0)$

$= \cos^2\alpha + \cos^2\alpha\ \cos^2 120^0 + \sin^2\alpha\ \sin^2 120^0 - 2\cos\alpha\ \cos 120^0 \sin\alpha\ \sin 120^0 + \cos^2\alpha\ \cos^2 120^0$
$+ \sin^2\alpha\ \sin^2 120^0 + 2\cos\alpha\ \cos 120^0 \sin\alpha\ \sin 120^0$

$= \dfrac{3\cos^2\alpha + 3\sin^2\alpha}{2} = \dfrac{3}{2}(1) = \dfrac{3}{2}.$

13. Correct option: (D)

$\cos\alpha + \cos\beta = a$, $\sin\alpha + \sin\beta = b$

$\alpha - \beta = 2\theta$

$(\cos\alpha + \cos\beta)^2 = a^2$

$\cos^2\alpha + \cos^2\beta + 2\cos\alpha\cos\beta = a^2 - - - - - - - - - - - - - - - - - (i)$

$(\sin\alpha + \sin\beta)^2 = b^2$

$\sin^2\alpha + \sin^2\beta + 2\sin\alpha\sin\beta = b^2 - - - - - - - - - - - - - - - - - (ii)$

Adding i and ii

$2 + \cos(\alpha - \beta) = a^2 + b^2$

$a^2 + b^2 = 2 + 2\cos^2\theta$

$\dfrac{\cos 3\theta}{\cos\theta} = \dfrac{4\cos^3\theta \times 3\cos\theta}{\cos\theta}$

$\dfrac{\cos 3\theta}{\cos\theta} = 4\cos^2\theta - 3$

$4\cos^2\theta = 3 - - - - - - - - - - - - - - - - - (iii)$

$a^2 + b^2 = 2 + (2\cos^2\theta - 1) = 2 + 4\cos^2\theta - 2$

$a^2 + b^2 = 4\cos^2\theta$

$a^2 + b^2 - 3 = 0,$ Since from (iii).

14. Correct option: (A)

$$\frac{\sin 2\theta}{1 - \cos 2\theta} = \frac{2\sin\theta\cos\theta}{2\sin^2\theta} = \frac{\cos\theta}{\sin\theta} = \cot\theta$$

15. Correct option: (B)

$$\sqrt{2 - 2\cos\theta} = \sqrt{2(1 - \cos\theta)} = \sqrt{2 \times 2\left(\frac{1 - \cos\theta}{2}\right)} = \sqrt{4\left(\frac{1 - \cos\theta}{2}\right)} = 2\sin\frac{\theta}{2}.$$

16. Correct option: (B)

$$\frac{\cot x}{\cot x - \cot 3x} + \frac{\tan x}{\tan x - \tan 3x} = \frac{\frac{1}{\tan x}}{\frac{1}{\tan x} - \frac{1}{\tan 3x}} + \frac{\tan x}{\tan x - \tan 3x}$$

$$= \frac{\frac{1}{\tan x}}{\frac{\tan x - \tan 3x}{\tan x \tan 3x}} + \frac{\tan x}{\tan x - \tan 3x} = \frac{\tan x - \tan 3x}{\tan x - \tan 3x} = 1.$$

17. Correct option: (D)

$$\frac{1}{\tan 3x - \tan x} - \frac{1}{\cot 3x - \cot x} = \frac{1}{\frac{\sin 3x}{\cos 3x} - \frac{\sin x}{\cos x}} - \frac{1}{\frac{\cos 3x}{\sin 3x} - \frac{\cos x}{\sin x}}$$

$$= \frac{\cos x\cos 3x}{\sin(3x - x)} - \frac{\sin 3x\sin x}{\sin(x - 3x)} = \frac{\cos x\cos 3x + \sin 3x\sin x}{\sin 2x}$$

$$= \frac{\cos(x - 3x)}{\sin 2x} = \frac{\cos(-2x)}{\sin 2x} = \frac{\cos 2x}{\sin 2x} = \cot 2x.$$

18. Correct option: a-R; b-Q; c-T; d-T

19. Correct Answer: 0

20. Correct Answer: 2

5. Mensuration

Solutions

Prism, Cube and Cuboid

LEVEL-I

1. Correct option: (B)

The lateral sides of the triangular prism are the three sides of the triangular base, each of which is equal to the corresponding side length given. In this case, the sides are 12cm, 16cm, and 20cm.

The total length of the lateral sides is the sum of these three side lengths:

$12cm + 16cm + 20cm = 48cm$

In addition to the lateral sides, we also have to consider the top and bottom bases of the triangular prism. Each base is a triangle with side lengths of 12cm, 16cm, and 20cm. To calculate the perimeter of each base, we add up these three side lengths:

$12cm + 16cm + 20cm = 48cm$

Since there are two bases, the total length of the base sides is:

$2 \times 48cm = 96cm$

Finally, to find the total length of the sides, we add the length of the lateral sides to the length of the base sides:

$48cm + 96cm = 144cm$

Therefore, the total length of the sides of the triangular prism is $144cm$.

2. Correct option: (A)

Area of equilateral triangle $= \frac{\sqrt{3}}{4} \times (side)^2$

Given equilateral triangle of side 8cm

Area of equilateral triangle $= \frac{\sqrt{3}}{4} \times (8)^2 = \frac{\sqrt{3}}{4} \times 64 = 16\sqrt{3}\ cm^2$

The volume of prism $= A \times h$.

Given height = 12cm, and Area = $16\sqrt{3}\ cm^2$.

The volume of prism = $16\sqrt{3}\ cm^2 \times 12\ cm = 192\sqrt{3}\ cm^3$.

Therefore, volume is $192\sqrt{3}\ c.c\ (cm^3)$.

3. Correct option: (B)

Edge of the cube = 8 cm

Total surface area $= 6a^2 = 6(8)^2 = 384\ cm^2$

Thus, the total surface area of the cube is $384\ cm^2$.

4. Correct option: (C)

Given, area of four walls of room are 12mts × 18mts × 5mts.

Area of four walls of room $= 2 \times h \times (l + b)$

Area of four walls of room $= 2 \times 5\ (12 + 18) = 10(30) = 300\ cm^2$

Therefore, the total area of the four walls of the room is 300 square centimetres.

5. Correct option: (D)

We know that,

Surface area of cube $= 6a^2$

Given, $600 = 6a^2$

$a^2 = \frac{600}{6} = 100$

$a = \sqrt{100} = 10$

$a = 10\ cm$.

Diagonal of cube = $\sqrt{3}a$
Substitute $a = 10\ cm$ in $\sqrt{3}a$.
Diagonal of cube = $\sqrt{3}(10) = 10\sqrt{3}cm$
Therefore, Diagonal of cube = $10\sqrt{3}cm$.

6. Correct option: (D)

let original edge of cube = $\boldsymbol{a}$,
now, New edge after doubling becomes,
New edge= $2a$
volume of new cube= $(edge)^{\mathbf{3}}$.
volume of new cube= $(2a)^{\mathbf{3}} = \mathbf{8}(a)^{\mathbf{3}}$**.**
Thus, the volume becomes 8 times the original volume.

7. Correct option: (C)

Given, two cubes have volume in ratio 1:27.
Let $x_1 and\ x_2$be edges of the cubes.
Then, volume of a cube $v = x^3$, where x is the edge.
Now,

$$\frac{v_1}{v_2} = \frac{1}{27}$$

$$\frac{{x_1}^3}{{x_2}^3} = \frac{1}{27}$$

$$\left(\frac{x_1}{x_2}\right)^3 = \frac{1}{27}$$

$$\frac{x_1}{x_2} = \sqrt[3]{\frac{1}{27}} = \frac{1}{3}.$$

We know, area of a cube $A = 6x^2$.
The ratio of Area = $\frac{A_1}{A_2} = \frac{6{x_1}^2}{6{x_2}^2} = \left(\frac{x_1}{x_2}\right)^2 = \left(\frac{1}{3}\right)^2 = \frac{1}{9}$.
Therefore, the ratio of area = 1: 9.

8. Correct option: (D)

Surface Area of cube = $6a^2$
rate = 13 paise/ cm^2.
Total cost = 343.98 Rs or 34398 paise
$\therefore Total\ area = \frac{Total\ cost}{Rate} = \frac{31398}{13} = 2646\ cm^2$
$Total\ area = 2646\ cm^2$
Surface Area of cube = $Total\ area$
$6a^2 = 2646\ cm^2$
$a^2 = \frac{2646}{6} = 441\ cm$
$\therefore a = 21cm$
Volume of cube= $(a)^{\mathbf{3}} = (21)^{\mathbf{3}} = 9261cm^3$.
Therefore, volume is $\mathbf{9261}cm^3$.

9. Correct option: (B)

Total surface area of a cube =96 cm^2
Total surface area of a cube = $6(Side)^2 = 96\ cm^2$
$\Rightarrow (Side)^2 = \frac{96}{6} = 16\ cm^2$.
$\Rightarrow Side = \sqrt{16} = 4\ cm$
$\Rightarrow Side = 4\ cm$

Therefore, side is 4 cm.

10. Correct option: (A)

Given: a triangular prism of sides 15cm, 20cm and 25cm and height 40cm.

$LSA = (a + b + c) \times h$

$LSA = (15 + 20 + 25) \times 40 = 60 \times 40 = 2400\ sq.cm.$

Therefore, $LSA = 2400\ sq.cm.$

11. Correct option: (D)

Lateral surface area of a cuboid $= 2(b \times h + l \times h) \quad \ldots.(1)$

Total surface area of a cuboid $= 2(l \times b + b \times h + l \times h) = 2(l \times b) + 2(b \times h + l \times h)$

$= 2 \times Area\ of\ the\ base + Lateral\ surface\ area$

Hence, $40 = 2 \times Area\ of\ the\ base + 26\ (on\ using\ (1))$

$\Longrightarrow 2 \times Area\ of\ the\ base = 14$

Therefore, $Area\ of\ the\ base = 7m^2$.

12. Correct option: (A).

Given, $l + b + h = 19cm$

Length of the diagonal of cube $\sqrt{l^2 + b^2 + h^2}$

Given, length of the diagonal =11cm

$\sqrt{l^2 + b^2 + h^2} = 11\ cm$

$=> \sqrt{l^2 + b^2 + h^2} = 11$

$=> l^2 + b^2 + h^2 = 11^2$

$=> (l + b + h)2 - 2(lb + bh + lh) = 121$

$=> 192 - 2(lb + bh + lh) = 121$

$=> 2(lb + bh + lh) = 240$

But $2(lb + bh + lh)$ is the total surface area of the cuboid.

Hence, total surface area of the cuboid $= 240cm^2$.

13. Correct option: (C)

Volume of cube= $(a)^3 = (7)^3 = \mathbf{343}\ cm^3$.

Therefore, volume is $\mathbf{343}\ cm^3$.

14. Correct option: (C)

The areas of the three adjacent faces of a rectangular box which meet in a point are known.

The product of these areas in equal to the square of the volume of the box.

Let length = l, breadth = b and height = h.

Then, Product of areas of 3 adjacent faces $= (lb \times bh \times lh) = (lbh)^2 = (volume)^2$.

15. Correct option: (B)

Ratio in the areas of three adjacent faces of a $cuboid = 2:3:4$

Volume = 9000 cm^3.

Let the area of faces be $2x, 3x, 4x$ and

Let $a, b, and\ c$ be the dimensions of the cuboid, then

$\therefore\ 2x = ab, 3x = bc, 4x = ca$

$\therefore\ ab \times bc \times ca = 2x \times 3x \times 4x$

$a^2b^2c^2 = 24x^3$

But $volume = abc = 9000\ cm^3$.

$24x^3 = (9000)^2$

$x^3 = \dfrac{81000000}{24} = 3375000$

$\therefore\ x = \sqrt[3]{3375000} = \sqrt[3]{(150)^3} = 150$

$\therefore\ ab = 300$

$bc = 450$

$ca = 600$

$\because\ abc = 9000$
Dividing one by one,
$c = 30, a = 20, b = 15$
$\therefore$ Shortest side = 15 cm.
Therefore, shortest side = 15 cm.

16. Correct option: (D)

Volume of cube = volume of cuboid $= 270 \times 100 \times 64 = 1728000\ cm^3$.
$\therefore$ Length of one side of the cube $= 120\ cm$
$\therefore$ Surface area of cube $= 6 \times 120 \times 120 = 86400\ cm^2$.
Therefore, Surface area of cube is $86400\ cm^2$.

17. Correct option: (C)

Volume of rectangular box $= 160 \times 100 \times 60\ cm^3$
Volume of one cubical block $= 20 \times 20 \times 20\ cm^3$
$\therefore$ Required number of cubical block $= \frac{160 \times 100 \times 60}{20 \times 20 \times 20} = 120\ blocks.$
Therefore, required answer is 120 blocks.

18. Correct option: (B)

Side of smallest cube $= L.C.M\ of\ 5, 10, 20 = 20cm$
Volume of the cube $= (20 \times 20 \times 20)cu.cm = 8000\ cu.cm$
Volume of the block$= (5 \times 10 \times 20)cu.cm = 1000\ cu.cm$
Number of blocks $= \frac{8000}{1000} = 8.$
Therefore, number of blocks is 8.

19. Correct option: (B)

Given,
$l = 2b\ and\ h = 3\ m$
Now, Area of four walls $= 108\ m^3$
$2(l + b)h = 108$
$18b = 108$
$b = 6\ m$
Therefore, $l = 2b = 12\ m$
Hence, volume of the cold storage $= lbh = 12 \times 6 \times 3 = 216\ m^3$.

20. Correct option: (B)

Base of right pyramid is square of $side\ (s) = 10\ cm$
Height of pyramid = 12 cm
slant height $= \sqrt{h^2 + \left(\frac{r^2}{2}\right)} = \sqrt{12^2 + 5^2} = 13$
$L.S.A = \frac{1}{2} \times perimeter\ of\ base \times slant\ height = \frac{1}{2} \times (4 \times 10) \times 13 = 260\ cm^2.$
$TSA = LSA + Area\ of\ base = 260 + 10^2 = 360\ cm^2.$
$Volume = \frac{1}{3} \times area\ of\ base \times height = \frac{1}{3} \times (10 \times 10) \times 12 = 400\ cm^2.$
Therefore, volume is $400\ cm^2$.

LEVEL-II

1. Correct option: (A)

$Volume\ of\ resulting\ metal\ cube = Volume\ of\ metal\ cube\ 1 + Volume\ of\ metal\ cube\ 2 + Volume\ of\ metal\ cube\ 3$

Volume of a cube $= a^3$
Let the side of the resulting cube be A
$A^3 = 3^3 + 4^3 + 5^3$

$=> A^3 = 27 + 64 + 125 = 216$

$\Rightarrow A^3 = 216$

$=> A = 6$

Hence, length of side of the resulting metal cube is 6cm.

Total surface area of a cube = $6a^2$

Total surface area of a cube = $6 \times (6)^2 = 216\ cm^2$.

2. Correct option: (B)

Let l,b,h be length, breadth & height of the cuboid.

According to question,

$x = lb, y = bh, z = lh$

$\therefore x \times y \times z = l^2b^2h^2$

$\Rightarrow xyz = (lbh)^2$

$\Rightarrow Volume = \sqrt{xyz}$

Therefore, $Volume\ is\ \sqrt{xyz}$.

3. Correct option: (C)

Here, l is the length of a diagonal.

V is volume of cube

a is side of the cube

Volume of cube $(V) = a^3 \quad ---$ (1)

We know that,

$\Rightarrow\ l = a^3$

$\Rightarrow\ l^3 = a\sqrt{3}$ [Taking cube on both sides]

$\Rightarrow\ l^3 = 3\sqrt{3}\,V$ [From (1)]

Therefore, $3\sqrt{3}\,V = l^3$

4. Correct option: (C)

Sum of all edges of a cube = 36 cm

No. of edge of a cube =12

$\therefore$ Length of its one edge $\frac{36}{12} = 3$.

Then $Volume = (edge)^3$

$V = (3)^3 cm^3 = 27\ cm^3$

Therefore, Volume is $27\ cm^3$.

5. Correct option: (D)

Area of a floor of a room = $15\ m^2$.

Height (h) = 4 m

$\therefore$ Volume of air in the room= $Floor\ area \times Height$

Volume of air in the room = $15\ m^2 \times 4\ m = 60\ m^3$.

Volume of air in the room = $60 \times 10 \times 10 \times 10\ dm^3$

Volume of air in the room = $60000\ dm^3$.

Therefore, Volume of air in the room is $60000\ dm^3$.

6. Correct option: (B)

Let the sides of the sheet be x cm and 5x cm.

Then, volume of the sheet = Volume of their on cube

$\Rightarrow x \times 5x \times 0.5 = 10 \times 10 \times 10$

$\Rightarrow x^2 = \frac{1000}{2.5} = \frac{10000}{25} = 400$

$\Rightarrow x^2 = 400$

$\Rightarrow x = 20$ cm

Now,

$\Rightarrow 5x = 5 \times 20 = 100\ cm$
$\therefore$ The sides of the sheer are 20 cm and 100 cm.

7. Correct option: (C)

Given, two cubes have volume in ratio 27:1.
Let $x_1 and\ x_2$ be edges of the cubes.
Then, volume of a cube $v = x^3$, where x is the edge.
Now,

$$\frac{v_1}{v_2} = \frac{27}{1}$$
$$\frac{{x_1}^3}{{x_2}^3} = \frac{27}{1}$$
$$\left(\frac{x_1}{x_2}\right)^3 = \frac{27}{1}$$
$$\frac{x_1}{x_2} = \sqrt[3]{\frac{27}{1}} = \frac{3}{1}.$$

Therefore, ratio of their edges is 3: 1.

8. Correct option: (A).

Volume of new cube formed $= 125 + 64 + 27 = 216$
Let the length of the cube = $\boldsymbol{l}$
Then, $216 = l^3$
$\therefore l = 6\ cm.$
Therefore, Volume of new cube formed $is\ 6\ cm.$

9. Correct option: (B)

Length of the resulting cuboid = $(12 + 12)\ cm\ = 24\ cm$
Breadth and height remaining the same i.e., 12 cm each
$Surface\ area\ = 2\ (lb + bh + lh)$
$= 2(24 \times 12 + 12 \times 12 + 24 \times 12) cm^2 = 1440\ cm^2.$
Therefore, $Surface\ area$ is 1440 cm^2.

10. Correct option: (B)

Let the length of side of a cube be 'a' (by cube) then
diagonal length $= \sqrt{3}a = 12\sqrt{3}$
$\Rightarrow a = 12$
Volume of 3 small cube = Volume of bid cube
$(3x)^3 + (4x)^3 + (5x)^3 = 12^3$

$$x^3 = \frac{1728}{216}$$

$x = 2.$
Now,
$3x = 3 \times 2 = 6.$
$4x = 4 \times 2 = 8.$
$5x = 5 \times 2 = 10.$
$\therefore sides\ length\ are\ 6,8,10\ cm$.

11. Correct option: (B)

Let their edges be a and b.
Then

$$\Rightarrow \frac{a^3}{b^3} = \frac{8}{27}$$
$$\Rightarrow \frac{a^3}{b^3} = \frac{2^3}{3^3}$$

$\Rightarrow \frac{a}{b} = \frac{2}{3}$

Now,

$\Rightarrow \frac{a^2}{b^2} = \frac{4}{9}$

$\Rightarrow \frac{6a^2}{6b^2} = \frac{4\times6}{9\times6}$

$\Rightarrow \frac{6a^2}{6b^2} = \frac{4}{9}$.

Ratio of their surface area is 4 : 9.

12. Correct option: (B)

Given, cube of edge 5cm

Surface area of cube = $6a^2$, where a is edge.

$6a^2 = 6 \times (5)^2 = 150\ cm^2$.

Now length of the edge of the cut cubes = 1 cm

Surface area of cube = $6b^2$, where b is edge.

$6b^2 = 6 \times (1)^2 = 6\ cm^2$.

Ratio = $\frac{\text{Surface area of cut cube}}{\text{Surface area of cube}} = \frac{6}{150} = \frac{1}{25} = 1:25$.

Therefore, the required answer is 1:25.

13. Correct option: (A)

Given, $l + b + h = 19cm$

Length of the diagonal of cube $\sqrt{l^2 + b^2 + h^2}$

Given, length of the diagonal =11cm

$\sqrt{l^2 + b^2 + h^2} = 11\ cm$

$=> \sqrt{l^2 + b^2 + h^2} = 11$

$=> l^2 + b^2 + h^2 = 11^2$

$=> (l + b + h)2 - 2(lb + bh + lh) = 121$

$=> 192 - 2(lb + bh + lh) = 121$

$=> 2(lb + bh + lh) = 240$

But $2(lb + bh + lh)$ is the total surface area of the cuboid.

Hence, total surface area of the cuboid = $240cm^2$.

14. Correct option: (A)

In a right pyramid, the slant height (l) is the hypotenuse of a right triangle formed by the height (h) and half of the base side ,$\frac{s}{2}$.

Given:

Height of the pyramid (h) = 12 cm

Side length of the square base (s) = 10 cm

To find the slant height (l), we can use the Pythagorean theorem:

$l^2 = h^2 + \left(\frac{s}{2}\right)^2$

Substituting the given values:

$l^2 = 12^2 + \left(\frac{10}{2}\right)^2$

$l^2 = 144 + 25$

$l^2 = 169$

Taking the square root of both sides:

$l = \sqrt{169}$

$l = 13$

Therefore, the slant height of the pyramid is 13 cm.

15. Correct option: (B)

To find the lateral surface area of a pyramid, we will use the formula:

$Lateral\ Surface\ Area\ = \left(\frac{1}{2}\right) \times\ perimeter\ of\ the\ base\ \times\ slant\ height$

Given: Side length of the square base (s) = 10 cm

Height of the pyramid (h) = 12 cm

To find the perimeter of the base, we can multiply the side length by 4 since the base is a square:

Perimeter of the base = $4 \times s$

Perimeter of the base = $4 \times 10\ cm$

Perimeter of the base = $40\ cm$

To find the slant height (l), we can use the Pythagorean theorem. In a right pyramid, the slant height is the hypotenuse of a right triangle formed by the height and half of the base side:

$l^2 = h^2 + \left(\frac{s}{2}\right)^2$

Substituting the given values:

$l^2\ =\ 12^2\ +\ \left(\frac{10}{2}\right)^2$

$l^2\ =\ 144\ +\ 25$

$l^2\ =\ 169$

Taking the square root of both sides:

$l\ =\ \sqrt{169}$

$l\ =\ 13$ cm

Now we can calculate the lateral surface area:

$Lateral\ Surface\ Area\ = \left(\frac{1}{2}\right) \times\ perimeter\ of\ the\ base\ \times\ slant\ height$

Lateral Surface Area = $\left(\frac{1}{2}\right) \times\ 40\ cm\ \times\ 13\ cm$

Lateral Surface Area = $20\ cm\ \times\ 13\ cm$

Lateral Surface Area = 260 cm²

Therefore, the lateral surface area of the pyramid is 260 cm².

16. Correct option: (C)

To find the volume of a pyramid, we can use the formula:

Volume = $\left(\frac{1}{3}\right) \times\ base\ area\ \times height$

Given: Side length of the square base (s) = 10 cm

Height of the pyramid (h) = 12 cm

The base area of a square is given by the formula:

$Base\ Area\ =\ side\ length\ \times\ side\ length$

Substituting the given value:

$Base\ Area\ =\ 10\ cm\ \times\ 10\ cm$

$Base\ Area\ =\ 100\ cm^2$

Now we can calculate the volume of the pyramid:

$Volume\ =\ \left(\frac{1}{3}\right) \times\ base\ area\ \times height$

$Volume\ =\ \left(\frac{1}{3}\right) \times\ 100\ cm^2\ \times 12\ cm$

$Volume\ =\ \left(\frac{1}{3}\right)\ \times\ 1200\ cm^3$

$Volume\ =\ 400\ cm^3$.

Therefore, the volume of the pyramid is 400 cubic centimeters (cm^3).

17. Correct Matching is: $a \to Q, b \to R, c \to S, d \to P$

If there ne a cuboid of length l,breadth b, height h, then

a) As the cuboid has six rectangular faces, the total surface area of the cuboid is calculated as follows:

Assume that, l, b, h be the length, breadth, and height of the cuboid respectively.

Therefore, the total surface area of the cuboid is 2 $(lh + lb + hb)$ square units.

b) The lateral surface area of a cuboid is given by $2(length + breadth) \times height.$

c) The total length of the edges, L, for a cuboid is $L = 4(l + b + h)$

d) The formula for calculating the Cuboid diagonal is quite simple: D = $\sqrt{l^2 + b^2 + h^2}$, where "D" is the length of the diagonal, "l" is the length of one side, "b" is the length of the second side, and "h" is the length of the third side.

18. Correct Answer is: 216 m^3.

Given, $l = 2b \; and \; h = 3\,m$

Now, Area of four walls = $108\ m^3$

$2(l + b)h = 108$

$18b = 108$

$b = 6\,m$

Therefore, $l = 2b = 12\,m$

Hence, volume of the cold storage = $lbh = 12 \times 6 \times 3 = 216\,m^3$.

19. Correct Answer is: 8 cu.cm

Given: the sum of all edges of a cube is 24cm,

Length of each edge = $\frac{24}{12} = 2\,cm.$

Volume of a cube = $2^3 = 8\,cu.cm.$

Therefore, the volume of cube is 8 cu.cm.

20. Correct Answer is: 4 cm.

Total surface area of a cube =96 cm^2

Total surface area of a cube = $6(Side)^2 = 96\,cm^2$

$\Rightarrow (Side)^2 = \frac{96}{6} = 16\,cm^2.$

$\Rightarrow Side = \sqrt{16} = 4\,cm$

$\Rightarrow Side = 4\,cm$

Therefore, side is 4 cm.

$\therefore$ Volume of cube = $(Side)^3 = (4)^3 = 64\,cm^3$.

STUDENT PRACTICE QUESTIONS

1. Correct option: (A)

It is given that the prism has a base like a rhombus whose diagonals are

Diagonal 1, d_1 = 28 cm

Diagonal 2, d_2 = 96 cm

So, the area of the base in the shape of a rhombus $\frac{1}{2} \times d_1 \times d_2 = \frac{1}{2} \times 28 \times 96 = 1344 cm^2$The height of the prism, h = 32 cm.

We know that,

The volume of a prism is given by,

$Volume = Area\,of\,base \times Height$

Substituting the above values in the formula, we get

$The\ volume\ of\ the\ prism = 1344 \times 32 = 43008\,cm^3$

Therefore, $the\ volume\ of\ the\ prism = 43008\,cm^3$.

2. Correct option: (C)
Let edge of a smaller cube be 1 m
∴ Edge of larger cube = 2 m
and Surface area of smaller cube $6 \times (1)^2 = 6m^2$.
∴ Surface area of larger cube $6 \times (2)^2 = 24m^2$
∴ % increase in surface area $= \frac{24-6}{6} \times 100 = 300\%$
Therefore, % increase in surface area $= 300\%$.
3. Correct option: (D).
Let the side of cube be $a\ cm$
We know that,
Diagonal of cube $= a\sqrt{3}$
According to the question,
$\Rightarrow a\sqrt{3} = 6\sqrt{3}$
$\Rightarrow a = 6$
Thus, the side of the cube is 6 cm
$\therefore\ Volume\ of\ cube\ = a^3 = (6)^3 = 216\ cm^3$
Hence, the volume of cube is $216\ cm^3$.
4. Correct option: (D)
Let the length of the edge of the be $'a'$.
Then according to the question, $a^3 = 6a^2$.
$a^3 = 6a^2$
$a = 6$.
Therefore, the value of a is 6.
5. Correct option: (C)
We have: $l = 16\ cm, b = 8\ cm, h = 6\ cm$
Total Surface area of the rectangular chalk $(Cuboidal\ in\ shape)\ = 2(lb + bh + lh)$
$= 2[16 \times 8 + 8 \times 6 + 6 \times 16]$
$= 2[128 + 48 + 96]$
$= 2 \times 272 = 544\ cm^2$.
Total Surface area of the rectangular chalk is $544\ cm^2$.
6. Correct option: (B)
Given it's diagonal $= 5\sqrt{5}$ cm
Diagonal $= \sqrt{l^2 + b^2 + h^2} = 5\sqrt{5}$
Squaring on both sides we get
$125\ =\ l^2 + b^2 + h^2$
Also, $T.S.A\ =\ 2(lb + bh + hl)$
Now, Take $l\ +\ b\ +\ h\ =\ 19$
Squaring on both sides
$(1 + b + h)^2\ =\ 19^2$
$l^2 + b^2 + h^2 +\ 2\ (\ lb +\ bh + hl)\ =\ 361$
$125\ +\ TSA\ =\ 361$
$TSA\ =\ 236\ cm^2$.
Therefore, T. S. A = $236\ cm^2$.
7. Correct option: (A).
Let the length, breadth and height of the box be l, b and h respectively.
Then, $Volume\ =\ lbh\ =\ \sqrt{(lbh)^2} = \sqrt{lb \times bh \times lh} = \sqrt{120 \times 72 \times 60} = 720cm^3$.
Therefore, Volume is $720cm^3$.

8. Correct option: (B)

Volume of cuboid = $12 \times 8 \times 3 = 288\, cm^3$.

Therefore, volume is 288 cu.cm.

9. Correct option: (B)

The length of the longest pole can be calculated by calculating the value of the diagonal of a room which is considered to be in the shape of a cuboid.

We know that, the Diagonal of the cuboid = $\sqrt{l^2 + b^2 + h^2}$.

The Diagonal of the cuboid = $\sqrt{12^2 + 8^2 + 9^2} = \sqrt{289} = 17\, m$

Therefore, the Diagonal of the cuboid is 17 m.

10. Correct option: (B)

Length of each side of cube = $\boldsymbol{5cm}$

Now, 3 cubes are joined end to end.

So, the length of the resulting cuboid will be $(l) = 5 + 5 + 5 = 15cm$

But, the breadth and height will remain the same.

So, Breadth (b)=5cm and Height (h)=5cm

Surface area of the cuboid = $2(lb + bh + hl)$

$= 2(15 \times 5 + 5 \times 5 + 5 \times 15)$

$= 2(75 + 25 + 75)$

$= 2 \times 175$

∴ Surface area of the cuboid = $350cm^2$.

11. Correct option: (A)

The length of the tank is three times the breadth: $L = 3B$

The breadth is 25 cm deep: $D = 25\, cm$

The tank holds 3000 liters of water:

We need to convert liters to cubic centimeters (cm^3) since the measurements are in centimeters.

$1\, liter = 1000\, cm^3$

$3000\, liters = 3000 \times 1000\, cm^3 = 3{,}000{,}000\, cm^3$

Now we can calculate the base area of the tank.

The volume of a rectangular tank is given by the formula:

$Volume = Length \times Breadth \times Depth$

Substituting the given values:

$3{,}000{,}000\, cm^3 = (3B) \times B \times 25\, cm.$

Simplifying the equation:

$3{,}000{,}000 = 75B^2$.

Dividing both sides by 75:

$40{,}000 = B^2$.

Taking the square root of both sides:

$B = \sqrt{40{,}000}$

$B \approx 200\, cm$

Since the length is three times the breadth:

$L = 3B = 3 \times 200\, cm = 600\, cm$

The base area of the tank can be calculated by multiplying the length and breadth:

$Base\ Area = Length \times Breadth$

$Base\ Area = 600\, cm \times 200\, cm = 120{,}000\, cm^2$

Therefore, the correct answer is A) 1200.

12. Correct option: (B)

Area of surface to be cemented $= 2 \times (l + b) \times h + (l \times b)$

i.e, area of four walls + area of the floor $= 2 \times (21) \times 4 + (106.25) = 274.25\, m^2$

$\therefore$ cost of cementing $= 24 \times 274.25 = Rs.6582$

Therefore, cost of cementing is $Rs.6582$.

13. Correct option: (B)

The diagonal of the room = $\sqrt{l^2 + b^2 + h^2} = \sqrt{12^2 + 7^2 + 5^2} = \sqrt{218}$.

14. Correct option: (B)

$Base\ area = lb = 12 \times 7 = 84\, m^2$.

Cost of flooring = Base area$\times$ $Rate = 84 \times 2 = Rs.168$

Therefore, cost of flooring is $Rs.168$.

15. Correct option: ()

The total area to be whitewashed

$= LSA + Area\ of\ roof$

$= 2(l + b)h + lb$

$= 2(12 + 7)5 = 274\ m^2$.

Therefore, cost of whitewashing $= 274 \times 3 = Rs.822$.

Therefore, cost of whitewashing $Rs.822$.

16. Correct Matching is: $a \rightarrow T, b \rightarrow P, c \rightarrow Q, d \rightarrow R$

a) The lateral surface area of a rectangular prism is the sum of the surface area of all its faces without the base of the rectangular prism.

$Lateral\ Surface\ Area = Perimeter\ of\ the\ base \times height$

b) The volume of a prism is the product of the area of the base and the height of the prism. $Prism\ volume\ (V) = B \times h$, where, B is the area of the base and h is the height of the prism.

c) If "P" is the base perimeter and "l" is the slant height, then the lateral surface area of the regular pyramid formula is given by

$Lateral\ Surface\ Area = \left(\frac{1}{2}\right) \times perimeter\ of\ the\ base \times slant\ height$

d) The volume of a triangular pyramid can be found using the formula $Volume = \left(\frac{1}{3}\right) \times A \times H$, where A = area of the triangle base, and H = height of the pyramid or the distance from the pyramid's base to the apex.

17. Correct Answer is: 180 cm^3. 2^3 =

Match box is a cuboid having its length(l), breadth(b) and height(h) as 4 cm, 2.5 cm and 1.5 cm respectively.

Volume of 1 match box

$= l \times b \times h = (4 \times 2.5 \times 1.5)cm3 = 15\ cm^3$

Hence, Volume of 12 such matchboxes $= (15 \times 12)cm^3 = 180\ cm^3$.

18. Correct Answer is: 6 cm.

Volume of new cube formed $= 125 + 64 + 27 = 216$

Let the $length\ of\ the\ cube = l$

$Then,\ 216 = l^3$

$\therefore l = 6\ cm$

Cylinder, Cone and Sphere

LEVEL-I

1. Correct option: (B)

The curved surface area of a right circular cylinder of the base radius 'r' and height 'h' is 2πrh

The curved surface area of the pillar = ?

Radius of the cylinder, r = 0.7 m

The Curved surface area of the pillar = $2\pi rh$

Curved surface area of the pillar $= 2 \times \frac{22}{7} \times 0.7\,m \times (1) = 2 \times 22 \times 0.1$

Curved surface area of the pillar $= 44 \times 0.1 = 4.4\ m^2$.

Therefore, the Curved surface area of the pillar is $4.4\ m^2$.

2. Correct option: (A)

Curved surface Area of cylindrical pillar = $2\pi rh$

Curved surface Area of cylindrical pillar $= 2 \times \frac{22}{7} \times 1.75 \times 15$

Curved surface Area of cylindrical pillar $= 2 \times 22 \times 0.25 \times 15$

Curved surface Area of cylindrical pillar = 165 m²

Now,

Cost of painting 1 m² curved surface of the pillar = $Rs.25$

Cost of painting 165 m² curved surface of the pillar $= Rs.25 \times 165 = Rs.4125.$

3. Correct option: (C)

Given, curved surface area =176 cm^2 and and base area 38.5 cm^2.

$\Rightarrow \pi r^2 = 38.5$

$\Rightarrow r^2 = 38.5 \times \frac{22}{7} = 12.25$

$\Rightarrow r = 3.5\,m.$

$Again, curved\ surface\ area\ = 176\ cm^2$

$\Rightarrow 2\pi rh = 176$

$\Rightarrow 2 \times \frac{22}{7} \times 3.5 \times h = 176$

$\Rightarrow h = 8.$

Volume = $\pi r^2 h = \frac{22}{7} \times 3.52 \times 8 = 308\ cm^3$

Therefore, Volume is 308 cm^3.

4. Correct option: (A)

Length of the rectangular paper will be height of the cylinder and width of the rectangular paper is circumference of the cylinder

$\therefore 2\pi r = 22cm$

$r = \frac{7}{2}$

$and\ h = 10\ cm$

volume of cylinder(V) = $\pi r^2 h$

$= \frac{22}{7} \times \left(\frac{7}{2}\right)^2 \times 10$

$= \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times 10$

$= 385\ cm^3$

Therefore, the volume of cylinder is 385 cm^3.

5. Correct option: (C)

⇒ Dimension of cuboid = $44cm \times 30cm \times 15cm$

⇒ Height of a cylinder (H)=28cm

According to the question,

⇒ Volume of a cuboid = Volume of cylinder.

⇒ $l \times b \times h = \pi \times r^2 \times H$

⇒ $44 \times 30 \times 15 = 722 \times (r^2) \times 28$

⇒ $44 \times 30 \times 15 = 22 \times 4 \times r^2$

⇒ $r^2 = \dfrac{44 \times 30 \times 15}{22 \times 4}$

⇒ $r^2 = 225\ cm^2$

∴ $r = 15\ cm.$

Therefore, radius is 15 cm.

6. Correct option: (C)

Let the height of X and Y be `h', and their radii be $r\ and\ 2r$ respectively.

Then, volume of $X\ =\ \pi r^2 h$ and

volume of Y = $\pi\ (2r)^2$, h = $\ 4\ \pi r^2 h$

$New\ height\ of\ X\ =\ 2h,$

$So, new\ volume\ of\ X\ = \pi r^2(2h) =\ 2\pi r^2 h$

$= \frac{1}{2}(4\pi r^2 h) = \frac{1}{2} volume\ of\ Y.$

Therefore, Volume of X becomes $Half\ volume\ of\ Y$.

7. Correct option: (D)

Wire is in the shape of a cylinder.

Let the radius of the original cylinder =r cm and height =h cm

Original volume = $\pi r^2 h$

New Radius after decrease = $3r$

Let the new height be H.

Given, $\pi r^2 h = \pi \left(\frac{r}{2}\right)^2 H$

$H = 9h$

Hence, the height becomes 9 times.

8. Correct option: (B)

Volume of earth dug out

$= \frac{22}{7} \times 7 \times 7 \times 10\ m^3 =\ 1540\ m^3$

Area of embankment

$= \dfrac{22}{7} \times ((28)^2 - (7)^2) m^2$

$= \dfrac{22}{7} \times ((28 + 7)(28 - 7)) m^2$

$= 2310\ m^2$

Height of embankment = $\frac{Volume}{Area}$

Therefore, Height of embankment = $\frac{1540}{2310}\ m = \frac{2}{3}\ m.$

9. Correct option: (C)

Circumference=12m and height (h)=πm Therefore

$2\pi r = 12\ or$

$r = \dfrac{12}{2\pi} = \dfrac{6}{\pi}$

$Now\, V = \pi r^2 h$

$= \pi \times \left(\frac{6}{\pi}\right)^2 \times \pi$

$= \frac{\pi \times 36 \times \pi}{\pi \times \pi} = 36m^3.$

Therefore, Volume is $36m^3$.

10. Correct option: (C).

Given, Diameter $d = 14cm$

$\therefore r = 214 = 7cm, l = 10cm$

Curved surface area of a cone= $\pi rl = \frac{22}{7} \times 7cm \times 10cm = 220cm^2$.

Therefore, Curved surface area of a cone is $220cm^2$.

11. Correct option: (B)

Curved surface area of cone painted $= \frac{35.20}{0.05} = 704\ cm^2$.

$\Rightarrow \boldsymbol{\pi r(r + l)} = 704\ cm^2.$

$\Rightarrow 704 = \pi r(r + l)$

$\Rightarrow 704 = \frac{22r(r + 25)}{7}$

$\Rightarrow \frac{704 \times 7}{22} = r^2 + 25r$

$\Rightarrow 224 = r^2 + 25r$

$\Rightarrow 0 = r^2 + 25r - 224$

$\Rightarrow 0 = r^2 + 32r - 7r - 224$

$\Rightarrow 0 = r(r + 32) - 7(r + 32)$

$\Rightarrow 0 = (r - 7)(r + 32)$

$\Rightarrow 0 = r - 7\ or\ 0 = r + 32$

$\Rightarrow r = 7\ or\ -32.$

Negative value can't be radius.

⇒ radius of cone = 7 cm

By Pythagoras Theorem,

$height\ of\ cone = (l^2 - r^2)^{\frac{1}{2}} = (25^2 - 7^2)^{\frac{1}{2}} = (625 - 49)^{\frac{1}{2}} = (576)^{\frac{1}{2}} = 24\ cm$

Volume of cone $= \frac{\pi r^2 h}{3} = \frac{22 \times 7 \times 7 \times 24}{3 \times 7} = 1232\ cubic\ cm$

Therefore, volume of cone is 1232 *cubic cm*.

12. Correct option: (C)

for cone

$r = 3cm$

$h = 5cm$

$v_1 = \frac{\pi r^2 h}{3}$

$v_1 = \frac{\pi \times (3)^2 \times (5)}{3}$

$v_1 = \frac{\pi \times 9 \times (5)}{3}$

$v_1 = \frac{\pi \times 45}{3}$

$v_1 = 15\pi$

for cylinder

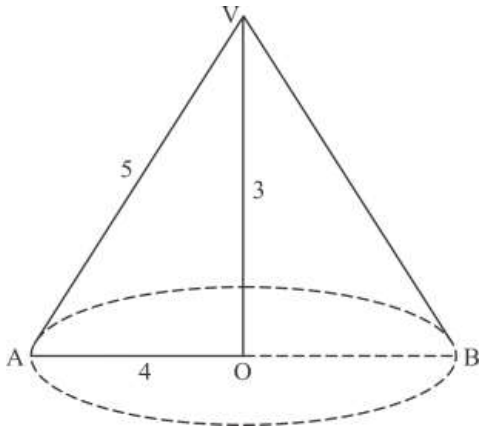
$r = 2cm$

$h = 4cm$

$v_2 = \pi r^2 h$

$v_2 = \pi \times (2)^2 \times (4)$
$v_2 = \pi \times 4 \times 4$
$v_2 = 16\pi$
$\frac{v_1}{v_2} = \frac{15\pi}{16\pi} = \frac{15}{16}$
$v_1 : v_2 = 15 : 16.$
Therefore, $v_1 : v_2 = 15 : 16.$

13. Correct option: (C).
Radius of cone VAOB, r = 4 cm
Height of cone VAOB, h = 3 cm



The volume of cone VAOB $= \frac{\pi \times (4)^2 \times (3)}{3} = \frac{\pi \times 16 \times 3}{3} = 16\pi\ c\ m^3$.
Therefore, volume of cone VAOB is $16\pi\ c\ m^3$.

14. Correct option: (D).
Let V_1 and V_2 are the volumes of cylinder and cone respectively.
Cylinder is melted to form a cone,
$\therefore V_1 = V_2$
$\Rightarrow \pi \times (8)^2 \times 2 = \frac{1}{3} \times \pi \times r^2 \times 6$ (where r is the radius of cone)
$\Rightarrow r^2 = (8)^2$
$\Rightarrow r = 8cm.$
Therefore, radius is 8 cm.

15. Correct option: (B)
Radius of a sphere $(r) = 14\ cm$
$\therefore$ Surface area of $sphere = 4\pi r^2 = 4 \times \frac{22}{7} \times 14 \times 14\ cm^2$
Surface area of $sphere = 2464\ cm^2$
Therefore, Surface area of $sphere\ is\ 2464\ cm^2$.

16. Correct option: (D)
Let the radius of the sphere before increase= $\boldsymbol{r}$.
Given, increased surface area = $352cm^2$.
Also, the surface area of the sphere before the increase is A= $4\pi r^2$
After increasing radius by 2 cm, the area is increased by $352cm^2$.
Applying the above conditions:
$\therefore 4\pi r^2 + 352 = 4\pi(r + 2)2$
$\Rightarrow 352 = 16\pi + 16\pi r$
$\Rightarrow 1 + r = \frac{352}{16\pi}$
$\Rightarrow 1 + r = 7$
$\Rightarrow r = 6cm.$

Therefore, Radius is 6 cm.

17. Correct option: (D)

Surface area of the sphere = $4\pi r^2$

The ratio of the surface area of A to the surface area of B is $\frac{A}{B} = \frac{(40)^2}{(10)^2} = \frac{1600}{100} = \frac{16}{1}$

Therefore, ratio of the surface area of A to the surface area of B is 16:1.

18. Correct option: (B)

Volume of sphere of radius r = $\frac{4}{3} \times \pi \times (r)^3$

Volume of sphere of radius r = $\frac{r}{3} \times 4\pi r^2$.

We know that,

Surface area of the sphere = $4\pi r^2$

Volume of sphere of radius r = $\frac{r}{3}$ ×surface area of sphere.

Hence, the volume of sphere of radius 'r' is obtained by multiplying its surface area by $\frac{r}{3}$.

19. Correct option: (C)

The sphere will have the diameter equal to the side of the cube as they are of the same height.

If the length of the side of the cube is a, then the diameter of the sphere =a unit

∴ Radius of the sphere = $\frac{a}{2}$

Volume of a sphere of radius 'r' = $\frac{4}{3} \times \pi \times (r)^3$

Hence, volume of this sphere = $\frac{4}{3} \times \pi \times \left(\frac{a}{2}\right)^3 = \frac{\pi a^3}{6}$

Volume of a cube of edge $a = a^3$

Ratio of their volumes = $\frac{\pi a^3}{6} : a^3$

$= \frac{\pi}{6} : 1 = \frac{\frac{22}{7}}{6} : 1 = \frac{22}{7 \times 6} : 1 = 11:21.$

1. Therefore, Ratio of their volumes is 11: 21.

20. Correct option: (A)

Given.

A cone, a hemisphere, and a cylinder stand on equal bases.

Hence radius of base of cone = radius of hemisphere = radius of cylinder=r

And,

They also have same height =h

Height of cone =h

Height of cylinder=h

Height of Hemisphere r=h (r is the radius of sphere)

Now,

Volume of cone $V_C = \frac{1}{3}\pi r^2 h$.....(1)

Volume of hemisphere $V_H = \frac{2}{3}\pi r^2 h$ .....(2)

Volume of cylinder $V_c = \pi r^2 h$.....(3)

From equation 1,2 and 3

The ratio of their volume

$V_C : V_H : V_c = \frac{1}{3}\pi r^2 h : \frac{2}{3}\pi r^2 h : \pi r^2 h$

$V_C : V_H : V_c = \frac{1}{3} : \frac{2}{3} : 1.$

$V_C : V_H : V_c = 1 : 2 : 3$

Therefore, the ratio of their volume $V_C : V_H : V_c = 1 : 2 : 3$.

LEVEL-II

1. Correct option: (B).

Let the length of the wire be h

Then,

$\pi \times \frac{3}{20} \times \frac{3}{20} \times h = \frac{4}{3}\pi \times 4 \times 4 \times 4$

$\Rightarrow h = \frac{4 \times 4 \times 4 \times 4 \times 20 \times 20}{3 \times 3 \times 3} cm$

$\Rightarrow h = \frac{102400}{27} cm$

$\Rightarrow h = 3792.5cm \qquad (1\,m = 100\,cm\ and\ 1\,cm = 0.01\,m)$

$\Rightarrow h = 37.9m.$

2. Correct option: (C)

Increase in volume of cylinder = volume of sphere

Let increase in the volume of cylinder be $\pi r^2 h$, where r is its radius and h is its height.

So, $\pi r^2 h = \frac{4}{3}\pi r^3$

$\Rightarrow 16\,\pi rh = \frac{4}{3}\pi r^3$

where radius of the cylinder=4cm and radius of the sphere=3cm

$\Rightarrow h = \frac{4}{3} \times \frac{27}{16} = \frac{36}{16} = \frac{9}{4}$

$\Rightarrow h = \frac{9}{4}$

Therefore, water rises by $\frac{9}{4}$cm in the vessel.

3. Correct option: (B)

Given : A Right circular cylinder

Height =14cm

curved surface area (CSA)=704sq.cm

Volume = ?

We know, Curved surface area of cylinder = $\boldsymbol{2\pi rh}$

$704 = 2\pi r \times 14$

$r = \frac{704}{2\pi r \times 14} = \frac{352}{14\pi}$

$r = \frac{176}{7\pi} \qquad \left(\pi = \frac{22}{7}\right)$

$r = \frac{176 \times 7}{7 \times 22} = 8$

$r = 8\,cm.$

Volume of a Right circle cylinder (v) = $\pi r^2 h$

$= \frac{22}{7} \times 8 \times 8 \times 14 cm^3 = 2816\ cm^3.$

Therefore**,** Volume of a Right circle cylinder is 2816 cm^3.

4. Correct option: (C)

Given: Circumference of the base= $\boldsymbol{2\pi r = 66\,cm}$

$\Rightarrow 2 \times \frac{22}{7} \times r = 66$

$\Rightarrow r = \frac{66 \times 7}{2 \times 22} = \frac{21}{2} = 10 \cdot 5$

$\Rightarrow r = 10 \cdot 5cm.$

Curved Surface area of a cylinder= $\boldsymbol{2\pi rh = 2640\ sq.cm}$

$\Rightarrow h = \frac{2640}{2\pi \times 10.5} = \frac{2640 \times 7}{2 \times 22 \times 10.5} = 40\ cm$

$\Rightarrow h = 40\ cm$

Volume of cylinder $= \pi r^2 h$

Volume of cylinder $= \frac{22}{7} \times 10.5 \times 10.5 \times 40 = 13860 cm^3$.

Therefore, Volume of cylinder is 13860 cm^3.

5. Correct option: (D)

It is given that

Diameter = 3mm = 0.3cm

Radius = $\frac{0{\cdot}3}{2}$ cm

We know that Number of lead shots = $\frac{\text{volume of cuboid}}{\text{volume of 1 lead shot}}$.

By substituting the values Number of lead shots = $\frac{12 \times 11 \times 9}{\frac{4}{3} \times \frac{22}{7} \times \left(\frac{0.3}{2}\right)^3}$.

On further calculation

Number of lead shots = $\frac{12 \times 11 \times 9}{\frac{4}{3} \times \frac{22}{7} \times \frac{0.0027}{8}}$

So we get Number of lead shots = 84000

Therefore, the number of lead shots are 84000.

6. Correct option: (C)

Volume of sphere $= \frac{4}{3}\pi r^3$

⇒r=7cm

vol. of sphere $= \frac{4}{3}\pi(7)^3 = \frac{1372}{3}\pi cm^3$__(1)

vol of cylinder $\pi R^2 h$__(2)

from eq (1) and (2)

$\frac{4}{3}\pi(7)(7)(7) = \pi(R)(R)\frac{7}{3}$

$R = 14cm$

Diameter of cylinder = 2 R = 2 × 14 = 28cm

Therefore, Diameter of cylinder is 28cm.

7. Correct option: (A)

T.S.A. of cylinder $= 2\pi rh + 2\pi r^2 = 2\pi \times \left(\frac{35}{100}\right) \times 1.25 + 2\pi \left(\frac{35}{100}\right)^2$

$\Rightarrow T.S.A = 3.52\ m^2$

cost of $1m^2$ is Rs 80

$\Rightarrow Total\ cost = 3.52 \times 80 = Rs.281.6$

Therefore, total cost is $Rs.281.6$.

8. Correct option: (D)

Diameter of circular well= $2n$

∴ Radius= $\frac{2}{2} = 1\ m$

Depth $(h) = 14\ m$

Volume of earth dug out $= \pi r^2 h$

Volume of earth dug out $= \frac{22}{7} \times 1 \times 1 \times 14 = 44\ m^3$

∴ Volume of earth dug out is $44\ m^3$.

9. Correct option: ()

C.S.A. of the cone = πrl

$119\frac{5}{7} cm^2 = \frac{838}{7} cm^2$

Base area= πr^2

$254\frac{4}{7} cm^2 = \frac{1782}{7} cm^2$

Now, $\pi \times r^2 = \frac{1782}{7} cm^2$

$\frac{22}{7} \times r^2 = \frac{1782}{7}\ cm^2$

$r^2 = \frac{1782 \times 7}{7 \times 22}\ cm^2$

$r^2 = \frac{1782}{22} = 81\ cm^2$

$r = 9\ cm$

Therefore, radius is 9 cm.

Now,

$\pi rl = \frac{838}{7} cm^2$

$\frac{22}{7} \times 9 \times l = \frac{838}{7} cm^2$

$l = \frac{838 \times 7}{7 \times 22 \times 9} cm^2$

$l = \frac{838}{198} cm^2$

$l = 4.232\ cm$

$h = \sqrt{r^2 + l^2}$

$h = \sqrt{(9)^2 + (4.232)^2} = \sqrt{81 + 18} = \sqrt{99} = 9.94\ = 10$

$h =\ 10\ cm$

Volume of the cone $= \frac{1}{3}\pi r^2 h$

$\frac{1}{3}\pi r^2 h = \frac{1}{3} \times \frac{22}{7} \times 9 \times 9 \times 10 =\ 848.57$

Therefore, Volume of the cone is 848.57 cm^3.

10. Correct option: (B)

Given, $r\ +\ h\ =\ 37\ cm$

Total surface area of the cylinder = 1628 cm2 (Given)

$\therefore 2\pi r(r + h) = 1628$

$\Rightarrow 2\ \pi r\ \times\ 37\ =\ 1628$

$\Rightarrow\ r\ =\ 7\ cm$

Now, $r\ +\ h\ =\ 37\ cm$

$\Rightarrow 7\ +\ h\ =\ 37$

$\Rightarrow\ h\ =\ 37\ -\ 7\ =\ 30\ cm$

Volume of the cylinder = $\pi r^2 h = \frac{22}{7} \times 7 \times 7 \times 30 =\ 4620\ cm^3$.

Therefore, Volume of the cylinder is $4620\ cm^3$.

11. Correct option: (B)

Let the radius of the cylinder is r and height is h

$\therefore$ Volume of the cylinder = $\pi r^2 h$

According to the question new radius is half of the initial radius

$\therefore$New radius=$\frac{r}{2}$

Height is h.

$\therefore$New volume of the cylinder $\pi \times \left(\frac{r}{2}\right)^2 \times h = \pi \times \frac{r^2}{4} \times h$

$\therefore$ The ratio of the new volume of the cylinder and initial volume of the original cylinder

$$= \frac{\pi \times \frac{r^2}{4} \times h}{\pi r^2 h} = \frac{1}{4}$$

$Therefore,\ Ratio\ = 1:4.$

12. Correct option: ()

Volume of spherical ball $V_s = \frac{4}{3}\pi(R^3 - r^3)$.....(1)

Radius = $\frac{d}{2}$

Radius $r = \frac{6}{2} = 3\ cm$

Radius $R = \frac{8}{2} = 4\ cm$

Volume of spherical ball $V_s = \frac{4}{3}\pi\ (4^3 - 3^3) = \frac{4}{3}\pi\ (64\ - 27) = \frac{4}{3}\pi\ (37) = \frac{148}{3}\pi\ cm^3$

Volume of spherical ball $V_s = \frac{148}{3}\pi\ cm^3$

Volume of cone $V_c = \frac{1}{3}\pi r^2 h$ .....(2)

Radius = $\frac{d}{2}$

Radius $= \frac{8}{2} = 4\ cm$

$V_c = \frac{1}{3}\pi(4)^2 h = \frac{16}{3}\pi h = \frac{16}{3}\pi h\ cm^3.$

$V_c = \frac{16}{3}\pi h\ cm^3.$

Now,

$V_s = V_c$

$\frac{148}{3}\pi\ = \frac{16}{3}\pi h$

$\frac{148}{16} = h$

$h = \frac{74}{8} = 9.25\ cm$

Therefore, the height of cone is 9.25 cm.

13. Correct option: (D)

Given the radius of the cylinder, r = 12 cm

It is also given that a spherical iron ball is dropped into the cylinder and the water level raised by 6.75 cm

Hence volume of water displaced = volume of the iron ball

Height of the raised water level, h = 6.75 m

Volume of water displaced = $\pi r^2 h$

$= \pi \times 12 \times 12 \times 6.75\ cm^3$

⇒ Volume of iron $ball\ = \pi \times 12 \times 12 \times 6.75\ cm^3 \rightarrow$ (1)

But, volume of iron ball = $\frac{4}{3}\pi r^3$---(2)

From (1) and (2) we get

$\frac{4}{3}\pi r^3 = \pi \times 12 \times 12 \times 6.75$

$r^3 = \pi \times 12 \times 12 \times 6.75 \times \frac{3}{4}$

$r^3 = 729$

$r^3 = 3^3$

r = 9

Thus the radius of the iron ball is 9 cm.

Passage-I

14. Correct option: (A)

Diameter of the vessel =4 cm

Radius of vessel =2 cm

Rise in water level =0.8 cm

Volume of the increased water level = $\pi r^2 h = \frac{22}{7} \times (2)^2 \times 0.8 = 10.06\, cu.cm$

$Volume\ of\ 300\ balls\ = 300 \times \frac{4}{3} \times \frac{22}{7} \times r^3$

$\Rightarrow r^3 = 10.06 \times \frac{1}{300} \times \frac{4}{3} \times \frac{22}{7}$

$\Rightarrow r^3 = 0.008$

$\Rightarrow r = 0.2$

So, the radius of each ball is 0.2 cm.

Diameter of each ball is $2 \times 0.2 = 0.4\, cm.$

Therefore, radius of each lead ball is 0.2 cm.

15. Correct option: (B)

Diameter of the vessel =4 cm

Radius of vessel =2 cm

Rise in water level =0.8 cm

Volume of the increased water level = $\pi r^2 h = \frac{22}{7} \times (2)^2 \times 0.8 = 10.06\, cu.cm$

Therefore, Volume of the increased water level is 10.06 cu.cm.

16. Correct option: (D)

Diameter of the vessel =4 cm

Radius of vessel =2 cm

Rise in water level =0.8 cm

Volume of the increased water level = $\pi r^2 h = \frac{22}{7} \times (2)^2 \times 0.8 = 10.06\, cu.cm$

$Volume\ of\ 300\ balls\ = 300 \times \frac{4}{3} \times \frac{22}{7} \times r^3$

$\Rightarrow r^3 = 10.06 \times \frac{1}{300} \times \frac{4}{3} \times \frac{22}{7}$

$\Rightarrow r^3 = 0.008$

$\Rightarrow r = 0.2$

So, the radius of each ball is 0.2 cm.

$Volume\ of\ 300\ balls\ = 300 \times \frac{4}{3} \times \frac{22}{7} \times r^3$

$Volume\ of\ 300\ balls\ = 300 \times \frac{4}{3} \times \frac{22}{7} \times (0.2)^3$

$= 300 \times \frac{4}{3} \times \frac{22}{7} \times 0.008$

$= \frac{211.2}{21}$

$Volume\ of\ 300\ balls = \ 10.05\, cu.cm$

Therefore $Volume\ of\ 300\ balls\ is\ 10.05\, cu.cm.$

17. Correct Matching is: $a \rightarrow R, b \rightarrow P, c \rightarrow T, d \rightarrow S$

a) If the base radii of two cylinders are equal, then the ratio of their curved surface area is r: R.

If the base radii of two cylinders are equal, then the ratio of their curved surface areas is directly proportional to the ratio of their heights. Therefore, the correct ratio would be r:R.

b) If the curved surface areas are equal, then the ratio of their volumes is $is\ r^2 + R^2$.

If the curved surface areas of two objects are equal, then the ratio of their volumes is given by $r^2 + R^2$. This is applicable for objects that have the same shape and only differ in size. The ratio of their volumes will be equal to the square of the ratio of their respective radii (or dimensions).

c) If the volumes of two objects are equal, then the ratio of their curved surface areas is h : H. This is known as the isoperimetric inequality. The isoperimetric inequality states that

among all objects with a given volume, a sphere has the minimum surface area. Therefore, if two objects have the same volume, their curved surface areas will have a similar ratio of h : H, where h represents the curved surface area of the object with a smaller surface area, and H represents the curved surface area of the object with a larger surface area.

d) If the heights are equal, then the ratio of their volumes is R : r.

If the heights of two cylinders are equal, then the ratio of their volumes is determined by the ratio of their base radii.

Therefore, $a \to R, b \to P, c \to T, d \to S$.

18. Correct Answer is: 12 cm

Radius of the sector, 'r' = 15 cm

Angle of the sector, 'x' = 216°

∴ Length of the arc, $l = \frac{x}{360} \times 2\pi r = \frac{216}{360} \times 2\pi r = \frac{3}{5} \times 2\pi r$

Perimeter of the base of the cone = Length of the arc

2πr of cone = $\frac{6}{5}$πr of the circle

Radius of the cone 'r' $= \frac{3}{5} \times 15 = 9$

Radius 'r' of the circle = slant height l of the cone = 9 cm

$height\ of\ cone = (l^2 - r^2)^{\frac{1}{2}} = (15^2 - 9^2)^{\frac{1}{2}}$

$= (225 - 81)^{\frac{1}{2}} = (144)^{\frac{1}{2}}$

$= (12^2)^{\frac{1}{2}} = 12\ cm$

Height of cone is 12 cm.

19. Correct Answer is: 24 cm.

$Radius\ of\ cone = (l^2 - h^2)^{\frac{1}{2}} = (25^2 - 7^2)^{\frac{1}{2}}$

$= (625 - 49)^{\frac{1}{2}} = (576)^{\frac{1}{2}}$

$= (24^2)^{\frac{1}{2}} = 24\ cm.$

Therefore, radius is 24 cm.

20. Correct Answer is: 5 cm

Given diameter of metallic cylinder=5 cm

So radius of cylinder=2.5 cm

And its height $\mathbf{3\frac{1}{3} = \frac{10}{3}\ cm}$

Then volume of cylinder = $\pi r^2 h = \pi \times (2 \cdot 5)^2 \times \frac{10}{3}\ cu.cm$

Let radius of sphere is r

Then volume of sphere =$\frac{4}{3}\pi r^3$

Then volume of cylinder =volume of sphere

$\therefore \frac{4}{3}\pi r^3 = \pi \times (2 \cdot 5)^2 \times \frac{10}{3}$

$\Rightarrow r^3 = \frac{2.5}{4} \times 2.5 \times 10 = \frac{62.5}{4} = 15.625 \Rightarrow r = 2.5cm$

$So\ diameter\ of\ sphere = 2 \times 2.5 = 5cm$

Therefore, diameter is 5 cm.

STUDENT PRACTICE QUESTIONS

1. Correct option: (D)

The area of the metal sheet required = The total surface area of the cylinder = $2\pi(r + h)$

$2\pi(r + h) = 2 \times \frac{22}{7} \times 7(7 + 10) = 44(17) = 748\ cm^2.$

Therefore, the area of metal sheet required is 748 cm^2.

2. Correct option: (B)

Given, height of cylinder = 6.5 cm and total surface area of cylinder = 220 cm^2.

$2\pi r(r + h) = 220\ cm^2$.

$r(r + 6.5) = 220 \times \frac{7}{2} \times 22$

$r^2 + 6.5r - 35 = 0$

$r^2 - 3.5r + 10r - 35 = 0$

$(r - 3.5)(r + 10) = 0$

$r = 3.5, -10$

Radius can not be negative, so take ,positive value of r ,

r = 3.5 cm

volume of cylinder(V) = $\pi r^2 h$

$= \frac{22}{7} \times (3.5)^2 \times 6.5$

$= \frac{1751.75}{7}$

$= 250.25\ cm^3$

Therefore, the volume of cylinder is 250.25 cm^3.

3. Correct option: (B)

We know, the base of a cone is circular.

Then, Area of a circle = πr^2=616.

$\Rightarrow r^2 = \frac{616 \times 7}{22} = 196$

$\Rightarrow r^2 = 196$

$=> r = 14cm.$

Also, total surface area of a cone = $\boldsymbol{\pi r(r + l)}$ where, r is the radius of the cone and l is the slant height.

For a cone, slant height $l = \sqrt{r^2 + h^2}$ where, h is the perpendicular height.

Hence, $l = \sqrt{(48)^2 + (14)^2} = \sqrt{2500}$

$\Longrightarrow l = 50\ cm.$

Hence, Total surface area of this cone, $= \frac{22}{7} \times 14 \times (14 + 50) = 2816$ cm^2.

Therefore, Total surface area of this cone is 2816 cm^2.

4. Correct option: (D)

The amount of canvas used to make the tent = Curved surface area of cylindrical part + Curved surface area of the conical part.

Curved Surface Area of a Cylinder of $Radius\ "R"\ and\ height\ "h"\ = 2\pi Rh$

Radius of the cylindrical part = $\frac{\text{Diameter}}{2} = \frac{9}{2}\ m$

Curved surface area of a $cone\ = \pi rl$, where r is the radius of the cone and l is the slant height.

Radius of the conical part = $\frac{\text{Diameter}}{2} = \frac{9}{2}$ m

Height of the conical part $= 10.8 - 4.8 = 6\ m$

For a cone, $l = \sqrt{r^2 + h^2}$, where h is the height.

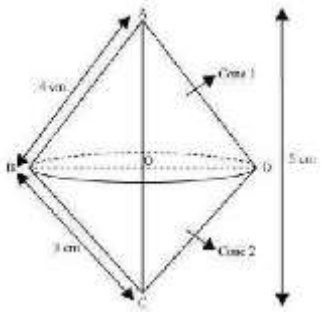
Hence, $l = \sqrt{(6)^2 + (4.5)^2}$

$\therefore l = 7.5\ m$

Hence, area of canvas $= 2 \times \pi \times 4.5 \times 4.8 + \pi \times 4.5 \times 7.5 = 76.95\pi = 241.84\ m^2$

Hence, area of canvas is 241.84 m^2.

5. Correct option: (B)



The double cone so formed by revolving this right-angled triangle ABC about its hypotenuse is shown in the figure.

Hypotenuse $AC = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$

$AC = 5\ cm$

$$\text{Area of } \Delta ABC = \frac{1}{2} \times AB \times AC$$

$\frac{1}{2} \times AC \times OB = \frac{1}{2} \times 4 \times 3$

$= \frac{1}{2} \times 5 \times OB = 6$

$OB = \frac{12}{5} = 2.4\ cm.$

$Volume\ of\ double\ cone\ = Volume\ of\ cone\ 1\ +\ Volume\ of\ cone\ 2$

$= \frac{1}{3} \times \pi r^2 h_1 + \frac{1}{3} \times \pi r^2 h_2 = \frac{1}{3} \times \pi r^2 (h_1 + h_2)$

$= \frac{1}{3} \times \pi r^2 (OA + OC)$

$= \frac{1}{3} \times 3.14 \times (2.4)^4 \times (5) = 30.14 cm^3.$

Therefore, $Volume\ of\ double\ cone\ is\ 30.14 cm^3$.

6. Correct option: (D)

let the Height of the $\boldsymbol{cone = h\ cm}$

Radius of the $\boldsymbol{cone = r\ \ cm}$

Then volume of the old cone= $\frac{1}{3} \times \pi \times r^2 \times h$

If the height and radius are double then

$Height = 2h$

$Radius = 2r$

Then new volume= $\frac{1}{3} \times \pi \times (2r)^2 \times 2h.$

⇒new volume $= \frac{1}{3} \times \pi \times 4r^2 \times 2h$

new volume $= 8 \times \frac{1}{3} \times \pi \times r^2 \times h$

Hence it is 8 times of the old volume.

7. Correct option: (B)

Let the radii of Cone1 and Cone2 are '$2r$ ' and 'r ' respectively.

⇒ Height of Cone1 and Cone2 are H_1 and H_2 respectively.

$\frac{1}{3} \times \pi \times (2r)^2 \times H_1 = \frac{1}{3} \times \pi \times (r)^2 \times H_2$

$\Rightarrow 4 \times (r)^2 \times H_1 = (r)^2 \times H_2$

$\Rightarrow 4 \times H_1 = H_2$

$\Rightarrow \frac{H_1}{H_2} = \frac{1}{4}$

∴ The ratio of the heights of two cones are 1 : 4.

8. Correct option: (A)

Let the radius of the globe be r.

The length of the equator of the globe means the circumference of the globe.

Circumference of the globe = 88cm

$2\pi r = 88$

$2 \times \frac{22}{7} \times r = 88$

$r = \frac{88 \times 7}{44} = 2 \times 7 = 14\ cm$

$r = 14\ cm$

The surface area of the globe is computed as,

∴Surface area = $4\pi r^2$

$4 \times \frac{22}{7} \times 14 \times 14 = 2464 cm^2$

Hence, the surface area is $2464 cm^2$.

9. Correct option: (A)

Diameter of spherical ball of lead = 3 cm

Radius(r)= $\frac{3}{2}$cm

Volume=$\frac{4}{3} \times \pi \times (r)^3 = \frac{4}{3} \times \pi \times \left(\frac{3}{2}\right)^3 = \frac{9}{2} \times \pi$

∴Volume=$\frac{9}{2} \times \pi\ cm^3$.

Diameter of spherical ball of lead = $\frac{3}{2}$ cm

Radius(r_1)= $\frac{3}{2}$cm

Volume=$\frac{4}{3} \times \pi \times (r)^3 = \frac{4}{3} \times \pi \times \left(\frac{3}{2}\right)^3 = \frac{9}{16} \times \pi$

∴Volume=$\frac{9}{16} \times \pi\ cm^3$

Diameter of second smaller ball = 2 cm

and radius (r_2)= $\frac{2}{2}$ = 1cm

∴Volume= $\frac{4}{3} \times \pi \times (r)^3 = \frac{4}{3} \times \pi \times (1)^3 = \frac{4}{3} \times \pi$

∴Volume= $\frac{4}{3} \times \pi\ cm^3$

Volume of third smaller ball

$= \frac{9}{2}\pi - \left(\frac{9}{16}\pi + \frac{4}{3}\pi\right)$

$= \left(\frac{9}{2} - \frac{9}{16} + \frac{4}{3}\right)\pi\ \ cm^3$.

$= \left(\frac{216-27-64}{48}\right)\pi\ \ cm^3$.

$= \left(\frac{125}{48}\right)\pi\ \ cm^3$.

∴ Radius of the third ball = $\left(\frac{volume \times 3}{4\pi}\right)^{\frac{1}{3}}$

$= \left(\frac{125\ \times 3}{48 \times 4\pi}\right)^{\frac{1}{3}} = \left(\frac{125}{64}\right)^{\frac{1}{3}} = \left(\left(\frac{5}{4}\right)^3\right)^{\frac{1}{3}} = \frac{5}{4}$

$= \frac{5}{4} = 1.25\ cm.$

$\boldsymbol{Diameter\ of\ third\ ball}\ = 2 \times radius$

$= 2 \times 1.25 = 2.5 cm$

∴ $\boldsymbol{Diameter\ of\ third\ ball\ is\ 2.5\ cm}$.

10. Correct option: (B)

Let r be the radius of single sphere.

Now, The volume of single sphere = sum of volume of three spheres

$\frac{4}{3}\times\pi\times(r)^3=\frac{4}{3}\times\pi\times(61)^3+\frac{4}{3}\times\pi\times(8)^3+\frac{4}{3}\times\pi\times(10)^3$

$\frac{4}{3}\times\pi\times(r)^3=\frac{4}{3}\times\pi\times(216+512+1000)$

$r^3=1728$

$r=12\,cm.$

Diameter $=20\times r=24\,cm.$

Therefore, the diameter $is\ 24\,cm.$

11. Correct option: (B)

We know the total surface area of a hemisphere = $3\pi r^2$

Here, r=14cm

Area of hemisphere = $3\pi r^2$

$\Rightarrow 3\times\frac{22}{7}\times 14$

$\Rightarrow 462\ cm^2$

Hence, the total surface area of a hemisphere of radius 14cm is 462 cm^2

12. Correct option: (A)

Given,

Radius of the hemisphere = Radius of the cone.

Now, Volume of the hemisphere $=\frac{2}{3}\pi R^3$ and

Volume of the cone $=\frac{1}{3}\pi R^2H$

Volume of the hemisphere = volume of the cone

$\frac{2}{3}\pi R^3=\frac{1}{3}\pi R^2$

$2R=H$

$H=2R$

Therefore, If 'H' is the height of the cone, then $H=2R$.

13. Correct option: (A)

Radius of the sector, 'r' = 15 cm

Angle of the sector, 'x' = 216°

$\therefore$ Length of the arc, $l=\frac{x}{360}\times 2\pi r=\frac{216}{360}\times 2\pi r=\frac{3}{5}\times 2\pi r$

Perimeter of the base of the cone = Length of the arc

2πr of cone = $\frac{6}{5}$πr of the circle

Radius of the cone 'r' $=\frac{3}{5}\times 15=9$

Radius 'r' of the circle = slant height l of the cone = 9 cm

$height\ of\ cone=(l^2-r^2)^{\frac{1}{2}}=(15^2-9^2)^{\frac{1}{2}}$

$=(225-81)^{\frac{1}{2}}=(144)^{\frac{1}{2}}$

$=(12^2)^{\frac{1}{2}}=12\ cm$

Height of cone is 12 cm.

Volume of the cone $=\frac{1}{3}\pi r^2h$

Volume of the cone $=\frac{1}{3}\times\frac{22}{7}\times(9)^2\times 12$

Volume of the cone = $1018.28\,cu.cm$

Therefore, Volume of the cone = $1018.28\,cu.cm$.

14. Correct option: (D)

The total surface area of a cone = $\boldsymbol{\pi r(r+l)}$ where, r is the radius of the cone and l is the slant height.

The total surface area of a cone = $\frac{22}{7} \times \mathbf{9(9+15)} = \frac{22}{7} \times 9(24) = \frac{4752}{7} = 678.58$

Therefore, the total surface area of a cone is 678.58 cm^2.

15. Correct option: (B)

Radius of the sector, 'r' = 15 cm

Angle of the sector, 'x' = 216°

∴ Length of the arc, $l = \frac{x}{360} \times 2\pi r = \frac{216}{360} \times 2\pi r = \frac{3}{5} \times 2\pi r$

Perimeter of the base of the cone = Length of the arc

2πr of cone = $\frac{6}{5}$πr of the circle

Radius of the cone 'r' $= \frac{3}{5} \times 15 = 9$

Radius 'r' of the circle = slant height l of the cone = 9 cm

$height\ of\ cone = (l^2 - r^2)^{\frac{1}{2}} = (15^2 - 9^2)^{\frac{1}{2}}$

$= (225 - 81)^{\frac{1}{2}} = (144)^{\frac{1}{2}}$

$= (12^2)^{\frac{1}{2}} = 12\ cm$

Height of cone is 12 cm.

For a cone, slant height $l = \sqrt{r^2 + h^2}$ where, h is the perpendicular height.

Hence, $l = \sqrt{(9)^2 + (12)^2} = \sqrt{81 + 144} = \sqrt{225}$

$\Rightarrow l = 15\ cm.$

Therefore, slant height of a cone is 15 cm.

16. Correct Matching is: $\boldsymbol{a \to Q, b \to P, c \to T, d \to S, R}$

a) The curved surface area is defined as the area of only curved surface, leaving the circular top and base.

Consider a cylinder having a height 'h' and base radius 'r'.

The curved surface area of a cylinder is given as **C.S.A.= $\boldsymbol{2\pi rh}$**.

b) The volume of a cylinder is the number of unit cubes (cubes of unit length) that can be fit into it. It is the space occupied by the cylinder as the volume of any three-dimensional shape is the space occupied by it.

Therefore, **volume of Cylinder = $\boldsymbol{\pi r^2 h}$**.

c) A cone's volume is defined as the amount of space or capacity it fills.

Volume of the cone $= \frac{1}{3}\boldsymbol{\pi r^2 h}$.

d) A sphere is the 3- D shape where the curved surface area equals to the total surface area of the figure. The curved surface area is the area in which only the area of the curved part is covered. The formula does not take into account the circular base.

The total surface area, on the other hand, is a combination of the curved area along with the area of the base The formula for the total surface area and the curved surface area of the sphere is **Surface area (TSA) = CSA = 4πr² square units.**

Therefore, : $a \to Q, b \to P, c \to T, d \to S, R.$

17. Correct Answer is: 7 cm.

Let radius of cylinder be r cm

Circumference of base of cylinder = Length of rectangle

$2\pi r = 44$

$r = \frac{44 \times 7 =}{2 \times 22} 7$

∴ Radius of the cylinder is 7 cm.

18. Correct Answer is: 1232 cubic cm.

Curved surface area of cone painted $= \frac{35.20}{0.05} = 704\ cm^2$.

$\Rightarrow \boldsymbol{\pi r(r+l)} = 704\ cm^2$.

$\Rightarrow 704 = \pi r(r+l)$

$\Rightarrow 704 = \frac{22r(r+25)}{7}$

$\Rightarrow \frac{704 \times 7}{22} = r^2 + 25r$

$\Rightarrow 224 = r^2 + 25r$

$\Rightarrow 0 = r^2 + 25r - 224$

$\Rightarrow 0 = r^2 + 32r - 7r - 224$

$\Rightarrow 0 = r(r+32) - 7(r+32)$

$\Rightarrow 0 = (r-7)(r+32)$

$\Rightarrow 0 = r - 7\ or\ 0 = r + 32$

$\Rightarrow r = 7\ or\ -32.$

Negative value can't be radius.

⇒ radius of cone = 7 cm

By Pythagoras Theorem,

$height\ of\ cone = (l^2 - r^2)^{\frac{1}{2}} = (25^2 - 7^2)^{\frac{1}{2}} = (625 - 49)^{\frac{1}{2}} = (576)^{\frac{1}{2}} = 24\ cm$

Volume of cone $= \frac{\pi r^2 h}{3} = \frac{22 \times 7 \times 7 \times 24}{3 \times 7} = 1232\ cubic\ cm$

Therefore, volume of cone is 1232 *cubic cm.*

19. Correct Answer is: 9

To find the sum of the digits in the length of the wire, we can use the formula for the volume of a sphere and the formula for the volume of a cylinder.

The volume of a sphere is given by the formula: Volume of sphere $= \frac{4}{3}\pi r^3$

where r is the radius of the sphere.

Given that the diameter of the copper sphere is 6 cm, the radius (r) is half of the diameter, so r = $\frac{6}{2}$= 3 cm.

Now, let's calculate the volume of the sphere: Volume sphere $= \frac{4}{3}\pi(3)^3 = \frac{4}{3}\pi(27) = 36\pi\ cm^3$

The volume of the sphere is equal to the volume of the wire, since the copper sphere is beaten and drawn into a wire.

The volume of a cylinder is given by the formula:

Volume of cylinder = $\pi r^2 h$, where r is the radius of the cylinder and h is the height (length) of the cylinder (the wire).

We are given that the diameter of the wire is 0.2 cm, so the radius (r) is 0.1 cm.

Let's substitute the values into the formula to find the length of the wire: $36\pi = \pi r^2 h$

$36 = 0.01 \times h$

$h = \frac{36}{0.01} = 3600\ cm$

The length of the wire is 3600 cm.

To find the sum of the digits in the length of the wire, we add up the individual digits: $3 + 6 + 0 + 0 = 9$.

Therefore, the sum of the digits in the length of the wire is 9.

20. Correct Answer is: 6 cm.

Volume of cone $= \frac{\pi r^2 h}{3}$.

Given base area of cone= πr^2 =38.5 cm^3.

Volume of cone = 77 cm^3.

$\frac{(38.5)h}{3} = 77$

$h = \frac{231}{38.5} cm.$

$h = 6\ cm$

$Therefore,$ height is 6 cm.

6. Progressions
Solutions

LEVEL-I

1. Correct option: (B)

The given AP is $20,25,30,...100$

The first term in series is $a=20$

Number of terms in AP is $t_n = a+(\mathrm{n}-1)\mathrm{d}$

d=common difference $= t_2 - t_1 = 5, t_n = 100$

$$t_n = a+(\mathrm{n}-1)\mathrm{d}$$
$$\Rightarrow 100 = 20+(\mathrm{n}-1)5$$
$$\Rightarrow \mathrm{n}-1=16 \Rightarrow \mathrm{n}=17$$

The correct Answer is 'B'

2. Correct option: (A)

To find the nth term in series is $t_n = a+(\mathrm{n}-1)\mathrm{d}$

$$\therefore t_{11} = 80 = a+10\mathrm{d}$$
$$\Rightarrow t_{16} = 40 = a+15\mathrm{d}$$
$$\Rightarrow t_{31} = x = a+30\mathrm{d}$$
$$-5d = 40 \Rightarrow d = -8$$
$$\therefore t_{11} = 80$$
$$\Rightarrow 80 = a-80 \Rightarrow a = 160$$
$$t_{31} = x = a+30\mathrm{d} = 160+30(-8) = -80$$

The correct Answer is 'A'

3. Correct option: (C)

$$a_1 = a_2 = 2$$
$$a_n = a_{n-1} - 1$$

Put n=3 in above equation implies

$$a_3 = a_{3-1} - 1 = a_2 - 1 = 2-1 = 1$$
$$a_4 = a_{4-1} - 1 = a_3 - 1 = 1-1 = 0$$
$$a_5 = a_{5-1} - 1 = a_4 - 1 = 0-1 = -1$$

The correct Answer is 'C'

4. Correct option: (D)

From the given data

The correct Answer is 'D'

5. Correct option: (B)

$\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in AP implies

$$\frac{1}{b} - \frac{1}{a} = \frac{1}{c} - \frac{1}{b} \quad ---(\mathrm{i})$$
$$(\mathrm{i})\,\mathrm{in}\left(\frac{1}{a}+\frac{1}{b}-\frac{1}{c}\right)\left(\frac{1}{b}+\frac{1}{c}-\frac{1}{a}\right) \Rightarrow$$
$$= \left(\frac{2}{a}-\frac{1}{b}\right)\left(\frac{2}{c}-\frac{1}{b}\right)$$
$$= \frac{4}{ac} - \frac{2}{b}\left(\frac{1}{a}+\frac{1}{c}\right) + \frac{1}{b^2}$$
$$= \frac{4}{ac} - \frac{3}{b^2}$$

The correct Answer is 'B'

6. Correct option: (D)

The first two digit no. divisible by 6 is 12 and the last two digit no. divisible by 6 is 96

So let a=12, d=6, $t_n = 96$

$$t_n = a + (\mathrm{n}-1)\,\mathrm{d}$$
$$\mathrm{a} = 12, \mathrm{d} = 6$$
$$\Rightarrow 96 = 12 + (\mathrm{n}-1)6$$
$$\Rightarrow 84 = (\mathrm{n}-1)6 \Rightarrow \mathrm{n} = 15$$

The correct Answer is 'D'

7. Correct option: (C)

$$a_1 = a, a_2 = b, a_n = c$$
$$\Rightarrow d = a_2 - a_1 = b - a$$
$$\therefore a_n = c \Rightarrow a_1 + (\mathrm{n}-1)\,\mathrm{d} = \mathrm{c}$$
$$\Rightarrow a + (\mathrm{n}-1)(\mathrm{b}-\mathrm{a}) = \mathrm{c}$$
$$\Rightarrow \mathrm{n} = \frac{b + c - 2a}{b - a}$$
$$\therefore s_n = \frac{n}{2}\left(a + a_n\right) = \frac{\left(a + c\right)(\mathrm{b} + \mathrm{c} - 2\,\mathrm{a})}{2(\mathrm{b}-\mathrm{a})}$$

The correct Answer is 'C'

8. Correct option: (C)

The simplified form of given AP is

$$\sqrt{2}, 2\sqrt{2}, 3\sqrt{2}, 4\sqrt{2}.....$$
$$a = \sqrt{2}, d = \sqrt{2}$$
$$s_n = \frac{n}{2}\left(2a + (\mathrm{n}-1)\,\mathrm{d}\right)$$
$$= \frac{n}{2}\left(2\sqrt{2} + (\mathrm{n}-1)\sqrt{2}\right) = \frac{n(\mathrm{n}+1)}{\sqrt{2}}$$

The correct Answer is 'C'

9. Correct option: (B)

The numbers between 100 and 500 that are divisible by 7 are 105, 112, 119, 126, 133, 140, 147... 483, 490, and 497.

Let such numbers be n.

Then, 497=105+(n−1)×7; or n=57497=105+(n−1)×7; or n=57.

The numbers between 100 and 500 that are divisible by 21 are 105, 126, 147... 483.

Let such numbers be m.

Then 483 = 105 + (m – 1) × 21; or m = 19.

Hence, the required number = n – m = 57 – 19 = 38.

The correct Answer is 'B'

10. Correct option: (B)

The AM between a, b is $\frac{a+b}{2}$

Given that AM is

$$\frac{a^n+b^n}{a^{n-1}+b^{n-1}}=\frac{a+b}{2}$$
$$\Rightarrow 2\left(a^n+b^n\right)=\left(a+b\right)\left(a^{n-1}+b^{n-1}\right)$$
$$\Rightarrow \left(a^{n-1}-b^{n-1}\right)\left(a-b\right)=0$$
$$\Rightarrow a^{n-1}-b^{n-1}=0$$
$$\Rightarrow \frac{a^{n-1}}{b^{n-1}}=1\Rightarrow\left(\frac{a}{b}\right)^{n-1}=\left(\frac{a}{b}\right)^0\Rightarrow n=1$$

The correct Answer is 'B'

11. Correct option: (A)

The sum of the series $1.2.3+2.3.4+3.4.5+.....$

So the nth term of AP will be $r(r+1)(r+2)$

$$r(r+1)(r+2)$$
$$=r(r^2+3r+2)$$
$$=r^3+3r^2+2r$$
$$S_n=\sum_{r=1}^{n}T_r$$
$$=\sum_{r=1}^{n}(r^3+3r^2+2r)$$
$$=\left[\frac{n(n+1)}{2}\right]^2+3\left[\frac{n(n+1)(2n+1)}{6}\right]+2\left[\frac{n(n+1)}{2}\right]$$
$$=\frac{n(n+1)(n+2)(n+3)}{4}$$

The correct Answer is 'A'

12. Correct option: (B)

Given GP is $2,2\sqrt{2},4,.....128$

First term a =2

Common factor $r=\frac{2\sqrt{2}}{2}=\sqrt{2}$

Last term l=128

$$l=ar^{n-1}$$
$$\Rightarrow 128=2\left(\sqrt{2}\right)^{n-1}$$
$$\Rightarrow \left(\sqrt{2}\right)^{n-1}=64=\left(\sqrt{2}\right)^{12}$$
$$\Rightarrow n-1=12$$
$$\Rightarrow n=13$$

The correct Answer is 'B'

13. Correct option: (C)

$$T_{m+n}=ar^{m+n-1}=p$$
$$T_{m-n}=ar^{m-n-1}=q$$
$$\therefore pq=(ar^{m+n-1})(ar^{m-n-1})$$
$$\Rightarrow T_m=pq=a^2r^{2m-2}=(ar^{m-1})^2$$
$$\Rightarrow ar^{m-1}=\sqrt{pq}$$

The correct Answer is 'C'

14. Correct option: (A)

$$2y = x + z$$
$$\Rightarrow e^{-y} = e^{-\left(\frac{x+z}{2}\right)}$$
$$\Rightarrow e^{-2y} = e^{-(x+z)}$$
$$\therefore 2y = (x+z)$$

The correct Answer is 'A'

15. Correct option: (B)

$$a = 1, a_3 + a_5 = 90$$
$$\Rightarrow ar^2 + ar^4 = 90$$
$$\Rightarrow a(r^2 + r^4) = 90$$
$$\Rightarrow r^2 + r^4 = 90$$
$$\therefore r^2 = \frac{-1 \pm \sqrt{1 - 4(-90)1}}{2} = 9$$
$$\Rightarrow r = \pm 3$$

Therefore common ratio of the GP is +3 or -3

The correct Answer is 'B'

16. Correct option: (D)

$a_1 = 3, a_n = 96, S_n = 189$

$\Rightarrow S_n = 189 = 3(r^{n-1})$

$\Rightarrow r^{n-1} = 32 \Rightarrow r^n = 32r$

$\therefore S_n = \frac{a(r^n - 1)}{r - 1} = 189$

$\Rightarrow \frac{3(32r - 1)}{r - 1} = 189 \Rightarrow r = 2$

$\because r^n = 32r$

$\Rightarrow 2^n = 64 \Rightarrow n = 6$

The correct Answer is 'D'

17. Correct option: (B)

From the given series value of $a = 3, r = \alpha$

Sum of infinite terms $S_\infty = \frac{a}{1-r}$

$$S_\infty = \frac{a}{1-r} = \frac{45}{8}$$
$$\Rightarrow \frac{3}{1-\alpha} = \frac{45}{8} \Rightarrow \alpha = \frac{7}{15}$$

The correct Answer is 'B'

18. Correct option: (C)

$$x^{\frac{1}{2}} x^{\frac{1}{4}} x^{\frac{1}{8}} x^{\frac{1}{16}} \ldots\ldots$$
$$= x^{\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\ldots\ldots}$$
$$\therefore \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \ldots\ldots$$
$$\Rightarrow S_\infty = \frac{a}{1-r} = \frac{\frac{1}{2}}{1-\frac{1}{2}} = 1$$

$$\therefore \frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\ldots\ldots=1$$
$$\Rightarrow x^{\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\ldots\ldots}=x^1=x$$

The correct Answer is 'C'

19. Correct option: (A, B)

The geometric mean of -9 and -16 is $gm=\sqrt{-9\times-16}=\sqrt{144}=\pm12$

The correct Answer is 'A'& 'B'

20. Correct option: (B)

$$2, a, 2^3, b, 2^5$$
$$\alpha=2, r=\frac{a}{2}$$
$$t_3=ar^2=8$$
$$\Rightarrow (2)\frac{a^2}{4}=8\Rightarrow a=4\Rightarrow r=2$$
$$\therefore t_2=a=\alpha r=(2)(2)=4$$
$$\therefore t_4=b=\alpha r^3=(2)(8)=16$$
$$\therefore ab=64$$

The correct Answer is 'B'

LEVEL-II

1. Correct option: (A)

7, 11, 15, 19….139 is in AP

$$a=7, d=4, t_n=139$$
$$\therefore t_n=a+(n-1)\,d=139$$
$$\Rightarrow 7+(n-1)4=139$$
$$\Rightarrow n=34$$

The correct Answer is 'A'

2. Correct option: (A)

$$a=2n-5, d=-3$$
$$t_{p+2}=a+(p+2-1)d$$
$$=2n-5+(p+2-1)-3$$
$$=2n-3p-8$$

The correct Answer is 'A'

3. Correct option: (C)

The common difference of first AP is d=4

The common difference of second AP is d=5

LCM of both common differences is 20

Also first common term in AP is 11 and n=10

$$a=11, d=4$$
$$t_{10}=11+(10-1)4$$
$$=191$$

The correct Answer is 'C'

4. Correct option: (C)

$$a_5+a_{20}=a_1+a_{24}$$
$$a_{10}+a_{15}=a_1+a_{24}$$
$$3(a_1+a_{24})=225\Rightarrow a_1+a_{24}=75$$
$$s_{24}=\frac{n}{2}(\mathrm{a+l})=\frac{24}{2}(a_1+a_{24})=900$$

The correct Answer is 'C'

5. Correct option: (A)

p ,q, r are in AP and the common difference among them is 'd'

$$2d = r - p$$

x ,y, z are in GP and the common ratio among them is 'r'

$$y^2 = xz$$
$$\Rightarrow y^{2d} = x^d z^d$$
$$\Rightarrow \frac{y^{(r-p)}}{x^{-(r-q)} z^{-(q-p)}} = 1$$
$$\Rightarrow x^{q-r} y^{r-p} z^{p-q} = 1$$

The correct Answer is 'A'

6. Correct option: (C)

$$2, x, y, z, k, 486 = a, ar, ar^2 ar^3 ar^4, ar^5$$
$$\therefore a = 2, ar^5 = 486$$
$$\Rightarrow r^5 = 243 \Rightarrow r = 5$$
$$\Rightarrow (2)\frac{a^2}{4} = 8 \Rightarrow a = 4 \Rightarrow r = 2$$
$$\therefore t_2 = ar = 8$$
$$\therefore t_3 = ar^2 = 16$$
$$\therefore t_4 = ar^3 = 32$$
$$\therefore t_5 = ar^4 = 64$$

The correct Answer is 'C'

7. Correct option: (B)

$$\frac{a}{1-r} = 20 \Rightarrow a = 20 - 2r$$
$$a^2 + (ar)^2 + (ar^2)^2 + \ldots.$$
$$a^2 + a^2 r^2 + \ldots.$$
$$\frac{a^2}{1-r^2} = 100$$
$$\Rightarrow a^2 = 100 - 100r^2$$
$$\Rightarrow (20 - 2r)^2 = 100 - 100r^2 (\because a = 20 - 2r)$$
$$\Rightarrow 5r^2 - 8r + 3 = 0$$
$$\therefore r = 1 (or) 0.6$$

if $r = 1$ *then* $a = 20(1 - 1) = 20 \times 0 = 0$ *That means every term is zero, so it can't have a sum of* 20,

So the value of $r = 0.6$, 5 times of common ratio implies $5 \times 0.6 = 3$

The correct Answer is 'B'

8. Correct option: (D)

$$16.2, 5.4, 1.8, \ldots$$
$$a = 16.2, r = \frac{5.4}{16.2} = \frac{1}{3}$$
$$S_n = \frac{a(1-r^n)}{1-r} = \frac{16.2(1-(1/3)^5)}{1-\frac{1}{3}}$$
$$= \frac{16.2(1-0.99)}{1-\frac{1}{3}} = 16.2(0.99) \times \frac{3}{2} = 24.2$$

The correct Answer is 'D'

9. **Correct option: (D)**

$$\sum_{n=1}^{\infty}\left(\frac{a}{1+a}\right)^n$$
$$=\left(\frac{a}{1+a}\right)^1+\left(\frac{a}{1+a}\right)^2+\left(\frac{a}{1+a}\right)^3+\ldots.$$
$$\therefore S_\infty=\frac{a(1-r^n)}{1-r}$$
$$=\frac{\frac{a}{1+a}(1-\left(\frac{a}{1+a}\right)^n)}{1-\frac{a}{1+a}}$$
$$=\frac{\frac{a}{1+a}}{1-\frac{a}{1+a}}$$
$$=\frac{\frac{a}{1+a}}{\frac{1}{1+a}}$$
$$=a$$

The correct Answer is 'D'

10. **Correct option: (A)**

$$\sum_{i=1}^{n}\sum_{j=1}^{i}\sum_{k=1}^{j}1$$
$$=\sum_{i=1}^{n}\sum_{j=1}^{i}j$$
$$=\sum_{i=1}^{n}\left(\frac{i(i+1)}{2}\right)=\frac{1}{2}\left[\sum_{i=1}^{n}i^2+\sum_{i=1}^{n}i\right]$$
$$=\frac{1}{2}\left[\frac{n(n+1)(2n+1)}{6}+\frac{n(n+1)}{2}\right]$$
$$=\frac{n(n+1)(n+2)}{6}$$

The correct Answer is 'A'

11. **Correct option: (B, D)**

sum of the first three consecutive terms of an A.P. is 9
sum of their squares is 35

$$a+a+d+a+2d=9$$
$$\Rightarrow 3a+3d=9$$
$$a+d=3\Rightarrow d=3-a$$
$$(a)^2+(a+d)^2+(a+2d)^2=35$$
$$a^2+9+(6-a)^2=35$$
$$a^2-6a+5=0$$
$$(a-1)(a-5)=0$$
$$a=1,a=5\Rightarrow d=2,-2$$
$$S_n=\frac{n}{2}[a+a+(n-1)d]$$
$$a=1,d=2$$
$$=\frac{n}{2}[a+a+(n-1)d]=n^2$$
$$a=5,d=-2$$
$$=\frac{n}{2}[a+a+(n-1)d]=n[6-n]$$

The correct Answer is 'B' & 'D'

12. **Correct option: (A, B, C, D)**

$$3k+4, 3k-7, k+12$$
$$\Rightarrow 2(3k-7) = 3k+4+k+12$$
$$\Rightarrow 6k-14 = 4k+16$$
$$\Rightarrow 2k = 30 \Rightarrow k = 15$$
$$\therefore 2k+5 = 35$$

The correct Answer is 'A','B','C'&'D'

13. **Correct option: (A, D)**

$$\frac{a+b}{2} = 20, \sqrt{ab} = 16$$
$$\Rightarrow a+b = 40, ab = 256$$
$$\Rightarrow (a-b)^2 = (a+b)^2 - 4ab = 1600 - 4(256) = 576$$
$$\therefore a-b = \pm 24$$
$$a = 8, b = 32$$
$$a = 32, b = 8$$

The correct Answer is 'A','D'

14. **Correct option: (B)**

$$5, 2, -1, -4, -7, \ldots\ldots$$
$$a = 5, d = -3$$
$$S_n = \frac{n}{2}[2a + (n-1)d]$$
$$= \frac{n}{2}[10 + (n-1) - 3]$$
$$= \frac{n}{2}[13 - 3n]$$

The correct Answer is 'B'

15. **Correct option: (A)**

$$S_n = \frac{n}{2}[2a + (n-1)d]$$
$$\Rightarrow 2S_n = 2an + n^2 d - nd$$
$$\Rightarrow 2S_n - n^2 d + nd = 2an$$
$$\Rightarrow a = \frac{2S_n - n^2 d + nd}{2n}$$

The correct Answer is 'A'

16. **Correct option: (B)**

$$S_n = \frac{n}{2}[2a + (n-1)d]$$
$$\Rightarrow 400 = \frac{16}{2}[2(5) + (16-1)d]$$
$$\Rightarrow 50 = 10 + 15d$$
$$\Rightarrow d = \frac{40}{15} = \frac{8}{3}$$

The correct Answer is 'B'

17. **Correct Matching is: a-Q; b-P, c-S, d-R**

a. $GM = \sqrt{25 \times 16} = \sqrt{400} = 20$

The correct Match is 'Q'

b. $(2x+2)^2 = (x)(3x+3)$
$\Rightarrow x^2 + 5x + 4 = 0$
$\Rightarrow (x+4)(x+1) = 0$
$a = -4 (substituting\ x\ as\ -1\ will\ make\ all\ terms\ zero)$
$\Rightarrow ar = 2(x+1) = 2(-4+1) = -6$

$$\Rightarrow r = \frac{3}{2}$$

$$t_4 = ar^3 = -4 \times \left(\frac{3}{2}\right)^3 = \frac{-27}{2}$$

The correct Match is 'P'

c. $([a-(a+d)]+[(a+2d)-(a+3d)]+\ldots+[a+2nd]) = -nd + a + 2nd = a + nd$ \

The correct Match is 'S'

d. $2 + 5 + 8 + 11\ldots..$

$a = 2, d = 2, S_n = 60100$

$S_n = \frac{n}{2}[a + a + (n-1)d]$

$\Rightarrow 60100 \times 2 = n(1 + 3n)$

$3n^2 + n - 120200 = 0$

$\Rightarrow (n - 200)(3n + 601) = 0$

$n = 200, \frac{-601}{3}$

The number of terms cannot be negative, so $n = 200$

The correct Match is 'R'

INTEGER TYPE

18. Correct Answer is: 1

$a_n = 128, S_n = 255, r = 2$

$\Rightarrow ar^{n-1} = 128$

$\Rightarrow a2^{n-1} = 128 \Rightarrow 2^n = \frac{128}{a}$

$S_n = \frac{a(r^{n-1})}{r-1} = 255$

$\Rightarrow \frac{a(2^{n-1})}{2-1} = 255 \Rightarrow a(2^{n-1}) = 255$

$\Rightarrow 256 - a = 255$

$\Rightarrow a = 1$

19. Correct Answer is: 5

$$AM = \frac{3 + \sqrt{2} + 7 - \sqrt{2}}{2} = \frac{10}{2} = 5$$

20. Correct Answer is: 7

4th term of first AP, $t_4 = -1 + 3d$

4th term of second AP, $t_4 = -8 + 3d$

Now, the difference between their 4th terms is i.e.,

$\Rightarrow -1 + 3d + 8 - 3d$

$\Rightarrow 7$

STUDENT PRACTICE QUESTIONS

1. Correct option: (C)

If 7th term and 13th terms of an A.P. are 34 and 64 respectively, then its 18th term is

7th term of A.P. is $a + 6d = 34$

13th term of A.P. is $a + 12d = 64$

Simplifying implies $6d = 30 \Rightarrow d = 5, a = 4$

18th term is $a + 17d = 4 + 85 = 89$

The correct Answer is 'C'

2. Correct option: (C)

Given series of AP is -17,-21,-25

nth term in AP is

$$t_n = a + (\mathrm{n}-1)\,\mathrm{d}$$
$$\mathrm{a} = -17, \mathrm{d} = -4$$
$$\Rightarrow t_n = a + (\mathrm{n}-1)\,\mathrm{d} = -17 + (\mathrm{n}-1) - 4$$
$$= -17 - 4\,\mathrm{n} + 4$$
$$= -4n - 13$$

The correct Answer is 'C'

3. Correct option: (D)

$$t_p = q, t_q = p, t_r = ?$$
$$t_p = a + (\mathrm{p}-1)\,\mathrm{d} = \mathrm{q}$$
$$t_q = a + (\mathrm{q}-1)\,\mathrm{d} = p$$
$$\Rightarrow t_p - t_q = d(\mathrm{p}-1-\mathrm{q}+1) = \mathrm{q}-\mathrm{p} \Rightarrow \mathrm{d} = -1$$
$$\therefore t_p \Rightarrow a + (\mathrm{p}-1)(-1) = \mathrm{q} \Rightarrow \mathrm{a} = \mathrm{q} + \mathrm{p} - 1$$
$$\mathrm{a} = q + p - 1, \mathrm{d} = -1$$
$$\Rightarrow t_r = a + (\mathrm{r}-1)\,\mathrm{d}$$
$$\Rightarrow q + p - r$$

The correct Answer is 'D'

4. Correct option: (C)

Let the common difference of given AP series be 'x'

This implies

$$b = a + x, c = a + 2x, d = a + 3x, e = a + 4x$$
$$\therefore a - 4b + 6c - 4d + e$$
$$= a - 4(\mathrm{a} + \mathrm{x}) + 6(\mathrm{a} + 2\,\mathrm{x}) - 4(\mathrm{a} + 3\,\mathrm{x}) + (\mathrm{a} + 4\,\mathrm{x})$$
$$= 8\,\mathrm{a} - 8\,\mathrm{a} + 16\,\mathrm{x} - 16\,\mathrm{x}$$
$$= 0$$

The correct Answer is 'C'

5. Correct option: (D)

$$S_n = \frac{n}{2}\left(2a + (\mathrm{n}-1)\,\mathrm{d}\right) = 200$$
$$n = 10, a = 5$$
$$\Rightarrow \frac{10}{2}\left(2(5) + (9)\,\mathrm{d}\right) = 200$$
$$\Rightarrow 10 + 9d = 40$$
$$\Rightarrow 9d = 30 \Rightarrow d = \frac{10}{3}$$

The correct Answer is 'D'

6. Correct option: (B)

General nth term can be rewrite as $\frac{1}{n(n+1)}$

$$\frac{1}{n(\mathrm{n}+1)} = \frac{n+1-n}{n(\mathrm{n}+1)} = \frac{1}{n} - \frac{1}{n+1}$$
$$\sum_{n=1}^{n}\left[\frac{1}{n} - \frac{1}{n+1}\right] = 1 + \frac{1}{2} + \frac{1}{3} + \ldots \frac{1}{n} - \frac{1}{2} - \frac{1}{3} - \frac{1}{4} - \ldots. - \frac{1}{n} - \frac{1}{n+1}$$
$$= 1 - \frac{1}{n+1}$$

The correct Answer is 'B'

7. Correct option: (C)

Given that

$$a_4 = p, a_7 = q, a_{10} = r$$
$$\Rightarrow a_4 = ar^3 = p$$
$$\Rightarrow a_7 = ar^6 = q$$
$$\Rightarrow a_{10} = ar^9 = r$$
$$\therefore (ar^6)^2 = (q)^2$$
$$\Rightarrow a^2 r^{12} = q^2$$
$$\Rightarrow (ar^9)(ar^3) = q^2$$
$$\Rightarrow pq = q^2$$

The correct Answer is 'C'

8. Correct option: (D)

$$9 + 99 + 999 + \ldots.$$
$$= ((10 - 1) + (100 - 1) + (1000 - 1) + \ldots.)$$
$$= \{(10 + 100 + 1000 + \ldots.) - (1 + 1 + 1 + \ldots\ldots)\}$$
$$\therefore 10 + 100 + 1000 + \ldots.$$
$$\therefore 1 + 1 + 1 + \ldots\ldots = n$$
$$\Rightarrow S_n = \frac{a(r^n - 1)}{r - 1}$$
$$a = r = 10$$
$$\Rightarrow S_n = \frac{10(10^n - 1)}{9}$$
$$\Rightarrow 9 + 99 + 999 + \ldots. = \frac{10(10^n - 1)}{9} - n$$
$$= \frac{10}{9}(10^n - 1) - n$$
$$= \frac{10^{n+1} - 10 - 9n}{9}$$

The correct Answer is 'D'

9. Correct option: (A)

$$\frac{9}{4}, g_1, g_2, g_3, \frac{4}{9}$$
$$a = \frac{9}{4}, ar = g_1, ar^2 = g_2, ar^3 = g_3, ar^4 = \frac{4}{9}$$
$$ar^4 = \frac{4}{9} \Rightarrow r^4 = \frac{16}{81} \Rightarrow r = \frac{2}{3}$$
$$\Rightarrow g_1 = ar = \frac{9}{4}.\frac{2}{3} = \frac{3}{2}$$
$$\Rightarrow g_2 = ar^2 = \frac{9}{4}.\frac{4}{9} = 1$$
$$\Rightarrow g_3 = ar^3 = \frac{9}{4}.\frac{8}{27} = \frac{2}{3}$$

The correct Answer is 'A'

10. Correct option: (D)

$$T_4 = ar^3 = 54$$
$$T_9 = ar^8 = 13122$$
$$\Rightarrow \frac{r^8}{r^3} = \frac{13122}{54} \Rightarrow r = 3$$
$$\therefore ar^3 = 54 \Rightarrow a = 2$$
$$\Rightarrow T_n = ar^{n-1} = 2.3^{n-1}$$

The correct Answer is 'D'

12. Correct option: (A)

$$a + ar = 5$$
$$\frac{a}{1-r} = 3 \Rightarrow a = 3(1-r)$$
$$3(1-r)(1+r) = \frac{5}{3}$$
$$r^2 = \frac{4}{9} \Rightarrow r = \frac{2}{3}$$
$$a = 3(1-r) = 1$$

The correct Answer is 'A'

14.Correct option: (B, C)

$$a = 54, d = -3, S_n = 513$$
$$S_n = \frac{n}{2}[2a + (n-1)d] = 513$$
$$\Rightarrow n^2 - 37n + 342 = 0$$
$$\Rightarrow (n-18)(n-19) = 0$$
$$\therefore n = 18,19$$

The correct Answer is 'B','C'

15.Correct option: (B)

$$G_1 G_2 \ldots . G_n$$
$$= (\sqrt{1 \times 1024})^n$$
$$= 2^{5n}$$
$$\Rightarrow 2^{5n} = 2^{45} \Rightarrow 5n = 45 \Rightarrow n = 9$$

The correct Answer is 'B'

16.Correct option: (B)

$$A_1 + A_2 ..... + A_n = 1025 \times 171$$
$$\therefore m(\frac{-2+1027}{2}) = 1025 \times 171$$
$$\Rightarrow m = 342$$

The correct Answer is 'B'

17. Correct option: (D)

$$r = (1024)^{\frac{1}{9+1}} = 2 (\because n = 9)$$
$$G_1 = 2, r = 2$$
$$G_1 + G_2 + \ldots + G_n$$
$$= \frac{2(2^9 - 1)}{2 - 1}$$
$$= 1022$$

The correct Answer is 'D'

18.**Correct Matching is: a-R: b-P; c-S; d-Q**

a.

$$\frac{x_1}{\frac{1}{3}} = \frac{x_2}{x_1} = \frac{9}{x_2}$$

$$\frac{x_1}{\frac{1}{3}} = \frac{9}{x_2} \Rightarrow x_1 = \frac{3}{x_2}$$

$$\frac{x_1}{\frac{1}{3}} = \frac{x_2}{x_1} \Rightarrow \frac{1}{3}x_2 = {x_1}^2$$

$$x_1 x_2 = 3 \Rightarrow x_2 = \frac{3}{x_1}$$

$$\frac{1}{3}x_2 = {x_1}^2 \Rightarrow \frac{1}{3}.\frac{3}{x_1} = {x_1}^2 \Rightarrow x_1 = 1$$

$$\therefore x_2 = \frac{3}{x_1} = 3$$

The correct Match is 'R'

b.

$$ar^2 = 4$$

$$a.ar.ar^2.ar^3.ar^4$$

$$= a^5 r^{10}$$

$$= (ar^2)^5 = 4^5 = 1024$$

The correct Match is 'P'

c.

$$\frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \ldots + \infty$$

$$S_\infty = \frac{a}{1-r} = \frac{\frac{1}{3}}{1-\frac{1}{3}} = \frac{1}{2}$$

The correct Match is 'S'

d.

$$S_n = \frac{1(3^n - 1)}{3 - 1}$$

$$\therefore \frac{1(3^n - 1)}{3 - 1} = 364$$

$$\Rightarrow 3^n - 1 = 728 \Rightarrow 3^n = 729 \Rightarrow n = 6$$

The correct Match is 'Q'

19. **Correct Answer is: 1**

$$2b = a + c$$

$$\Rightarrow b + b = a + c$$

$$\Rightarrow b - c = a - b$$

$$\Rightarrow \frac{b-c}{a-b} = 1$$

20. **Correct Answer is: 2**

$$S_n = 3n^2 - n$$

$$\therefore a = S_1 = 3(1)^2 - 1 = 2$$

7. Statistics and Probability

Solutions

LEVEL-I

1. Correct option: (B)

Particular item occurs multiple times in a class interval is called as Frequency

The correct Answer is 'B'

2. Correct option: (B)

The groups into which large data is condensed are called classes.

The correct Answer is 'B'

3. Correct option: (A)

Range of data = Maximum value -Minimum value

$\therefore 98.3-89=9.3$

The correct Answer is 'A'

4. Correct option: (C)

From the bar graph it is evident that the number of students born in the months of October and November is '4' respectively

And hence the difference between two numbers is zero

Also from the same bar graph it is evident that the number of students born in the months of March and April is '3' respectively

And hence the difference between two numbers is zero

The correct Answer is 'C'

5. Correct option: (C)

$$2,3,5,7,11,13,17,19,23,29$$

$$n = 10$$

$$\frac{n}{2} = 5^{th} term = 11$$

$$\frac{n}{2} + 1 = 6^{th} term = 13$$

$$Median = \frac{11+13}{2} = 12$$

The correct Answer is 'C'

6. Correct option: (C)

A frequency polygon, is drawn by joining the midpoints of all the top edges of the rectangles in a histogram by straight lines

The correct Answer is 'C'

7. Correct option: (B)

The given class intervals are 10-20, 20-30. In these class intervals the value 20 is lies in the class interval 20-30.

The correct Answer is 'B'

8. Correct option: (A)

11, 13, 15, 19, p + 2, p + 4, 30, 35, 39, 46

The median of given observations arranged in ascending order is 25.

$$Median = \frac{(p+2)+(p+4)}{2} = 25$$

$$\Rightarrow p + 3 = 25$$

$$\Rightarrow p = 22$$

The correct Answer is 'A'

9. Correct option: (B)

Range is not a measure of central tendency

The correct Answer is 'B'

10. Correct option: (C)

The given frequency range varies from 14 to 112

The class intervals for the given set of data are

$$13-22,23-32,33-42,43-52,53-62$$
$$63-72,73-82,83-92,93-102,103-112$$

Hence, number of class interval = 10

The correct Answer is 'C'

11. Correct option: (B)

Width of each class = 2.5

No. of classes = 9

Lower class boundary of the lowest class = 10.6

∴ Upper class limit of highest class= 10.6 + 9×2.5

= 10.6 + 22.5 = 33.1

The correct Answer is 'B'

12. Correct option: (C)

Ogive is the graphical representation of cumulative frequency.

The correct Answer is 'C'

13. Correct option: (A)

Let the 'n' observations be $X_1, X_2 .....X_n$

Let $\overline{X}$ be the mean of n observations, then we have

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$
$$\Rightarrow \sum_{i=1}^{n} X_i = n\overline{X}$$

Multiply a constant k to each of the observations. Then the observations become $kX_{i,} .....kX_n$

If $\overline{Y}$ be the mean of n observations, then we have

$$\overline{Y} = \frac{1}{n}\sum_{i=1}^{n} kX_i$$
$$= \frac{k}{n}\sum_{i=1}^{n} X_i$$
$$= k.\frac{1}{n}\sum_{i=1}^{n} X_i$$
$$= k\overline{X}$$

The correct Answer is 'A'

14. Correct option: (A,B)

In set 2, 2, 2, 2, 4

Mean = $\frac{2+2+2+2+4}{5} = 2.4 \approx 2$

Median is the 3rd term which is equal to 2

Since 2 occurs maximum number of times mode =2

In the set 1, 3, 3, 3, 5

Mean = $\frac{1+3+3+3+5}{5} = 3$

Median is the 3rd term which is equal to 3

Since 3 occurs maximum number of times mode =3

The correct Answer is 'A' & 'B'

15. Correct option: (B)

The given set is 2, 2, 4, 5, and 12

Mean = $\frac{2+2+4+5+12}{5}=5$

Median is the 3rd term which is equal to 4

Since 2 occurs maximum number of times mode =2

Mean > Mode

The correct Answer is 'B'

16. Correct option: (A)

Mean – Mode = 3 (Mean - Median)

If we solve the known relationship between the mean, median and mode, we get,

Mean – Mode = 3 Mean – 3 Median

On transposing Mean from the LHS to the RHS, we get,

– Mode = 3 Mean – Mean – 3 Median

– Mode = 2 Mean – 3 Median

On transposing -3 Median from RHS to the LHS, we get,

3 Median – Mode = 2 Mean

It can be also written as, 2 Mean = 3 Median – Mode

The correct Answer is 'A'

17. Correct option: (A)

The algebraic sum of deviation of a set of n values from A.M. is zero. This is one of the mathematical properties of arithmetic mean.

The correct Answer is 'A'

18. Correct option: (D)

$$\frac{x_1+x_2+....+x_n}{n}=a$$

$$\Rightarrow x_1+x_2+....+x_n=a.n$$

$$\Rightarrow \frac{x_1+\alpha+x_2+\alpha....+x_n+\alpha}{n}$$

$$=\frac{x_1+x_2+....+x_n+n\alpha}{n}$$

$$=\frac{na+n\alpha}{n}$$

$$=a+\alpha$$

The correct Answer is 'D'

19. Correct option: (A)

Arranging numbers in ascending order 17, 26, 29, 32, 33, 34, 45, 56, 60

Here n = 9 which is odd.

So Median =((n+1)/2) term

Median = ((9+1)/2) term

Median = (10/2) term

Median = 5th term

Median = 33

Hence median is 33.

If 26 is replaced by 62, new set of numbers in ascending order is shown below.

17, 29, 32, 33, 34, 45, 56, 60, 62

Here n = 9 which is odd.

So Median =((n+1)/2) term

Median = ((9+1)/2) term
Median = (10/2) term
Median = 5th term
Median = 34
Hence median is 34.
The correct Answer is 'A'

20. Correct option: (A)

$$mean = \frac{Sum of all observations}{total no of observations}$$
$$= \frac{1+4+3+3+4+2+2+3+3+5}{10}$$
$$= \frac{30}{10}$$
$$= 3$$

The correct Answer is 'A'

LEVEL-II

1. Correct option: (A)

$$Class\ mark = \frac{Lower\ limit + Upper\ limit}{2}$$
$$\Rightarrow 25 = \frac{Lower\ limit + 40}{2}$$
$$\Rightarrow Lower\ limit = 10$$

The correct Answer is 'A'

2. Correct option: (A)

Lower boundary means the lower limit of the class interval.
The lower limit of 20−30 is 20.
The correct Answer is 'A'

3. Correct option: (C)

The first 'n' natural numbers are 1, 2, 3, 4… n

Sum of first n natural numbers $= \frac{n(n+1)}{2}$

Arithmetic Mean

$$= \frac{n(n+1)}{\frac{2}{n}}$$
$$= \frac{n(n+1)}{2n}$$
$$= \frac{(n+1)}{2}$$

The correct Answer is 'C'

4. Correct option: (A)

The mean weight of children is $\text{mean weight of the children} = \frac{\text{TotalWeight (in kg)}}{\text{No. of children}}$

$$\Rightarrow \text{mean weight of the children} = \frac{(40).(2)+(41).(3)+(42).(6)+(43).(2)+44.(5)+45.(5)}{2+3+6+2+5+5}$$
$$= \frac{986}{23}$$
$$= 42.86kg$$

The correct Answer is 'A'

5. Correct option: (A)

$$\text{mean of data} = \frac{\sum F.X}{\sum F}$$

$$\Rightarrow \text{mean weight of the children} = \frac{(4).(11)+(8).(2)+(6).(3)+(7).(y)}{11+2+3+y}$$

$$\Rightarrow 5.3 = \frac{78+7y}{16+y}$$

$$= 4$$

The correct Answer is 'A'

6. Correct option: (B)

$$Mode = l + \left[\frac{h \times (f - f_1)}{(2f - f_1 - f_2)}\right]$$

$$l = 10.5, f = 10, f_1 = 4, f_2 = 6, h = 15 - 11 = 4$$

$$\therefore Mode = 10.5 + \left[\frac{4 \times (10-4)}{(2 \times 10 - 4 - 6)}\right] = 12.9 \approx 13$$

The correct Answer is 'B'

7. Correct option: (B)

Class Interval	Mean of class interval	Frequency	Deviation	fd
100-104	102	15	110-102=8	120
104-108	106	20	4	80
108-112	110	32	0	0
112-116	114	18	-4	-72
116-120	118	15	-8	-20

$$\text{Mean} = \frac{108+112}{2} = 110$$

The correct Answer is 'B'

8. Correct option: (C)

Class interval having more frequency 32 is (108-112)

And hence class width is 112-108=4

The correct Answer is 'C'

9. Correct option: (B)

Class Interval	Mean of class interval	Frequency	Deviation	fd
100-104	102	15	110-102=8	120
104-108	106	20	4	80
108-112	110	32	0	0
112-116	114	18	-4	-72
116-120	118	15	-8	-20

$$\therefore Assumed mean, a = 110,$$
$$Class\ width,\ h\ =\ 4$$
$$and\ total\ observation,\ N\ =\ 100$$
$$By\ assumed\ mean\ method,$$
$$\text{Mean} = a + \frac{\sum fd}{N}$$
$$= 110 + \frac{-8}{100} = 109.9 \approx 109$$

The correct Answer is 'B'

10. Correct Matching is: a - s, b – r, c – q, d – p,

a. The most frequent value in data is known as Mode

The correct Answer is 'S'

b. The mean cannot be calculated from the graph.

The correct Answer is 'R'

c. Ogive is used to determine median of a continuous data.

The correct Answer is 'Q'

d. Standard deviation is not a measure of central tendency

The correct Answer is 'P'

11. Correct Answer is: 6

$$median = x$$
$$mode = 3.median - 2mean$$
$$\Rightarrow 55.2 = 3x - 2(53.4)$$
$$\Rightarrow 3x = 55.2 + 2(53.4)$$
$$\Rightarrow x = 54 \Rightarrow \frac{x}{9} = 6$$

12. Correct Answer is: 10

If M is the mid-value of class interval 20 − 30 then

$$M = \frac{20 + 30}{2} = 25$$
$$\therefore \frac{M}{2.5} = \frac{25}{2.5} = 10$$

13. Correct Answer is: 6

If the difference between the mode and median is 2, then the difference between the median and mean is _____

$$mode - median = 2$$
$$mode = 3.median - 2mean$$
$$\Rightarrow 55.2 = 3x - 2(53.4)$$
$$\Rightarrow 3x = 55.2 + 2(53.4)$$
$$\Rightarrow x = 54 \Rightarrow \frac{x}{9} = 6$$

STUDENT PRACTICE QUESTIONS

1. Correct option: (D)

The first 5 whole numbers are

$$0, 1, 2, 3, 4$$
$$\frac{0+1+2+3+4}{5} = \frac{10}{2} = 5$$

The correct Answer is 'D'

2. Correct option: (A)

From above data, we see that modal shirt size is 39 as it has the maximum frequency.

The correct Answer is 'A'

3. Correct option: (B)

Class width =20-15=5

Here a=20, h=5

Lower limit of the class $20-\frac{5}{2}=17.5$

Upper limit of the class $20+\frac{5}{2}=22.5$

The correct Answer is 'B'

4. Correct option: (B)

$Number\ of\ boys\ in\ the\ school\ =\ 90$

$Mean\ marks\ of\ boys\ =45\%$

$Total\ marks\ of\ 90\ boys\ =90\times45=4050$

$Number\ of\ girls\ in\ the\ school\ =\ 30$

$Mean\ marks\ of\ girls\ =70\%$

$\therefore Total\ marks\ of\ 30\ girls\ =30\times70=2100$

$Total\ marks\ secured\ by\ 90\ boys\ and\ 30\ girls=\ 4050\ +\ 2100\ =\ 6150$

$Number\ of\ boys\ and\ girls\ who\ have\ appeared\ for\ exam\ from\ the\ school=\ 90\ +\ 30\ =\ 120$

$\therefore\ Average\ marks\ \%\ of\ school=\frac{6150}{120}=51.25\%$

The correct Answer is 'B'

5. Correct option: (C)

In statistics, a central tendency (or measure of central tendency) is a central or typical value for a probability distribution. Colloquially, measures of central tendency are often called averages.

The correct Answer is 'C'

6. Correct option: (C)

The arithmetic mean of 30, 36, 39, 23 and 27.

$$\frac{30+36+39+23+27}{5}$$
$$=\frac{155}{5}$$
$$=31$$

The correct Answer is 'C'

7. Correct option: (C)

$$\frac{a_1+a_2+a_3+a_4+a_5+a_6+a_7+a_8}{8}=18$$
$$\frac{a_9+a_{10}+a_{11}+a_{12}+a_{13}+a_{14}+a_{15}+a_8}{8}=20$$
$$\frac{a_1+a_2+a_3+a_4+a_5+a_6+a_7+a_8+\ldots.a_{15}}{15}=19$$
$$\therefore a_1+a_2+a_3+a_4+a_5+a_6+a_7+a_8+\ldots.a_{15}=(18+20)\times8$$
$$\Rightarrow 19\times15+a_8=38\times8$$
$$\therefore a_8=19$$

The correct Answer is 'C'

8. Correct option: (B)

Let x and y be the lower and upper class limit of a continuous frequency distribution

Now, mid - point of a class

$$\Rightarrow \frac{x+y}{2} = m$$

$$x = lower\ class\ limit$$

$$y = l, upper\ class\ limit$$

$$\therefore \frac{x+l}{2} = m \Rightarrow x = 2m - l$$

$$\Rightarrow x = 2m - l$$

The correct Answer is 'B'

9. Correct option: (B)

The mid value of the class =10

Width of the interval=6

$$Lower\ \lim it = \text{midvalue} - \frac{width}{2}$$

$$= 10 - \frac{6}{2} = 7$$

The correct Answer is 'B'

10. Correct option: (D)

A frequency polygon is constructed by plotting frequency of the class interval and mid value of the class.

The correct Answer is 'D'

11. Correct option: (A)

Let the total wages of laborers working in shift 1 is x

Let the total wages of laborers working in shift 2 is y

Let the total wages of laborers working in shift 3 is z

Mean wages of 150 laborers = $\frac{x+y+z}{a+b+c}$

Where a, b and c is the number of laborers working is shift 1, 2 and 3 respectively

Mean wages =114 Rs

$$\therefore \frac{x+y+z}{a+b+c} \Rightarrow$$

$$\frac{x+y+z}{150} = 114 \Rightarrow x+y+z = 17100$$

Mean wages of laborers in shift 1 is $\frac{x}{a} = 121.5 \Rightarrow x = 121.5 \times 60 = 7290$

Mean wages of laborers in shift 2 is $\frac{y}{b} = 107.5 \Rightarrow x = 107.5 \times 40 = 4310$

$$x+y+z = 17100$$

$$\Rightarrow z = 17100 - 7290 - 4310 = 5500$$

Let the mean of laborers in shift 3=c

$$\therefore m = \frac{z}{c} = \frac{5500}{50} = 110$$

Then mean wage of laborer working in third shift=Rs 110

The correct Answer is 'A'

12. Correct option: (A)

$$x_1, x_2, ...x_{75}$$

$$mean = \frac{x_1 + x_2 + ... + x_{75}}{75} = 25$$

$$\Rightarrow x_1 + x_2 + ... + x_{75} = 1875$$

$$Mean\ of\ \ new\ numbers = \frac{\frac{x_1}{5} + \frac{x_2}{5} + ... + \frac{x_{75}}{5}}{75}$$

$$= \frac{x_1 + x_2 + ... + x_{75}}{5 \times 75}$$

$$= \frac{25 \times 75}{5 \times 75} = 5$$

The correct Answer is 'A'

13. Correct option: (B)

The even numbers between 50 and 60 are 52, 54, 56 and 58.

The odd numbers between 50 and 60 are 51, 53, 55, 57 and 59.

The AM of even numbers implies $\frac{52+54+56+58}{4} = 55$

The AM of odd numbers implies $\frac{51+53+55+57+59}{5} = 55$

The difference between two means is zero

The correct Answer is 'B'

14. Correct option: (D)

Here, total score in 10 innings= 60 × 10 = 600

After 11th innings, mean score raised to 62.

So, 62 × 11 = 682

Score of 11th innings = 682 – 600 = 82

The correct Answer is 'D'

15. Correct option: (D)

Arranging in ascending order, we get -3,-3,-1,0,2,2,2,5,5,5,5,6,6,6

Here n = 14 (which is even)

$$\therefore Median = \frac{1}{2}\left[\frac{n}{2} term + (\frac{n}{2} + 1) term\right]$$

$$= \frac{1}{2}\left[7^{th} term + 8^{th} term\right]$$

$$= \frac{1}{2}[2 + 5] = 3.5$$

The correct Answer is 'D'

16. Correct option: (D)

The mean of 20 numbers is 40

$$Mean = \frac{n_1 + n_2 + \ldots + n_{20}}{20}$$

$$\Rightarrow \frac{n_1 + n_2 + \ldots + n_{20}}{20} = 40$$

$$\Rightarrow n_1 + n_2 + \ldots + n_{20} = 40 \times 20$$

$$\therefore mean = \frac{(n_1 - 5) + (n_2 - 5) + \ldots + (n_{20} - 5)}{20}$$

$$= \frac{(n_1 + n_2 + \ldots + n_{20}) - 100}{20}$$

$$= \frac{800 - 100}{20}$$

$$= 35$$

The correct Answer is 'D'

17. Correct option: (D)

Given that average of a, b, c, d, e is 28

Mean of a, c, e is 24

$$\frac{a+b+c+d+e}{5}=28$$

$$\Rightarrow a+b+c+d+e=140$$

$$\frac{a+c+e}{3}=24$$

$$\Rightarrow a+c+e=72$$

$$\therefore b+d+72=140 \Rightarrow b+d=140-72=68$$

$$\therefore \frac{b+d}{2}=34$$

The correct Answer is 'D'

18. Correct option: (D)

$$n=50, \overline{X}=250$$

$$\overline{X}=\frac{1}{n}\sum X_i$$

$$\Rightarrow \sum X_i = 50\times 250 = 12500$$

$$Therefore,\ incorrect\ value\ of \sum X_i = 12500$$

$$correct\ value\ of \sum X_i = 12500-115+165=12550$$

$$correct\ \text{mean}=\frac{12550}{50}=251$$

The correct Answer is 'D'

19. Correct option: (A)

$$\therefore \frac{p+p+2+p+4+p+6+p+8p-2+p-4+p-6+p-8}{9}=10$$

$$\Rightarrow \frac{9p}{9}=10$$

$$\Rightarrow p=10$$

$$The\ mean\ of\ least\ 5\ terms\ are$$

$$\frac{p+p-2+p-4+p-6+p-8}{5}=\frac{5p-20}{5}$$

$$\Rightarrow \frac{5(10)-20}{5}=6$$

The correct Answer is 'A'

20. Correct option: (A)

Mean of 6 numbers=20

Sum of 6 numbers= 120.

New mean=15.

Sum of 5 numbers =75.

Deleted number =120−75 = 45.

The correct Answer is 'A'

Probability

LEVEL-I

1. Correct option: (B)

When a die is rolled total number of events -6 {1, 2, 3, 4, 5, 6}

The favorable cases for even number -3 {2, 4, 6}

$$probability = \frac{no.offavoriblecases}{totaln\,o\,ofcases} = \frac{3}{6} = \frac{1}{2}$$

The correct Answer is 'B'

2. Correct option: (C)

When two die are rolled total number of events $=6^2=36$

The favorable case is getting a single score of '7'

Total number of favorable cases is {(1, 6), (2, 5), (3, 4) (4, 3) (5, 2) (6, 1)}

$$probability = \frac{no.offavoriblecases}{totaln\,o\,ofcases} = \frac{6}{36} = \frac{1}{6}$$

The correct Answer is 'C'

3. Correct option: (A)

When two die are rolled total number of events $=6^2=36$

The favorable case is getting equal numbers

Total number of favorable cases is {(1, 1), (2, 2), (3, 3) (4, 4) (5, 5) (6, 6)}

$$probability = \frac{no.offavoriblecases}{totaln\,o\,ofcases} = \frac{6}{36} = \frac{1}{6}$$

The correct Answer is 'A'

4. Correct option: (A)

When a die is rolled twice total number of events $=6^2=36$

The favorable case is getting doublet

Total number of favorable cases is {(1, 1), (2, 2), (3, 3) (4, 4) (5, 5) (6, 6)}

$$probability = \frac{no.offavoriblecases}{totaln\,oofcases} = \frac{6}{36} = \frac{1}{6}$$

The correct Answer is 'A'

5. Correct option: (B)

When a die is rolled twice total number of events $=6^2=36$

The favorable case is getting both are even numbers

Total number of favorable cases is {(2, 2), (2, 4) (2, 6) (4, 2) (4, 4) (4, 6) (6, 2) (6, 4) (6, 6)}

$$probability = \frac{no.offavoriblecases}{totaln\,oofcases} = \frac{9}{36} = \frac{1}{4}$$

The correct Answer is 'B'

6. Correct option: (A)

The sum on the uppermost faces of the dice is an odd number. Hence exactly one of the two numbers on the uppermost face is 2.

So the different possible combinations

are (2,3),(2,5),(2,7),(2,11),(2,13),(3,2),(5,2),(7,2),(11,2) and (13,2)

Total required combinations =10

$$probability = \frac{no.offavoriblecases}{totaln\,oofcases} = \frac{10}{36} = \frac{5}{18}$$

The correct Answer is 'A'

7. Correct option: (A)

When a two dies is rolled total number of events $=6^2=36$

The favorable case is getting the dots on the second die is less than that on the first die is

Total number of favorable cases is {(2, 1), (3, 1) (3, 2) (4, 1) (4, 2) (4, 3) (5, 1) (5,2) (5, 3), (5, 4) , (6,1), (6, 2), (6, 3), (6, 4) (6, 5}

$$probability = \frac{no.offavoriblecases}{tota\ln oofcases} = \frac{15}{36} = \frac{5}{12}$$

The correct Answer is 'A'

8. Correct option: (B)

When three dies is rolled total number of events $=6^3=216$

The favorable case is the sum of the numbers on them is 6

Total number of favorable cases is {(1, 1, 4)(1, 4, 1)(1,2,3),(1, 3, 2) (2, 3, 1)(2, 1, 3) (2, 2, 2) (3, 2, 1) (3, 1, 2) (4, 1,1)}

$$probability = \frac{no.offavoriblecases}{tota\ln oofcases} = \frac{10}{216} = \frac{5}{108}$$

The correct Answer is 'B'

9. Correct option: (B)

When three dies is rolled total number of events $=6^3=216$

The favorable case is getting triplet

Total number of favorable cases is {(1,1,1) (2,2,2) (3,3,3) (4,4,4) (5,5,5) (6,6,6)}

$$probability = \frac{no.offavoriblecases}{tota\ln oofcases} = \frac{6}{216} = \frac{1}{36}$$

The correct Answer is 'B'

10. Correct option: (A)

The total no .of cards is 52

The favorable case is getting king

Total number of favorable cases is 4

$$probability = \frac{no.offavoriblecases}{tota\ln oofcases} = \frac{4}{52} = \frac{1}{13}$$

The correct Answer is 'A'

11. Correct option: (B)

A leap year has 366 days or 52 weeks and 2 odd days. The two odd days can be {Sunday,Monday},{Monday,Tuesday},{Tuesday,Wednesday},{Wednesday,Thursday},{Thursday,Friday},{Friday,Saturday},{Saturday,Sunday}. So there are 7possibilities out of which 2 have a Sunday.

So the probability of 53 Sundays is 2/7

The correct Answer is 'B'

12. Correct option: (A)

There are 365 days in a non-leap year.365 days =52 weeks +1 day

Which means, every day of the week will come at least 52 times in a year. To have 53 Tuesdays in a year, the remaining one day should be Tuesday.

Since, the remaining one day can be Sunday, Monday, Tuesday, Wednesday, Thursday, Friday or Saturday.

So the probability of 53 Tuesdays is 1/7

The correct Answer is 'A'

13. Correct option: (D)

Probability of selecting a month = $\frac{1}{12}$

Probability of getting Friday is = $\frac{1}{7}$

Hence required probability $\frac{1}{12} \times \frac{1}{7} = \frac{1}{84}$

The correct Answer is 'D'

14. Correct option: (C)

When a Card is drawn at random from a pocket of 100 cards numbered 1 to 100 then the number of events will be 100

The favorable case is drawing a number which is a square is

Therefore the number of squares between 1to 100 is 10

$$probability = \frac{no.offavoriblecases}{tota\ln oofcases} = \frac{10}{100} = \frac{1}{10}$$

The correct Answer is 'C'

15. Correct option: (D)

S is a sample space $S = x \in N : 1 \le x \le 100$ this implies that in Sample space there are 99 events are there

$$E = \{x : (x+1)(x-1) \in S\}$$
$$= x^2 - 1$$
$$\therefore xcanbe 2,3,4,5,6,7,8,9,10$$

So that only x will be in 'S'

So the favorable cases are '9'

$$probability = \frac{no.offavoriblecases}{tota\ln oofcases} = \frac{9}{99} = \frac{1}{11}$$

The correct Answer is 'D'

16. Correct option: (A)

A number n is chosen at random from 1,2,3,4... 1000 therefore there are 1000 events

Favorable case is that n is a number that leaves remainder 1 when divided by 7 is

There are 142 multiples for 7 less than 1000; therefore number of favorable cases is 142

$$probability = \frac{no.offavoriblecases}{tota\ln oofcases} = \frac{142}{1000} = \frac{71}{500}$$

The correct Answer is 'A'

17. Correct option: (B)

Since x and y each can take values from 0 to 10, so the total number of ways of selecting x and y is 11×11=121.

Now, |x-y| >5 ⇒ x - y < - 5 or x-y > 5,

when x - y > 5, the favorable cases are {(6,0), (7,0) (7,1) (8,0) (8,1) (8,2) (9,0), (9,1), (9,2), (9,3),(10,0), (10,1), (10,2), (10,3), (10,4)} total 15

The reverse of the above sets are favorable cases for x - y <- 5 and hence another 15 are there in favorable cases

Therefore total favorable cases are 30

$$probability = \frac{no.offavoriblecases}{tota\ln oofcases} = \frac{30}{121}$$

The correct Answer is 'B'

18. Correct option: (C)

19. Correct option: (A)

$$P(A) = \frac{3}{4}, P(B) = \frac{2}{5}, P(A \cap B) = \frac{1}{4}$$
$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
$$= \frac{3}{4} + \frac{2}{5} - \frac{1}{4}$$
$$= \frac{9}{10}$$
$$= 0.9$$

The correct Answer is 'A'

20. Correct option: (C)

$$P(A\cup B)=0.65, P(A\cap B)=0.15$$
$$P(\overline{A})+P(\overline{B})=1-P(A)+1-P(B)$$
$$=2-(P(A)+P(B))$$
$$P(A\cup B)=P(A)+P(B)-P(A\cap B)$$
$$P(A)+P(B)=P(A\cup B)+P(A\cap B)$$
$$\therefore 2-(P(A)+P(B))=2-P(A\cup B)-P(A\cap B)$$
$$=2-0.65-0.15$$
$$=1.2$$

The correct Answer is 'C'

LEVEL-II

1. Correct option: (A)

$$P(A\cup B)=0.8, P(A\cap B)=0.3$$
$$P(\overline{A})+P(\overline{B})=1-P(A)+1-P(B)$$
$$=2-(P(A)+P(B))$$
$$P(A\cup B)=P(A)+P(B)-P(A\cap B)$$
$$P(A)+P(B)=P(A\cup B)+P(A\cap B)$$
$$\therefore 2-(P(A)+P(B))=2-P(A\cup B)-P(A\cap B)$$
$$=2-0.8-0.3$$
$$=0.9$$

The correct Answer is 'A'

2. Correct option: (C)

$$P(A\cup B)=0.7, P(A\cap B)=0.2$$
$$P(\overline{A})+P(\overline{B})=1-P(A)+1-P(B)$$
$$=2-(P(A)+P(B))$$
$$P(A\cup B)=P(A)+P(B)-P(A\cap B)$$
$$P(A)+P(B)=P(A\cup B)+P(A\cap B)$$
$$\therefore 2-(P(A)+P(B))=2-P(A\cup B)-P(A\cap B)$$
$$=2-0.7-0.2$$
$$=1.1$$

The correct Answer is 'C'

3. Correct option: (B)

$$P(A\cup B)=1-P(\overline{A}\cap\overline{B})$$
$$\therefore P(\overline{A}\cap\overline{B})=1-P(A\cup B)$$
$$=1-0.14$$
$$=0.86$$

4. Correct option: (B)

$$P(A\cap B)=\frac{1}{2}, P(\overline{A}\cap\overline{B})=\frac{1}{3}, P(A)=p, P(B)=2p$$
$$P(A\cup B)=P(A)+P(B)-P(A\cap B)$$
$$P(A\cup B)=1-P(\overline{A}\cap\overline{B})=1-\frac{1}{3}=\frac{2}{3}$$
$$\Rightarrow\frac{2}{3}=3p-\frac{1}{2}$$
$$\Rightarrow 3p=\frac{7}{6}\Rightarrow p=\frac{7}{18}$$

The correct Answer is 'B'

5. Correct option: (A)

If A, B are two mutually exclusive events such that $P(\mathrm{A}\cap\mathrm{B})=0$

$$P(\mathrm{A}\cup\mathrm{B})=P(\mathrm{A})+P(\mathrm{B})-P(\mathrm{A}\cap\mathrm{B})$$
$$\therefore P(\mathrm{A}\cup\mathrm{B})=P(\mathrm{A})+P(\mathrm{B})$$
$$=0.5+0.3$$
$$=0.8$$
$$\therefore P(\mathrm{A}\cup\mathrm{B})=1-\mathrm{P}(\overline{A}\cap\overline{B})$$
$$\therefore \mathrm{P}(\overline{A}\cap\overline{B})=1-P(\mathrm{A}\cup\mathrm{B})$$
$$=1-0.8$$
$$=0.2$$

The correct Answer is 'A'

6. Correct option: (B)

Let A be the event that the card drawn is spade,
B is the event that card drawn is king and S is the sample space,
When a card is drawn from a pack of cards.

$n(\mathrm{S})={}^{52}C_1=52, n(\mathrm{A})={}^{13}C_1=13, n(\mathrm{B})={}^{4}C_1=4$

$n(\mathrm{A}\cap\mathrm{B})={}^{1}C_1=1$

$\therefore P(\mathrm{A})=\dfrac{n(\mathrm{A})}{n(\mathrm{S})}=\dfrac{13}{52}=\dfrac{1}{4}$

$\therefore P(\mathrm{B})=\dfrac{n(\mathrm{B})}{n(\mathrm{S})}=\dfrac{4}{52}=\dfrac{1}{13}$

$\therefore P(\mathrm{A}\cap\mathrm{B})=\dfrac{n(\mathrm{A}\cap\mathrm{B})}{n(\mathrm{S})}=\dfrac{1}{52}$

$\therefore P(\mathrm{A}\cup\mathrm{B})=\dfrac{13}{52}+\dfrac{4}{52}-\dfrac{1}{52}=\dfrac{16}{52}=\dfrac{4}{13}$

The correct Answer is 'B'

7. Correct option: (C)

When two die are rolled total number of events $=6^2=36$

The favorable case is getting sum of the numbers at most 10

Total number of un favorable cases is {(5, 6), (6, 5), (6, 6)}

Therefore total number of favorable cases will be 36-3=33

$$probability=\frac{no.offavoriblecases}{tota\ln oofcases}=\frac{33}{36}=\frac{11}{12}$$

The correct Answer is 'C'

8. Correct option: (A)

When two die are rolled total number of events $=6^2=36$

The favorable case is getting sum as a prime number

Total number of favorable cases is {(1,1), (1, 2), (1,4) (1, 6) (2,1) (2,3) (2,5) (3,2) (3, 4)(4,1) (4,3) (5, 2) (5,6) (6,1) (6,5)}

$$probability=\frac{no.offavoriblecases}{tota\ln oofcases}=\frac{15}{36}=\frac{5}{12}$$

The correct Answer is 'A'

9. Correct option: (A)

A throws 16

B throws more than 16, means B throws, 17 or 18.

Total number of outcomes on throwing 3 dices = 6×6×6=216

Favorable outcomes = {(5,6 ,6), (6,5,6), (6,6,5) (6,6,6)}

No. of favorable outcomes = 4

Probability of B throwing a number greater than 16=4/216=1/54

The correct Answer is 'A'

10. Correct option: (B)

If 3 dices are thrown simultaneously then total outcomes will be 216

Favorable outcomes = 20

Probability of event =20/216=5/54

The correct Answer is 'B'

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Correct option: (A,B,C,D)

A.

$$P(E_1) = 1 - P(RRR)$$
$$= 1 - \left[\frac{1}{2}\frac{2}{4}\frac{3}{5}\right] = 0.9$$

B.

$$P(E_2) = 3P(BRR)$$
$$= 3\left[\frac{2}{3}\frac{1}{4}\frac{2}{5}\right] = 0.5$$

C.

$$P(E_3) = \frac{P(RRR)}{P(RRR) + P(BBB)}$$
$$= \frac{0.1}{0.1 + 0.4}$$
$$= 0.2$$

D.

$$P(E_4) = 1 - P(BBB)$$
$$= 1 - 0.4$$
$$= 0.6$$

12. Correct option: (A, B)

13.Correct option: (A, C, D)

$P(\mathrm{E}_1) = \frac{1}{4}, P\left(\frac{\mathrm{E}_2}{\mathrm{E}_1}\right) = \frac{1}{2}, P\left(\frac{\mathrm{E}_1}{\mathrm{E}_2}\right) = \frac{1}{4}$

$P\left(\frac{\mathrm{E}_2}{\mathrm{E}_1}\right) = \frac{1}{2} = \frac{P(\mathrm{E}_2 \cap \mathrm{E}_1)}{P(\mathrm{E}_1)}$

$\Rightarrow P(\mathrm{E}_2 \cap \mathrm{E}_1) = \frac{1}{8}$

$P\left(\frac{\mathrm{E}_1}{\mathrm{E}_2}\right) = \frac{1}{4} = \frac{P(\mathrm{E}_1 \cap \mathrm{E}_2)}{P(\mathrm{E}_2)} \Rightarrow P(\mathrm{E}_2) = \frac{1}{2}$

$\therefore P(\mathrm{E}_1 \cap \mathrm{E}_2) = P(\mathrm{E}_2)P(\mathrm{E}_1) = \frac{1}{8}$

Therefore E_1 and E_2 are Independent events.

$\therefore P(\mathrm{E}_2) = 2 \times P(\mathrm{E}_1)$

also $P(\mathrm{E}_1)^2 = P(\mathrm{E}_2) \times P(\mathrm{E}_1 \cap \mathrm{E}_2)$

$\therefore P(\mathrm{E}_1 \cap \mathrm{E}_2), P(\mathrm{E}_2), P(\mathrm{E}_1)$ *are in G.P.*

The correct Answer is 'A' 'C' and 'D'

14. Correct option: (B)

P (both of them are selected) = P (A ∩ B) = P (A) × P (B)

$$P(\mathrm{A}) = \frac{1}{3}, P(\mathrm{B}) = \frac{1}{4}$$

$$\therefore P(\mathrm{A} \cap \mathrm{B}) = \mathrm{P}(\mathrm{A})\,\mathrm{P}(\mathrm{B}) = \frac{1}{12}$$

The correct Answer is 'B'

15. Correct option: (B)

P (none of them are selected) = P (A' ∩ B') = P (A') × P (B')

$$P(\mathrm{A}') = 1 - P(\mathrm{A}) = \frac{2}{3}, P(\mathrm{B}') = 1 - P(\mathrm{B}) = \frac{3}{4}$$

$$\therefore P(\mathrm{A}' \cap \mathrm{B}') = \mathrm{P}(\mathrm{A}')\,\mathrm{P}(\mathrm{B}') = \frac{1}{2}$$

The correct Answer is 'B'

16. Correct option: ()

P (one of them are selected) = P (A ∩ B') + P (B ∩ A')

$$P(A)\,P(B') + P(B)\,P(A')$$

$$= \frac{1}{3}.\frac{3}{4} + \frac{1}{4}\frac{2}{3}$$

$$= \frac{1}{6} + \frac{1}{4}$$

$= \frac{5}{12}$

17. Correct Matching is: a-S; b-P; c-Q; d-R

$$P(\mathrm{A}) = \frac{3}{4}, P(\mathrm{B}) = \frac{2}{5} P(\mathrm{A} \cap \mathrm{B}) = \frac{1}{4}$$

$$P(\mathrm{A} \cup \mathrm{B}) = P(\mathrm{A}) + P(\mathrm{B}) - P(\mathrm{A} \cap \mathrm{B})$$

$$= \frac{3}{4} + \frac{2}{5} - \frac{1}{4}$$

$$= \frac{9}{10}$$

The Correct Match is 'S'

$$P(\mathrm{A}) = \frac{1}{4}, P(\mathrm{B}) = \frac{1}{2}, P(\mathrm{A} \cup \mathrm{B}) = \frac{5}{8} P(\mathrm{A} \cap \mathrm{B}) = \frac{1}{4}$$

$$P(\mathrm{A} \cup \mathrm{B}) = P(\mathrm{A}) + P(\mathrm{B}) - P(\mathrm{A} \cap \mathrm{B})$$

$$\frac{5}{8} = \frac{1}{4} + \frac{1}{2} - P(\mathrm{A} \cap \mathrm{B})$$

$$\therefore P(\mathrm{A} \cap \mathrm{B}) = \frac{3}{4} - \frac{5}{8} = \frac{1}{8}$$

The Correct Match is 'P'

$$P(\mathrm{A}) = \frac{1}{4}, P(\mathrm{B}) = \frac{1}{2}, P(\mathrm{A} \cap \mathrm{B}) = \frac{1}{5}$$

$$P(\overline{A} \cap \overline{B}) = 1 - P(\mathrm{A} \cup \mathrm{B})$$

$$= 1 - \frac{1}{4} - \frac{1}{2} - \frac{1}{5}$$

$$= \frac{9}{20}$$

The Correct Match is 'Q'

$$P(A\cup B)=\frac{3}{4}, P(A\cap B)=\frac{1}{5}$$
$$P(\overline{A})+P(\overline{B})=1-P(A)+1-P(B)$$
$$=2-\{P(A)+P(B)\}$$
$$=2-\{P(A\cup B)+P(A\cap B)\}$$
$$=2-\{\frac{3}{4}+\frac{1}{5}\}$$
$$=\frac{21}{20}$$

The Correct Match is 'R'

18. Correct Answer is:

$$P(A\cap B')=P(A\cup B)-P(B)=\frac{1}{30}$$
$$P(A\cup B)=\frac{1}{6}$$
$$\Rightarrow P(B)=\frac{1}{6}-\frac{1}{30}=\frac{4}{30}$$
$$\therefore 10.P(B)=\frac{4}{3}$$

19. Correct Answer is: 7

A and B throws a die

Ways that B's throw always greater than A's is

If A=1 then B=1, 2, 3, 4, 5, 6 $P_1=\frac{1}{6}\times\frac{6}{6}$ $P(\text{cluborheart})=\frac{13}{52}+\frac{13}{52}-\frac{6}{52}=\frac{20}{52}=\frac{5}{13}$

If A=2 then B=2, 3, 4, 5, 6 $P_2=\frac{1}{6}\times\frac{5}{6}$

If A=3 then B=3, 4, 5, 6 $P_3=\frac{1}{6}\times\frac{4}{6}$

If A=4 then B=4, 5, 6 $P_4=\frac{1}{6}\times\frac{3}{6}$

If A=5 then B=5, 6 $P_5=\frac{1}{6}\times\frac{2}{6}$

If A=6 then B=6 $P_6=\frac{1}{6}\times\frac{1}{6}$

Total probability= $\frac{1}{36}(6+5+4+3+2+1)=\frac{7}{12}$

$12p=12\times\frac{7}{12}=7$

20. Correct Answer is: 3

The number of determinant of the second order that can be formed with 0, 1=2×2×2×2=16

Since each of the four places of elements can be filled in 2 ways: by 0 or 1

∴n(S) =16

Let E= the events of getting non-negative determinant.

Then E′= the event of getting negative determinant.

Clearly negative determinant of second order that can be made with 0,1 are

$$\begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix}$$

$$\therefore P(E') = \frac{n(E')}{n(S)} = \frac{3}{16}$$

$$\therefore \frac{p}{16} = \frac{3}{16} \Rightarrow p = 3$$

STUDENT PRACTICE QUESTIONS

1. Correct option: (B)

When two die are rolled total number of events $=6^2=36$

The favorable case is getting ten in a single throw

Total number of favorable cases is {(4, 6), (5, 5), (6, 4)}

$$probability = \frac{no.offavoriblecases}{tota\ln oofcases} = \frac{3}{36} = \frac{1}{12}$$

The correct Answer is 'B'

2. Correct option: (A)

When a die is rolled twice total number of events $=6^2=36$

The favorable case is the numbers on them are different i.e. they must not be equal numbers

Total number of not favorable cases is {(1, 1), (2, 2), (3, 3) (4, 4) (5, 5) (6, 6)}

$$\text{Non favorable } probability = \frac{no.ofnonfavoriblecases}{tota\ln oofcases} = \frac{6}{36} = \frac{1}{6}$$

$$\therefore favorableprobablity = 1 - nonfavorableprobablity$$

$$\Rightarrow 1 - \frac{1}{6} = \frac{5}{6}$$

The correct Answer is 'A'

3. Correct option: (C)

When a die is rolled twice total number of events $=6^2=36$

The favorable case is getting an even number and an odd number

Total number of favorable cases is {(1, 2), (1, 4) (1, 6) (2, 1) (2, 3) (2, 5) (3, 2) (3, 4) (3, 6) (4,1) (4,3), (4,5) (5,2) (5,4), (5,6) , (6,1), (6,3), (6,5)}

$$probability = \frac{no.offavoriblecases}{tota\ln oofcases} = \frac{18}{36} = \frac{1}{2}$$

The correct Answer is 'C'

4. Correct option: (A)

When three dies is rolled total number of events $=6^3=216$

The favorable case is the sum of the numbers on them is 16

Total number of favorable cases is {(5,5,6) (5,6,5) (6,5,5) (6,6,4) (4,6,6) (6,4,6)}

$$probability = \frac{no.offavoriblecases}{tota\ln oofcases} = \frac{6}{216} = \frac{1}{36}$$

The correct Answer is 'A'

5. Correct option: (B)

The total number of cards = 52. Symbols that cards contain are 'Heart', 'Diamond', 'Spade', 'Club'. Every symbol has 13 cards.

Number of favorable outcomes i.e. 'a card of heart' is 13.

$$probability = \frac{no.offavoriblecases}{tota\ln oofcases} = \frac{13}{52} = \frac{1}{4}$$

The correct Answer is 'B'

6. Correct option: (D)

The 2 odd days may be the combination of Sunday & Monday, Monday & Tuesday, Tuesday & Wednesday, Wednesday & Thursday, Thursday & Friday, Friday & Saturday or Saturday & Sunday.

Therefore, the total number of possible outcomes or elements of a sample space is 7.

probability of 2 Odd days not to be Monday & Tuesday, Tuesday & Wednesday, Wednesday & Thursday, Thursday & Friday or Friday & Saturday

Expected events of A = {Sun & Mon}, {Wed & Thu}, {Thu & Fri}, {Fri & Sat}, {Sat & Sun} = 5

So the probability of 52 Tuesdays is 5/7

The correct Answer is 'D'

7. Correct option: (D)

Probability of 1 Odd day not to be Monday

The sample space S = {Sun, Mon, Tue, Wed, Thu, Fri, Sat}

Expected events of A = {Sun, Tue, Wed, Thu, Fri, Sat}

P (A) = {Sun, Tue, Wed, Thu, Fri, Sat}/{Sun, Mon, Tue, Wed, Thu, Fri, Sat}

P(A)=6/7=0.86

6/7 or 0.86 is probability for 52 Mondays in a non-leap year.

The correct Answer is 'D'

8. Correct option: (D)

When a number is picked at random from 1 to 900 then the number of events will be 900

The favorable case is drawing a number which is a square is

Therefore the number of squares between 1to 900 is 30

$$probability = \frac{no.offavoriblecases}{tota\ln oofcases} = \frac{30}{900} = \frac{1}{30}$$

The correct Answer is 'D'

9. Correct option: (B)

$$P(\mathrm{A}) = \frac{2}{7}, P(\mathrm{B}) = \frac{3}{5}, P(\mathrm{A} \cap \mathrm{B}) = 0$$

$$P(\mathrm{A} \cup \mathrm{B}) = P(\mathrm{A}) + P(\mathrm{B})$$

$$= \frac{2}{7} + \frac{3}{5}$$

$$= \frac{31}{35}$$

The correct Answer is 'B'

10. Correct option: (B)

$$P(\mathrm{A}) = \frac{1}{4}, P(\mathrm{B}) = \frac{1}{2}, P(\mathrm{A} \cup \mathrm{B}) = \frac{5}{8}$$

$$\frac{5}{8} = \frac{1}{4} + \frac{1}{2} - P(\mathrm{A} \cap \mathrm{B})$$

$$P(\mathrm{A} \cap \mathrm{B}) = -\frac{5}{8} + \frac{3}{4}$$

$$= \frac{1}{8}$$

The correct Answer is 'B'

11. Correct option: (C, D)

$$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)\,P(B)}{P(B)} = P(B)$$

$$P((A \cup B)') = P(A' \cap B') = P(A')\,P(B')$$

The correct Answer is 'C' & 'D'

12. Correct option: (B, C, D)

$$\text{If M and N are independent then}$$

$$P(\mathrm{M} \cap N) = \mathrm{P(M)\,P(N)}$$

$$P(\mathrm{M} \cap \overline{N}) = \mathrm{P(M)} - \mathrm{P(M} \cap N)$$

$$= \mathrm{P(M)} - P(\mathrm{M})\,\mathrm{P(N)}$$

$$= \mathrm{P(M)\,P}(\overline{N})$$

$$\therefore \mathrm{M}, \overline{N} \text{ are independent}$$

$$P(\overline{M} \cap \overline{N}) = 1 - \mathrm{P(M} \cup \mathrm{N)}$$

$$= 1 - \mathrm{P(M)} - \mathrm{P(N)} - \mathrm{P(M} \cap \mathrm{N)}$$

$$= P(\overline{M})P(\overline{N})$$

$$\therefore \overline{M}, \overline{N} \text{ are independent}$$

The correct Answer is 'B' 'C' and 'D'

13. Correct option: (A)

If 2 dices are thrown simultaneously then total outcomes will be 36

Favorable outcomes = (1,5) (2,4)(3,3)(3,5)(4,2)(4,4)(5,1)(5,3)(6,2)

Probability of event =10/36=5/18

The correct Answer is 'A'

14. Correct option: (A)

Probability of getting a total of 10:

To get a total of 10 with two dice, you can have the following combinations: (4, 6), (5, 5), and (6, 4). There are three favorable outcomes.

Probability of getting a total of an even number:

An even total can be obtained in multiple ways: (2, 2), (2, 4), (2, 6), (4, 2), (4, 4), (4, 6), (6, 2), (6, 4), and (6, 6). There are nine favorable outcomes.

$$\frac{3}{36} + \frac{9}{36} = \frac{1}{3}$$

The correct Answer is 'A'

15. Correct Matching is: a-R; b-Q; c-P; d-S

A

Let A: event of drawing a king $P(\mathrm{A}) = \frac{4}{52} = \frac{1}{13}$

B: event of drawing a spade = $P(\mathrm{B}) = \frac{13}{52} = \frac{1}{4}$

$P(A \cap B) = \frac{1}{52}$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{1}{13} + \frac{1}{4} - \frac{1}{52}$$

$$= \frac{4}{13}$$

The Correct Match is 'R'

B

The number of cards in a pack = 52

The number of the kings in a pack = 4

If we draw one card at random from the 52 cards,

Then the probability of getting a king = number of kings/number of all the cards i.e. probability of getting a king = $\frac{4}{52}$

The number of the queens in a pack = 4

If we draw one card at random from the 52 cards,

Then the probability of getting a queen = number of queens/number of all the cards

i.e. probability of getting a queen = $\frac{4}{52}$

The probability of getting a king or a queen out of all the cards = probability of getting a king out of all the cards + probability of getting a queen out of all the cards $\frac{4}{52}+\frac{4}{52}=\frac{2}{13}$

Hence the probability of getting a king or a queen out of 52 cards is $\frac{2}{13}$

The Correct Match is 'Q'

C

A=the drawn card is a spade; B= the drawn card is a heart.

$P(\text{A})=\frac{13}{52}$ (As there are 13 spade cards in a fair deck of 52 cards)

$P(\text{B})=\frac{13}{52}$ (As there are 13 heart cards in a fair deck of 52 cards)

Events A and B are *mutually exclusive events.

As by addition theorem of probability, P (A or B) = P (A) + P (B) for mutually exclusive events.

P (spade or heart) = P (A U B) = $\frac{13}{52}+\frac{13}{52}=\frac{1}{2}$

The Correct Match is 'P'

D

There are 13 cards in each suit, so there are 26 cards that are either hearts or clubs.

However, we have double counted the cards that are both a heart and a club (i.e., the Ace, King, Queen, Jack, and 10 of hearts and clubs).

The probability of drawing a heart or a club is then:

P(heart or club) = $=P(\text{cluborheart})=\frac{13}{52}+\frac{13}{52}-\frac{6}{52}=\frac{20}{52}=\frac{5}{13}$

The Correct Match is 'S'

16. Correct Answer is:

$(x^4-y^4)=(x^2+y^2)\,(x+y)\,(x-y)$

Let us consider five sets $(5k, 5k+1, 5k+2, 5k+3, 5k+4)$.

Case I: $(x-y)$ *is divisible by* 5.

Elements are to be selected from the same set, 5. ${}^{n}C_{2}$

Case II: $x+y$ *is divisible by* 5 *but* $x-y$ *is not divisible by* 5

It can be done from $(5k+1, 5k+4)$ *or* $(5k+2, 5k+3)$.

Total ways $=2n^2$ $\ldots.(2)$

$Case\ III\colon\ x^2+y^2\ is\ divisible\ by\ 5\ but\ x-y\ or\ x+y\ is\ not\ divisible\ by\ 5 \quad \ldots(3)$

$We\ can\ select\ one\ number\ either\ from(5k+1,5k+4)\ and\ the\ second\ number\ from\ (5k+2,5k+3).$

$Hence, the\ number\ of\ cases\ = 2n\times 2n = 4n^2 \quad \ldots\ldots\ldots\ldots\ldots(3)$

$Total\ number\ of\ \ favourable\ cases\ = 5\frac{n(n-1)}{2}+6n^2 = \frac{17n^2-5n}{2}$

$Total\ number\ of\ cases\ = 5\frac{n(n-1)}{2}$

$Pr\ o\ bailty = \frac{17n-5}{25n-5}$

$here, n=3 \Rightarrow Pr\ o\ bailty(p) = \frac{46}{70} = \frac{23}{35}$

$\therefore \frac{p}{23} = \frac{1}{35}$

17. Correct Answer is:1

$P(A\cap B) = \frac{1}{12}, P(\overline{A}\cap\overline{B}) = \frac{1}{2}, P(A) = \frac{1}{m}$

$P(\overline{A}\cap\overline{B}) = P(\overline{A}).P(\overline{B}) = \frac{1}{2}$

$\Rightarrow [1-P(A)].[1-P(B)] = \frac{1}{2}$

$\Rightarrow 1-P(A)-P(B)+P(A).P(B) = \frac{1}{2}$

$P(A\cap B) = P(A)\,P(B) = \frac{1}{m}P(B) = \frac{1}{12}$

$\therefore P(B) = \frac{m}{12}$

$\Rightarrow 1-\frac{1}{m}-\frac{m}{12}+\frac{1}{12} = \frac{1}{2}$

$\Rightarrow 1+\frac{1}{12}-\frac{1}{2} = \frac{1}{m}+\frac{m}{12}$

$\Rightarrow \frac{7}{12} = \frac{m^2+12}{12m}$

$\Rightarrow 12m^2+144-24m = 0$

$\Rightarrow m = 1\pm\sqrt{-11}$

$\therefore [m] = 1$

$\Rightarrow \left[\frac{m}{3}\right] = \frac{1}{3}$

8. Matrices

Solutions

LEVEL – I

1. Correct option: (A)

Given that $A = \begin{bmatrix} 2 & 3 & 1 \\ 0 & -1 & 5 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 & -1 \\ 0 & -1 & 3 \end{bmatrix}$

We have to find the matrix $2A - 3B$

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 0 & -1 & 5 \end{bmatrix} \Rightarrow 2A = \begin{bmatrix} 4 & 6 & 2 \\ 0 & -2 & 10 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 2 & -1 \\ 0 & -1 & 3 \end{bmatrix} \Rightarrow 3B = \begin{bmatrix} 3 & 6 & -3 \\ 0 & -3 & 9 \end{bmatrix}$$

Therefore $2A - 3B =$

$$= \begin{bmatrix} 4 & 6 & 2 \\ 0 & -2 & 10 \end{bmatrix} - \begin{bmatrix} 3 & 6 & -3 \\ 0 & -3 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 1 \end{bmatrix}$$

Then the correct option is A)

2. Correct option: (A)

Given that

$$A = \begin{bmatrix} 2 & 1 & 3 & -1 \\ 1 & -2 & 2 & -3 \end{bmatrix}, B = \begin{bmatrix} 2 & 1 & 0 & 3 \\ 1 & -1 & 2 & 3 \end{bmatrix}$$

We are given $2A + 3B - 5C = 0$

Then

$$C = \frac{2A + 3B}{5}$$

We have to find the matrix C

$$A = \begin{bmatrix} 2 & 1 & 3 & -1 \\ 1 & -2 & 2 & -3 \end{bmatrix} \Rightarrow 2A = 2\begin{bmatrix} 2 & 1 & 3 & -1 \\ 1 & -2 & 2 & -3 \end{bmatrix}$$

$$B = \begin{bmatrix} 2 & 1 & 0 & 3 \\ 1 & -1 & 2 & 3 \end{bmatrix} \Rightarrow 3B = 3\begin{bmatrix} 2 & 1 & 0 & 3 \\ 1 & -1 & 2 & 3 \end{bmatrix}$$

Therefore $2A + 3B =$

$$= \begin{bmatrix} 10 & -5 & 6 & 7 \\ 5 & -7 & 10 & 3 \end{bmatrix}$$

So

$$\frac{2A + 3B}{5} = \begin{bmatrix} \frac{10}{5} & -\frac{5}{5} & \frac{6}{5} & \frac{7}{5} \\ \frac{5}{5} & -\frac{7}{5} & \frac{10}{5} & \frac{3}{5} \end{bmatrix} = \begin{bmatrix} 2 & -1 & \frac{6}{5} & \frac{7}{5} \\ 1 & -\frac{7}{5} & 2 & \frac{3}{5} \end{bmatrix}$$

Then the correct option is A)

3. Correct option: (C)

Given that $\begin{bmatrix} x-3 & 2y-8 \\ z+2 & 6 \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ -2 & a-4 \end{bmatrix}$

We have to find the value of x, y, z, a so we can use the property of equal matrix

$$x - 3 = 5 \Rightarrow x = 8$$

$$2y - 8 = 2 \Rightarrow y = 5$$

$$z + 2 = -2 \Rightarrow z = -4$$

$$6 = a - 4 \Rightarrow a = 10$$

Then the correct option is C)

4. Correct option: (A)

Matrix A: 3×5 (3 rows, 5 columns).

Matrix B: 2×3 (2 rows, 3 columns).

The order of BA: 2×5 (2 rows, 5 columns).

Then the correct option is C)

5. Correct option: (D)

Given that $m[-3 \quad 4] + n[4 \quad -3] = [10 \quad -11]$

We have to find the value of $3m + 7n$

Using the property of the matrix we can write

$$-3m + 4n = 10$$
$$4m - 3n = -11$$

Solving both of the equation

$$m = -2, n = 1$$

Therefore $3m + 7n = -6 + 7 = 1$

Then the correct option is D)

6. Correct option: (D)

Given that

$$U = [2 \quad -3 \quad 4], V = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}, X = [0 \quad 2 \quad 3], Y = \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix}$$

We have to find the value of $UV + XY$

Using Row-Column multiplication

$$UV = [2 \quad -3 \quad 4] \times \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} = [6 - 6 + 4] = [4]$$

$$XY = [0 \quad 2 \quad 3] \times \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix} = [0 + 4 + 12] = [16]$$

Therefore

$$UV + XY = [4] + [16] = [20]$$

Then the correct option is D)

7. Correct option: (C)

Given that

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

We have to find the value of A^2

Using Row-Column multiplication

$$A^2 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$
$$= \begin{bmatrix} 2 & 0 & 2 \\ 0 & 0 & 0 \\ 2 & 0 & 2 \end{bmatrix} = 2 \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix} = 2A$$

Therefore

$$A^2 = 2A$$

Then the correct option is C)

8. Correct option: (D)

Given that

$$A = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, B = \begin{bmatrix} -5 & 4 & 0 \\ 0 & 2 & -1 \\ 1 & -3 & 2 \end{bmatrix}$$

We have to find the value of $A \times B$

To determine if we can multiply matrices A and B (AB), we need to check if the number of columns in matrix A matches the number of rows in matrix B. Matrix A has 1 column, and matrix B has 3 rows, so they are not compatible for matrix multiplication because the number of columns in A (1) does not match the number of rows in B (3).

Then the correct option is D)

9. Correct option: (D)

Given that $A = \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}, B = \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix}, (A+B)^2 = A^2 + B^2 + 2AB$

We have to find the value of a, b

Using Row-Column multiplication we can calculate

$$(A+B)^2 = \begin{bmatrix} 1+a & 0 \\ 2+b & 0 \end{bmatrix} \times \begin{bmatrix} 1+a & 0 \\ 2+b & 0 \end{bmatrix} = \begin{bmatrix} (1+a)^2 & 0 \\ (1+a)(2+b) & 0 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} -1 & -2 \\ 4 & -1 \end{bmatrix}$$

$$B^2 = \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix} \times \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix} = \begin{bmatrix} a^2+b & a-1 \\ ab-b & b+1 \end{bmatrix}$$

$$A \times B = \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix} \times \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix} = \begin{bmatrix} a-b & 2 \\ 2a+b & 1 \end{bmatrix}$$

$$2[A \times B] = \begin{bmatrix} (2a-2b) & 4 \\ 2(2a+b) & 2 \end{bmatrix}$$

Therefore

$$A^2 + B^2 + 2AB = \begin{bmatrix} -1+a^2+b+(2a-2b) & 2+(a-1) \\ (ab-b)+4+2(2a+b) & 2+b \end{bmatrix}$$

$$\begin{bmatrix} (1+a)^2 & 0 \\ (1+a)(2+b) & 0 \end{bmatrix} = \begin{bmatrix} -1+a^2+b+(2a-2b) & 2+(a-1) \\ (ab-b)+4+2(2a+b) & 2+b \end{bmatrix}$$

$$\Rightarrow a = -1, b = -2$$

Then the correct option is D)

10. Correct option: (A)

Given that

$$A = \begin{bmatrix} 1 & -1 \\ 3 & 4 \\ -1 & 5 \end{bmatrix}, B = \begin{bmatrix} -1 & 2 \\ 3 & 1 \end{bmatrix}$$

We have to find the value of AB

Using Row-Column multiplication we can calculate

$$AB = \begin{bmatrix} 1 & -1 \\ 3 & 4 \\ -1 & 5 \end{bmatrix} \times \begin{bmatrix} -1 & 2 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} -1-3 & 2-1 \\ -3+12 & 6+4 \\ 1+15 & -2+5 \end{bmatrix}$$

$$= \begin{bmatrix} -4 & 1 \\ 9 & 10 \\ 16 & 3 \end{bmatrix}$$

Therefore

$$AB = \begin{bmatrix} -4 & 1 \\ 9 & 10 \\ 16 & 3 \end{bmatrix}$$

Then the correct option is A)

11. Correct option: (C)

Given that $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

We have to find the value of $A^2 - (a+d)A$

Using Row-Column multiplication we can calculate

$$A^2 = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \times \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a^2+bc & ab+bd \\ ca+cd & bc+d^2 \end{bmatrix}$$

$$(a+d)A = \begin{bmatrix} a^2+ad & ba+bd \\ ac+dc & ad+d^2 \end{bmatrix}$$

Therefore

$$A^2 - (a+d)A = \begin{bmatrix} a^2+bc & ab+bd \\ ca+cd & bc+d^2 \end{bmatrix} - \begin{bmatrix} a^2+ad & ba+bd \\ ac+dc & ad+d^2 \end{bmatrix}$$

$$= \begin{bmatrix} bc-ad & 0 \\ 0 & bc-ad \end{bmatrix} = (bc-ad)\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = (bc-ad)I$$

Then the correct option is C)

12. Correct option: (B)

Given that $A = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{bmatrix}$

We have to find the value of A^{100}

Using Row-Column multiplication we can calculate

$$A^2 = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} \times 2 & 1 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \frac{3}{2} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} \times 3 & 1 \end{bmatrix}$$

So we can conclude that the

$$A^n = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} \times n & 1 \end{bmatrix}$$

Therefore

$$A^{100} = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} \times 100 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 50 & 1 \end{bmatrix}$$

Then the correct option is B)

13. Correct option: (B)

1. A symmetric matrix has the property that $A^T = A$, where A^T is the transpose of A. This means that the matrix is equal to its own transpose, and the elements are symmetric with respect to the main diagonal.

2. A skew-symmetric matrix has the property that $A^T = -A$, where A^T is the transpose of A. This means that the transpose of the matrix is equal to its negative, and the elements are antisymmetric with respect to the main diagonal.

$A = A^T$ (Symmetric property)

$A = -A^T$ (Skew-symmetric property)

Combining these two equations, we get: $A = -A$

To have A = -A, the only possible solution is that all elements of A are zero (null matrix). Therefore, if A is both symmetric and skew-symmetric, it is a null matrix.
Then the correct option is B)

14. Correct option: (*)

Given that $A = \begin{bmatrix} 3 & 1 & -1 \\ 0 & 1 & 2 \end{bmatrix}$

We have to find the type of AA^T

$$A^T = \begin{bmatrix} 3 & 0 \\ 1 & 1 \\ -1 & 2 \end{bmatrix}$$

Using Row-Column multiplication we can calculate

$$AA^T = \begin{bmatrix} 3 & 1 & -1 \\ 0 & 1 & 2 \end{bmatrix} \times \begin{bmatrix} 3 & 0 \\ 1 & 1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 11 & -1 \\ -1 & 5 \end{bmatrix}$$

So we can conclude that none of the option is correct.

15. Correct option: (C)

Given that $A = skew - symmetric\ matrix\ with\ 3 \times 3$

Let

$$A = \begin{bmatrix} 0 & a & b \\ -a & 0 & m \\ -b & -m & 0 \end{bmatrix}$$

As $A = -A^T => a_{ij} = -a_{ji} => for\ diagonal\ elements => a_{ii} = -a_{jj} => a_{ii} = 0$

Therefore

$$|A| = \left| \begin{bmatrix} 0 & a & b \\ -a & 0 & m \\ -b & -m & 0 \end{bmatrix} \right|$$

$$|A| = 0(0 + m^2) - (-a)\big(0 - (-bm)\big) + (-b)(am - 0) = abm - abm = 0$$

So we can conclude that

$$trace(A) = |A|$$

Then the correct option is C)

16. Correct option: (A)

Given that

$$A = \begin{bmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{bmatrix} = 2 \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Therefore

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{bmatrix} => A^2 = 4 \begin{bmatrix} 3 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{bmatrix}$$

$$A^2 = 12 \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} => A^2 \times A = 12 \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \times 2 \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$A^3 = 24 \times 3 \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

So we can conclude that

$$A^3 - 35A = 24 \times 3\begin{bmatrix}1 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 1\end{bmatrix} - 35 \times 2\begin{bmatrix}1 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 1\end{bmatrix} = (72-70)\begin{bmatrix}1 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 1\end{bmatrix}$$

$$= 2\begin{bmatrix}1 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 1\end{bmatrix} = A$$

Then the correct option is A)

17. Correct option: (A)

Given that

$$A = \begin{bmatrix}0 & \alpha\\ 0 & 0\end{bmatrix}$$

Then

$$A + I = \begin{bmatrix}0 & \alpha\\ 0 & 0\end{bmatrix} + \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} = \begin{bmatrix}1 & \alpha\\ 0 & 1\end{bmatrix}$$

So

$$(A+I)^2 = \begin{bmatrix}1 & \alpha\\ 0 & 1\end{bmatrix} \times \begin{bmatrix}1 & \alpha\\ 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 2\alpha\\ 0 & 1\end{bmatrix}$$

$$(A+I)^3 = \begin{bmatrix}1 & 2\alpha\\ 0 & 1\end{bmatrix} \times \begin{bmatrix}1 & \alpha\\ 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 3\alpha\\ 0 & 1\end{bmatrix}$$

$$(A+I)^n = \begin{bmatrix}1 & n\alpha\\ 0 & 1\end{bmatrix}$$

Therefore

$$(A+I)^{50} = \begin{bmatrix}1 & 50\alpha\\ 0 & 1\end{bmatrix}$$

So we can write

$$(A+I)^{50} - 50A = \begin{bmatrix}1 & 50\alpha\\ 0 & 1\end{bmatrix} - \begin{bmatrix}0 & 50\alpha\\ 0 & 0\end{bmatrix} = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$

$$\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} = \begin{bmatrix}a & b\\ c & d\end{bmatrix} \Rightarrow a = 1, b = 0, c = 0, d = 1$$

$$\Rightarrow a + b + c + d = 2$$

Then the correct option is A)

18. Correct option: (A)

Given that

$$A = \begin{bmatrix}\cos\theta & \sin\theta & 0\\ -\sin\theta & \cos\theta & 0\\ 0 & 0 & 1\end{bmatrix}$$

Therefore

$$A^T = \begin{bmatrix}\cos\theta & -\sin\theta & 0\\ \sin\theta & \cos\theta & 0\\ 0 & 0 & 1\end{bmatrix}$$

So we can conclude that

$$AA^T = \begin{bmatrix}\cos\theta & \sin\theta & 0\\ -\sin\theta & \cos\theta & 0\\ 0 & 0 & 1\end{bmatrix} \times \begin{bmatrix}\cos\theta & -\sin\theta & 0\\ \sin\theta & \cos\theta & 0\\ 0 & 0 & 1\end{bmatrix}$$

$$= \begin{bmatrix} \cos^2\theta + \sin^2\theta & 0 & 0 \\ 0 & \cos^2\theta + \sin^2\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

$$A^T A = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos^2\theta + \sin^2\theta & 0 & 0 \\ 0 & \cos^2\theta + \sin^2\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

Therefore

$$AA^T = A^T A = I$$

Then the correct option is A)

19. Correct option: (A)

Given that

$$A = \begin{bmatrix} x & 1 & 4 \\ -1 & 0 & 7 \\ -4 & -7 & 0 \end{bmatrix}, A = -A^T$$

The given condition shows that matrix A is skew-symmetric matrix and diagonal elements of a skew symmetric matrix is always equal to zero.

Therefore $x = 0$

Then the correct option is A)

20. Correct option: (A)

Given that

$$A = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix}, B = \begin{bmatrix} x & 1 \\ y & -1 \end{bmatrix}, (A^2 + B^2) = (A + B)^2$$

Then

$$(A + B) = \begin{bmatrix} 1 + x & 0 \\ 2 + y & -2 \end{bmatrix}$$

$$(A + B)^2 = \begin{bmatrix} 1 + x & 0 \\ 2 + y & -2 \end{bmatrix} \times \begin{bmatrix} 1 + x & 0 \\ 2 + y & -2 \end{bmatrix} = \begin{bmatrix} (1 + x)^2 & 0 \\ (1 + x)(2 + y) - 2(y + 2) & 4 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix} \times \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$B^2 = \begin{bmatrix} x & 1 \\ y & -1 \end{bmatrix} \times \begin{bmatrix} x & 1 \\ y & -1 \end{bmatrix} = \begin{bmatrix} x^2 + y & x - 1 \\ xy - y & y + 1 \end{bmatrix}$$

Therefore

$$(A^2 + B^2) = (A + B)^2$$

$$\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} + \begin{bmatrix} x^2 + y & x - 1 \\ xy - y & y + 1 \end{bmatrix} = \begin{bmatrix} (1 + x)^2 & 0 \\ (1 + x)(2 + y) - 2(y + 2) & 4 \end{bmatrix}$$

$$\begin{bmatrix} x^2 + y - 1 & x - 1 \\ xy - y & y \end{bmatrix} = \begin{bmatrix} (1 + x)^2 & 0 \\ (1 + x)(2 + y) - 2(y + 2) & 4 \end{bmatrix}$$

$$=> x - 1 = 0 => x = 1, y = 4$$

Then the correct option is A)

LEVEL-II

1. Correct option: (D)

Given that $A = \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix}, A^2 = B$

Then

$$A^2 = \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix} \times \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} \alpha^2 & 0 \\ \alpha + 1 & 1 \end{bmatrix}$$

Therefore

$$\begin{bmatrix} \alpha^2 & 0 \\ \alpha + 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix}$$

$$\alpha^2 = 1 \; and \; \alpha + 1 = 5 \; => \alpha \; has \; no \; real \; values$$

Then the correct option is D)

2. Correct option: (A)

Given that $A^2 = 2A - I, n \neq 2$

Then

$$A^3 = A \times A^2 = A(2A - I) = 2A^2 - A = 2(2A - I) - A = 4A - 2A - A = 3A - 2I$$

$$A^4 = A \times A^3 = A(3A - 2I) = 3A^2 - 2A = 3(2A - I) - 2A = 6A - 3I - 2A = 4A - 3I$$

Therefore

$$A^n = nA - (n - 1)I$$

Then the correct option is A)

3. Correct option: (D)

Given that

$$\begin{bmatrix} 1 & x & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 2 \\ 0 & 5 & 1 \\ 0 & 3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ x \end{bmatrix} = 0$$

Then

$$\begin{bmatrix} 1 & x & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 2 \\ 0 & 5 & 1 \\ 0 & 3 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 5x + 6 & 4 + x \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 5x + 6 & 4 + x \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ x \end{bmatrix} = [1 + 5x + 6 + x(4 + x)] = [x^2 + 9x + 7]$$

So we can write

$$[x^2 + 9x + 7] = [0] => x^2 + 9x + 7 = 0$$

Using quadratic formula

$$x = \frac{-9 \pm \sqrt{53}}{2}$$

Then the correct option is D)

4. Correct option: (B)

Given that $A = m \times (n + 5), B = m \times (11 - n)$

Also given that

$$AB \; and \; BA \; exist$$

From the matrix multiplication we can say that if $AB \; and \; BA$ exist therefore

$$=> AB => n + 5 = m, BA => m = 11 - n$$

So we can write $n + 5 = 11 - n => 2n = 6 => n = 3, m = 5 + 3 = 8$

Therefore we can say that AB and BA are of order 8×8 and 3×3

Then the correct option is B)

5. Correct option: (C)

Given that

$$A=\begin{bmatrix}1 & 2 & 2\\ 2 & 1 & 2\\ 2 & 2 & 1\end{bmatrix}$$

Then

$$A^3-4A^2-6A=A(A^2-4A-6I)$$

$$A^2-4A-6I=\begin{bmatrix}1 & 2 & 2\\ 2 & 1 & 2\\ 2 & 2 & 1\end{bmatrix}\times\begin{bmatrix}1 & 2 & 2\\ 2 & 1 & 2\\ 2 & 2 & 1\end{bmatrix}-4\begin{bmatrix}1 & 2 & 2\\ 2 & 1 & 2\\ 2 & 2 & 1\end{bmatrix}-6\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$$

$$=>\begin{bmatrix}9-4-6 & 8-8 & 8-8\\ 8-8 & 9-4-6 & 8-8\\ 8-8 & 8-8 & 9-4-6\end{bmatrix}=(-1)\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}=-I$$

So we can write

$$A^3-4A^2-6A=A(A^2-4A-6I)=A(-I)=-A$$

Then the correct option is C)

6. Correct option: (B)

Given that

$$A=\begin{bmatrix}a^2 & ab & ac\\ ab & b^2 & bc\\ ac & bc & c^2\end{bmatrix},a^2+b^2+c^2=1$$

Then

$$A^2=\begin{bmatrix}a^2 & ab & ac\\ ab & b^2 & bc\\ ac & bc & c^2\end{bmatrix}\times\begin{bmatrix}a^2 & ab & ac\\ ab & b^2 & bc\\ ac & bc & c^2\end{bmatrix}$$

$$=>\begin{bmatrix}a^2(a^2+b^2+c^2) & ab(a^2+b^2+c^2) & ac(a^2+b^2+c^2)\\ ab(a^2+b^2+c^2) & b^2(a^2+b^2+c^2) & bc(a^2+b^2+c^2)\\ ac(a^2+b^2+c^2) & bc(a^2+b^2+c^2) & c^2(a^2+b^2+c^2)\end{bmatrix}$$

$$=\begin{bmatrix}a^2 & ab & ac\\ ab & b^2 & bc\\ ac & bc & c^2\end{bmatrix}=A$$

So we can write

$$A^2=A$$

Then the correct option is B)

7. Correct option: (C)

Given that

$$3A=\begin{bmatrix}1 & 2 & 2\\ 2 & 1 & -2\\ x & 2 & y\end{bmatrix},AA'=I=>A=\frac{1}{3}\begin{bmatrix}1 & 2 & 2\\ 2 & 1 & -2\\ x & 2 & y\end{bmatrix}$$

Then we have to find $x+y$

$$AA'=\frac{1}{3}\begin{bmatrix}1 & 2 & 2\\ 2 & 1 & -2\\ x & 2 & y\end{bmatrix}\times\frac{1}{3}\begin{bmatrix}1 & 2 & x\\ 2 & 1 & 2\\ 2 & -2 & y\end{bmatrix}$$

$$=> \frac{1}{9}\begin{bmatrix} 9 & 0 & x+4+2y \\ 0 & 9 & 2x+2-2y \\ x+4+2y & 2x+2-2y & x^2+4+y^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

So we can write

$$x+2y+4=0, 2x+2-2y=0 => x=-2, y=-1$$

Then the correct option is C)

8. Correct option: (C)

Given that $A = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$

Then

$$A^2 = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix} = 2A$$

$$A^3 = A \times A^2 = (2^2)A$$

Therefore

$$A^{n+1} = 2^n A = 2^n \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

Then the correct option is C)

9. Correct option: (B)

Given that

$$A = \begin{bmatrix} 4 & 1 & 0 \\ 1 & -2 & 2 \end{bmatrix}, B = \begin{bmatrix} 2 & 0 & -1 \\ 3 & 1 & 4 \end{bmatrix}, C = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, (3B-2A)C + 2X = 0$$

Then

$$X = -\frac{1}{2}(3B-2A)C$$

Therefore

$$X = -\frac{1}{2}\left(\left(3\begin{bmatrix} 2 & 0 & -1 \\ 3 & 1 & 4 \end{bmatrix} - 2\begin{bmatrix} 4 & 1 & 0 \\ 1 & -2 & 2 \end{bmatrix}\right) \times \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}\right) = -\frac{1}{2}\left(\begin{bmatrix} -2 & -2 & -3 \\ 7 & 7 & 8 \end{bmatrix} \times \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}\right) = -\frac{1}{2}\begin{bmatrix} -3 \\ 13 \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{3}{2} \\ \frac{13}{2} \end{bmatrix}$$

Then the correct option is B)

10. Correct option: (B)

Given that

$$A = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}$$

Then

$$A^T = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$$

We have to find

$$(2A)\left(\frac{1}{4}A^T\right) = \frac{1}{2}(AA^T)$$

So

$$A \times A^T = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix} \times \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix} = \begin{bmatrix} \cos^2\alpha + \sin^2\alpha & 0 \\ 0 & \cos^2\alpha + \sin^2\alpha \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$=> \frac{1}{2}AA^T = \frac{1}{2}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} = \begin{bmatrix}\frac{1}{2} & 0\\ 0 & \frac{1}{2}\end{bmatrix}$$

Then the correct option is B)

11. Correct option: (A, C, D)

Given that

$$A = \begin{bmatrix}0 & 2\beta & \gamma\\ \alpha & \beta & -\gamma\\ \alpha & -\beta & \gamma\end{bmatrix}, A\ is\ orthogonal$$

Then $AA^T = I$

Putting the matrix

$$= \begin{bmatrix}0 & 2\beta & \gamma\\ \alpha & \beta & -\gamma\\ \alpha & -\beta & \gamma\end{bmatrix} \times \begin{bmatrix}0 & \alpha & \alpha\\ 2\beta & \beta & -\beta\\ \gamma & -\gamma & \gamma\end{bmatrix} = \begin{bmatrix}4\beta^2+\gamma^2 & 2\beta^2-\gamma^2 & -2\beta^2+\gamma^2\\ 2\beta^2-\gamma^2 & \alpha^2+\beta^2+\gamma^2 & \alpha^2-\beta^2-\gamma^2\\ -2\beta^2+\gamma^2 & \alpha^2-\beta^2-\gamma^2 & \alpha^2+\beta^2+\gamma^2\end{bmatrix}$$

$$= \begin{bmatrix}4\beta^2+\gamma^2 & 2\beta^2-\gamma^2 & -2\beta^2+\gamma^2\\ 2\beta^2-\gamma^2 & \alpha^2+\beta^2+\gamma^2 & \alpha^2-\beta^2-\gamma^2\\ -2\beta^2+\gamma^2 & \alpha^2-\beta^2-\gamma^2 & \alpha^2+\beta^2+\gamma^2\end{bmatrix} = \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$$

So we can write

$$2\beta^2 - \gamma^2 = 0 => 2\beta^2 = \gamma^2$$

$$\alpha^2 - \beta^2 - \gamma^2 = 0 => \alpha^2 = 3\beta^2 = \frac{3}{2}\gamma^2$$

$$\alpha^2 + \frac{\alpha^2}{3} + \frac{2}{3}\alpha^2 = 1 => 2\alpha^2 = 1 => \alpha = \pm\frac{1}{\sqrt{2}}$$

$$\beta = \pm\frac{1}{\sqrt{6}} => \alpha = \pm\sqrt{3}\beta$$

Then the correct option is A, C, D)

12. Correct option: (A, B, C)

Given that

$$A = \begin{bmatrix}1 & 0\\ 2 & 1\end{bmatrix}, A^{2012} = \begin{bmatrix}a & b\\ c & d\end{bmatrix}$$

Then

$$A^2 = \begin{bmatrix}1 & 0\\ 2 & 1\end{bmatrix} \times \begin{bmatrix}1 & 0\\ 2 & 1\end{bmatrix} = \begin{bmatrix}1 & 0\\ 4 & 1\end{bmatrix}$$

$$A^3 = A \times A^2 = \begin{bmatrix}1 & 0\\ 2 & 1\end{bmatrix} \times \begin{bmatrix}1 & 0\\ 4 & 1\end{bmatrix} = \begin{bmatrix}1 & 0\\ 6 & 1\end{bmatrix}$$

So we can write

$$A^n = \begin{bmatrix}1 & 0\\ 2n & 1\end{bmatrix}$$

Therefore

$$A^{2012} = \begin{bmatrix}1 & 0\\ 2\times 2012 & 1\end{bmatrix} = \begin{bmatrix}1 & 0\\ 4024 & 1\end{bmatrix} = \begin{bmatrix}a & b\\ c & d\end{bmatrix}$$

So

$$a = 1, b = 0, c = 4024, d = 1$$

Therefore we can conclude

$$a = d, a + b + c + d = 4026, a^2 + b^2 + d^2 = 2$$

Then the correct option is A, B, C)

13. Correct option: (*)

Given that

$$A = \begin{bmatrix} 5 & 7 & 3 \\ 1 & 5 & 2 \\ 3 & 2 & 1 \end{bmatrix}, A^3 - kA^2 + lA = I$$

Then

$$A^2 = \begin{bmatrix} 5 & 7 & 3 \\ 1 & 5 & 2 \\ 3 & 2 & 1 \end{bmatrix} \times \begin{bmatrix} 5 & 7 & 3 \\ 1 & 5 & 2 \\ 3 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 41 & 76 & 32 \\ 16 & 36 & 15 \\ 20 & 33 & 14 \end{bmatrix}$$

$$A^3 = A \times A^2 = \begin{bmatrix} 377 & 731 & 307 \\ 161 & 322 & 135 \\ 175 & 333 & 140 \end{bmatrix}$$

None of the given option is correct.

14. Correct option: (B)

Given that

$$A = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, B = [2 \quad 3 \quad 2]$$

Then

Using $(A+B)^T = A^T + B^T \; for \; A, B \; of \; same \; type$

$$(A + B^T)^T = A^T + B^{T^T} = A^T + B \; as (A, B^T \; has \; same \; type)$$

$$(A + B^T)^T = A^T + B$$

So we can write

$$= [1 \quad 0 \quad -1] + [2 \quad 3 \quad 2] = [3 \quad 3 \quad 1]$$

Then the correct option is B)

15. Correct option: (A)

Given that

$$A + 2B = \begin{bmatrix} 2 & -4 \\ 1 & 6 \end{bmatrix}, A^T + B^T = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$$

Then

Using $(A+B)^T = A^T + B^T \; for \; A, B \; of \; same \; type$

$$A^T + B^T = (A + B)^T = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$$

$$((A + B)^T)^T = A + B = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}^T = \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix}$$

So we can write

$$(A + 2B) = (A + B) + B = \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix} + B$$

Therefore

$$\begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix} + B = \begin{bmatrix} 2 & -4 \\ 1 & 6 \end{bmatrix} => B = \begin{bmatrix} 2 & -4 \\ 1 & 6 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 1 & -4 \\ -1 & 7 \end{bmatrix}$$

So we can conclude

$$A = \begin{bmatrix} 2 & -4 \\ 1 & 6 \end{bmatrix} - 2B = \begin{bmatrix} 2 & -4 \\ 1 & 6 \end{bmatrix} - \begin{bmatrix} 2 & -8 \\ -2 & 14 \end{bmatrix} = \begin{bmatrix} 0 & 4 \\ 3 & -8 \end{bmatrix}$$

Then the correct option is A)

16. Correct option: (A)

Given that

$$A = \begin{bmatrix} 1 & 5 & 3 \\ 2 & 4 & 0 \\ 3 & -1 & -5 \end{bmatrix}, B = \begin{bmatrix} 2 & -1 & 0 \\ 0 & -2 & 5 \\ 1 & 2 & 0 \end{bmatrix}$$

Then

$$3A - 4B^T$$

Here

$$B^T = \begin{bmatrix} 2 & -1 & 0 \\ 0 & -2 & 5 \\ 1 & 2 & 0 \end{bmatrix}^T = \begin{bmatrix} 2 & 0 & 1 \\ -1 & -2 & 2 \\ 0 & 5 & 0 \end{bmatrix}$$

Therefore

$$3A - 4B^T = \begin{bmatrix} 3 & 15 & 9 \\ 6 & 12 & 0 \\ 9 & -3 & -15 \end{bmatrix} - \begin{bmatrix} 8 & 0 & 4 \\ -4 & -8 & 8 \\ 0 & 20 & 0 \end{bmatrix} = \begin{bmatrix} -5 & 15 & 5 \\ 10 & 20 & -8 \\ 9 & -23 & -15 \end{bmatrix}$$

Then the correct option is A)

17. Correct Matching is: a-S, b-P, c-T, d-Q

a) $\begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix} \times \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$

Using matrix multiplication and $\sqrt{-1} = i \Rightarrow i^2 = -1$

$$= \begin{bmatrix} 0 & i^2 \\ -i^2 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

a) -→(S)

b) $\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

Using matrix multiplication

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

b) -→(P)

c) $\begin{bmatrix} 1 & 4 \\ 2 & 9 \end{bmatrix} \times \begin{bmatrix} 9 & -4 \\ -2 & 1 \end{bmatrix}$

Using matrix multiplication

$$= \begin{bmatrix} 9-8 & -4+4 \\ 18-18 & -8+9 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

c) -→(T)

d) $\begin{bmatrix} 1 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 2 & 1 \\ 1 & 0 \\ 3 & 2 \end{bmatrix}$

Using matrix multiplication

$$= \begin{bmatrix} 5 & 3 \end{bmatrix}$$

d) -→(Q)

18. Correct Answer is: (2)

Given that

$$\begin{bmatrix} 3 & -2 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 11 \\ -5 \end{bmatrix}$$

Then we have to find $3x + 7y =$

Solving

$$\begin{bmatrix} 3 & -2 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3x-2y \\ 2y-x \end{bmatrix}$$

Therefore

$$\begin{bmatrix} 3x-2y \\ 2y-x \end{bmatrix} = \begin{bmatrix} 11 \\ -5 \end{bmatrix} => 3x-2y = 11, 2y-x = -5$$

Solving both of the equation

$$x = 3, y = -1$$

So value of

$$3x + 7y = 9 - 7 = 2$$

19. Correct Answer is: (3)

Given that $5A = \begin{bmatrix} 3 & -4 \\ 4 & x \end{bmatrix}, AA^T = A^T A = I$

Then we have to find

$$x =$$

Solving

$$A = \begin{bmatrix} \frac{3}{5} & -\frac{4}{5} \\ \frac{4}{5} & \frac{x}{5} \end{bmatrix}$$

Therefore

$$A^T = \begin{bmatrix} \frac{3}{5} & \frac{4}{5} \\ \frac{-4}{5} & \frac{x}{5} \end{bmatrix}$$

So we can write

$$\begin{bmatrix} \frac{3}{5} & -\frac{4}{5} \\ \frac{4}{5} & \frac{x}{5} \end{bmatrix} \times \begin{bmatrix} \frac{3}{5} & \frac{4}{5} \\ \frac{-4}{5} & \frac{x}{5} \end{bmatrix} = \begin{bmatrix} 1 & \frac{12}{25} - \frac{4x}{25} \\ \frac{12}{25} - \frac{4x}{25} & \frac{16+x^2}{25} \end{bmatrix} = I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Therefore

$$12 - 4x = 0 => x = 3$$

20. Correct Answer is: (9)

Given that $trace(A) = 3$

Then we have to find the $trace(3A)$

As we know that trace of a matrix is sum of its diagonal elements so sum of diagonal elements become 9 for a matrix if matrix is multiplied by 3.

Therefore the new trace of the matrix 3A=9

STUDENT PRACTICE QUESTIONS

1. Correct option: (A)

Given that

$$A + B = \begin{bmatrix} 2 & 3 & 1 \\ 6 & -1 & 5 \end{bmatrix}, A - B = \begin{bmatrix} 2 & 1 & -1 \\ 0 & -1 & 3 \end{bmatrix}$$

We have to find the matrix B

$$B = \frac{(A+B) - (A-B)}{2}$$

$$(A+B) - (A-B) = \begin{bmatrix} 0 & 2 & 2 \\ 6 & 0 & 2 \end{bmatrix}$$

Therefore B

$$\frac{(A+B)-(A-B)}{2}=\frac{1}{2}\begin{bmatrix}0 & 2 & 2\\ 6 & 0 & 2\end{bmatrix}=\begin{bmatrix}0 & 1 & 1\\ 3 & 0 & 1\end{bmatrix}$$

Then the correct option is A)

2. Correct option: (C)

Given that $$A=\begin{bmatrix}ab & b^2\\ -a^2 & -ab\end{bmatrix}$$

We have to find the value of A^2

Using Row-Column multiplication

$$A^2=\begin{bmatrix}ab & b^2\\ -a^2 & -ab\end{bmatrix}\times\begin{bmatrix}ab & b^2\\ -a^2 & -ab\end{bmatrix}$$
$$=\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$$

Therefore

$$A^2=\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$$

Then the correct option is C)

3. Correct option: (C)

Given that

$$A=\begin{bmatrix}0 & c & -b\\ -c & 0 & a\\ b & -a & 0\end{bmatrix},B=\begin{bmatrix}a^2 & ab & ac\\ ab & b^2 & bc\\ ac & bc & c^2\end{bmatrix}$$

We have to find the value of $A\times B$

Using Row-Column multiplication

$$A\times B=\begin{bmatrix}0 & c & -b\\ -c & 0 & a\\ b & -a & 0\end{bmatrix}\times\begin{bmatrix}a^2 & ab & ac\\ ab & b^2 & bc\\ ac & bc & c^2\end{bmatrix}$$
$$=\begin{bmatrix}0+abc-abc & 0+cb^2-b^2c & 0+bc^2-bc^2\\ -ca^2+0+a^2c & -abc+0+abc & -ac^2+0+ac^2\\ a^2b-a^2b+0 & ab^2-ab^2+0 & abc-abc+0\end{bmatrix}=\begin{bmatrix}0 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}=O$$

Therefore

$$A\times B=O$$

Then the correct option is C)

4. Correct option: (A)

Given that

$$A=\begin{bmatrix}-1 & 0\\ 0 & 2\end{bmatrix}$$

We have to find the value of A^3-A^2

Using Row-Column multiplication we can calculate

$$A^2=\begin{bmatrix}-1 & 0\\ 0 & 2\end{bmatrix}\times\begin{bmatrix}-1 & 0\\ 0 & 2\end{bmatrix}=\begin{bmatrix}1 & 0\\ 0 & 4\end{bmatrix}$$
$$A^3=A\times A^2=\begin{bmatrix}-1 & 0\\ 0 & 2\end{bmatrix}\times\begin{bmatrix}1 & 0\\ 0 & 4\end{bmatrix}=\begin{bmatrix}-1 & 0\\ 0 & 8\end{bmatrix}$$

Therefore

$$A^3-A^2=\begin{bmatrix}-1 & 0\\ 0 & 8\end{bmatrix}-\begin{bmatrix}1 & 0\\ 0 & 4\end{bmatrix}=\begin{bmatrix}-2 & 0\\ 0 & 4\end{bmatrix}=2\begin{bmatrix}-1 & 0\\ 0 & 2\end{bmatrix}=2A$$

Then the correct option is A)

5. Correct option: (D)

Given that

$$A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$$

We have to find the value of A^n

Using Row-Column multiplication we can calculate

$$A^2 = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} \times \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 5 & -8 \\ 2 & -3 \end{bmatrix} = \begin{bmatrix} 1+2\times 2 & -4\times 2 \\ 2 & 1-2\times 2 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} \times \begin{bmatrix} 5 & -8 \\ 2 & -3 \end{bmatrix} = \begin{bmatrix} 1+2\times 3 & -4\times 3 \\ 3 & 1-2\times 3 \end{bmatrix}$$

So we can conclude that the

$$A^n = \begin{bmatrix} 1+2n & -4n \\ n & 1-2n \end{bmatrix}$$

Therefore

$$A^n = \begin{bmatrix} 1+2n & -4n \\ n & 1-2n \end{bmatrix}$$

Then the correct option is D)

6. Correct option: (B)

Given that

$$A = \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix}$$

Using Row-Column multiplication we can calculate

$$A^2 = \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix} \times \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

As $A^2 = I$

So we can conclude that the given matrix is Involuntary matrix.

Then the correct option is B)

7. Correct option: (A)

Given that $A = square\ matrix$

Let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Then

$$A' = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

Therefore

$$A + A' = \begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} 2a & c+b \\ b+c & 2d \end{bmatrix}$$

$$A - A' = \begin{bmatrix} a & b \\ c & d \end{bmatrix} - \begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} 0 & b-c \\ c-b & 0 \end{bmatrix}$$

$$AA' = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \times \begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} a^2+b^2 & ac+bd \\ ac+bd & c^2+d^2 \end{bmatrix}$$

So we can conclude that

$$A + A' = (A + A')^T, (A - A') = -(A - A')^T$$

Therefore $A + A'$ is symmetric matrix.

Then the correct option is A)

8. Correct option: (C)

Given that $A = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$

Then

$$A^2 = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix} \times \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 9+2 & 6+2 \\ 3+1 & 2+1 \end{bmatrix} = \begin{bmatrix} 11 & 8 \\ 4 & 3 \end{bmatrix}$$

$$aA = \begin{bmatrix} 3a & 2a \\ a & a \end{bmatrix}$$

$$bI = \begin{bmatrix} b & 0 \\ 0 & b \end{bmatrix}$$

Therefore

$$A^2 + aA + bI = \begin{bmatrix} 11 & 8 \\ 4 & 3 \end{bmatrix} + \begin{bmatrix} 3a & 2a \\ a & a \end{bmatrix} + \begin{bmatrix} b & 0 \\ 0 & b \end{bmatrix} = \begin{bmatrix} 11+3a+b & 8+2a \\ 4+a & 3+a+b \end{bmatrix}$$

Given

$$A^2 + aA + bI = 0 => \begin{bmatrix} 11+3a+b & 8+2a \\ 4+a & 3+a+b \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$=> 4 + a = 0 => a = -4, 3 + a + b = 0 => 3 - 4 + b = 0 => b = 1$$

So we can conclude that

$$a = -4, b = 1$$

Then the correct option is C)

9. Correct option: (B)

Given that $A = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$

Then

$$A^2 = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \times \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$$

$$A^3 = A^2 \times A = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$$

Therefore

$$A^n = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$$

So $A + A^2 + A^3 + \cdots + A^n = A + A + .. (n\ terms) = nA$

Then the correct option is B)

10. Correct option: (A)

Given that

$$A = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}, B = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, C = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}, i = \sqrt{-1}\ or\ i^2 = -1$$

Then

$$A^2 = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix} \times \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix} = \begin{bmatrix} i^2 & 0 \\ 0 & i^2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = -I$$

$$B^2 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = -I$$

$$C^2 = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix} \times \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix} = \begin{bmatrix} i^2 & 0 \\ 0 & i^2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = -I$$

Therefore

$$A^2 = B^2 = C^2 = -I$$

Then the correct option is A)

11. Correct option: (A, B, D)

Given that

$$A = \begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$

Then

$$trace(A) = sum\ of\ diagonal\ elements = -1$$

$$A^2 = \begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^3 = A \times A^2 = \begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$

$$A^n = A(n = odd), A^n = I(n\ is\ even)$$

Therefore we can write

$$A^{2012} = I => trace(A^{2012}) = 3$$

As we prove that

$$A^2 = I$$

Then A will be involuntary matrix.

Then the correct option is A, B, D)

12. Correct option: (B, C)

Given that

$$[x \quad 4 \quad -1] \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 2 \\ 0 & 2 & 4 \end{bmatrix} \begin{bmatrix} x \\ 4 \\ -1 \end{bmatrix} = 0$$

Then

$$[x \quad 4 \quad -1] \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 2 \\ 0 & 2 & 4 \end{bmatrix} = [2x + 4 \quad x - 2 \quad 4]$$

$$[2x + 4 \quad x - 2 \quad 4] \begin{bmatrix} x \\ 4 \\ -1 \end{bmatrix} = [2x^2 + 4x + 4x - 8 - 4] = [2x^2 + 8x - 12]$$

So we can write

$$[2x^2 + 8x - 12] = [0] => x^2 + 4x - 6 = 0$$

Using quadratic formula

$$x = \frac{-4 \pm \sqrt{40}}{2} = -2 \pm \sqrt{10}$$

Then the correct option is B, C)

13. Correct option: (B)

Given that

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, A^2 = O$$

Then

$$A^2 - (a + d)A + (ad - bc)I$$

$$A^2 = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \times \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a^2 + bc & ab + bd \\ ac + cd & cb + d^2 \end{bmatrix}$$

$$(a+d)A = \begin{bmatrix} a^2+ad & ab+bd \\ ac+dc & d^2+ad \end{bmatrix}$$
$$(ad-bc)I = \begin{bmatrix} ad-bc & 0 \\ 0 & ad-bc \end{bmatrix}$$

Therefore

$$A^2-(a+d)A+(ad-bc)I =$$
$$= \begin{bmatrix} a^2+bc & ab+bd \\ ac+cd & cb+d^2 \end{bmatrix} - \begin{bmatrix} a^2+ad & ab+bd \\ ac+dc & d^2+ad \end{bmatrix} + \begin{bmatrix} ad-bc & 0 \\ 0 & ad-bc \end{bmatrix}$$
$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O$$

Then the correct option is B)

14. Correct option: (B)

Given that

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, A^2 = O$$

Then

$$A^2 = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \times \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a^2+bc & ab+bd \\ ac+cd & cb+d^2 \end{bmatrix} = O$$
$$c(a+d) = 0 => a+d = 0$$

Therefore

$$trace(A) = a+d = 0$$

Then the correct option is B)

15. Correct option: (D)

Given that

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, A^2 = O$$

Then

$$(I+A)^2 = (I+2A)$$
$$(I+A)^3 = (I+3A) => (I+A)^n = (I+nA)$$

Therefore

$$(I+A)^{100} = I+100A$$

Then the correct option is D)

16. Correct Matching is: a-R, b-P, c-P, d-P

a) $A = \begin{bmatrix} 1 & 1 & -2 \\ 0 & 1 & 4 \\ 2 & 3 & 1 \end{bmatrix} then\ A^3-3A^2-5A$

Using matrix multiplication

$$A^2 = \begin{bmatrix} 1 & 1 & -2 \\ 0 & 1 & 4 \\ 2 & 3 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 1 & -2 \\ 0 & 1 & 4 \\ 2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} -3 & -4 & 0 \\ 8 & 13 & 8 \\ 4 & 8 & 9 \end{bmatrix}$$
$$A^3 = \begin{bmatrix} 1 & 1 & -2 \\ 0 & 1 & 4 \\ 2 & 3 & 1 \end{bmatrix} \times \begin{bmatrix} -3 & -4 & 0 \\ 8 & 13 & 8 \\ 4 & 8 & 9 \end{bmatrix} = \begin{bmatrix} -3 & -7 & 0 \\ 24 & 45 & 44 \\ 22 & 39 & 33 \end{bmatrix}$$

Therefore

$$A^3-3A^2-5A =$$

$$= \begin{bmatrix} -3 & -7 & 0 \\ 24 & 45 & 44 \\ 22 & 39 & 33 \end{bmatrix} - 3\begin{bmatrix} -3 & -4 & 0 \\ 8 & 13 & 8 \\ 4 & 8 & 9 \end{bmatrix} - 5\begin{bmatrix} 1 & 1 & -2 \\ 0 & 1 & 4 \\ 2 & 3 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 5 & 10 \\ 0 & 6 & 20 \\ 10 & 15 & 6 \end{bmatrix} - 5\begin{bmatrix} 1 & 1 & -2 \\ 0 & 1 & 4 \\ 2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

a) -→(R)

b) $B = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix} then\ (B-2I)(B-3I)$

Therefore

$$B - 2I = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ -1 & -1 \end{bmatrix}$$

$$B - 3I = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix}$$

Therefore

$$(B-2I)(B-3I) = \begin{bmatrix} 2 & 2 \\ -1 & -1 \end{bmatrix} \times \begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O$$

b) -→(P)

c) $C = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix} then\ C(C-3I)(C-15I)$

Therefore

$$C - 3I = \begin{bmatrix} 5 & -6 & 2 \\ -6 & 4 & -4 \\ 2 & -4 & 0 \end{bmatrix}$$

$$C - 15I = \begin{bmatrix} -7 & -6 & 2 \\ -6 & -8 & -4 \\ 2 & -4 & -12 \end{bmatrix}$$

$$(C-3I)(C-15I) = \begin{bmatrix} 5 & -6 & 2 \\ -6 & 4 & -4 \\ 2 & -4 & 0 \end{bmatrix} \times \begin{bmatrix} -7 & -6 & 2 \\ -6 & -8 & -4 \\ 2 & -4 & -12 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 10 \\ 10 & 20 & 20 \\ 10 & 20 & 20 \end{bmatrix}$$

Therefore

$$C(C-3I)(C-15I) = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix} \times \begin{bmatrix} 5 & 10 & 10 \\ 10 & 20 & 20 \\ 10 & 20 & 20 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O$$

c) -→(P)

d) $D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix} then\ (D-I)(D-2I)(D-3I) =$

Therefore

$$D - I = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 2 \end{bmatrix}$$

$$D-2I=\begin{bmatrix}-1 & 0 & 0\\ 0 & 0 & 1\\ 1 & 0 & 1\end{bmatrix}$$

$$D-3I=\begin{bmatrix}-2 & 0 & 0\\ 0 & -1 & 1\\ 1 & 0 & 0\end{bmatrix}$$

Therefore

$$(D-I)(D-2I)(D-3I)=$$

$$=\begin{bmatrix}0 & 0 & 0\\ 0 & 1 & 1\\ 1 & 0 & 2\end{bmatrix}\times\begin{bmatrix}-1 & 0 & 0\\ 0 & 0 & 1\\ 1 & 0 & 1\end{bmatrix}\times\begin{bmatrix}-2 & 0 & 0\\ 0 & -1 & 1\\ 1 & 0 & 0\end{bmatrix}$$

$$=\begin{bmatrix}0 & 0 & 0\\ 1 & 0 & 2\\ 1 & 0 & 2\end{bmatrix}\times\begin{bmatrix}-2 & 0 & 0\\ 0 & -1 & 1\\ 1 & 0 & 0\end{bmatrix}=\begin{bmatrix}0 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}=O$$

d) -→(P)

17. Correct Answer is: (0)

Given that

$$A=\begin{bmatrix}x & 1\\ 1 & 0\end{bmatrix}, A^2=I$$

Then we have to find

$$x=$$

Solving

$$=\begin{bmatrix}x & 1\\ 1 & 0\end{bmatrix}\times\begin{bmatrix}x & 1\\ 1 & 0\end{bmatrix}=\begin{bmatrix}x^2+1 & x\\ x & 1\end{bmatrix}$$

Therefore

$$\begin{bmatrix}x^2+1 & x\\ x & 1\end{bmatrix}=I=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}\Rightarrow x=0$$

So value of

$$x=0$$

18. Correct Answer is: (3)

Given that

$$A=\begin{bmatrix}1 & 2 & 3\\ 2 & 4 & 5\\ x & 5 & 6\end{bmatrix}, A^T=A$$

Then we have to find

$$x=$$

Solving

$$A^T=\begin{bmatrix}1 & 2 & x\\ 2 & 4 & 5\\ 3 & 5 & 6\end{bmatrix}$$

Therefore

$$\begin{bmatrix}1 & 2 & 3\\ 2 & 4 & 5\\ x & 5 & 6\end{bmatrix}=\begin{bmatrix}1 & 2 & x\\ 2 & 4 & 5\\ 3 & 5 & 6\end{bmatrix}\Rightarrow x=3$$

So value of

$$x=3$$